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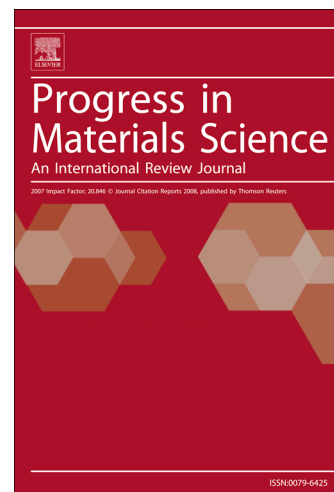
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Magnetocaloric effect: from materials research to refrigeration devices

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The magnetocaloric effect and its most straightforward application, magnetic refrigeration, are topics of current interest due to the potential improvement of energy efficiency of cooling and temperature control systems, in combination with other environmental benefits associated to a technology that does not rely on the compression/expansion of harmful gases. This review presents the fundamentals of the effect, the techniques for its measurement with consideration of possible artifacts found in the characterization of the samples, a comprehensive and comparative analysis of different magnetocaloric materials, as well as possible routes to improve their performance. An overview of the different magnetocaloric prototypes found in literature as well as alternative applications of the magnetocaloric effect for fundamental studies of phase transitions are also included.

Contents

1. Introduction	4
2. Thermodynamics of the Magnetocaloric Effect	8
2.1. Thermodynamic relations	8
2.2. Figures of merit of the magnetocaloric effect	12
2.3. Thermodynamic cycles	14
2.3.1. Carnot cycle	14
2.3.2. Ericsson cycle	15
2.3.3. Brayton cycle	16
2.3.4. AMR (Active Magnetic Regenerator) cycle	16
3. Experimental methods for the characterization of the magnetocaloric effect	17
3.1. Direct measurements	18
3.2. Indirect measurements	25
3.3. Dynamic magnetocaloric effect	30
4. Models of the magnetocaloric effect	32
4.1. Thermodynamic models	33
4.2. First principle models	35
5. Magnetocaloric materials	36
5.1. Crystalline materials	36
5.1.1. Rare earth (RE) containing crystalline materials	36
5.1.2. Rare earth free crystalline materials	116
5.2. Amorphous materials	170
5.2.1. Rare earth based amorphous materials	171
5.2.2. Transition metal based amorphous materials	173
6. Enhancing the magnetocaloric effect through material processing techniques	177
6.1. Multiphase materials and composites	178
6.2. Influence of fabrication techniques	179
6.2.1. Rapid solidification	179
6.2.2. Powder metallurgy	180
6.2.3. Liquid reactive sintering and Thermally Decomposition and Recombination process	181
6.3. Nanostructures	181
6.3.1. Nanocrystalline materials	182
6.3.2. Magnetocaloric nanoparticles	183
6.3.3. Nanostructured films (or thin films)	185
6.3.4. Self-assembled nanostructures	187
6.4. Special treatment conditions	188
6.4.1. Hydrogenation	188
6.4.2. Annealing	189
6.4.3. Porous/metal foam structuring	190
7. Fundamental studies in magnetism using magnetocaloric characterization	191
7.1. Critical phenomena	191
7.2. Analysis of the order of phase transitions	193
8. Refrigerator devices	194
8.1. Low temperature devices	196
8.2. Devices working close to room temperature	206
8.3. General considerations about system design	224

8.4. Other caloric refrigeration technologies	225
9. Conclusions and future outlook.....	227
10. Acknowledgements	228
11. References.....	238

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1. Introduction

Energy efficiency and sustainability are priority topics for most research funding agencies worldwide and are recurrent topics in different mass media. In fact, according to data from Lawrence Livermore National Laboratory 61 % of the estimated USA energy consumption in 2015 corresponds to rejected energy, while only the remaining 39 % is actually employed in energy services. Values are comparable for other developed countries, like 63% rejected energy in 2011 in UK or Spain. These data indicate that it is not only necessary to focus on the primary energy source and avoid our dependence on non-renewable energy sources but also to dedicate large research efforts for improving the efficiency of energy conversion. In particular, most renewable energy sources have to be converted into electricity before their final use, and in USA 67% of the conversion process in 2015 resulted in wasted energy [1]. In many cases, magnetic materials play a relevant role in energy conversion into electricity. This constitutes a driving force for the improvement of magnetic materials for energy applications [2], with some current examples covered in a recent Viewpoint Set of Scripta Materialia [3].

Among the different final uses of energy in the residential and commercial sectors, refrigeration and air conditioning account for a relevant fraction of electricity use, with numbers varying from country to country due to their different climate. Recent data from EIA [4] estimate that 87% of USA households are furnished with air-conditioners and ~114 million units account for an annual energy demand of 186 billion kWh of electricity. There is no doubt that our way of life relies on our capability of cooling food and controlling the temperature of our living and working environments. If developing countries adopt similar trends in cooling habits, there can be a 50-fold increase in the

demand of air conditioners [5]. In the European Union, heating and cooling in buildings and industry account for half of the EU's energy consumption and in order to fulfil the EU's climate and energy goals, the heating and cooling sector must drastically reduce their energy consumption and cut their use of fossil fuels [6]. Magnetic refrigeration can play a relevant role in this effort to improve the energy efficiency of temperature and climate control. It is expected that these refrigerators based on magnetocaloric effect (MCE) will be energetically more efficient than the current ones based on the compression and expansion of gasses.

In fact, magnetic refrigeration is a timely topic of research. The discovery of magnetic materials that exhibit a remarkable change in their temperature when they are adiabatically magnetized close to room temperature [7] has produced a surge in the number of publications on MCE in the last two decades. Simultaneously, as these materials enable the possibility of designing magnetic refrigerators with operation temperatures close to room temperature, there has been a noticeable increase in the development of magnetic refrigerator prototypes [8]. This scientific and technological interest is mainly due to the environmental friendliness of magnetic refrigeration, which is energetically more efficient than the conventional refrigerator devices and does not require the use of ozone depleting or greenhouse effect related gasses. Room temperature magnetic refrigeration had been just a promise since Brown designed and built the first successful magnetic refrigerator prototype [9], with devices restricted to research laboratories. However, the situation has begun to change, as a domestic appliance of a wine cooler based on this principle of operation, developed by Haier, Astronautics Corporation of America and BASF, was presented in 2015 edition of the Las Vegas Consumer Electronics Show (CES2015) [10].

As this is a topic that encompasses material science and current trends in technology, we are convinced that a comprehensive overview, which analyzes materials properties in combination with the main characteristics of the models representing them and the basic operation principles of the devices, will facilitate the search of better materials with enhanced applicability. This will also enable technologists to have the tools for appropriate material selection and implementation of the appropriate models of materials properties for their simulations.

We will start this review with an overview of the thermodynamics associated to the MCE and its different thermodynamic cycles that can be used in magnetic refrigeration, since these two aspects justify the strategies used in the search for new materials. Subsequently, the experimental characterization techniques will be presented, making a distinction between those that directly measure the adiabatic temperature change or the isothermal magnetic entropy change, and those that infer these properties from indirect measurements. In this aspect, it is extremely important to clarify and, to some extent, quantify the different artifacts that can appear during the measurement procedure and analysis of the results. Such information relating the attributions of these artifacts allows experimentalists to reconcile the results obtained from direct and indirect techniques. In addition, it is important to know the different models that can represent these properties and this will be presented as two main types: first principles modeling and thermodynamic models. The revision of current magnetocaloric materials will be categorized into their microstructures and compositions. There could be a third classification criterion based on the order of the phase transition but it would not be compatible with compositional classification because minor stoichiometric changes in the materials might alter the nature of their phase transitions. Crystalline and amorphous materials will be considered independently as their properties differ in a great extent

even qualitatively. Within these families, we will also classify the materials as rare earth containing and rare earth free, in agreement with the current concerns to avoid the use of strategic materials in energy-related applications. It will also be shown that materials fabrication and processing techniques, which enable the presence of multiple phases, refinement of the microstructure or alteration in the stoichiometry of the phases, will result in different, sometimes optimized, magnetocaloric properties. Next, it is worth dedicating a part of the review to present an alternative application of the magnetocaloric characterization of materials. This includes performing fundamental studies regarding the phase transitions involved, determining critical exponents and establishing the order of the phase transition even in cases for which purely magnetic methods are not applicable. The last part of the review will be devoted to the main application of the magnetocaloric effect, which has been the reason for its enormous interest nowadays: magnetic refrigeration. A comparison of existing refrigerator prototypes is presented placing the current materials' research in a proper context: devices need good magnetocaloric materials and magnetocaloric research would not have attained its current level without the possibility of implementing magnetic refrigerators. Conclusions and future outlook are then presented.

There have been previous review papers on magnetocalorics [2, 8, 11-21]. Some of them dealt with a specific aspect of the topic (theoretical models, prototypes, operation temperature close to room temperatures, etc.); others focus on specific types of materials. In particular, a recent book focuses mostly on magnetocaloric energy conversion and the principle of operation of different devices [22], which makes a difference with previous books, more dedicated to materials science [23]. We will refer to these reviews when appropriate. However, for the sake of completeness we will

compile the main information of the different families of magnetocaloric materials, models and methods in the present review.

2. Thermodynamics of the Magnetocaloric Effect

Although books on magnetocaloric effect already include an introduction on thermodynamics [22, 23], it is necessary to include in this review a brief description of thermodynamic relations and how to define the magnetocaloric response of a material from them. In addition, the different figures of merit that allow us to compare materials performance have to be defined, as they will be used throughout this review. This chapter ends with a succinct description of the different thermodynamic cycles that can be used in magnetic refrigerator devices, as those will ultimately control the way in which materials are used.

2.1. Thermodynamic relations

The magnetic effect is introduced in thermodynamic systems via the performed magnetic work, using the principles of energy conservation:

$$dU = dQ - dW \quad (1)$$

where U , Q and W are the internal energy, the heat flux and the performed work, respectively (d denotes that the heat flux and work are not state functions). In a quasistatic process, the different work terms are expressed as functions of different intensive magnitudes, Y_i , and its extensive counterpart, X_i , in the form of:

$$dW = \sum_i \pm Y_i dX_i \quad (2)$$

where the index i indicates the various types of work performed on the system. For example, mechanical work is expressed as $\delta W_{mec} = p dV$ (where p is pressure and V is volume) and magnetic work, $\delta W_{mag} = -\mu_0 H dm$ (where H is magnetic field, μ_0 is the free-space magnetic permeability and m is the magnetic moment). Considering a quasistatic heat flux ($\delta Q = T dS$, where T is temperature and S is entropy) in a closed system, the energy conservation principle can be expressed as:

$$dU = T dS - p dV + \mu_0 H dm \quad (3)$$

where the internal energy is a function of entropy, volume and magnetic moment: $U(S, V, m)$. Legendre transformations can be used to obtain different thermodynamic potentials as a function of other parameters. Normally, the experimental studies are performed at constant pressure while varying the temperature and the magnetic field, leading to Gibbs free energy as the most useful potential:

$$G(T, p, H) = U - TS + pV - m\mu_0 H \quad (4)$$

and its differential form:

$$dG = V dp - S dT - m\mu_0 dH. \quad (5)$$

From this equation, entropy and magnetic moment can be obtained as:

$$S = - \left(\frac{\partial G}{\partial T} \right)_{p, H} \quad (6)$$

$$m = - \frac{1}{\mu_0} \left(\frac{\partial G}{\partial H} \right)_{p, T}. \quad (7)$$

The equality in the second derivate (Maxwell relations) gives the following relation:

$$\left(\frac{\partial S}{\partial H}\right)_{p,T} = \mu_0 \left(\frac{\partial m}{\partial T}\right)_{p,H}. \quad (8)$$

This thermodynamic introduction can be further extended to obtain an analytical expression for MCE, which is defined as the reversible temperature change produced upon a magnetic field change in an adiabatic process. Hence, if we express the entropy as a function of magnetic field and temperature:

$$dS = \left(\frac{\partial S}{\partial T}\right)_{p,H} dT + \left(\frac{\partial S}{\partial H}\right)_{p,T} dH \quad (9)$$

and considering $dS = 0$, we get:

$$dT = -\mu_0 \left[\left(\frac{\partial S}{\partial T}\right)_{p,H} \right]^{-1} \left(\frac{\partial S}{\partial H}\right)_{p,T} dH. \quad (10)$$

From the definition of the heat capacity (C) at a constant magnitude x (in our case $x = p, H$):

$$C_x = \left(\frac{dQ}{dT}\right)_x = T \left(\frac{dS}{dT}\right)_x \quad (11)$$

and combined with the Maxwell relation (8), we obtain:

$$dT = -\mu_0 \frac{T}{C_{p,H}} \left(\frac{\partial m}{\partial T}\right)_{p,H} dH \quad (12)$$

while the total temperature change due to a magnetic field change is:

$$\Delta T_{ad} = -\mu_0 \int_{H_I}^{H_F} \frac{T}{C_{p,H}} \left(\frac{\partial m}{\partial T}\right)_{p,H} dH \quad (13)$$

where H_F and H_I are the final and initial magnetic fields, respectively. Although it might seem that the temperature change of the sample might depend on its total magnetic moment, we have to notice that heat capacity is also an extensive variable, which eliminates the dependence of ΔT_{ad} on the size of the sample. In addition, MCE is also commonly characterized as the magnetic entropy change due to magnetic field change in an isothermal process ($dT = 0$). Then (9) takes the form of:

$$dS = \left(\frac{\partial S}{\partial H} \right)_{p,T} dH = \mu_0 \left(\frac{\partial m}{\partial T} \right)_{p,H} dH \quad (14)$$

where the total magnetic entropy change due to a magnetic field change is expressed as:

$$\Delta S_M = \mu_0 \int_{H_I}^{H_F} \left(\frac{\partial m}{\partial T} \right)_{p,H} dH, \quad (15)$$

where the sign of the derivative determines the sign of ΔS_M . The case in which ΔS_M is negative is known as direct MCE while if it is positive it is denoted as inverse MCE.

Furthermore, magnetocaloric magnitudes can be correlated to heat capacity (11) by expressing the total entropy at a constant magnetic field as:

$$S_H(T) = S_0 + \int_0^T \frac{C_{p,H}}{T} dT. \quad (16)$$

where S_0 is the zero entropy term. From this, the magnetocaloric magnitudes are obtained as [24, 25]:

$$\Delta S_M(T, \Delta H) = [S_{H_F}(T) - S_{H_I}(T)]_T \quad (17)$$

$$\Delta T_{ad}(T, \Delta H) = [T_{H_F}(S) - T_{H_I}(S)]_S \quad (18)$$

wherein the entropy curves have to be inverted as $T_H(S)$ to calculate the adiabatic temperature change.

It can be observed in the previous expressions of MCE (equations (13) and (15)) that its maximum appears when an abrupt change of magnetization exist, i.e. the region of interest will be close to a phase transition, either of first or second order type, which implies a change in the magnetic moment.

2.2. Figures of merit of the magnetocaloric effect

In order to find magnetocaloric materials with optimal properties, we need to define the figures of merit used for the comparison. The most intuitive one is the adiabatic temperature change, ΔT_{ad} , which is the temperature change of the material upon the adiabatic application of magnetic field, eq. (13). However, magnetic entropy change, ΔS_M , eq. (15) is easier to determine though less straightforward than the former magnitude.

The amount of heat that can be transferred between the cold and hot reservoirs, namely refrigerant capacity (RC) [26] (or relative cooling power by other authors, RCP) is defined as:

$$RC(H) = \int_{T_{cold}}^{T_{hot}} \Delta S_M(T, H) dT, \quad (19)$$

where T_{cold} and T_{hot} are the temperatures of the reservoirs. Different approximations in the calculation of the integral give rise to the diverse definitions of RC available in the literature [27, 28]. One typical approximation is that the temperature span between the reservoirs corresponds to the full width at half maximum (FWHM) of the peak, which could lead to numerous problems. For example, for extremely broad but short peaks

their corresponding RC values would be large despite that such a shallow peak cannot be used for practical applications. If the peak magnitude of the adiabatic temperature change or the magnetic entropy change are too small, their RC would have no meaning from the practical point of view regardless of its practical definition. RC_{FWHM} approximates the integral by the product of the peak entropy change, ΔS_M^{pk} , times the full width at half maximum, δT_{FWHM} ; RC_{Area} corresponds to the calculation of the area under the curve; and RC_{WP} corresponds to the area of the largest rectangle, which can be inscribed under the curve [26]. In cases when hysteretic losses are relevant and that will make the material heat up, a net RC upon the subtraction of these losses is required [28].

In addition, there are other figures of merit proposed for magnetocaloric materials, whereby magnetocaloric magnitudes are referred to the materials' cost [29]. One very recent suggestion is the coefficient of refrigerant performance, CRP , which compares the net work in a reversible cycle with the positive work on the refrigerant [30]. This implies a ratio of the reversible refrigerant capacity divided by the magnetic work performed on the magnetic refrigerant, which requires the knowledge of the magnetization curve at the transition temperature. Though this metric is highly advantageous since it removes the effect of hysteresis and it is adimensional, it is usually not possible to calculate it from literature data as the experimental magnetization curves are less commonly included in publications (or not with large enough resolution). Therefore, unlike RC , which can be mostly calculated from literature values, CRP requires additional measurements in order to evaluate it. Hence, this review covers material comparisons based on the typical magnitudes of ΔS_M^{pk} , ΔT_{ad} and RC , and not CRP .

2.3. Thermodynamic cycles

In general, a magnetic refrigerator uses a magnetocaloric material to absorb the heat of a low temperature load (cold exchanger) and discharges heat to a high temperature sink (hot exchanger). In turn, the load is cooled with repetitive cycles of these processes. These refrigerators usually consist of a working material (magnetocaloric), a variable magnetic field generator system, hot and cold heat exchangers and a heat transfer system (using fluid or gas). Moreover, a regenerator can be included in the device. A regenerator is a thermal device that transfer heat between the different parts of the cycle, increasing the temperature span of the refrigerator. The regenerator alternatively absorbs and releases heat from/to the heat transfer fluid during different stages of the process. The regenerator is an important component of magnetic refrigerators devices due to the low temperature change of the MCE with respect to the conventional gas systems.

The refrigerators operate using the combination of various thermodynamic processes such as adiabatic, isothermal and isofield magnetizations. Consequently, the principal cycles used in a magnetic refrigerator include the Carnot, Ericsson, Brayton or AMR (active magnetic regenerator) cycles.

2.3.1. Carnot cycle

The Carnot cycle is the paradigmatic cycle for magnetic refrigeration as its processes involve the magnitudes used to characterize the MCE of the material (ΔS_M and ΔT_{ad}).

In addition, it is the most efficient cycle working between two thermal sources. The cycle consists of two adiabatic and two isothermal processes (represented in Figure 1-a).

While ramping to the maximum applied magnetic field, two processes occur (1→2): at (1), the temperature of the magnetic refrigerant increases from T_{cold} to T_{hot} under adiabatic conditions and subsequently at (2), the material isothermally exchanges heat

with the high temperature sink, $Q_{hot} = T_{hot} \Delta S_M (T_{hot})$. Next, the field is decreased to its initial value, leading to a decreased temperature in the material (from T_{hot} to T_{cold}) during an adiabatic process at (3) and then a following heat absorption from the low temperature load at (4), $Q_{cold} = T_{cold} \Delta S_M (T_{cold})$.

In addition, the coefficient of performance (COP), i.e. the ratio between the refrigeration capacity, $\dot{Q}_{cold}$, and the power input, $\dot{W}$, (heat lifted/work done), when expressed in the Carnot cycle corresponds to:

$$\text{COP}_{\text{Carnot}} = \frac{T_{cold}}{T_{hot} - T_{cold}}. \quad (20)$$

The efficiency (η) of a refrigerator is defined as the ratio of its COP to the Carnot COP.

2.3.2. Ericsson cycle

The Ericsson cycle consists of two isothermal and two isofield processes (represented in Figure 1-b). During the isofield processes, heat regeneration is required. At (1), the magnetocaloric material is subjected to an increased magnetic field isothermally whereby its heat is discharged to the high temperature sink. Then at constant high field at (2), the sample is cooled down to T_{cold} by the heat absorption of the regenerator. Subsequently at (3), the material is demagnetized isothermally, absorbing heat from the load. Finally, the regenerator discharges heat to the material at (4), increasing the temperature of the material and closing the cycle.

For an ideal Ericsson cycle, if ΔS_M is a constant value ($S_H(T)$ curves are parallel), the performance of the cycle is optimal (the second law of thermodynamics is fulfilled as an equality and the maximum efficiency is attained).

2.3.3. Brayton cycle

The Brayton cycle consists of two adiabatic and two isofield processes (represented in Figure 1-c). Under adiabatic conditions at (1), a temperature change of the material is induced by varying the magnetic field. Subsequently, at constant magnetic field in (2), the material placed in contact with thermal fluid decreases its temperature (further cooling is achieved using a regenerator though it is unnecessary). Then, the magnetocaloric material is demagnetized adiabatically at (3), leading to a decrease in its temperature. Subsequently at constant low field at (4), the temperature of the material increases while the material is in contact with the thermal fluid (4). The Brayton cycle has a lower refrigeration capacity but larger heat rejection in comparison to Ericsson cycle.

2.3.4. AMR (Active Magnetic Regenerator) cycle

In an AMR cycle, the magnetocaloric material operates as both the refrigerant and regenerator. The AMR cycle was introduced by Steyert [31] and with the exception of the Carnot cycle, is the most efficient one for magnetic refrigeration at room temperature. The transfer fluid is required to pass through the magnetocaloric material acting as a regenerator. The AMR cycle consists of two adiabatic and two isofield processes (during which the heat regeneration occurs). The material is placed in contact at the two extreme ends of the cold and hot exchangers at T_{cold} and T_{hot} , respectively. In the beginning, the transfer fluid is at the cold side at T_{cold} and as the material is adiabatically magnetized, its temperature at the cold side increases to $T_{cold} + \Delta T_{cold}$ while in the hot side to $T_{hot} + \Delta T_{hot}$. Next, the fluid flows through the material, increasing its temperature from T_{cold} to $T_{hot} + \Delta T_{hot}$ (hotter than the high temperature sink). Then, at the hot side, the fluid discharges heat to the sink decreasing temperature to T_{hot} . Subsequently, when the material is adiabatically demagnetized, its temperature

decreases to $T_{cold} - \Delta T_{cold}$ at the cold side and to $T_{hot} - \Delta T_{hot}$ at the hot side. Then, as the fluid passes through the material, its temperature reduces from T_{hot} to $T_{cold} - \Delta T_{cold}$ (lower than the temperature of the cold load). The fluid then absorbs heat from the cold load, increasing its temperature up to T_{cold} (closing the cycle).

3. Experimental methods for the characterization of the magnetocaloric effect

Typically, the characterization of MCE parameters are classified as direct and indirect methods according to how the parameters are obtained. One straightforward example of direct characterization techniques is a temperature sensor on the magnetocaloric material that is subjected to a magnetic field change, which directly determines the ΔT_{ad} of the sample. Contrastingly, indirect characterization methods require heat capacity and/or magnetization measurements data to determine ΔT_{ad} of the magnetocaloric material. Today, the indirect techniques are employed more commonly than the direct characterization methods, and one possible reason could be calorimeters and magnetometers are more commonly available in many laboratories (as compared to instrumentations measuring MCE parameters). However, the direct methods offer advantageous solutions to understand the performance of the magnetocaloric material, such as dynamic magnetocaloric properties of the material, namely dependence of $\Delta T_{ad}(T, H)$ on magnetic field change rate, which can only be obtained from the direct measurements. Furthermore, the challenges of accurately determining MCE in materials undergoing the first-order magnetic phase transition using indirect techniques still persist today. Current reported experimental apparatus, which characterized MCE parameters directly or indirectly, are discussed in the following pages.

3.1. Direct measurements

As indicated above, attractive candidates for promising magnetocaloric materials today are among those exhibiting first order magnetostructural transitions (FOPT) at transition temperatures near room temperatures. Their transition temperatures are field-dependent, hence leading to the possibility of inducing the transition by varying the magnetic fields, which produced large MCE properties (commonly termed as giant MCE). Nevertheless, the magnetostructural transitions are typically accompanied by magnetic and thermal hysteresis, raising reservations about their effective performances. Due to this hysteresis, the practice to appropriately determine the ΔS_M of FOPT materials from their magnetization data had raised many disagreements in regards to the equations used for computation as well as the way to perform the isothermal magnetization measurements (for further details, the reader is referred to the following section on indirect measurements). Therefore, the direct MCE measurements of such GMCE materials could serve as a better alternative to the open challenges in evaluating their material performances, for e.g. considering their dynamic properties and hysteretic influences.

The measurement of ΔT_{ad} had been recommended to offer a direct performance evaluation of magnetocaloric materials, instead of ΔS_M because the latter is an indirect and non-measurable parameter that only roughly estimates a quantitative MCE value [32]. Several types of experimental apparatus to directly measure ΔT_{ad} had been custom-built, and are typically classified depending on the procedure to apply the variable magnetic field to the sample (pulsed magnetic fields or static magnetic fields combined with moving samples), as well as the method for measuring the temperature change (either contact or non-contact, like the thermoacoustic method). The reader is referred to [23] for an overview of the most traditional methods.

Moreover, a commercial equipment for direct ΔT_{ad} measurements, named Magnetocaloric Measuring Setup (MMS), is available today and was also presented by Spichkin et al. (see refs. [33, 34]). The setup comprises of a Halbach permanent magnet field source (up to 1.7 T), ΔT measuring system, the measuring and control systems of both temperature and magnetic field, measuring insert (to hold the specimen), cryogenic Dewar and control computer. This allows the characterization of the dynamic mode of ΔT_{ad} , in addition to performing multi-cycling of the measurements during the periodic variation of the magnetic field, in order to mimic the protocol followed in real practical magnetic refrigerators. One of the challenges for measuring with this type of setup is the necessity of having a large thermal mass of the sample in comparison to the sample holder. In the case of low mass samples, like melt-spun ribbons, recalibration with a standard, with a similar shape, could be needed [35].

Another concern in the performance evaluation of magnetocaloric materials arises when they are applied in practical magnetic refrigerators and subjected to accumulative cycles of ΔH , which operates in alternating (AC) magnetic fields. As the efficiencies of the refrigerators improve by increasing the working cycle frequency, this leads to alarming concerns regarding the field frequency dependence of MCE as well as the evolution of heat exchange process. The latter could be resolved by increasing the surface area of the magnetocaloric refrigerant material (i.e. thin films or ribbons) to optimize the heat transfer with the heat exchange fluid. However, the requirement of a dynamic characterization of the magnetocaloric properties remains. An experimental setup to directly measure ΔT_{ad} in modulating small amplitude magnetic fields was developed, with the aim to rectify the conventional laborious measurements like using high power inputs from large magnetic fields as well as the challenges of accurately characterizing samples with small size and mass [36]. The configuration enabled ΔT_{ad} detection using

a lock-in amplifier and in small AC magnetic field generated by an electromagnet (up to 500 Oe and a frequency of 0.2-2 Hz). The authors then further developed another instrumental design to measure MCE in AC fields of frequencies up to 50 Hz, in order to investigate the AC field-frequency dependence of MCE [37]. This was attained by a custom-built alternating magnetic field source consisting of a set of four permanent magnets (0.62 T each) fixed onto a disk, which then, when rotated via stepper motor, generated AC magnetic fields with a relatively high frequency (up to 50 Hz).

As the influence of cyclic ΔH in experimental MCE data offers vital information about the material performance, it is important to evaluate that of FOPT materials, reason being their MCE can be larger during the first application of ΔH (or when characterized in quasistatic conditions) in comparison to the corresponding values measured in further runs of ΔH . For this reason, Skokov et al. developed a custom-built device to directly characterize the ΔT_{ad} of FOPT materials, with experimental protocols considering both the cyclic field dependence and temperature dependence of ΔT_{ad} [38]. The device configuration included Halbach-cylinder-configured permanent magnets (up to 1.93 T), type-T thermocouple directly in contact with the specimen and a passive adiabatic screen to minimize the radiative heat exchange between the specimen and holder. Dynamical studies of FOPT materials such as LaFeSi-based, Heusler type (Ni-Mn-In, Ni-Mn-In-Co), characterized by this setup had been reported [38, 39]. These directly measured ΔT_{ad} results of Ni-Mn-In when combined with the zero-field entropy curve, enabled the determination of a large reversible ΔS_M in a fast mode (this is not possible by conventional standard calorimetric measurements alone). The cyclic field dependence of ΔT_{ad} in LaFeSi-type magnetocaloric alloys, characterized using this setup and experimental protocol, had reported a ~17% reduction in ΔT_{ad} upon field cycling.

Furthermore, apparatus designs to directly measure ΔT_{ad} in pulsed high magnetic fields (up to 55 T) with surface modifications to the test specimens were also reported [40, 41]. Kohama et al. experimentally characterize C_p and ΔT_{ad} in 250 millisecond pulsed magnetic fields (up to 50 T from their report in ref. [40]) using an own-built calorimeter cell that comprised of a deposited NiCr-film (~10 nm thick) heater on one side of the sample and bare-chip resistive thermometers. The authors reported that the film indicated slight temperature and magnetic field dependences unlike thermometers: its measured magnetoresistance was 0.1% up to 45 T, which is 50x smaller than that of the resistive thermometer. They also used a high effective impedance system to generate a constant AC current during the pulse and a four-contact AC method with a digital lock-in system for the characterization of resistance of the thermometer and ΔT_{ad} of $\text{Sr}_3\text{Cr}_2\text{O}_8$ samples but proposed that the film deposited heater is less appropriate for metallic samples due to their high conductivities. In contrast, Kihara et al. fabricated a resistive film thermometer (a patterned $\text{Au}_x\text{Ge}_{1-x}$ alloy film of 100 nm thick) directly on the surface of the sample to measure its ΔT_{ad} in pulse magnetic fields up to 55 T, which could minimize the C_p of the sensor and maximize the thermal conductance between the sample and the sensor [41]. The tunable temperature dependence of resistivity of the film thermometer by its alloy composition enabled an adjustable sensitivity depending on the desired temperature range for measurements. A CaF_2 layer (150 nm thick) was developed, using vacuum evaporation technique, to electrically isolate the film thermometer from the sample. The resistance measurement of the film thermometer was performed using the standard DC four-probe practice or AC technique at 50 Hz using a digital lock-in with a time constant of 20 μs .

As previously mentioned, to prevent that the thermal mass of the temperature sensor (incorporated in many MCE measurement setups) could limit the applicability for high

frequency measurements, non-contact ΔT_{ad} measurement methods were then implemented. Such instrumentation lowers the time constant of ΔT_{ad} characterization as well as improves the accuracy of the measurements [42, 43]. One example, using the magneto-modulation technique (at >50 Hz and non-contact thermal radiation detection by HgCdTe-infrared detector), could perform highly sensitive low magnetic field ΔT_{ad} measurements of magnetocaloric micro- and thin-film specimens of FOPT (metamagnetic La-Fe-Co-Si) and SOPT materials [43]. Temperature-dependent measurements of the magnetocaloric peak in thin films were performed, presenting different magnetic field-dependences of the peak shapes that could be analytically expressed as interpolations between H_{ext}^2 -behavior (low magnetic fields) and $H_{ext}^{\frac{2}{3}}$ -behavior (high magnetic fields).

Recently, a non-contact direct ΔT_{ad} measurement instrumentation, which operated based on the mirage effect (an optical beam deflection technique used in photothermal spectroscopy to measure thermal diffusivity and the thermo-optical absorption spectrum of materials) was developed to characterize the response of magnetocaloric materials to fast magnetic field pulses that represent high frequency operating conditions [44]. Using this setup to characterize for the direct ΔT_{ad} of Fe₂P-based compounds reveals that they exhibit excellent dynamic response and magnetocaloric properties, enabling their applicability. Its protocol adopts a simple and versatile measurement method as well as short measuring time, extending its applications to characterize low mass samples (i.e. ribbons, foils and thin films) with similar heat capacities as the sensors. Furthermore, the reduced measurement time constant to less than one millisecond enabled that the material's response under extreme operating conditions (1 T at a frequency of about 150 Hz) could be measured, which was a reported breakthrough.

In addition, direct MCE measurements were also extended to the characterization of ΔS_M , not only ΔT_{ad} experiments, of magnetocaloric materials using calorimetric techniques. Differential scanning calorimeter (DSC) is proposed as an appropriate study of FOPT as they directly measure the heat flux and that enabled the determination of latent heat of the transition [45]. Direct ΔS_M determination in field-induced transition had been evaluated from the calorimetric measurement curves of a purpose-built DSC which is maintained at a constant temperature during the sweeping of the magnetic field [45]. This was attained due to the induction of FOPT upon increasing the magnetic field whilst fixing the temperature of the calorimeter at above that of the peak of the transition ($H=0$) [46]. The calorimeter operates under an applied field of up to 5 T for a temperature range of 10-300 K. In addition, this technique is based on employing differentially connected configuration of Peltier cells as heat flux sensors and its overall design could adapt to any cryostat equipped with a superconducting magnet. Adapted from this design, another custom-built calorimeter was developed, which was reported to directly characterize $\Delta S_{\text{Isothermal}}$ for comparing to the barocaloric effect properties of Ni-Mn-In materials (a shape-memory alloy with FOPT) [46]. In contrast to the previously mentioned version, this device design could be placed within the poles of electromagnet, operates under magnetic fields up to 1 T and temperature control range of 200-350 K.

Another Peltier calorimetric technique, exploiting active use of Peltier heat, was also reported to directly characterize ΔS_M using an electromagnet as the magnetic field source (up to 2.3 T) [47]. In this case, the power-type Peltier cells were arranged in a sandwich configuration to actively function as heat pump and heat flow sensor, enabling measurements recorded using a digital feedback control. The authors reported that the measured entropy change as a function of magnetic field at the critical transition

temperatures of a FOPT material revealed superposition of two contributions, one associated with the latent heat of the FOPT (which is hysteretic) and another with the conventional MCE of the phases without hysteresis. This active use of Peltier cells in calorimeters to perform direct MCE measurements avoided the unfavorable introduction of time delays by the sample from the passive techniques. However, the former is still limited by the time constants of the measuring cells, leading the authors to construct another design to investigate the dynamics of the symmetric differential arrangement of Peltier cells with theoretical heat flux diffusion models in the calorimeter [48]. Hence, a symmetric arrangement was used for the advanced setup. In this case, two types of Peltier cells were utilized: miniaturized ones for measuring heat flux while power-type cells for temperature regulation. The setup is capable of characterizing for specific C_p under constant magnetic field and ΔS_M induced by the variable magnetic field below room temperature. The authors demonstrated that their developed models, through this calorimeter setup, could enable kinetic corrections on the influence of time constants of the measuring cells, from the agreement of ΔS_M results obtained from the direct isothermal measurements and from the integration of measured C_p data. Another rather similar setup was reported by Nielsen et al. [49]. The modifications include using Halbach-type magnets as magnetic field source and fully automated systematic resetting function of the thermal and magnetic histories of the sample for measurements.

Another calorimetric technique for direct ΔS_M and ΔT_{ad} measurements using own-built adiabatic calorimetry instrumentation had been reported [48]. The setup included specially designed cryocooler from Cryogenic Limited and modified commercial adiabatic calorimeters (from Termis Ltd.) to directly characterize C_p , ΔS_M and ΔT_{ad} at magnetic fields (up to 9 T). Taking into consideration the influences of thermal and magnetic histories of the magnetic samples (especially FOPT materials), the authors

developed appropriate experimental measurement protocols during increasing and decreasing the magnetic fields respectively. This led them to attain consistent ΔS_M results for a given sample, exhibiting FOPT with hysteresis, obtained from different techniques (direct measurements and indirect determination from magnetization and C_p measurements). As a result, this could address the open questions regarding the peak values of ΔS_M (computed from magnetization data) larger than the corresponding values calculated from C_p or direct thermal measurements for the same sample [50, 51]. Moreover, the same characterization protocols were demonstrated to be also applicable to direct ΔT_{ad} measurements of FOPT materials using the same calorimeter, giving consistent ΔT_{ad} results that were determined directly or computed indirectly from C_p data. The materials studied using this adiabatic calorimetry instrumentation include Gd, Gd-Si-Ge-based materials, Mn-Co-As and Ni-Co-Mn-Sn alloys [51-54].

3.2. Indirect measurements

In general, indirect methods of MCE determination involved using magnetization data to calculate $\Delta S_M(T, H)$ and C_p data to compute $\Delta T_{ad}(T, H)$ and $\Delta S_M(T, H)$. As mentioned earlier and in many review papers associated to MCE, the calculation of $\Delta S_M(T, H)$ using one of the Maxwell's relations, strictly speaking, is only applicable to SOPT cases due to the infinite term of $\partial M / \partial T$ at FOPT. Hence, ΔS_M of the latter could be determined using the Clausius-Clapeyron equation instead. However, the large artifacts found in the determination of ΔS_M from magnetization data of FOPT materials are not due to this divergence but to the phase coexistence that takes place while the transition is taking place. The indirect characterization of MCE is performed using magnetometers and calorimeters as experimental techniques.

Magnetometry, which is readily accessible in numerous laboratories, has been the most common method used to map out $M(H, T)$ of magnetocaloric samples to indirectly

determine $\Delta S_M(T, H)$. The $M(H, T)$ measurements are typically time consuming, especially in the case of FOPT. Thus, a conventional vibrating sample magnetometer (VSM) is well suited to perform these tedious measurements within the range applicable for room temperature magnetocaloric devices ($\Delta H=0-2$ T; $\Delta T=200-350$ K) as compared to using a helium-cooled superconducting magnet (though it could offer higher magnetic fields up to 10 T and wider temperature range down to few kelvins). In the VSM method, the vibration of the sample within the constant uniform magnetic field led to an induced voltage measured in the pickup coils, which is used to determine the magnetic moment of the sample. It is sensitive enough to characterize samples of small mass and volume (starting from less than one cubic millimeter) and of any shape (bulk, plates, fragments, powders, etc.) though care has to be taken in order to avoid artifacts emerging from the demagnetizing field in the sample [55]. In general, it measures bulk magnetization and yields specific magnetization when divided by sample mass, which then calculates ΔS_M per unit mass. In order to convert the results to magnetic entropy change per unit volume, the density of the sample should be known. This is a trivial problem for bulk alloys, but can pose some difficulties for powder samples, porous materials, etc. being the reason why most of the magnetocaloric literature quotes the values per unit mass.

Basically, there are two methods to perform $M(T, H)$ measurements for MCE determination: (1) iso-field $M(T)$ measurement curves for different values of applied field, or (2) isothermal $M(H)$ measurement curves at different temperatures. For the first technique, $\partial M/\partial T$ from the experimental results could be numerically integrated to determine ΔS_M , while in the second method the integrated area between two magnetization isotherms is used to compute ΔS_M at intermediate temperature. A tradeoff should be found between the smoothness of the obtained ΔS_M curves (the larger the

temperature step used in the experiments, the lower the noise introduced in the curves by numerical derivation) and the temperature resolution of the curves (a smaller temperature increment will produce denser curves).

Regarding the speed of measurement, it is typically faster to keep the temperature constant and sweep the magnetic field (method 2 above) than varying the temperature for different constant values of applied magnetic field (method 1). Therefore, if we only focus on speed, isothermal $M(H)$ curves would be the method of choice. This method will give appropriate results for materials with a SOPT with no further precaution.

However, the coexistence of phases in FOPT imposes certain restrictions on the way that the isothermal curves should be measured [50, 56] and the application of Maxwell relation to isothermal magnetization curves gives unphysical magnetic entropy change spikes unless the sample is properly prepared before each measurement. To avoid these spurious results, two different approaches can be followed: either measuring isofield magnetization curves or preparing the sample to be in a single-phase state before each isothermal measurement [57]. A detailed description of the measurement protocols can be found in a recently published application note [58], and an example of their application to LaPrFeSi in [59]. In summary, $M(T)$ measurements can give reliable results, while $M(H)$ measurements require to saturate the transformation by heating to a temperature well above/below the transition to make it complete in the presence/absence of magnetic field, prior to reaching the desired measuring temperature and acquiring the $M(H)$ curve. The presence or absence of applied field during this procedure of erasing the memory of the material (analogous to saturation before making a $M(H)$ hysteresis loop for conventional ferromagnets) depends on the nature of the transition. For example, if the transition shifts to higher temperatures upon the

application of magnetic field, the erasing procedure should include stages of heating the sample in zero field or cooling it in high field. This is the case for LaFeSi-type alloys.

It is important to note that magnetocaloric materials, by definition, change their temperature when subjected to a variant magnetic field adiabatically. Therefore, artifacts might appear when the ramp rates of magnetic field are too fast [60, 61].

As for calorimetric technique for indirect determination of MCE, $C_p(H)$ is needed to assess the performance of a magnetocaloric material, however most commercial calorimeters could only measure the temperature dependence of the specific heat in the absence of magnetic field for a given sample and adapting them to measure in magnetic fields can be sophisticated. Basically, there are two reported varieties of calorimeters measuring C_p for magnetocaloric assessment: heat pulse type (the temperature of the sample is measured upon a known amount of heat supplied for a definite time interval while a constant temperature of the calorimeter is maintained during the measurement) and scanning types (whereby the temperature of the calorimeter is continuously changed while measuring the heat flux between a thermal block and the sample). Most of these utilize two heat flow sensors so that one can behave as a reference and enable the calorimeter to work differentially, and are known as DSC, which are the current custom-built designs specifically for MCE measurements (due to their shorter measurement durations and better precision to characterize C_p near FOPT) [62].

Nonetheless, heat pulse calorimeters were typically used for MCE characterization (e.g. a completely automatic design rapidly measuring C_p within ~3-350 K up to 10 T was reported by Pecharsky et al. [63]) before a customized DSC, using Peltier cells as heat flow sensors, was demonstrated to obtain highly reproducible C_p data with high sensitivity [64]. Many devices which adopted a similar approach were developed and some examples were mentioned in previous section that directly characterize MCE

(whereas the original idea in using Peltier cells as heat flow sensors in [64] performs measurements of C_P and an approximation of heat release accompanying a magnetization process). An example of a customized DSC based on Peltier elements as heat flow sensors was developed by Jeppesen et al., which could indirectly determine MCE [65]. It operated in a temperature range of 200-340 K up to 1.8 T, giving a signal/noise ratio of 10^2 - 10^3 , and its configuration could be further used with other types of magnets (permanent magnets, superconducting coils) and in a wider temperature range.

A contrasting approach of a thin film nanocalorimeter based on a commercial microchip module (comprised of submicron silicon nitride membrane with a film-thermopile and a resistive film-heater) and AC calorimetry had been reported [66, 67]. It was shown capable of measuring C_P of low-mass samples (micrograms) within 5–300 K up to 8 T.

Furthermore, it is worth mentioning that to calculate the total entropy of the material from heat capacity data, it is necessary to perform measurements at temperatures close to zero kelvin and to know the zero entropy term (see equation (16)). If the zero entropy term is assumed field independent, it has no influence on $\Delta S_M(T)$ and $\Delta T_{ad}(T)$ curves.

On the other hand, the necessity to reach temperatures close to zero kelvin would preferably be avoided. It has been found that starting the measurement at certain temperature (relatively close to the transition temperature) the magnetocaloric magnitudes are the same as those starting from zero kelvin. This temperature is the

same for both $\Delta S_M(T)$ and $\Delta T_{ad}(T)$ [68]. Furthermore, this had been theoretically studied using numerical calculations of the Bean and Rodbell model, and experimentally investigated applicable to SOPT materials, like Gd-Zn [68] and Gd [69], and FOPT materials, like $\text{Gd}_5\text{Si}_2\text{Ge}_2$ [69]. This proposed method of identifying the optimal start temperature is based on the requirement that magnetocaloric curves go to

zero at their high temperature tails. In this way, when starting from this optimal temperature, the magnetocaloric curve is the same as that starting from 0 K, minimizing the error.

Both calorimetric and magnetic measurement techniques, when using the proper measurement protocols, are able to provide reliable values for the magnetocaloric characteristics of materials. The analysis of error propagation [24] indicates that, depending on the sample and on experimental conditions, there are circumstances in which heat capacity techniques can provide lower absolute errors than results obtained from magnetization measurements (e.g. for ErAl_2 in ref. [24]), while other cases provide comparable values of the total error (e.g. Gd metal in the same reference). Therefore, for each experiment, the actual contributions to the errors in the magnetocaloric magnitudes should be determined.

3.3. Dynamic magnetocaloric effect

An important point to consider in the selection of a magnetocaloric material for its application in a real refrigerator device is its operation in cycles of cooling and heating, magnetization and demagnetization. Theoretical estimates of the frequency limit in the operations of these devices indicate values in the order of 100 Hz [70]. Although this frequency is much larger than the one used nowadays, it gives us a warning that even if the quasistatic characterization of materials is an important starting point, the ultimate comparison among magnetic refrigerants should be based on their cyclic and dynamic performances.

Unfortunately, most of the published literature focus on the quasistatic characterization of samples. In the case of indirect characterization methods, like those based on magnetization measurements, they are by definition quasistatic, as the whole set of

$M(T,H)$ curves has to be acquired before calculating the magnetic entropy change. As previously mentioned for first order phase transitions, the incomplete transformation of phases would give rise to artifacts when determining ΔS_M , which prevents the direct realization of cycles. However, it has recently been shown that the cyclic response of the material can be extracted from the quasistatically determined $\Delta S_M(T)$ curves as the intersection of the heating and cooling curves [59].

For the direct measurement of adiabatic temperature change, the procedure to get its cyclic response is less complicated as the measurement protocol can mimic the cycles performed in the refrigerator device and directly measure the cyclic changes of temperature. They might be, however, smaller than those predicted for the whole transformation since the loops undergone by the material in a refrigerator device might not accomplish the complete transformation. Nevertheless, there are procedures to predict this amount of reduction of the performance metrics [59].

Another important parameter of the magnetocaloric materials for their eventual applications is the influence of the rate of magnetic field variation on their ΔS_M and ΔT_{ad} values. Experimentally, the most straightforward way to produce a fast variation of magnetic field is using pulsed facilities. In that way, sweep rates up to 1500 T/s can be achieved [44, 71, 72]. In the case of Heusler alloys, the analysis of the dynamic behavior reveals that the nucleation process of the emerging phase is rather slow as compared to the movement of the phase boundary, posing a limitation for cycles above 100 Hz [72]. However, for the typical frequencies used nowadays in magnetic refrigeration, dynamical effects do not seem to be of a concern.

4. Models of the magnetocaloric effect

Some of the new magnetocaloric materials found in the literature are discovered through a process of trial and error. In some cases, the search of materials was based on the crystallographic structure of phases, the magnitude of their magnetization, etc., and has led to an initial discovery of a promising material, which could be subsequently refined for better properties. However, as in many other branches of science, the exhaustive search approach for solving the problem usually leads to a limited success, when results are compared to the efforts (time, costs, etc.) required to identify new magnetocaloric materials. In order to find new potential magnetocaloric materials, it is necessary to understand the mechanisms of large MCE (or even small MCE) reported in already known alloys and compounds. The models developed for their magnetothermal responses can facilitate the discovery of modifications, which would even further enhance their performance. In addition, models developed for a known material can be useful for predicting its response under different excitation conditions. If the model is properly developed, the time hassle in validation experiments can be minimized as only selected excitation conditions should be measured; subsequently, the performance of a device could be predicted by using the model.

The classification of models used in this field of research can be made between thermodynamic models and first principle models. Thermodynamic models are less capable for the identification of new materials but are extremely useful to evaluate the response of known materials (and some compositional variations) under multitude of excitation conditions. Therefore, they are also suitable to be introduced in models of the performance of magnetic refrigerators. First principle models are generally more suitable to predict the behavior of unknown alloys and compounds, streamlining the search for new compositions but that requires a detailed knowledge of the possible

phases that will be present in the material. In this section, we will outline the main characteristics of the most representative cases of both families of models.

4.1. Thermodynamic models

Thermodynamic models utilize macroscopic variables to describe the change of magnetization with field and temperature. In many cases, this is made via equations of state (EOS), which link the magnitudes M , H and T , and the variables that describe the material under study are identified by fitting the experimental measurements to those EOS. Alternatively, Preisach models have been developed to reproduce the hysteretic behavior of FOPT [73].

For SOPT, the EOS can be obtained from a power expansion of the thermodynamic potential following the formalism of Landau [74]. Depending on the number of terms that we keep in the power expansion, the description can be almost accurate including corrections to the first order mean field approximation. Another possibility is to use a heuristic approach to tailor a known EOS to the material by including additional features known to be fulfilled in it. An example of this type of EOS is that of Arrott-Noakes [75], wherein the critical behavior with exponents different from those of the mean field ones was incorporated:

$$H^{\frac{1}{\gamma}} = a(T - T_c)M^{\frac{1}{\gamma}} + bM^{\frac{1}{\beta} + \frac{1}{\gamma}} \quad (21)$$

where a and b are characteristic parameters of the material under study and β and γ are the critical exponents. In this way, we can control the universality class to which the material belongs. This EOS has been extensively used for the study of SOPT magnetocaloric materials [76-79] and was the initial basis on which the scaling of magnetocaloric response and its universal curve was founded [76, 80]. The

phenomenological construction of the universal curve, which was also extended to ΔT_{ad} [33], has been used to predict the behavior of samples in experimental conditions that were not available in the laboratory (for e.g. larger magnetic fields or temperatures not within the experimental limits of the device).

For FOPT materials, the most popular EOS in the field of magnetocaloric research is the one from Bean and Rodbell [81]. This was initially developed to study the first order transition in MnAs compound. Based on experimental results, the phenomenological dependence of the exchange interaction on the inter-atomic spacing was considered and was introduced into the dependence of critical temperature on the volume change. With a control parameter η , the order of the transition from SOPT ($0 \leq \eta < 1$) to FOPT ($\eta > 1$) can be modified. Recently, it has been shown that the case $\eta = 1$ corresponds to the tricritical point, i.e. to the separation of first and second order phase transitions, and the critical exponents coincide with those associated to that special point [82]. This model has been successfully applied to a variety of magnetocaloric materials beyond the original MnAs [83-86].

There are some limitations to be considered when using these models based on an EOS for describing the magnetocaloric response of materials. The main one is that the material must be in technical saturation as EOS relates state functions, which are dependent on the state of the system and not the route followed to achieve it. Therefore, if there is a multi-domain configuration (e.g. for small fields), the magnetization of the sample will depend on how its different domains are configured, which is dependent on the history of magnetization and temperature that the sample followed. Therefore, results obtained from the model of the magnetic entropy change under small fields has to be analyzed with precaution.

Moreover, thermomagnetic hysteresis of FOPT materials can be further characterized using FORC (first order reversal curves). It provides a comprehensive fingerprint of the $M(T,H)$ curves under excitation conditions that are not limited to saturation loops although it cannot be considered as a thermodynamic model of the material under study. Once the FORC distribution is obtained, it can model the behavior of the material under different minor loops of temperature and field. In particular, there is a growing interest on the T-FORC (temperature-FORC) characterization of magnetocaloric materials [87] and its use to determine how the hysteretic behavior is affected by temperature and magnetic field.

4.2. First principle models

The use of first principles calculations of exchange coupling energies and magnetic moments of magnetocaloric materials has become a usual procedure in the study of first order phase transition materials [88-94].

In most cases, modelling is performed with a two-stage approach [90, 95]. Firstly, first principles calculations, using Density Functional Theory, are used to predict the zero temperature properties of the material; subsequently, the obtained magnitudes are used as inputs in thermomagnetic models, such as the Bean & Rodbell model [81], to extract information about the values of magnetization at finite temperatures, magnetic entropy, magnetostructural transition temperatures, and MCE.

More recently, the disordered local moment formalism, which is a model for metals above their Curie temperatures [96], has been used to study the temperature dependent properties and MCE of different materials, like FeRh [97, 98], Lanthanide intermetallics [99] and LaFeSi [100].

In general, a good agreement is found between the experimental results and the outcome of the models. However, it has to be considered that such models require a fair amount of details of the studied material, which somehow limits the predictive power to materials relatively similar to those previously analyzed.

5. Magnetocaloric materials

5.1. Crystalline materials

5.1.1. Rare earth (RE) containing crystalline materials

Magnetocaloric behavior of pure lanthanide elements can be found in the review paper on magnetocaloric materials published in 2000 by Gschneidner and Pecharsky [11].

Among them, Gd is the only pure metal with a Curie temperature close to room temperature. For intra-lanthanides we also suggest to visit that review paper [11]. Slight improvement of MCE response in Gd has been reported by small addition of Nb [101] or Fe [102].

Small additions of B to Gd in $\text{Gd}_{1-x}\text{B}_x$ ($x=0$ to 0.07) preserved the parent hexagonal structure while slightly increasing the cell volume. For $x=0.02$, the alloy displayed $\Delta S_M = -2.8 \text{ J kg}^{-1} \text{ K}^{-1}$ ($\mu_0 \Delta H = 1 \text{ T}$) at $T_C = 298 \text{ K}$ and retained a constant ΔS_M at higher B contents [103]. Unlike for arc-melted samples, in $\text{Gd}_{1-x}\text{B}_x$ melt spun ribbons, the formation of GdB_2 and Gd_2B_5 is prevented even for $x=0.20$, keeping T_C constant but causing a continuous decrease of MCE properties with the increasing B content [104].

Earlier in 1997, Gschneidner et al. [105] studied $\text{Gd}_{1-x}\text{C}_x$ solid solutions, which exhibit hexagonal structure and displayed constant MCE values for minor C additions (up to $x=0.025$) [106] and a slight increase of T_C . However, the crystallization of an impurity Gd_2C phase occurred, which had a $T_C = 623 \text{ K}$ and a rhombohedral CdCl_2 -type structure.

With higher C content, T_C increased while $|\Delta S_M|$ progressively decreased. Theoretical studies using Monte Carlo simulations [107] showed an agreement of increasing T_C for C additions only up to $x=0.09$, but predicted higher values of magnetic entropy change, which could be ascribed to the presence of Gd_2C impurities observed experimentally [105, 106]. Partial substitution of Dy for Gd in $Gd_{0.98-x}Dy_xC_{0.02}$ tuned T_C between 295 K and 275 K for $x=0$ and $x=0.2$ respectively [108].

H impurities in GdH_x (x from 0 to 0.15) slightly increased T_C and broadened the magnetic transition [109]. Within this compositional range, ΔT_{ad} decreased from 4.9 K to 4.3 K at 1.78 T as the hydrogen content increased. In addition, the hydrogenated samples exhibit an increased electrical resistivity, which is an additional advantage.

Additions of Mn [110] or In [111] to hexagonal Gd decreased T_C without significant effects on MCE. However, $GdMn_2$ phase was formed in $Gd_{1-x}Mn_x$, which amount increases as Mn content increases in the alloy for $x=0.05-0.2$, while for $Gd_{1-y}In_y$, a solid solution free of impurities was obtained up to $y=0.04$ [112].

Ga substitution in $Gd_{100-x}Ga_x$ also led to a decrease of T_C . Jayaraman et al. [113] produced $Gd_{100-x}Ga_x$ ($x=0, 5, 10, 15$, and 20 at. %) by conventional melting as well as rapid quenching, and in both cases α -Gd and Gd_5Ga_3 phases were found.

Recently, Provenzano et al. [114] reported a clear enhancement in both RC and ΔS_M with 10 at. % substitution of Fe and Co for Gd. As Fe and Co are insoluble in Gd hexagonal phase, they are located at the grain boundaries of Gd grains, forming a $(Fe,Co)_2Gd$ phase, which exhibits a higher T_C than that of Gd.

Room temperature MCE has been also studied for other Gd-M binary compounds such as $Gd_{0.85}Y_{0.15}$, $Gd_{0.85}Tb_{0.15}$ (peak temperature of MCE decreases and $\Delta T_{ad} \sim 2$ K at 1 T)

[115], $\text{Gd}_{0.75}\text{Zn}_{0.25}$ [115, 116] systems. Partial substitution of Pr or Nd for Gd in $\text{Gd}_{1-x}\text{RE}_x$ ($x \leq 0.05$) shows equivalent, close to linear, decreases of T_C as x increased [117]. Composites of $\text{Gd}_{1-x}\text{Ho}_x$ alloys ($x=0, 0.09$ and 0.20 with mass ratios $0.59, 0.17$ and 0.24 , respectively) have been recently proposed to optimize flat MCE response in the region between $265\text{--}293\text{ K}$, adequate for Ericsson refrigeration cycle [118]. $\text{Gd}_{1-x}\text{Y}_x$ (x from 0 to 0.4) alloys produced by arc-melting, exhibit some Gd and Y residual phases. Y substitution in the studied range linearly reduces T_C of this SOPT from 296 to 196 K , whereas $|\Delta S_M|$ decreases from 5.43 to $4.24\text{ J kg}^{-1}\text{K}^{-1}$ for $x=0\text{--}0.4$ at 3 T [119]. The summarized influential studies of various elemental additions on T_C and ΔS_M of hexagonal Gd is presented in Figure 2.

Other results are focused on low dimensional systems such as nanocrystalline Gd [120], Gd/W thin film heterostructures [121] and Gd/Ti multilayers [122, 123]. Ultrathin films of Dy [124] and Ho [125] show a strong enhancement of MCE for low field changes when the thickness of the film is below the helix period whereby the helical structure is not formed. Dy nanoparticles and coated Dy nanocapsules have been also studied [126]. Superparamagnetic behavior is observed due to the small particle size (3.6 nm , smaller than the helix period). $\Delta S_M > 2\text{ J kg}^{-1}\text{K}^{-1}$ is obtained from 7.5 to 105 K when field changes from 1 to 5 T . Novel routes to process Gd can be found in the recent literature: films deposition [127] or upon subjected to severe plastic deformation [128]. Recently, Dudek et al. [129] proposed a different approach to enhance MCE using magneto-auxetic systems predicting that mechanical stimulus could lead to large magnetic entropy changes. Caloric and multicaloric effects have been recently revised by Crossley et al. [130], Starkov et al. [131], and, particularly in multiferroic systems, by Flerov et al. [132] and Balli et al. [133].

In this section, several rare earth containing families of materials, whose MCE have deserved the interest of the research community, will be described in the following order: Laves phases, $\text{La}[\text{Fe}(\text{SiAl})]_{13}$ compounds, $\text{Gd}_5(\text{Si,Ge})_4$ compounds, ThCr_2Si_2 -type compounds, ferromagnetic Lanthanum manganites and other RE containing compounds. As a general rule, the different compounds described will be classified by their crystal structures. The different references collected in this review usually describe this characteristic of their studied samples. However, further information for RE containing intermetallics could be found in [134].

5.1.1.1. *Laves phases*

Laves phases, named after the German crystallographer Fritz H. Laves [135], present a stoichiometry AB_2 with a structural configuration mainly due to a close packing structure of spherical atoms instead of chemical bonding. These phases crystallize in three types of structures [136]: cubic MgCu_2 -type and hexagonal MgZn_2 and MgNi_2 structures (Figure 3). The AB_2 intermetallic compounds with interesting magnetocaloric properties exhibit the cubic phase (space group $\text{Fd-}3\text{m}$), whereby A is a rare earth metal (including Sc and Y) and B mainly is a transition metal (other elements as Al, Ga, Ge etc. can be found). However, there are other cases in which non-cubic Laves phases are reported to display MCE, such as ternary compounds of $\text{Hf}_{1-x}\text{Ta}_x\text{Fe}_2$ [137, 138], GdFeAl [139], DyCoAl [140] and DyFeAl ($-\Delta S_M = 6.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 130 K and 5 T and the critical exponents were determined) [141] whereby they crystallize in MgZn_2 -type structure (space group $\text{P6}_3/\text{mmc}$). $\text{Hf}_{0.83}\text{Ta}_{0.17}\text{Fe}_2$ presents a FOPT and GMCE ($|\Delta S_M| = 18 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1 T and 225 K), which are ascribed to AFM-FM transition [137]. However, this value is in contradiction with the results previously obtained by Herbst et al. at slightly lower fields [142] ($|\Delta S_M| = 1.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H = 0.9 \text{ T}$ and 225 K). On the other hand, Diop et

al. [143] reported an intermediate value for $\text{Hf}_{0.825}\text{Ta}_{0.175}\text{Fe}_2 \sim 5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=1 \text{ T}$ but below 200 K, despite that the peak extends to higher temperatures at higher fields. Recent results [144] show that the presence of supercooled AFM and superheated FM states leads to unconventional thermal effects in $\text{Hf}_{0.82}\text{Ta}_{0.18}\text{Fe}_2$. Dong et al. studied partial substitution of Co for Fe [145] and Zr for Hf in $(\text{Hf,Ta})\text{Fe}_2$ [146]. Both substitutions led to a decrease of the transition temperatures but Co reduces MCE, while Fe enhances it.

As shown in Figure 3, RE atoms in the cubic MgCu_2 -type structure form a diamond lattice while *B* atoms surround the RE atoms forming a tetrahedra. In those cases where transition metal is present, itinerant electron magnetism from TM can also occur in addition to the magnetic sublattice formed by RE. This phenomenon has been systematically studied for Co containing Laves phases. In RECo_2 compounds, the two magnetic sublattices either couple parallel for light RE or antiparallel for heavy RE. Therefore, Laves phases with Pr, Nd and Sm are ferromagnets ($T_C=40, 98$ and 220 K respectively) while with Gd, Tb, Dy, Ho and Er are ferrimagnets (FiM) ($T_C=406, 231, 138, 78$ and 32 K respectively) [147-151]. Moreover, FiM HoCo_2 ($T_C=78 \text{ K}$) and FM NdCo_2 ($T_C=98 \text{ K}$) undergo a spin reorientation transition, which is FOPT, at 16 and 42 K respectively [148]. Mudryk et al. [152], combining magnetic and in situ X-ray diffraction experiments, have recently shown that the magnetic moment of Co in HoCo_2 was reduced below its spin reorientation transition and its PM-FiM transition was assisted by itinerant electron metamagnetism.

Laves phases can present FOPT or SOPT. In the case that itinerant electron metamagnetic transition occurs in Laves phases, GMCE has been reported and a related review can be found in reference [153]. One example is RECo_2 system. Khmelevskiy and Mohn [154] developed a model predicting the order of the magnetic transition in

RECo₂ using the lattice parameter by considering magnetovolume effects and spin fluctuations. For large lattice parameters (>0.722 nm), SOPT occurs as the Co atom carries a magnetic moment whereas for small lattice parameters (<0.705 nm), Co has no magnetic moment. Between those limits, itinerant electron metamagnetic transition (and thus FOPT) can be induced after the induction of magnetic moments in the transition metal sublattice from the molecular field of the RE sublattice. However, there are two exceptions in this series: TbCo₂ and TmCo₂. Both exhibit SOPT despite their corresponding lattice parameters (0.721 nm and 0.714 nm respectively) are within the theoretical limits of the model. In the case of TbCo₂, their high T_C (>200 K) can lead to spin fluctuations effects. In TmCo₂, $T_C=4$ K but its molecular field is assumed to be very weak and insufficient to induce metamagnetism in the Co sublattice. In addition, there is a good agreement of the predictions of Khmelevskiy and Mohn with the experimental results of element substitutions and pressure application in RECo₂. E.g., element substitution leads to an increase of the lattice parameter in RECo₂, which tunes T_C to higher temperatures but at the expense of its FOPT. Similarly, pressure application reduces the cell volume and yields a decrease of transition temperatures [153]. Due to the interplay of magnetic transition and cell parameters, it is worth mentioning that lattice parameters of the different pseudo-binary alloys follows linearity or Vegard law (Figure 4).

The first studies on MCE of Laves phases were reported by Nikitin and Tishin for HoCo₂ ($\Delta T_{ad}>5$ K at $\mu_0\Delta H=6$ T and $T<100$ K) [155]. It is worth mentioning that although $|\Delta S_M|$ increases with ΔH in MCE materials with FOPT, it is mostly the broadening of the MCE peak that significantly enhances RC. On the other hand in SOPT materials, both their ΔS_M and RC clearly enhances with ΔH . This can be observed from the relative change of ΔS_M and RC_{FWHM} shown in Figure 5, using the

data from Ref. [156]. The universal curve describing the MCE behavior for SOPT has been successfully applied for several Laves phases undergoing such transition: $\text{Tb}(\text{Co,Fe})_2$ [157, 158], other RECo_2 [148] and $(\text{Er,Dy})\text{Al}_2$ [159]. Von Ranke and co-workers, using molecular field approach and accounting for the influence of crystalline electrical field [160], reproduced the experimental MCE data of REAl_2 ($\text{RE}=\text{Dy}$ and Er). Later they considered a Heisenberg Hamiltonian model to account for short range interactions [161] and Álvarez et al. [162] included the magnetic fluctuations in the analysis. The model was also extended to RECo_2 systems with FOPT, considering that the localized spins of RE ions are immersed in an effective subsystem of itinerant electrons from the 3d band of Co atoms [163]. In the frame of these models, they can reproduce the anomalous effect found in some Laves aluminide phases [164] and the change from FOPT to SOPT [165]. Hysteretic losses are ascribed to FOPT (see for example Ref. [166]).

In general, most of the Laves phases that are attractive for MCE exhibit transition temperatures below 100 K, while those with transition temperatures above 100 K typically comprises of Co as TM (although they could also display low transition temperatures). The relationship between $|\Delta S_M|$ and T_C (measured at $\mu_0\Delta H=5$ T) of this family of alloys is presented in Figure 6. In addition, studies on pseudo-binary compounds are also reported for $(\text{Ho,Tb})\text{Co}_2$ [167], $(\text{Ho,Gd})\text{Co}_2$ [168], $(\text{Er,Dy})\text{Co}_2$ [166, 169, 170], $(\text{Er,Dy,Gd})\text{Co}_2$ [171], $(\text{Er,Ho})\text{Co}_2$ [166], $(\text{Ho,Dy})\text{Co}_2$ [172], $(\text{Tb,Dy})\text{Co}_2$ [173], $(\text{Tb,Dy,Er})\text{Co}_2$ [174], $(\text{Tb,Ce})\text{Co}_2$ [175], and more recently for $(\text{Tb,Er})\text{Co}_2$ and $(\text{Tb,Ho})\text{Co}_2$ [176]. Transition temperatures in Laves family can be tuned to above 300 K. However, $|\Delta S_M|$ decreases exponentially with transition temperature, regardless the order of the transition, as shown in Figure 6. As a general rule for this family, as the Curie temperature increases, FOPT changes to SOPT.

Minor partial substitutions of non-magnetic elements for Co in DyCo_2 or GdCo_2 were reported to shift transition temperatures to higher values as observed for $\text{Dy}(\text{Co}_{1-x}\text{Al}_x)_2$ [177], whereby its T_C increased from 143 K ($x=0$) to 206 K ($x=0.1$) and changing from FOPT for $x \leq 0.02$ to SOPT for $x \geq 0.04$. Another example is $\text{Gd}(\text{Co},\text{Al})_2$ [149], which T_C initially increased from 406 K ($x=0$) to 418 K for $x=0.05$, then decreased to 357 K for $x=0.4$, and its lattice parameters increased to above 0.7259 nm and exhibits SOPT. Results on $\text{Dy}(\text{Co}_{1-x}\text{Ga}_x)_2$ were also reported [178] which T_C increases up to 196 K for $x=0.2$, although some discrepancies are found with the data supplied by Singh et al. [153]. The latter authors also observed a decrease of T_C from 164 K to 154 K for $x=0.075$ to $x=0.15$ in $\text{Dy}(\text{Co}_{1-x}\text{Si}_x)_2$ [153]. However, higher Si concentrations weaken the magnetic interaction leading to a decrease in the Curie temperature. A table-like behavior has been recently reported in multiphase GdCoAl alloys, containing GdCo_2 phase [179]. The Al substitution for Co in $\text{Ho}(\text{Co}_{1-x}\text{Al}_x)_2$ Laves phase yields itinerant electron metamagnetism and positive magnetovolume effect at low Al concentration, increasing transition temperatures from 94 to 159 K as x increases from 0.01 to 0.1 [180]. $\text{GdCo}_x\text{Al}_{2-x}$ crystallizes in a mixture of MgCu_2 - and MgZn_2 -type phases but adopts a single phase transition at 77 K. With annealing, ΔS_M enhances but RC reduces [181].

Substitution of Fe for Co in $\text{Dy}(\text{Co}_{1-x}\text{Fe}_x)_2$ [182] and $(\text{Tb},\text{Dy},\text{Ho})\text{Co}_{1.8}\text{Fe}_{0.2}$ [183] alloys changes FOPT to a SOPT despite their lattice parameters are within the defined limits of the model from Khmelevskiy and Mohn. However, Curie temperature is enhanced close to room temperature and thus the same argument explaining the second order character of the magnetic transition of TbCo_2 can be assumed for these Fe containing alloys. In the case of $\text{Er}(\text{Co}_{1-x}\text{Fe}_x)_2$ series [184], T_C increases from 32 K to 360 K for $x=0$ to $x=0.2$, with a significant decrease of MCE due to the change of the character of

the transformation from a FOPT to a SOPT. Partial Fe substitution for Co in $\text{Tb}(\text{Co}_{1-x}\text{Fe}_x)_2$ [158] shifted the transition temperatures to room temperature and reduced $|\Delta S_M|$ from ~ 7 to $\sim 4 \text{ J kg}^{-1} \text{ K}^{-1}$ ($\mu_0 \Delta H = 5 \text{ T}$) when x increases from 0 to 0.1, despite resulting in a broader curve. In this case, a structural transition from Fd-3m to R-3m is coupled to the magnetic transformation [158]. Partial substitution of Co for Fe in SmCo_2 increases the Curie temperature. The spin reorientation transition reported for the Fe-free compound is not observed in $\text{SmCo}_{1.8}\text{Fe}_{0.2}$. Its maximum magnetic entropy change is modest at its Curie temperature of 300 K ($\Delta S_M = -0.20 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 1.5 \text{ T}$) [151]. Annealing the alloys led to 40% enhancement in the maximum entropy change but not in their refrigerant capacities.

Ni substitution for Co in $\text{Dy}(\text{Co}, \text{Ni})_2$ [185] led to a decrease of Curie temperature, which is ascribed to the reduced magnetic exchange. Their lattice parameters were retained relatively constant. Progressive Ni substitution for Co in $\text{Gd}(\text{Co}, \text{Ni})_2$ [186] lowered both the lattice parameter and Curie temperature, unlike the behavior earlier described for RECo_2 alloys. Partial substitution of Mn for Co has been also studied: e.g. $\text{Gd}(\text{Co}, \text{Mn})_2$ [187] (its increased Mn content reduced T_C while $|\Delta S_M|$ slightly increased and SOPT was observed) and $\text{RECo}_{1.8}\text{Mn}_{0.2}$ [188] (it exhibits $T_C = 140, 211$ and 308 K for Dy, Ho and Tb, respectively, although $|\Delta S_M|$ decreases with Curie temperature, as generally observed in Figure 6). Mn additions in off-stoichiometric ErCo_2Mn_x [189] were reported changing the transition character from FOPT for $x=0$ to SOPT. Moreover, when x increased from 0 to 0.6, the Curie temperature increased from 35 K to 212 K. However, a large amount of $\text{Th}_6\text{Mn}_{23}$ -type impurity phase was observed for higher Mn additions. B additions in off-stoichiometric DyCo_2B_x also change FOPT to SOPT and increase the transition temperatures [190]. The formation of vacancies in $\text{Er}_{1-x}\text{Co}_2$ leads

to an increase of transition temperatures (from 32 to 46 K as x increases from 0 to 0.03) but at the expense of a reduced MCE response [191].

In the case of RENi_2 Laves phases, some deviation from the ideal MgCu_2 -type structure is generally reported and most of them exhibit a cubic superstructure (space group $F\bar{4}3m$), which unit cell is double of that of the original cubic Laves phase, and where there is a regular arrangement of vacancies of RE sites [192]. Recently, different RENi_2 alloys have been proposed to form composite materials to obtain table-like MCE responses [193]. Unlike for Laves phases containing Co, Mn or Fe, no magnetic moment is expected for Ni in these alloys despite a value of $\sim 0.2 \mu_B$ has been reported in GdNi_2 [194]. For Mn additions in off-stoichiometric GdNi_2Mn_x , MgCu_2 structure was retained at low x values but an additional phase with PuNi_3 -type structure was formed at higher x values. However, the compositional limits differ from the results from Gerasimov et al. [195] and those of Aryal et al. [196]. Magnetic moment of Ni/Mn atoms increases from 1.5 to $3.2 \mu_B$ as x increases from 0.5 to 1.5 and the best MCE properties observed among the series for non-mixed samples is $-\Delta S_M = 3.9 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\sim 175 \text{ K}$ and $\text{RC}_{\text{FWHM}} = 485 \text{ J/kg}$ at 5 T [196]. The off-stoichiometric TbNi_2Mn_x ($x \leq 1.10$) compound also crystallizes in the MgCu_2 -type structure [197] and has a SOPT at 150 K, leading to $|\Delta S_M| = 3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T. Cwik et al. have studied several pseudo-binary Laves compounds including $\text{Y}_x\text{Ho}_{1-x}\text{Ni}_2$ [198], $\text{Lu}_x\text{Ho}_{1-x}\text{Ni}_2$ [199], $\text{La}_x\text{Ho}_{1-x}\text{Ni}_2$ [200], $\text{Sc}_x\text{Ho}_{1-x}\text{Ni}_2$ [201], $\text{Tb}_x\text{La}_{1-x}\text{Ni}_2$ [202] and $\text{Gd}_x\text{Sc}_{1-x}\text{Ni}_2$ [192]. Generally, their transition temperatures are very low and only those of $\text{Gd}_x\text{Sc}_{1-x}\text{Ni}_2$ series are above 30 K. In the $\text{Gd}_x\text{Sc}_{1-x}\text{Ni}_2$ series, Curie temperature decreases from 75.4 to 13.6 K as x decreases from 1 to 0.2 and MCE decreases with x (ΔT_{ad} decreases from 5.8 to 3.9 K at $\mu_0 \Delta H = 4.2 \text{ T}$ as x decreases from 1 to 0.7) [192].

For other Co-free Laves phases, their transition temperatures typically lie near to or even above 100 K. For example, the T_C of $Gd(Fe_x,Al_{1-x})_2$ [203] initially decreases from 175 K ($x=0$) to 100 K ($x=0.15$) then increases up to 190 K for $x=0.4$ (its ΔS_M behaves inversely to T_C). However, for $GdFe_{0.3}(Al,B)_{1.7}$ [204] no significant effect on T_C was observed with respect to B-free alloy. Another example is $EuRh_{1.2}Zn_{0.8}$ [205] alloy, which exhibits a SOPT, lattice parameter of 0.76858 nm and $\Delta S_M = -7 \text{ J kg}^{-1} \text{ K}^{-1}$ at 7 T and $T_C = 95 \text{ K}$. Partial substitutions of Ce for Gd in $Gd_{1-x}Ce_xAl_2$ reduce the Curie temperature from 163 K ($x=0$) to 37 K ($x=0.75$) and ΔS_M values followed an opposite trend to the general described one decreasing from 3.82 ($x=0$) to 2.04 $\text{J kg}^{-1} \text{ K}^{-1}$ ($x=0.75$) at 2 T [206]. Alho et al. [207] theoretically calculated the MCE response in $Gd_{1-x}Dy_xAl_2$ and found a reduction of T_C to $\sim 65 \text{ K}$ for $x=1$ with a continuous increase of $|\Delta S_M|$ with Dy content although the simulated values are larger than experimental results. $REAl_2$ ($RE=Pr$ and Nd) compounds order ferromagnetically at 32 K for Pr and 77 K for Nd, showing at 3 T a $-\Delta S_M = 10.2$ and $3.74 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively [208]. Recently Gil et al. [209] theoretically studied the anisotropic magnetocaloric effect in $HoAl_2$ at 20 K. They ascribed this MCE to the spontaneous spin reorientation transition from $[110]$ to $[100]$ direction axis as temperature increases. Other theoretical studies reported by Ribeiro et al. include $Er_{1-x}Tb_xAl_2$ [210] and $Tb_{1-x}Gd_xAl_2$ [211] alloy series. Partial substitution of Er for Pr in $Pr_{1-x}Er_xAl_2$ reduces the transition temperatures (and lattice parameter) and exhibit a maximum $\Delta T_{ad} = 5 \text{ K}$ at 5 T for $x=0.95$ [212]. Only $Pr_{0.6}Er_{0.4}Al_2$ alloy exhibits a sharp peak in the heat capacity experiments and even shows interesting magnetoresistance properties, and two MCE peaks at ~ 7 and 30 K [213].

In the case of giant magnetoresistive $HoCu_2$ with a Néel temperature at 10.5 K (SOPT) the alloy shows a large $\Delta S_M = -19.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T [214]. Gao et al. [215] studied the $TbMn_{1.8}Fe_{0.2}$ compound, which undergoes a SOPT of the Tb sublattice at 80 K with

$\Delta S_M = -9.84 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 3.5 \text{ K}$ at $\mu_0 \Delta H = 5 \text{ T}$. For $\text{ErAl}_{2-x}\text{Fe}_x$ series, although cubic Laves phase is found for the extreme compositions, MgZn_2 -type structure is observed in the range $0.875 \leq x \leq 1.25$ [216]. For $x \sim 1.6$, T_C is tuned to near room temperature while a local maximum in $|\Delta S_M|$ is observed for $x \sim 1.2$ but not exceeding $3 \text{ J kg}^{-1} \text{ K}^{-1}$ for 2 T . More details on aluminides crystallizing in cubic Laves phase can be found in the review of Gschneidner and Pecharsky [11].

5.1.1.2. *La[Fe(SiAl)]₁₃*

Mixing of La and Fe does not form any intermetallic compound but their binary phase diagram shows an immiscible system. However, minor additions of Si or Al can yield the formation of a fcc NaZn_{13} -type structure (space group Fm-3c) containing 112 atoms per unit cell: 8 FeI, 96 FeII and 8 La positions, Si atoms are located in FeII sites [217] (e.g. see Figure 1 in Ref. [218]). This intermetallic compound exhibits very interesting magnetic behavior and also GMCE, which was reported in 2001 by Hu et al. [219] for $\text{LaFe}_{11.4}\text{Si}_{1.6}$ ($-\Delta S_M \sim 20 \text{ J kg}^{-1} \text{ K}^{-1}$ at 210 K and $\text{RC}_{\text{FWHM}} = 530 \text{ J/kg}$ at $\mu_0 \Delta H = 5 \text{ T}$). The GMCE was ascribed to an itinerant electron metamagnetic transition, which is FOPT. The phenomenology of the transition of this family of alloys and their corresponding MCE was reviewed in 2009 by Shen et al. [15]. For those alloys of this family with a FOPT, sharp volume shrinkage occurs as temperature increases above the magnetic transition without changing the symmetry. Moreover, several microstructural techniques, including X-ray [220], neutron diffraction [217, 218] and Mössbauer spectroscopy [221-224], have shown the coexistence of two NaZn_{13} -type lattices in a $\pm 2 \text{ K}$ range around T_C [15]. Recent studies combining ab initio calculations and nuclear

resonant inelastic x-ray scattering show that the changes in the vibrational density of states across this FOPT imply an anomalous magnetoelastic softening [225].

The itinerant electron metamagnetic transition can be lost upon increasing Si content or other compositional effects, which then lead to SOPT [12] reducing MCE but also implying negligible hysteresis losses. Typical curves of MCE for this family of alloys can be found in Ref. [226]. FOPT associated with lattice expansion implies large strains in the bulk samples that lead to their degradation after few cycles. Lyubina et al. [227, 228] could overcome this problem by producing porous samples which could reduce grain boundaries. Although $\sim 20\%$ reduction in ΔT_{ad} is observed in the porous samples, both its thermal and magnetic hystereses are reduced compared to their bulk counterparts.

Figure 7 shows the field dependence of the relative ΔS_M and RC_{FWHM} values of $LaFe_{11.8}Si_{1.2}$ (FOPT) and $LaFe_{11.2}Si_{1.8}$ (SOPT) from Ref. [226]. As previously shown for Laves phases, the increase of the maximum applied field does not enhance the relative ΔS_M of alloys that exhibit FOPT as much as it does for those with SOPT. However, the MCE peak significantly broadens with field, yielding a similar relative enhancement in RC for FOPT and SOPT. SOPT data can be better represented using the power law predictions from the universal curve analysis [229] than using mean field values.

Nevertheless, De Medeiros and Oliveira [230] could find a good agreement of the experimental MCE data and calculations (using a microscopic model under mean field approach and a phenomenological consideration of the magnetovolume coupling) of $LaFe_{13-x}Si_x$ with FOPT.

The improvement of MCE in this family of alloys has drawn attention to two important factors: tuning transition temperatures to room temperature and enhancing the purity of phases.

In regards to tuning the transition for room temperature applications, several ways have been proposed: changing the Fe/Si ratio, partial substitution of Fe for Co or introducing interstitial elements (mainly H, although C, B and N have been also studied). Generally, when the transition temperatures shift to higher temperatures, a decrease of MCE could be accompanied. For example, as the Si/Fe ratio increases in $\text{LaFe}_{13-x}\text{Si}_x$, T_C increases from 190 to 240 K when x increases from 1.3 to 2.2, but $|\Delta S_M|$ at $\mu_0\Delta H=5$ T decreases from 29 to below $10 \text{ J kg}^{-1}\text{K}^{-1}$, respectively, and the transition changes from FOPT to SOPT for compositions around $x\sim 1.6$ [231-234]. Moreover, MCE deteriorates as the magnetic moment of Fe atoms decreases with increasing Si content. As the average magnetic moment of Fe in $\text{LaFe}_{13-x}\text{Si}_x$ compounds typically shows a linear behavior [231], the average magnetic moment of Fe atoms $\langle m_{Fe} \rangle$ can be estimated as:

$$\langle m_{Fe} \rangle = \frac{31.1 - 5.2x}{13 - x} \mu_B$$

On the other hand, Co substitution for Fe does not affect the magnetic moment and thus the Curie temperature can be shifted to higher temperatures in a more efficient way than decreasing Fe/Si ratio [231, 234-236]. This behavior can be observed in Figure 8 by comparing the black and red lines. However, the first order character of the transition converts to SOPT when the amount of Co substituting Fe increases. Conversely, when substituting Co for Si, $\alpha\text{-Fe}(\text{Si},\text{Co})$ phase is stabilized [237]. Regarding other transition metals substituted for Fe, Pathak et al. [238] studied 1 at % Ni, Cu, Cr and V for Fe in $\text{La}(\text{Fe}_{0.99}\text{X}_{0.01})_{11.4}\text{Si}_{1.6}$, reporting a general deterioration of MCE. This MCE weakening is also confirmed from the recent results of minor substitutions of Cr, Mn, Co and Ni for

Fe reported by Gercsi et al. [239]. They report that the magnetic moments of Cr and Mn couple in an antiparallel manner to that of Fe, which is not the case for Co and Ni whereby they form parallel coupling to Fe. However, they also observe that with V substitution for Fe, an 11 K increase in T_C with a significant decrease in hysteresis losses are observed. For partial substitutions of Mn for Fe [240-242], T_C decreases and the MCE peak broadens with increasing Mn content. Partial substitution of Cr for Fe also leads to a decrease in T_C , unlike partial Ni [242] or Cu [243] substitution for Fe.

Partial substitutions of RE for La has been studied for Ce [244-248], Nd [249, 250], Pr [231, 251-255], Gd [256] and Er [257]. Generally, the decrease in Curie temperature is compensated with an enhancement of MCE. Both ΔS_M and T_C are related following a similar behavior than that found in the ternary $\text{La}(\text{Fe},\text{Si})_{13}$ series. There is a roughly linear decrease of $|\Delta S_M|$ as T_C increases, regardless the FOPT or SOPT character of the transformation (see Figure 8).

The substitution of Ce for La in $\text{La}_{1-y}\text{Ce}_y(\text{Fe}_{0.86}\text{Si}_{0.14})_{13}$ enables the change of SOPT ($y=0$) to FOPT for $y=0.3$, and consequently, MCE is clearly enhanced with a shift of about 30 K to lower temperatures [245]. For Nd substitutions ($0 < y < 0.4$ in $\text{La}_{1-y}\text{Nd}_y(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}$), the lattice parameter linearly decreases with Nd concentration but its T_C does not change monotonically (between 205 and 215 K). Its $|\Delta S_M|$ is reduced from 13 to 10 $\text{J kg}^{-1} \text{K}^{-1}$ at 5 T from $y=0$ to 0.3, but the MCE peak broadens and leads to an enhancement of RC_{FWHM} (from 340 to 400 J/kg from $y=0$ to 0.3) [249]. The influence of Pr substitutions is similar to those of Ce substitutions, whereby the Curie temperatures are shifted to lower values and enhanced $|\Delta S_M|$ is attained. A 37 % increase in both $|\Delta S_M|$ and RC is reported in $\text{La}_{0.5}\text{Pr}_{0.5}(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}$ compared to Pr-free alloy, whereas T_C decreases about 10 K [252]. Moreover, the lattice parameter

decreases linearly with Pr content. Recently, Fujita et al. [258] reported that the hysteresis of $\text{La}(\text{FeSi})_{13}$ compounds is reduced by controlling the Fermi level and magnetovolume effect with partial substitutions of Al and Pr. In $\text{La}_{0.7}\text{Pr}_{0.3}\text{Fe}_{11.4-x}\text{Cr}_x\text{Si}_{1.6}$ compound, the FOPT is preserved up to $x=0.34$. When $x=0$ increases to $x=0.06$, $-\Delta S_M(5\text{ T})$ reduces from $17\text{ J kg}^{-1}\text{K}^{-1}$ (at 197 K) to $12\text{ J kg}^{-1}\text{K}^{-1}$ (at 180 K) and corresponding RC_{FWHM} decreases from 390 J/kg to 365 J/kg. At $x=0.34$, $-\Delta S_M(5\text{ T})$ increases to $17.5\text{ J kg}^{-1}\text{K}^{-1}$ (at 195 K) and RC_{FWHM} also increases to 420 J/kg (with $\Delta T_{\text{ad}}=8.4\text{ K}$) [259].

Both α -Fe and LaFeSi impurities were observed. For Gd substitutions in $\text{La}_{0.80}\text{Gd}_{0.20}\text{Fe}_{11.4}\text{Si}_{1.6}$, $\Delta S_M(5\text{ T})$ enhances ($-21\text{ J kg}^{-1}\text{K}^{-1}$) but with a lower Curie temperature (200 K), and also presents a $\text{RC}_{\text{FWHM}}=400\text{ J/kg}$ [256]. Similar to Nd substitutions, the substitution of Er in $\text{La}_{1-y}\text{Er}_y\text{Fe}_{11.44}\text{Si}_{1.56}$ yields a reduction of the lattice parameter, converting FOPT to SOPT for $y=0.3$ [257]. For $y=0.1$, $|\Delta S_M|$ decreases 25 % with respect to $y=0$ despite retaining its FOPT character and a relatively constant Curie temperature. When Ca substitutes for La [260], the general relationship between ΔS_M and T_C is observed and $|\Delta S_M|$ decreases as T_C increases as it occurs for Fe/Si and La/RE series. Bao et al. [261] studied the viability of using La-Ce-Pr-Nd mischmetal instead of purified RE to produce $\text{La}_{0.7}(\text{Ce,Pr,Nd})_{0.3}\text{Fe}_{11.6}\text{Si}_{1.4}$ and its hydrides (see below the effect of interstitials). Their main goal is the cost reduction of the starting materials in producing NaZn_{13} -phase with similar MCE parameters than those of $\text{La}_{0.7}\text{Ce}_{0.3}\text{Fe}_{11.6}\text{Si}_{1.4}$ system. However, a higher amount of LaFeSi phase is observed in the alloy containing mischmetal. Large magnetic hysteresis of 56 J/kg, in $\text{La}_{0.5}\text{Pr}_{0.5}\text{Fe}_{11.4}\text{Si}_{1.6}$ compound ($|\Delta S_M|=28.1\text{ J kg}^{-1}\text{K}^{-1}$ at 1.5 T and $T_C=186\text{ K}$) decreases upon grounding to micrometric particles, whereas its thermal hysteresis (1 K for the bulk) is almost eliminated. Authors attribute the reduction of the hysteresis to the

decrease of internal strains and improvement of heat transfer due to increase in surface to volume ratio [20].

The presence of interstitials in this family of alloys plays a particularly important role in compositional effects (sometimes appearing as the negative pressure effect), which generally increases Curie temperature. Among them, H is the most important but N, C and B have been studied too. Since 2002, hydrides are known to shift the FOPT to higher temperatures without depreciating ΔS_M after the works of Fujita et al. [262] and Chen et al. [263, 264]. Several advantages were observed in hydrogenated samples with respect to H-free ones [265-267], such as smaller thermal hysteresis (~ 1 K [264, 265]) and an enhanced ΔT_{ad} (7 K at $\mu_0 \Delta H = 2$ T [262] or 15.4 K at 5 T [268]) as the heat capacity remains almost unchanged [265]. Similar results have been found for hydrides of alloys with partial substitution of RE for La [252, 267, 269].

Neutron diffraction studies [217] conclude that H are located within the octahedron, formed by 2 La and 4 FeII (4f interstitials) as it occurs for C [270]. Recent in situ neutron studies describe the kinetics of H absorption in $\text{LaFe}_{11.44}\text{Si}_{1.56}\text{C}_x$ ($x=0$ and 0.2) using Johnson-Mehl-Avrami-Kolmogorov theory, following the occupation of deuterium in 24d octahedral interstitial [271]. They found that C hinders H absorption, slowing down the kinetics and reducing 24d occupancy from ~ 0.04 to ~ 0.025 .

Mössbauer studies show a decrease of the magnetic moment in Mn doped alloys [272] and an increase in hyperfine field at 4.2 K after hydrogenation [223] without changes in the PM spectra. This concludes that an isotropic volume expansion due to the incorporation of H to the lattice is responsible for the large increase of T_C (from 195 to 330 K after 1.6 atoms of H per formula unit in $\text{La}(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}$) and attributed to magnetovolume effect (which is typical of itinerant electron metamagnetic transitions).

The kinetics of hydriding also has been recently studied by Wang et al. [273]. Jia et al.

[274] found that partial substitution of Ce for La plays the opposite role to H interstitials although the crystallographic positions are completely different.

Hydrogenation process was studied by Wang et al. [275] as well as homogenization of hydrogenated samples [276], also studied by Ma et al. [277] and more recently by Wang et al. [273] (who studied the kinetics of hydrogen absorption and desorption) and Baumfeld et al. [278], who also found that the timescale of H diffusion in partially hydrogenated samples lies in the order of hours. H saturation could be obtained after 1 h at hydrogen-filling pressure of 65 bar in $\text{LaFe}_{11.44}\text{Si}_{156}$. To incorporate interstitial hydrogen to the alloy, Mandal et al. [279, 280] successfully used reactive milling as an alternative route to the conventional annealing in hydrogen atmosphere. Lyubina et al. [281] have proposed electrolytic hydriding in the FM state (when the cell volume is ~1 % larger and thus a higher diffusion of H is allowed) as an economic and safe procedure to produce hydrides, with the advantage of performing the experiments at low temperatures. Fujieda et al. [282] showed that high MCE values were retained when hydrogenation was performed on single phase systems. However, in inhomogeneous samples (with phase impurities) MCE reduces after hydriding due to the inhomogeneous distribution of H, which leads to a T_C distribution within the sample. Recently, Bez et al. [283] proposed a novel hydrogen-decrepitating-sintering-hydrogenation process, which consists of three steps: (1) a starting La-Fe-Si ingot is hydrogen-decrepitated at 423 K; (2) milled material is sintered at 1423 K; and (3) hydrogenating the sample during slow cooling from 773 K. They reduced the time duration of treatment from days to few hours.

However, the main drawback of these hydrides is instability [284]. Teixeira et al. [285] found that a small addition of C increases the temperature of hydrogen desorption from 460 K (C-free) to 540 K in samples with 0.2 atoms of C per formula unit. In this line,

Zhao et al. [286] found a reduced rate of hydrogen absorption and release upon little C additions. Barcza et al. [287] reported that H-saturated samples remained stable while unsaturated hydrides are unstable close to their Curie temperatures and decompose into two types of samples (hydrogen rich and hydrogen poor phases), which adversely deteriorates MCE. To obtain stable hydrides with tunable Curie temperatures, they produced fully saturated $\text{LaFe}_{11.74-y}\text{Mn}_y\text{Si}_{1.26}\text{H}_z$ alloy series ($z \sim 1.53$, implying hydrogen saturation). Basso et al. [288] recently reported the deterioration of MCE with increasing Mn content but this is only noticed for samples with transitions below room temperature, while T_C can be tuned to ~ 300 K for $x=0.25$ without noticeable effect on MCE. These results agree with the analyses of Bean-Rodbell model, which describe well the experimental results with FOPT (parameter $\eta > 1$ for $x=0.25$) [289]. Similar qualitative results, using smaller η values, were also obtained by Piazzzi et al. [290]. No isotope effect has been observed for this family of magnetocaloric materials [291].

Pressure application, in contrast to the effect of interstitials, leads to a decrease of the transition temperatures but while MCE enhances for hydrided samples, it deteriorates for H-free counterparts [292].

In the case of incorporating nitrogen into $\text{La}(\text{FeSi})_{13}$ lattice, both their lattice parameter and T_C increase. However, the FOPT converts to SOPT in N-containing samples and significantly decreases MCE [293].

Using C interstitials in $\text{LaFe}_{11.6}\text{Si}_{1.4}\text{C}_x$ was proposed as an alternative to H by Chen et al. in 2003 [294], with the aim to develop a system of a higher chemical stability than hydrides. They found an increase of T_C from 195 to 250 K after adding 0.6 C atoms per formula unit. However, FOPT is only maintained for lower C content ($x \sim 0.2$) with a ~ 50 % decrease in ΔS_M after $x=0.6$, also in agreement with results reported by Teixeira

et al. [285]. The presence of C interstitials in $\text{La}_{0.5}\text{Pr}_{0.5}\text{Fe}_{11.5}\text{Si}_{1.5}\text{C}_x$ [295] leads to an increase of both Curie temperature and lattice parameter as x increases from 0 to 0.3 with ~75 % reduction of hysteresis losses. On the other hand, <15 % reduction of $|\Delta S_M|$ (~30 $\text{J kg}^{-1}\text{K}^{-1}$ for $\mu_0\Delta H=5$ T) is observed for $x=0.3$ compared to C-free alloy. Although $\text{LaFe}_{11.2}\text{Co}_{0.7}\text{Si}_{1.1}\text{C}_x$ ($x=0.1-0.3$) presents SOPT, the RC for these systems are larger than that of $\text{Gd}_5\text{Ge}_2\text{Si}_2$ exhibiting a FOPT [296]. Shen et al. [297] combined Co and C additions and obtained $\Delta S_M=-11.6 \text{ J kg}^{-1}\text{K}^{-1}$ at 295 K and 5 T with negligible hysteresis since the transformation changes to SOPT.

B additions were also studied as elemental substitutions for Fe [298] and as additional element to occupy interstitial sites [299-301]. The former case leads to an increase of both Curie temperature and lattice parameter with B increase for a constant Si content, whereas thermal hysteresis (2 K for B-free samples) becomes negligible with B substitution. The latter case leads to an increase in T_C (7-8 K) and lattice parameter with a boron content of 0.1 to 0.5 in the formula unit. The B interstitials also reduce both thermal and magnetic hystereses.

Although there is a general decrease in $|\Delta S_M|$ with the transition temperatures, the MCE peak also generally broadens with increasing transition temperature. Thus, the reduction of RC with transition temperatures is smeared out with respect to that of ΔS_M in this family of alloys (Figure 9).

NaZn_{13} -type structures with Al instead of Si was, in fact, firstly reported with MCE data among the LaFe_{13} family of alloys [302, 303]. $\text{La}(\text{Fe}_{0.98}\text{Co}_{0.02})_{11.7}\text{Al}_{1.3}$ exhibits a SOPT with $\Delta S_M=-11 \text{ J kg}^{-1}\text{K}^{-1}$ and $T_C=200$ K at $\mu_0\Delta H=5$ T, which positively compares with the results of Gd. Studies on $\text{La}(\text{Fe}_x\text{Al}_{1-x})$ series ($0.86\leq x\leq 0.89$) [304] show a gradual change from FM to weak AFM state. When Co substitutes for Fe, Curie temperatures

increase for both Si- and Al-containing alloys [232]. Although MCE decreases with increasing Co content in the Si-containing LaFe_{13} alloys (ascribed to the conversion of FOPT to SOPT), >100 % enhancement of MCE is obtained in $\text{La}(\text{Fe}_{1-x}\text{Co}_x)\text{Al}_{1.6}$ for $x=0.04$ and even remains unchanged with further increasing Co content (due to the increase in magnetic moment as deduced from the increase of hyperfine magnetic field [232]). In these aluminides, the effects of RE substitution for La (i.e. Pr and Nd) on their MCE, show that MCE enhances while Curie temperatures decrease with respect to the values of the parent system [305].

When interstitials are introduced to these Al-containing compounds, e.g. hydrogen additions, their T_C has the same behavior as those of the Si-containing alloys and furthermore ΔS_M are improved [306]. Carbon additions in Al-containing alloys also yield similar effects as those of Si containing alloys, in which T_C increases while $|\Delta S_M|$ slightly decreases as C content increases [221-223, 307]. B additions in $\text{LaFe}_{11.5}\text{Al}_{1.5}\text{B}_x$ were studied by Zhang et al. [308] and Xu et al [307]. The original FOPT AFM-to-PM changes to SOPT FM-to-PM with B additions. For $x=0.1, 0.2$ and 0.3 , the alloys exhibit $-\Delta S_M=7.64, 7.38$ and $7.25 \text{ J kg}^{-1}\text{K}^{-1}$ (3 T) at 178, 185 and 189 K respectively [308]. Other studies compare the effect of B and H interstitials in these systems [309].

As mentioned earlier, obtaining the single phase system in LaFe_{13} alloys is another concern in the research community. Palstra et al. [310] studied the stabilization of NaZn_{13} -type phase in $\text{LaFe}_{13-x}\text{Si}_x$ ($1.3 \leq x \leq 2.6$) alloys after annealing 10 days at 1173 K but could only obtain single phased samples in the range of $1.56 \leq x \leq 2.47$. To exhibit FOPT, a lower Si content is essential. However, this causes difficulties to attain single phased $\text{La}(\text{Fe},\text{Si})_{13}$ samples and the formation of impurities, such as α -Fe phase (~5 wt.% is generally obtained after annealing considered as successful ones) and other La-

rich phases like LaFeSi, still persists even after long durations of annealing at high temperatures (several days or even more than one month). This must be considered when comparing different results due to inhomogeneous compositions and, moreover, multiphase behavior rather than single phase behavior should be expected [311]. In the case of $\text{LaFe}_{13-x}\text{Al}_x$, the phase stability of NaZn_{13} exhibits a wider compositional range ($1.0 \leq x \leq 7.0$) with LaFe_4Al_8 only formed for higher Al contents and α -Fe for lower Al concentrations [15].

To optimize the homogeneity of the samples, several procedures are followed: varying the compositions, optimizing the annealing temperatures, ultrafast cooling from the melt and high energy milling. For compositional variations, it is reported that RE partial substitutions for La in $(\text{La,RE})\text{Fe}_{13-x}\text{Si}_x$ produces a single phase for $x=1.6$ but not for lower x values when $\text{RE}=\text{Ce}$ [312], and when $\text{RE}=\text{Nd}$, the phase fraction of α -Fe linearly decreases from 6.0 to 1.2 wt.% as x increases from 1.2 to 1.8. However, when $\text{RE}=\text{Gd}$ in $(\text{La,RE})\text{Fe}_{13-x}\text{Si}_x$, [256] the amount of α -Fe significantly decreases and the presence of LaFeSi is negligible for 0.2 Gd atoms per formula unit. With partial substitutions of Ca for La, the phase fraction of the La-rich impurity phase decreases while that of α -Fe phase increases instead [260]. Partial Co substitutions for Fe also lead to decreasing the amount of deleterious α -Fe phase [232, 297] but the opposite is observed when substituting Co for Si [237]. A minor α -Fe phase in $\text{La}(\text{Fe}_{1-x}\text{Mn}_x)_{13}$ for $x>0.02$ still persists even after annealing 30 days at 1300 K [240]. In fact, Mn substitutions in both $\text{La}(\text{FeSi})_{13}$ and their hydrides avoid the formation of co-existent phases. Composites with different Mn content provide a constant MCE extended over a large temperature span [313]. In addition, single phase also can be obtained by adding C interstitials in $\text{La}(\text{Fe,Si})_{13}$ (for ~ 0.2 C atoms per formula unit) [285, 314, 315]. Furthermore, off-stoichiometric compositions (excess La concentration) also have been

proposed as an effective way of reducing impurities [316-318]. A surplus of Fe in $\text{La}(\text{Fe}_x)_{11.6}\text{Si}_{1.4}$ shows that the amount of LaFeSi impurity phase depletes to nearly zero for $x=1.02$ [319]. However, Lai et al. [320] reported that the order of transition changed in their off-stoichiometric samples when La/Fe ratio decreases and a larger amount of $\alpha\text{-Fe}(\text{Si})$ type phase was formed compared to that of the stoichiometric composition. On the other hand, the phase fraction of LaFeSi significantly decreases with decreasing La/Fe ratio. It is worth noticing that off-stoichiometric compositions could appear due to the additions of excess La to compensate any losses upon arc-melting, as reported in [240]. On the other hand, the off-stoichiometric compositions with La excess are also reported to yield the NaZn_{13} -type phase and even La_5Si_3 phase, which does not significantly affect MCE [321, 322].

Liu et al. made a systematic microstructural study of $\text{LaFe}_{11.8}\text{Si}_{1.2}$ and $\text{LaFe}_{11.6}\text{Si}_{1.4}$ alloys as a function of annealing temperatures and duration [323]. They suggest an intermediate phase with an eutectoid-type lamellae morphology previous to the formation of the NaZn_{13} -type phase. The optimum annealing temperatures differ between the two studied compositions, decreasing from 1373 K to 1323 K as Si content increases from 1.2 to 1.4 at per formula unit. The optimum annealing duration was found to be 7 days, in which ~90 % of the desired phase could be formed. The minimum annealing duration explored was 1 day from this work. However, optimization has been also reported after high temperature annealing for short times: 91.4 vol.% of NaZn_{13} -type phase was reported to be formed in $\text{LaFe}_{11.6}\text{Si}_{1.4}$ alloy after annealing 1 h at 1573 K an arc melted sample with a matrix of 79.9 vol.% of $\alpha\text{-Fe}$ and 20.1 vol.% of LaFeSi phases in [324]. Kolat et al. [325] optimized the time duration of annealing the same composition at high temperature of 1473 K in the range from 5 min to 120 h and reported the formation of a single phase of NaZn_{13} structure after annealing

for 30 min. Xiang et al. [326] has studied the optimization of annealing conditions of $\text{LaFe}_{11.5}\text{Si}_{1.5}$ alloy within the range from 1223 K (5 h) to 1673 K (0.5 h) and found a significant amount of NaZn_{13} -type phase upon annealing at 1423 K. Within the temperature range from 1423 to 1573 K, the formation of α -Fe and LaFeSi phases rapidly decreases but LaFeSi phase with the presence of La_5Si_3 phase are detected upon annealing at higher temperatures, i.e. 1673 K. The optimum annealing conditions for this alloy were reported at 1523 K for 3-5 h. Better single phased samples are reported to be produced at ~ 1500 K, probably due to the formation of NaZn_{13} -type phase from the reactions between γ -Fe and La-rich phase [327].

Rapid quenching has been proposed for minimizing the annealing time durations because a better compositional homogenization of the sample is attained. Several studies on melt-spun ribbons [233, 328-333], strip casting [334, 335] or on melt-extraction [336] can be found in the literature. Liu et al. reported that an almost single phased systems in $\text{La}(\text{Fe},\text{Co})_{11.4}\text{Si}_{1.6}$ is obtained by annealing for 20 min at 1273 K [233]. They also found for as-quenched samples, an increased formation of NaZn_{13} -type phase with increasing Si content and wheel speed [329], in agreement with Lyubina et al. [331]. Moreover, melt-spun samples exhibit a significant reduction of magnetic and thermal hysteresis [330]. Hou et al. studied the microstructures [332] and magnetic domain structures [333] of both the wheel and air surfaces of the melt-spun ribbons. Copper mold casting method has been also found to reduce the optimum annealing duration from weeks to few hours [301]. In addition, high energy ball milling has been also used to decrease the annealing duration down to ~ 30 min to obtain monophasic systems [337, 338]. Recently, Patissier and Paul-Boncour [339] used spark plasma sintering (30 min at 1473 K) for ball milled powder (without annealing) and achieved 89 wt.% of the NaZn_{13} -type structure. The presence of C from contamination is

considered as the reason for their larger Curie temperatures as compared to those obtained from conventional annealing.

Mechanical stability of the system is also a drawback for this FOPT family leading to virgin effects [340]. However, Yuan et al. [341] observed irreversible microstructural changes (microcracks, etc.) even in $\text{La}(\text{Fe}_{0.94}\text{Co}_{0.06})_{11.8}\text{Si}_{1.2}$, which exhibits SOPT, but only above 1000 cycles. A 10 % deterioration in the MCE response was observed after 8000 cycles. Enhanced mechanical behavior has been attained by embedding the samples in soft materials, such as epoxy resin [342-345], polymer [346-348] or, more recently, in composites with a metallic amorphous [349] or tin matrices [350]. Also recently, Shao et al. [351] found that the presence of $\alpha\text{-Fe}(\text{Co},\text{Si})$ impurities contributes to enhancing the mechanical stability even with respect to polymer bonded samples. Core-shell $\text{LaFe}_{11.65}\text{Si}_{1.35}/\text{Cu}$ spark plasma sintered powder with improved mechanical properties and corrosion resistance has been also reported [352]. Corrosion resistance has been studied for $\text{La}(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}$ [353], $\text{La}(\text{FeMnSi})_{13}$ [354], and $\text{La}(\text{Fe},\text{Co})_{13-x}\text{Si}_x$ series and compared to Gd [355]. Recent results show that corrosion resistance in these alloys enhances with partial substitutions of Co for Fe, and particularly C additions [356].

5.1.1.3. $\text{Gd}_5(\text{Si},\text{Ge})_4$ phases

In 1997, Pecharsky and Gschneidner [7] discovered that $\text{Gd}_5\text{Si}_2\text{Ge}_2$ intermetallic compound underwent a magnetostructural FOPT upon heating from orthorhombic Gd_5Si_4 -type (space group Pnma) to monoclinic structure (space group $\text{P2}_1/\text{a}$). Both binary Gd_5Si_4 and Gd_5Ge_4 compounds (Gd_5Si_4 - and Sm_5Ge_4 -type structures respectively) exhibit orthorhombic structures. A representation of these three different structures can be found in Ref. [357]. Both orthorhombic structures have Pnma space group, but while dimers Ge-Ge bonds are present between the different slabs in Gd_5Si_4 -

type structure, they are absent in Sm_5Ge_4 -type structure [357]. The two different structures can be distinguished from the different ratio of the lattice parameters. The discovery of $\text{Gd}_5\text{Si}_2\text{Ge}_2$ system was a key point in directing the research focus to room temperature magnetocaloric effect and several specific reviews devoted to the structure and properties of this particular system have been reported [358-361]. The magnetostructural transition is ascribed to the stabilization of monoclinic phase, which can be only achieved, for ternary $\text{Gd}_5(\text{Si}_x\text{Ge}_{1-x})_4$ systems, for x ranging within $0.24 < x < 0.5$ [362]. However, the equivalence between temperature and hydrostatic and chemical pressure is reported to affect the polymorphism of $\text{Gd}_5\text{Si}_2\text{Ge}_2$ system [363]. A recent phase diagram around $\text{Gd}_5\text{Si}_2\text{Ge}_2$ composition is supplied by Melikhov et al. [364].

The transformation would imply a sharp change in the magnetization (and thus a GMCE) when the critical temperature, T_{Critical} , (at which the free energies of both orthorhombic and monoclinic phases are equal) lies between the Curie temperatures of these two structures: $T_C^{\text{M}} < T_{\text{Critical}} < T_C^{\text{O}}$, where T_C^{M} is the Curie temperature of the monoclinic phase and T_C^{O} is that of the orthorhombic phase (Figure 10). The GMCE should be enhanced as the difference between T_C^{O} and T_{Critical} increases.

The magnetic properties of $\text{Gd}_5(\text{Si}_x\text{Ge}_{1-x})_4$ system are composition dependent [357]. As Si content decreases, the Curie temperature of the simple ferromagnet linearly decreases from 335 K ($x=1$) to 295 K ($x \sim 0.5$). This Si concentration agrees with the upper limit of Si content for the formation of monoclinic distortion in the parent orthorhombic Gd_5Si_4 structure, as mentioned earlier (ternary $\text{Gd}_5(\text{Si}_x\text{Ge}_{1-x})_4$ systems for $0.24 < x < 0.5$). At this stage, when the temperature decreases, the system initially orders ferromagnetically but with low net magnetic moment and, at lower temperatures, converts to another FM phase through a FOPT. The rate of the reduction of both transition temperatures, as Si

content decreases, is faster than that of decreasing T_C for alloys with $x > 0.5$. For compositions below $x = 0.24$ (the lower compositional limit for stable monoclinic structure), orthorhombic structure forms again and the transition temperatures decrease, which yields a FiM system at ~ 125 K with a FOPT from FiM to FM state at lower temperatures. The transition temperature ascribed to this transformation also decreases as Si content decreases [365]. Si-free compound is AFM at temperatures below 130 K, which is also preserved at low fields down to ~ 2 K, while at higher magnetic fields above 1 T, a field induced structural transition occurs below 30 K [366].

Concerning MCE, the reference composition of this alloy family is $\text{Gd}_5\text{Ge}_2\text{Si}_2$, which was discovered by Pecharsky and Gschneidner in 1997 [7]. This system was reported with GMCE after the first GMCE material, FeRh alloy, and it exhibits a lower thermal hysteresis (~ 2 -5 K) than that of the latter [367]. $\text{Gd}_5\text{Ge}_2\text{Si}_2$ shows $\Delta S_M = -18.5 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 15$ K, both at 276 K at 5 T. MCE was evidently enhanced in Ge-enriched samples but their transition temperatures shift to lower values: $\text{Gd}_5(\text{Ge}_{1-x}\text{Si}_x)_4$ shows, at 5 T, $|\Delta S_M| \sim 70 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C \sim 150$ K, $RC_{FWHM} \sim 700 \text{ J/kg}$ for $x = 0.25$ while for $x = 0.5$ $|\Delta S_M| \sim 20 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C \sim 300$ K and $RC_{FWHM} \sim 500 \text{ J/kg}$ [365].

Theoretical models have been developed to describe the barocaloric and magnetocaloric effects of this compound. Melikhov et al. [368] used a phenomenological model, based on Jiles-Atherton theory (FM) and Curie Weiss law (PM), to describe its FOPT.

Alvaranega et al. [369] considered the magnetic field dependence of the entropy of phonons in a Hamiltonian model and include Zeeman, magnetoelastic and exchange interactions. Pires et al. reviewed the exotic Griffiths-like phase appearing in $\text{RE}_5(\text{Ge}_{1-x}\text{Si}_x)_4$ compounds [370] using $\text{RE} = \text{Gd}$ and Tb as examples. Von Moos et al. [371] developed a non-equilibrium Preisach-type model to study the hysteresis in $\text{Gd}_5\text{Ge}_2\text{Si}_2$ compound.

Figure 11 shows the ΔS_M values of this alloy family as a function of the transition temperature at $\mu_0\Delta H=5$ T. The maximum values considered in the figure are $\sim 40 \text{ J kg}^{-1} \text{ K}^{-1}$. Although values as high as $70 \text{ J kg}^{-1} \text{ K}^{-1}$ were reported in earlier publications, they were just observed at sharp peaks related to the inappropriate use of Maxwell relation for ΔS_M determination in these FOPT materials in the region of phase coexistence[372-375].

Compositional effects have been reported for $\text{Gd}_5\text{Si}_{1.985}\text{Ge}_{1.985}\text{X}_{0.03}$, whereby $\text{X}=\text{Fe}$, Ni , Co , Cu , Al , Ga or C , [376] and $\text{Gd}_5\text{Si}_2\text{Ge}_{1.9}\text{X}_{0.1}$, whereby $\text{X}=\text{Co}$, Cu , Ga , Mn , Al , Bi and Sn [377]. In general, their GMCE are eliminated and Curie temperatures are shifted to higher values (~ 300 K), except for minor amount of Ga substitution in which the GMCE is preserved till further increased Ga content whereby the former effects are resulted [376, 378]. Analogously to Ga , small substitutions of Sn [379] retain FOPT and enhance both Curie temperature and MCE ($\text{Gd}_5\text{Si}_{1.935}\text{Ge}_{2.035}\text{Sn}_{0.03}$ exhibits a $T_C=277$ K and a ΔS_M which is twice as large as that of $\text{Gd}_5\text{Si}_{1.95}\text{Ge}_{2.05}$ whose $T_C=260$ K). Minor Pb substitutions for Ge and Si in $\text{Gd}_5\text{Si}_2\text{Ge}_2$, also preserve FOPT and thus GMCE [380]. Regarding hysteresis losses, they reduce significantly upon 0.1 addition per formula unit of Cu , Ga , Mn and Al [377] as previously observed for Fe [28]. However, 0.1 addition of Sn or Bi per formula unit has a negligible effect on MCE and the associated hysteresis losses. Single phase was observed after substituting these two elements, unlike substituting other elements that form multiphase systems [377]. However, Li et al. succeeded in enhancing the MCE of $\text{Gd}_5\text{Si}_{1.95-x}\text{Ge}_{2.05-x}\text{Sn}_{2x}$ ($x\leq 0.05$) [381] and this is also confirmed by other studies [382]. It has been shown that little Mn content in $\text{Gd}_5\text{Si}_{2.05-x}\text{Ge}_{1.95-x}\text{Mn}_{2x}$ ($2x=0, 0.03$ and 0.08) preserves monoclinic structure (with some 5:3 type impurity phase) and GMCE [383]. Similar effects are obtained with In substitution [384]. Partial substitutions of Nb in $\text{Gd}_5\text{Si}_{2-x}\text{Ge}_{2-x}\text{Nb}_{2x}$ alloys [385] are

reported to beneficially increase Curie temperatures and MCE as x increases up to $x=0.05$ ($\Delta S_M = -9.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C \sim 295 \text{ K}$ and $\mu_0 \Delta H = 2 \text{ T}$, which is a $\sim 50 \%$ enhancement compared to Nb-free alloy), however, a low amount of detrimental Gd_5Si_3 phase has been found.

Fe substitution has been shown to be more effective in reducing hysteresis losses upon substitution for Ge than for Si in $\text{Gd}_5\text{Ge}_2\text{Si}_2$, being almost zero for $\text{Gd}_5\text{Ge}_{1.75}\text{Si}_2\text{Fe}_{0.25}$ [386]. Further increased amount of Fe substitution leads to destabilization of 5:4 stoichiometry and the appearance of 5:3 structures, which deteriorate MCE [387].

Complete substitution of Sb for Ge in $\text{Gd}_5\text{Si}_2\text{Sb}_2$ [388] leads to the formation of a ferromagnet, crystallizing in Sm_5Ge_4 -type structure with $T_C = 240 \text{ K}$, and with moderate MCE of $|\Delta S_M| < 5 \text{ J kg}^{-1} \text{ K}^{-1}$ even at 7 T . One advantage in this material is the almost flat shape of the $\Delta S_M(T)$ curve, which could indicate a high RC but the reported data prevents further estimation. Similarly, in $\text{Gd}_5\text{Si}_{2.75}\text{P}_{1.25}$ alloy, orthorhombic Sm_5Ge_4 -type structure is formed with a Curie transition at 177 K and $\Delta S_M = -7.8 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 5 \text{ T}$ [389]. Chernyshov et al. [390] studied $\text{Gd}_5\text{Ge}_{4-x}\text{Sb}_x$ system and found the formation of $\text{Tm}_5\text{Sb}_2\text{Si}_2$ -type phase (Cmca space group) for high Sb content but Sm_5Ge_4 -type structure for $x=0.7$, and 1, exhibiting FOPT and GMCE at $\sim 40 \text{ K}$.

The additions of O to $\text{Gd}_5\text{Si}_{1.95}\text{Ge}_{2.05}$ deteriorate MCE in various ways [391]. Low oxygen content leads to the stabilization of orthorhombic phase and heterogeneous samples show a reduction of half of the MCE values, although having flat curves ($\Delta S_M \sim 7 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\text{RC}_{\text{FWHM}} > 420 \text{ J/kg}$ at $\mu_0 \Delta H = 5 \text{ T}$). At higher oxygen content, the alloy decomposes into 1:1 (CrB-type structure) (80-90 %) and 5:3 (Mn_5Si_3 -type structure) stoichiometries ($\sim 10 \%$). Hydrogen additions also stabilize the orthorhombic phase but loses the GMCE even with its FOPT preserved after 0.1 atoms of H per

formula unit [392]. The unit cell volume increases and T_C decreases as the Gd-Gd distance increases with the increase of H content [393].

Both purity and homogeneity play crucial roles in MCE [394]. The presence of untransformed orthorhombic phase and impurity phases deteriorates MCE, particularly, AFM hexagonal Gd_5Si_3 phase. Annealing at high temperatures leads to an evident enhancement of ΔS_M and ΔT_{ad} compared to those of arc-melted samples without post-annealing. This has been ascribed to the rearrangement of Ge and Si in their corresponding atomic sites of the monoclinic structure as well as the removal of retained polymorphic orthorhombic phase [395]. Generally, for rapid quenched systems, a higher cooling rate from the melt also enhances MCE [396, 397]. Moreover, it is particularly worth mentioning that it is more beneficial to use highly pure Gd instead of commercial one [398, 399] as the former leads to a clear enhancement of the MCE in $Gd_5Si_2Ge_2$ compound [395].

Several studies have been devoted to partial or complete substitution of other RE for Gd. Concerning the binary alloys, RE_5Si_4 (RE=Gd, Tb, Dy, Ho, Er) compounds are FM systems with Curie temperatures ranging from 336 K (for Gd) to 25 K (for Er) [400]. On the other hand, RE_5Ge_4 compounds are AFM systems with Néel temperatures ranging from 47 K (for Gd) to 25 K (for Er). The structure of $RE_5Si_{4-x}Ge_x$ systems depends on the Ge content: for high Ge content, orthorhombic structure exists from $x=4$ to $x \sim 1.3, 2.6, 3.2$ and 2.9 when RE=Dy, Tb, Gd and Pr respectively; monoclinic structure exists from the former values down to $x \sim 0.75, 1.3, 2.0$ and 1.5 for RE=Dy, Tb, Gd and Pr respectively. Finally, orthorhombic structure is observed for RE=Dy, Tb, and Gd; and tetragonal Zr_5Si_4 -type structure ($P4_12_12$ space group) for Pr [401]. In the case of RE=Nd, Magen et al. [402] stabilized monoclinic structure for $x=1.6$ and Yang et al. [403] reported that as-cast samples, with $x=1.2$ and 2 , crystallize into a mixture of

monoclinic and orthorhombic structures but crystallize into tetragonal Zr_5Si_4 -type structure after annealing at 1273 K. For $\text{RE}=\text{Sm}$, the formation of orthorhombic phases is reported for Si- and Ge-rich compositions while that of monoclinic phase is reported for $\text{Sm}_5\text{Si}_2\text{Ge}_2$ compound [404].

Substitutions of Tb for Gd has been studied in $\text{Gd}_x\text{Tb}_{5-x}\text{Si}_4$ in the complete range $0 \leq x \leq 5$ [400]. The volume of the unit cell linearly decreases (and density increases) with increasing Tb content while T_C linearly changes with Tb content from 346 K for Gd_5Si_4 to 223 K for Tb_5Si_4 . On the other hand, ΔS_M does not significantly change (slightly enhanced from -8 to $-9 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$ as T_C decreases) which is ascribed to exhibiting a SOPT. Morellón et al. [405] reported higher absolute values of ΔS_M of $\text{Tb}_5\text{Si}_x\text{Ge}_{4-x}$ ($x=0, 2$ and 4) for $x=4$, whereby the alloy exhibits a SOPT. In the case of $x=0$, Tb_5Ge_4 crystallizes in orthorhombic Pnma space group, and exhibits a complex canted AFM structure below 91 K and two peaks in its $\Delta S_M(T)$ curves, at 60 K and 95 K ($-3.7 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$). Similarly to Gd containing intermetallic, $\text{Tb}_5\text{Si}_2\text{Ge}_2$ crystallizes in monoclinic $\text{P}112_1/a$ space group and shows GMCE, which is ascribed to a magnetostructural FOPT with $\Delta S_M=-21.8 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\sim 110 \text{ K}$ and $\mu_0\Delta H=5 \text{ T}$. Thuy et al. [406] found that $\text{Tb}_5\text{Si}_3\text{Ge}_1$ compound exhibits monoclinic phase with a transition temperature of 215 K and $\sim 50\%$ decrease in $|\Delta S_M|$ with respect to that of $\text{Tb}_5\text{Si}_2\text{Ge}_2$. Lower values of $|\Delta S_M|$ reported by Thuy et al. with respect to those values reported by Morellón et al. could be due to the absence of post-annealing in Thuy's samples leading to the presence of tetragonal $\text{Tb}(\text{Si},\text{Ge})$ impurity phase. Small partial substitution of Tb for Gd in $\text{Gd}_{5-x}\text{Tb}_x\text{Si}_{1.72}\text{Ge}_{2.28}$ [407] can preserve the first order magnetostructural character of the transition while T_C decreases and $|\Delta S_M|$ increases with increasing Tb content (up to $\sim 25 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=2 \text{ T}$ and $T_C \sim 220 \text{ K}$). The preservation of

magnetostructural FOPT in $\text{Gd}_{3.7}\text{Tb}_{1.3}\text{Si}_x\text{Ge}_{4-x}$ ($x=1.72, 2$ and 2.02) has been confirmed by Deng et al. [408], obtaining GMCE at the corresponding transition temperatures and further enhancing MCE after annealing. In $\text{Tb}_4\text{GdSi}_{2.035}\text{Ge}_{1.935}\text{Mn}_{0.03}$, FOPT is preserved at around 143 K ($\Delta S_M = -22.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T) but besides the formation of monoclinic phase, impurity phases such as orthorhombic phase and Gd_5Si_3 type phase are also found [409].

Xie et al. [410] studied the effect of partial substitution of Dy for Gd in the pseudo-binary compound $\text{Gd}_{5-x}\text{Dy}_x\text{Si}_4$, which crystallizes in Sm_5Ge_4 -type orthorhombic structure. A SOPT is observed and Curie temperature progressively decreases from 338 to 140 K (T_C of parent Gd_5Si_4 and Dy_5Si_4 binary compounds respectively). ΔS_M values of $\sim -2.5 \text{ J kg}^{-1} \text{ K}^{-1}$ ($\mu_0\Delta H = 0.95 \text{ T}$), very similar to that of pure Gd, are retained along the compositional series. In the pseudo-binary $\text{Dy}_5\text{Si}_{4-x}\text{Ge}_x$ compounds, the formation of monoclinic structure occurs for the compositional range around $x=1$ [411]. For $x=0.5$ and 1, two transition temperatures are observed, one at low temperature ($\sim 15 \text{ K}$ for both) and the second one is a SOPT at 132 K for $x=0.5$ but FOPT at 66 K for $x=1$ (in agreement with the presence of monoclinic structure). However, the presence of the low temperature transition reduces MCE compared to that of the compound with Gd and shows similar transition temperatures [411]: $\Delta S_M = -9.6$ and $-21 \text{ J kg}^{-1} \text{ K}^{-1}$ (-32 at the spike) for $x=0.5$ and 1 respectively at $\mu_0\Delta H = 5 \text{ T}$.

$\text{Ho}_5(\text{Si}_{0.8}\text{Ge}_{0.2})_4$ compound does not exhibit a FOPT although it crystallizes in the monoclinic structure at room temperature. This compound orders ferromagnetically at 50 K and exhibits a Griffiths phase like behavior between 50 and 100 K. Their MCE results are $\Delta S_M \sim -3 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\text{RC}_{\text{FWHM}} \sim 90 \text{ J/kg}$ at 2 T [412]. Binary Ho_5Si_4 compound undergoes a SOPT from FM to PM state at 76 K and a spin reorientation

transition at ~15 K. Their MCE results at ~76 K are $\Delta S_M = -14.8 \text{ J kg}^{-1} \text{ K}^{-1}$, $\Delta T_{ad} = 6.1 \text{ K}$ and $RC_{FWHM} \sim 435 \text{ J/kg}$ at $\mu_0 \Delta H = 5 \text{ T}$ [413].

When Er completely substitutes for Gd in $\text{Er}_5\text{Si}_x\text{Ge}_{4-x}$ [414], the crystallization of monoclinic phase occurs in the range of $3.5 \leq x \leq 3.9$. Ordering temperatures are below 30 K and moderate MCE parameters are found ($|\Delta S_M| < 12 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 5 \text{ T}$). In the case of partial substitution of Er for Gd in $\text{Er}_{5-x}\text{Gd}_x\text{Si}_4$ ($x=0.50$ and 1.25) compounds, complex magnetic behavior was found [415]. Orthorhombic Gd_5Si_4 -type structure is formed and the transition to monoclinic, found in Gd free compound at ~230 K, is suppressed. Linear increase of the three lattice parameters following Vegard law is found. Two transition temperatures are found: to AFM at relatively high temperatures (61 and 101 K for $x=0.1$ and 0.25 respectively) and to FM at low temperatures (27 and 46 K for $x=0.5$ and 1.25 respectively). ΔS_M at the high temperature transitions are ~-12 and $-8 \text{ J kg}^{-1} \text{ K}^{-1}$ for $x=0.5$ and 1.25 respectively.

Tm_5Ge_4 compound has been recently studied by Zou et al. [416] and it crystallizes in orthorhombic Sm_5Ge_4 -type structure, being AFM with two transitions at 13 and 21 K.

The effects of light RE substitution for Gd in the MCE of $\text{Gd}_5\text{Si}_{4-x}\text{Ge}_x$ alloys have also been studied. In the case of $\text{Ce}_5\text{Si}_{4-x}\text{Ge}_x$ [417], its crystal structure is tetragonal Zr_5Si_4 -type (space group $P4_12_12$) in the range of $0 \leq x < 2.15$, monoclinic $\text{Gd}_5\text{Si}_2\text{Ge}_2$ -type structure at $x \sim 2.2$ and orthorhombic Sm_5Ge_4 -type structure in the range of $2.4 < x \leq 4$.

Unlike Gd compounds, the Ce_5Si_4 compound is AFM and the Ce_5Ge_4 compound, FM. The latter exhibits the maximum magnetic entropy change of the series ($\Delta S_M = -11.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C = 11.5 \text{ K}$ and 5 T).

$\text{Gd}_{5-x}\text{Pr}_x\text{Si}_4$ ($x=0, 0.5, 0.75$ and 1.0) [400] crystallizes in Gd_5Si_4 -type orthorhombic structure and has a SOPT at the T_C . Their Curie temperatures, magnetic moments and

$|\Delta S_M|$ decrease with increasing Pr content (T_C reduces from ~320 to ~290 K, magnetic moments decrease from 7.85 to 7.00 μ_B and $|\Delta S_M|$ lowers from 8.2 to 6.1 $\text{J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5$ T as x increases from 0.5 to 1).

While Nd_5Si_4 and Nd_5Ge_4 crystallize in tetragonal Zr_5Si_4 -type and Sm_5Ge_4 -type structures respectively, monoclinic $\text{Gd}_5\text{Si}_2\text{Ge}_2$ -type structure can be stabilized in $\text{Nd}_5\text{Si}_2\text{Ge}_2$ compound although phase coexistence occurs. The maximum $|\Delta S_M|$ value of the series is obtained at the FiM transition temperature of Ge-free compound, 71 K. However, recent results on partial substitution of Nd for Gd in $\text{Gd}_{5-x}\text{Nd}_x\text{Si}_2\text{Ge}_2$ ($x=0.05$, 0.1 and 0.2) series, show that monoclinic phase only crystallizes for $x=0.05$ whereas the other two compounds crystallize in Gd_5Si_4 -type phase, changing from FOPT to SOPT for $x\geq 0.1$ [418]. Although transition temperature can be enhanced in orthorhombic $\text{Nd}_5\text{Si}_3\text{Ge}$ up to 124 K, ~23 % decrease in $|\Delta S_M|$ is observed [411].

Partial substitution of Y for Gd in $\text{Gd}_4\text{Y}_1\text{Si}_{1.9}\text{Ge}_{2.1}$ [419] preserves the monoclinic structure and reduces its ordering temperature to 149 K although the magnetic moment per Gd atom is enhanced to 7.87 μ_B compared to Y-free alloys. A FOPT from FM to PM state occurs at this temperature accompanied by a GMCE ($\Delta S_M=-28$ $\text{J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5$ T).

Small partial substitution of Zr for Gd in $\text{Gd}_{5-x}\text{Zr}_x\text{Si}_2\text{Ge}_2$ alloys was proposed to reduce the amount of detrimental AFM hexagonal Gd_5Si_3 -type phase [420]. The amount of desired 5:4 type phase enhances with increasing Zr concentration, from 70 % ($x=0$) to 85 % ($x=0.05$). However, higher Zr additions led to a reduced fraction of 5:4 phase. Moreover, the presence of Zr changes 5:4 monoclinic structure (in Zr-free alloy) into a 5:4 orthorhombic one, thus converting FOPT to SOPT. Nevertheless, although $|\Delta S_M|$ decreases from 7.8 to 5.5 $\text{J kg}^{-1}\text{K}^{-1}$ (2 T) from $x=0$ to 0.05, a three-fold enhancement of

RC_{FWHM} is attained (from ~50 to above 150 J/kg). For Zr-containing compounds, their Curie temperatures monotonically decrease from 292 to 272 K from $x=0.05$ to $x=0.20$.

The MCE of Re_5Sn_4 ($Re=Tb, Dy, Ho, Er$ [421]) has been also studied and the alloys exhibit low transition temperatures (22, 15 and 8 K for the AFM-PM transition of the alloys with Dy, Ho and Er respectively). When $Re=Tb$ in Re_5Sn_4 , two magnetic transitions occur at 54 and 84 K ascribed to spin reorientation and FM-PM respectively. Their $\Delta S_M(T)$ curves show an inverse MCE behavior for all the studied samples at low temperatures and a direct MCE at higher temperatures. Their major phase adopts the Sm_5Ge_4 -type structure (with an enhanced phase fraction after annealing) but up to 18 wt. % of Re_5Sn_3 and $Re_{11}Sn_{10}$ impurity phases were detected.

Application of pressure decreases the MCE of $Gd_5Si_2Ge_2$ and shifts their transition temperatures to higher values [422]. However, a significant broadening is reported, which resulted in >50% enhancement in RC for $Gd_5Ge_2Si_2$ prepared under the pressure application of 4.22 kbar [423]. For Tb_5Si_4 compound, the pressure application enhances its MCE [424] transforming from conventional MCE to a GMCE. However, the application of pressure induced a mixed phase in Gd_5Ge_4 compound [425], whereby its FM phase increases with pressure, resulting in higher transition temperatures. This is due to the progressive decrease of FOPT (AFM-FM) contribution to MCE while the contribution of SOPT (FM-PM) to MCE progressively increases.

Thin films of $Gd_5Si_{1.3}Ge_{2.7}$ show a broader FOPT ($|\Delta S_M|=8.83 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T), which reduces its magnetic hysteresis [426], while annealing significantly decreases its MCE response [427]. For multilayer systems [428], about 10% reduction of the onset field that is necessary to launch the metamagnetic transition in $Gd_5Si_{1.5}Ge_{2.5}$ was attained by evaporating thin layers of pure Fe (0.1-0.2 μm) on both polished faces of this alloy. Yue

et al. [429] produced three component layers of $\text{Gd}_5\text{Si}_2\text{Ge}_2/\text{Gd}/\text{Gd}_5\text{Si}_{1.85}\text{Ge}_{2.15}$ and achieved a constant adiabatic temperature change in a wide temperature range from 240 to 310 K, which indicates that optimization could be attained by tailoring layer thickness. Thin films of Gd_5Si_4 (350 nm) protected with Pt developed by Hadimani et al. [430] show multiphase character (including Gd_5Si_4 and Gd_5Si_3) and their transitions lay close to those of the bulk system, which is not the case for ball milled powders despite same composition [431].

Ball milling of $\text{Gd}_5\text{Si}_2\text{Ge}_2$ -type powder leads to a large decrease in $|\Delta S_M|$, MCE peak temperature and a broadening of the transition [432, 433]. This is characteristic of the MCE of powdered samples produced after high energy milling. Trevizoli et al. [434] reported that pulverized bulk samples prior to compaction show that only slight changes to their MCE were observed with respect to those of the bulk counterparts. Short millings lead to various effects in $\text{RE}_5\text{Si}_{1.3}\text{Ge}_{2.7}$ systems, depending on $\text{RE}=\text{Gd}$ or Tb . A $\sim 35\%$ decrease in the MCE of $\text{RE}_5\text{Si}_{1.3}\text{Ge}_{2.7}$ is observed for $\text{RE}=\text{Gd}$, while $\sim 23\%$ enhancement in MCE is obtained when $\text{RE}=\text{Tb}$ [435].

New applications are also explored by embedding $\text{Gd}_5\text{Si}_2\text{Ge}_2$ particles in a piezoelectric polymer [436]. Upon applying a magnetic field, an electrical voltage is induced as a consequence of the strain induced due to the mechanical interaction between the magnetic and the piezoelectric phases.

5.1.1.4. ***RE-TM₂-X₂***

These intermetallic compounds crystallize in the body-centered tetragonal ThCr_2Si_2 -type structure (space group $I4/mmm$) formed by atomic layers stacked with a sequence of RE-X-TM-X-RE ($\text{X} = \text{Si, Ge or B}$ and $\text{TM} = \text{transition metal}$, being Mn the only element carrying a magnetic moment). In Mn -containing $\text{RE-TM}_2\text{-X}_2$, its magnetic

behavior is critically dependent on the Mn-Mn intralayer distance, d_{Mn-Mn} , leading to their complex magnetic phase diagrams. Therefore, different magnetic structures can be found concerning Mn-Mn interlayer (*ie*) and intralayer (*ia*) interactions [437]: I) FM exchange-coupled *ie* and AFM coupled *ia* ($d_{Mn-Mn} > 0.287$ nm for germanides) leading to a canted FM structure; II) both *ie* and *ia* AFM coupled, leading to a canted AFM structure ($0.287 \text{ nm} > d_{Mn-Mn} > 0.284$ nm for germanides); III) AFM structure without canting occurring for lower values of d_{Mn-Mn} ($d_{Mn-Mn} < 0.284$ nm for germanides). Moreover, the interactions with RE moments would lead to more complex structures. One such example could be found in Figure 1 of Ref. [437], whereby the transition concerning the ordering of RE moments appears at low temperatures, while at higher temperature multiple magnetic transitions and re-entrant ferromagnetism can be observed.

The MCE ascribed to this rich magnetic phenomenology has been studied and their reported results are shown in Figure 12 and Figure 13 ($|\Delta S_M|$ and ΔT_{ad} respectively). In general, MCE response (ΔT_{ad} and ΔS_M) decreases with increasing transition temperature and GMCE is observed below 50 K. MCE character can change from inverse to direct in a relative small temperature range as observed in $\text{Sm}_{1-x}\text{Gd}_x\text{Mn}_2\text{Ge}_2$ [438, 439] and $\text{SmMn}_2\text{Ge}_{2-y}\text{Si}_y$ although exhibiting modest results [440]. In this case, the largest MCE ($\Delta S_M = -3 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 5 \text{ T}$ for $y=2$) was obtained at the Sm ordering temperature (which increases from $\sim 40 \text{ K}$ to $> 100 \text{ K}$ from $y=2$ to 0). Furthermore, the application of pressure has a considerable effect tuning the transition temperatures of these systems and also increasing the stability range of the AFM order [441]. In addition, MCE ascribed to the low temperature transitions enhances with increasing pressure through the contraction of lattice parameters but no results were reported for the transitions occurring at room temperature.

$\text{La}_{0.5}\text{Pr}_{0.5}\text{Mn}_2\text{Si}_2$ compound [442] exhibits a crossover from inverse to direct MCE close to room temperature at ~ 250 K, with $|\Delta S_M| \sim 2 \text{ J kg}^{-1} \text{ K}^{-1}$ at 220 K at $\mu_0 \Delta H = 5$ T, while values above 280 K are not reported. Similar MCE curves are found for $\text{La}_{1-x}\text{Ce}_x\text{Mn}_2\text{Si}_2$ [443] but explored in a wider temperature range. Their inverse MCE at low temperatures changes into direct MCE at room temperature, with $\Delta S_M \sim -3 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 7$ T.

Fe substituted LaMn_2Ge_2 alloys are reported with $\Delta S_M = -1 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 1.8$ T around room temperature and their transition temperatures decreasing with substituting Fe [444]. This is also included in the review paper by Brück et al. [445].

GMCE has been observed for $\text{NdMn}_2\text{Ge}_{0.4}\text{Si}_{1.6}$ ($\Delta S_M = -15 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 5$ T) associated to a FOPT from AFM to FM state at 36 K [446]. $\text{PrMn}_{0.4}\text{Fe}_{0.6}\text{Ge}_2$ also shows a large MCE of $\Delta S_M \sim -11 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T for the ordering temperature of Pr at ~ 25 K, whereas a modest $|\Delta S_M|$ of $\sim 1 \text{ J kg}^{-1} \text{ K}^{-1}$ (inverse MCE) was measured at the Néel transition at ~ 140 K (ascribed to interlayer Mn interaction) and at Curie transition (~ 165 K) with a crossover at 160 K [447]. MCE has been also studied for $\text{Nd}(\text{Mn}_{1-x}\text{Fe}_x)_2\text{Ge}_2$ ($x=0.1$ and 0.2) compounds [448]. For $x=0.1$, a weak transition at 33 K (Nd sublattice), a spin reorientation at 188 K and a Curie temperature of 290 K are observed. For $x=0.2$, the compound exhibits a similar behavior as SmMn_2Ge_2 , i.e. four magnetic transitions: Besides a weak ordering transition at ~ 33 K for the Nd sublattice, a FM behavior is also observed at low temperatures up to 95 K, which becomes AFM at temperatures up to 185 K. FM appears again at higher temperatures up to the Curie temperature (235 K). Modest MCE parameters are obtained: $\Delta S_M = -2.35$ and $-1.84 \text{ J kg}^{-1} \text{ K}^{-1}$ (both at $\mu_0 \Delta H = 5$ T) for $x=0.1$ and 0.2 respectively. Similarly, for $x=0$, the compound presents a spin reorientation at 208 K and a Curie temperature at 328 K with $\Delta S_M = -1.10 \text{ J kg}^{-1} \text{ K}^{-1}$ at

$\mu_0\Delta H=2$ T [449]. Partial Cu substitution for Mn in $\text{NdMn}_{2-x}\text{Cu}_x\text{Si}_2$ ($x\leq 1$) yields higher FM ordering temperature of the Nd-sublattice (from 36 to 100 K), changing from FOPT to SOPT for $x\geq 0.6$, with a large reduction of MCE: $\Delta S_M=-27$ J kg⁻¹K⁻¹ ($x=0$) to 0.5 J kg⁻¹K⁻¹ ($x=1$), both at 5 T [450]. On the other hand, $\text{NdMn}_{1.7}\text{Cr}_{0.3}\text{Si}_2$ presents three magnetic transitions at 380 K (from PM to intralayer AFM), 320 K (to canted AFM) and 42 K (FOPT to canted FM). At this temperature, 42 K, it exhibits $-\Delta S_M=15.9$ J kg⁻¹K⁻¹ and $\Delta T_{ad}=5$ K, both at 5 T [451]. $\text{Nd}_{0.2}\text{Gd}_{0.8}\text{Mn}_2\text{Ge}_2$ compound exhibits a transition from FiM to AFM at 262 K due to lattice contraction [452] and retained its AFM state down to 142 K. Moreover, a compensation temperature is observed at 82 K while an inverse MCE ($\Delta S_M=+0.68$ J kg⁻¹K⁻¹ at $\mu_0\Delta H=1$ T) is observed around 262 K and a direct (and larger) MCE around 142 K ($\Delta S_M=-0.81$ J kg⁻¹K⁻¹ at $\mu_0\Delta H=1$ T). TmMn_2Si_2 shows a large reversible MCE ($\Delta T_{ad}=10.1$ K, $|\Delta S_M|=22.7$ J kg⁻¹K⁻¹ and $\text{RC}_{\text{FWHM}}=250$ J/kg at 5 T) ascribed to a SOPT at 5.5 K [453]. In the case of TbMn_2Si_2 compound [454], two magnetic transitions are observed at higher temperatures (50 and 64 K respectively). While the former transition remains independent of the applied field, the latter shifts to higher temperatures. An estimated value of $\Delta T_{ad}=7.5$ K was obtained using the specific heat at zero-field. Moreover, it exhibits $\Delta S_M=-16$ J kg⁻¹K⁻¹ and $\text{RC}_{\text{FWHM}}=396$ J/kg at 5 T due to the two successive FOPTs. Results are in agreement with those of Maji et al. [455], who also explain the significant exchange bias field observed at 5 K due to the presence of AFM and FM domains. Compounds with ThCr_2Si_2 -type structure but without Mn content have also been studied: e. g. ErRu_2Si_2 (GMCE of $\Delta S_M=-20$ J kg⁻¹K⁻¹ at $\mu_0\Delta H=7$ T has been found at very low transition temperature of 5.5 K [456] associated to the field induced metamagnetic transition from AFM to FM state), and GdRu_2Si_2 ($\Delta S_M\sim -7$ J kg⁻¹K⁻¹ at $\mu_0\Delta H=7$ T and its transition temperature at ~ 45 K) [457].

Li and Nishimura [458] studied MCE of different RE-(Ni,Co)₂B₂(C) borides, which also exhibit I/4mmm space group with LuNi₂B₂C-type structure. Some of these intermetallics exhibit a single second order FM to PM transition at low temperatures: e.g. PrCo₂B₂ ($\Delta S_M = -4.9 \text{ J kg}^{-1} \text{ K}^{-1}$ (2 T) at $T_C = 18 \text{ K}$ [459]) and NdCo₂B₂ ($\Delta S_M = -4.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (2 T) at $T_C = 27 \text{ K}$ [460]). However, for Dy_{0.9}Tm_{0.1}Ni₂B₂C [461], which is a superconductor below 4.5 K, it exhibits GMCE ascribed to field induced metamagnetic transition from AFM to FM state at $T \sim 9.2 \text{ K}$ with $\Delta S_M = -19.1 \text{ J kg}^{-1} \text{ K}^{-1}$ (7 T). The same research group [462] has reported similar GMCE values for GdCo₂B₂, with $\Delta S_M = -21.5 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 18.9 \text{ K}$, both at $\sim 25 \text{ K}$ and $\mu_0 \Delta H = 7 \text{ T}$. Fe substitution in ErNi_{2-x}Fe_xB₂C lowers the magnetic transition and reduces the hysteresis with respect to $x=0$. In this series, inverse MCE is observed at low fields (ascribed to AFM behavior) but changes to direct MCE at higher fields. The ΔS_M reported for $x=0, 0.1$ and 0.2 are -14.5, 12.7 and $10.6 \text{ J kg}^{-1} \text{ K}^{-1}$ (7 T) respectively [463]. Besides ThCr₂Si₂-type and CeMg₂Si₂-type (P4/mmm space group) phases, a new superstructure was observed for NdMn_xCr_{2-x}Si₂C series [464], which shows a $\Delta S_M = -26.0 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T for $x=1.7$. This corresponds to a 63% enhancement of MCE with respect to the interstitial-free composition described above [451].

EuFe₂As₂ exhibits a giant reversible anisotropic magnetocaloric response with a small hysteresis for a FOPT with $|\Delta S_M| = 14.6$ and $10.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 20 K and 5 T for in- and out-of-plane directions of the applied magnetic field respectively [465]. DyNi_{2-x}TM_xB₂C (TM=Co and Cr; $x=0.1$ and 0.2) shows that substitutions of Cr or Co for Ni decrease transition temperature and reduce magnetic hysteresis. Its AFM state at low temperature leads to an inverse MCE, which changes to direct MCE at high fields $> 2 \text{ T}$ due to a field induced metamagnetic FOPT. Their $|\Delta S_M|$ values decrease with increasing x [466]. In addition, RECu₂X₂ (RE=Ho, Er; X=Si, Ge) systems with AFM-PM

transition were also studied: e.g. ErCu_2Si_2 ($-\Delta S_M = 22.8 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2.5 K and 5 T [467]) and HoRu_2Si_2 ($-\Delta S_M = 9.11 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T and 19 K associated to a spin-flip metamagnetic transition [468]). Paramanik et al. [469] have recently proposed a method to generate magnetic field/temperature phase diagram for the latter compound based on the MCE response at the transitions.

5.1.1.5. *Ferromagnetic Lanthanum Manganites*

Manganites for magnetocaloric purposes are, in general, FM perovskite manganites, with the formula $\text{La}_{1-x-y}\text{RE}_x\text{A}_y\text{Mn}_{1-z}\text{TM}_z\text{O}_3$ (where RE is a rare earth which partially or totally substitutes lanthanum, A is an element of IA or IIA group, with $y \sim 0.3$ and TM is a transition element, which partially substitutes Mn). Although the parent LaMnO_3 compound is semiconducting and orders antiferromagnetically at 150 K, ferromagnetism is observed after the formation of mixed valence in Mn ions via double exchange mechanism between Mn^{+4} and Mn^{+3} . Mixed valence can be achieved by several ways, such as partial substitution of a non-trivalent ion for rare earth, excess of oxygen [470] or La deficiency [471]. However, the magnetic phase diagrams of these lanthanum manganites as a function of the composition are relatively complicated. Taking $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ as an example (found in Ref. [472]), at low temperatures, this system shows several states as x increases from 0 to 1: AFM with A-type phase (antiparallel planes), FM, again A-type AFM, AFM but C-type (antiparallel alignment of chains) and finally G-type AFM (antiparallel alignment of all magnetic neighbors). In $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ series, upon cooling down from PM phase, a single SOPT to an ordered magnetic structure is attained, except for about $0.5 < x < 0.65$, whereby SOPT from PM to FM phase occurs prior to a FOPT from FM to AFM phase. For about $0.8 < x < 0.95$, the magnetic transition is coupled to a structural transition and thus FOPT occurs. In addition, as x increases from 0 to 1, the structures of PM phases also change from

orthorhombic (two phases with space group Pbnm structures) to rhombohedral (R-3c), tetragonal (I4/mcm) and cubic (Pm-3m) [472]. Moreover, for about $0.5 < x < 0.65$, the FOPT from FM to AFM transition also implies a change from tetragonal to another orthorhombic structure (Fmmm).

Ferromagnetism is favored in metallic manganites due to the enhancement of double exchange interactions by electron transfer. However, the hopping integral gets null due to the antiparallel spins between neighboring Mn and thus AFM manganites are generally insulator systems [473]. Moreover, there are many types of magnetism observed for manganites including AFM insulators, FM insulators, AFM metals, glassy insulators, more complex canted magnetic structures and mixed phase states. The small differences among the possible ground states, which are due to competitions among different interactions, are the responsible for this wide variety of phenomenologies. These interactions are listed in Ref. [470]: crystal field splitting between e_g and t_{2g} electrons, kinetic energy of e_g electrons, Hund coupling between e_g and t_{2g} electrons, Jahn-Teller character of Mn^{+3} and Heisenberg magnetic coupling between nearest neighbors.

In the case of FM manganites, tailoring of Curie temperature in a wide range from ~ 150 to ~ 375 K can be obtained either upon substitution of a divalent ion (Ca^{2+} , Ba^{2+} , Sr^{2+} , Pb^{2+} ...) or a monovalent ion (Na^{1+} , K^{1+} , Ag^{1+} , Li^{1+} ,...) for La^{3+} (see Table S5 of the Supplementary Material and e.g. [474] and references therein) or by an excess of oxygen [475]. In addition, the ground state of the manganites are affected by the average radii of the cations at the different sites. This can be tuned by partial substitution of La^{3+} by a trivalent rare earth, Y^{3+} or Bi^{3+} , or in La-free Pr or Nd manganites [474]. Figure 14 shows the Curie temperatures of different manganites (data taken from [474]) as a function of the tolerance factor t , which is defined as:

$$t = \frac{\langle R_{La} \rangle + R_O}{\sqrt{2}(\langle R_{Mn} \rangle + R_O)}$$

where $\langle R_{La} \rangle$ is the average radii of La-site atoms, $\langle R_{Mn} \rangle$ is the average radii of Mn-site atoms and R_O is the radius of the O^{2-} . A correlation factor $r=0.64$ is found, which is highly significant using statistical criteria for ~100 points. Moreover, deviations from the nominal stoichiometry in the experimental samples must also affect the data dispersion (e.g. charged vacancies will affect Mn^{+4} concentration).

Another possible method to tune T_C is via substitution of Mn by other element, generally TM but semimetals or metalloids can also be used. This generally leads to a deterioration of FM character as FM double exchange mechanism in $Mn^{3+}-O^{2-}-Mn^{4+}$ groups competes with AFM superexchange mechanism when Mn is substituted by other transition metal. Figure 15 shows the Curie temperatures of (La-A)-(Mn-TM)- O_3 as a function of TM content (where A is a divalent ion). Regarding TM substitutions for Mn in manganites, the tunable Curie temperature per 1 at. % substitution of TM is presented in Figure 16. A similar T_C dependence was found by Ghosh et al. after substituting 5 % of TM (from Cr to Zn) [476]. Conversely, the results from Nisha et al. [477] using V substitution for Mn differ, showing an opposite behavior of increasing T_C with V content. However, the latter study reported multiphase character of their samples.

Another exception to the generic rule of lowered Curie temperature with TM substitutions for Mn is found in $La_{0.7}Te_{0.3}Mn_{1-x}Co_xO_3$ series, in which Curie temperature increases from 160 to 220 K as x increases from 0 to 0.5 [478].

The first MCE studies of manganites appeared in 1996 [479-481]. Figure 17 shows ΔS_M and RC_{FWHM} (1 T) estimated from the data supplied by Phan and Yu [474]. No significant correlation is found with the tolerance factor and thus with the Curie temperature. However, it is worth noticing that off-stoichiometric systems should affect

the magnetic properties and could explain the dispersion of the data. Several factors apart from compositional changes can affect MCE: grain size [482-485], chemical order [486-489], sintering temperature [490-492] or pressure application [493-495]. For some compositions, the magnetic entropy change ascribed to the charge-ordered metamagnetic transition is also studied [496, 497]. Field dependence of magnetocaloric properties in manganites have been studied [148, 498-500] and SOPT systems exhibit the predicted power law dependence with the magnetic field change for ΔS_M and RC_{FWHM} , as shown in Figure 18 using the data reported by [480]. Tricritical point is claimed as the explanation of the critical exponents found in $La_{0.7}Ca_{0.3}Mn_{0.91}Ni_{0.09}O_3$ [501] ($\Delta S_M = -7.1 \text{ J kg}^{-1} \text{ K}^{-1}$ at 199.4 K and 4 T). Using the scaling law for the field dependence of ΔS_M , a very low exponent $n=0.55$ is obtained and indicates the tricritical behavior. Recently, Abassi et al. [502] considered a phenomenological description of temperature dependence of magnetization for $La_{0.7}Ba_{0.1}Ca_{0.23}MnO_3$, which could nicely describe its MCE at SOPT. Phase coexistence (both orthorhombic phases with Pnma and Fmmm space groups) in $La_{0.5}Ca_{0.5-x}Sr_xMnO_3$ series ($0 \leq x \leq 0.5$) was studied using the field dependence of MCE [503]. The correlation of magnetocaloric and transport properties have also been pointed in doped manganites [504].

Higher $|\Delta S_M|$ values were reported for samples exhibiting a FOPT (e.g. $5.27 \text{ J kg}^{-1} \text{ K}^{-1}$ for 1.5 T in $La_{0.7}Ca_{0.3}MnO_3$ [505]). Recently, Bez et al. [506] tried to clarify the transformation character of $La_{0.67}Ca_{0.33}MnO_3$ compound with a systematic study of the presence of hysteresis using different measurement protocols. A FOPT was found but the experimental hysteresis is extrapolated to zero for an ideal infinitely slowly heating/cooling rate. Inhomogeneities and compositional variations affecting the transition temperatures are claimed to be responsible for this behavior. The close proximity of the Curie (263 K), and metal-insulator (265 K) transitions in

$\text{La}_{0.7}\text{Ca}_{0.21}\text{Ag}_{0.09}\text{MnO}_3$ compound [507] yields a large MCE in this orthorhombic system, whereby $|\Delta S_M| = 4.8 \text{ J kg}^{-1} \text{ K}^{-1}$ for 1 T. Unfortunately, the authors did not conduct experiments to determine the order of the transitions involved. Turcaud et al. [508] compare the experimental data of $\text{La}_{1-x}\text{Ca}_x\text{MnO}_3$ series with the predictions from Bean-Rodbell model, to quantify the effects of distributed transition temperatures and other inhomogeneities on the entropy change. A study of Ca deficiency in $\text{La}_{0.8}\text{Ca}_{0.2-x}\text{MnO}_3$ shows a transition from orthorhombic Pnma to rhombohedral R-3c structure, with increase of the unit cell volume and the transition temperature from 240 K ($x=0$) to 260 K ($x=0.2$) without ΔS_M significantly affected [509]. Quenched disorder yields changes from FOPT to SOPT in the so-called rounding process. This was observed for $\text{La}_{0.7}\text{Ca}_{0.3-x}\text{Ba}_x\text{MnO}_3$ series [510] with FOPT for $x < 0.05$ while SOPT for $x > 0.075$. Within these values, FOPT also changes to SOPT at magnetic fields above 4 T.

Low dimensionality manganites, such as nanoparticles and nanotubes produced by sol-gel technique [511-515], have recently received attention due to their broader MCE peaks. Their grain size reduction suppresses the long range order and instead aids the formation of metastable phases [516]. Das et al. [517] have recently transformed AFM bulk $\text{La}_{0.48}\text{Ca}_{0.52}\text{MnO}_3$ to a FM system, with an enhanced MCE response upon reducing the particle size to nanometer scale. The use of different substrates can also affect the transition temperatures in $\text{Sm}_{0.55}\text{Sr}_{0.45}\text{MnO}_3$ due to the strains in the lattice [518].

Composite systems of manganites and other oxides have also been proposed as candidates for magnetic refrigeration, with the general advantage of an extended temperature range for table-like MCE responses [519-533].

Although the studied compositions are generally (RE-A)-Mn-O₃ (A is a metal), some studies on (RE-A)₃-Mn₂-O₇ double perovskite manganites can be found [492, 534-537].

Moreover, inverse and direct MCEs have been reported in $\text{PrBaMn}_2\text{O}_6$, which exhibits a tetragonal superlattice perovskite structure [538]. Balli et al. have recently characterized the MCE properties of a $\text{La}_2\text{MnNiO}_6$ monocrystal [539], using simulations from a mean field model. $\text{LaSrMn}_{2-x}\text{Fe}_x\text{O}_5$ ($0 \leq x \leq 0.5$) series has been studied and Fe substitution decreases Curie temperature (SOPT) from 365 K ($x=0$) to 330 K ($x=0.5$) ($-\Delta S_M = 0.75 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T for $x=0.2$) [540]. In Mn- and RE-free double perovskite $\text{Ba}_2\text{CrMo}_{1-x}\text{W}_x\text{O}_6$ compound, similar values for MCE have been obtained: $\Delta S_M = -1.6 \text{ J kg}^{-1}\text{K}^{-1}$ at 285 K and $\mu_0\Delta H = 1 \text{ T}$ for $x=0.5$ [541]. Recently, the giant anisotropy of the HoMn_2O_5 aids a large $\Delta S_M = -12.4 \text{ J kg}^{-1}\text{K}^{-1}$ by simply rotating the single crystal in a field of 7 T [542].

5.1.1.6. Y-Fe alloys

Y can be found in the literature among the RE elements. Therefore, the following discussion is [1] placed in the sections concerning RE-containing alloys.

De Oliveira [543] reported the experimental results of pure Fe, Co and Ni, which reasonably fit with those of Nikitin et al. [544] and described the itinerant electron YFe_2 and YFe_3 systems. YFe_2 crystallizes in the cubic Laves phase (MgCu_2 type structure) [545] and has a magnetic ordering temperature of 540 K. Upon hydrogen additions, the cubic structure is not observed. For $\text{YFe}_2(\text{H}_{1-y}\text{D}_y)_{4.2}$, it crystallizes in a monoclinic structure below 340 K, with a strong isotope effect on its magnetic properties [546]. The MCE of $\text{YFe}_2\text{D}_{4.2}$ is $|\Delta S_M| > 10 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 5 \text{ T}$ at temperatures between 90 and 110 K, below the metamagnetic transition at 127 K. Hydrides of other Laves phases also present monoclinic structure, such as HoMn_2H_x [547] but not in the case of EuRh_2H_x [205] or YMn_2H_x and other RMn_2H_6 compounds [548].

YFe_3 crystallizes in PuNi_3 -type structure [549] (space group $R\bar{3}m$) and exhibits a Curie temperature of 570 K. The magnetization of both compounds YFe_2 and YFe_3 emerges

from Fe atoms. Slightly lower ΔS_M values are observed for the Fe-richest compound but, in both cases, are considered moderate MCE: between 0.4 and 0.5 J kg⁻¹ K⁻¹ at 1.58 T field change.

Y₂Fe₁₇ crystallizes in Th₂Ni₁₇-type structure with a Curie temperature of 295 K and $\Delta S_M \sim -5$ J kg⁻¹ K⁻¹ at $\mu_0 \Delta H = 5$ T [550]. Its melt-spun ribbons show higher Curie temperature of ~ 310 K, with $\Delta S_M = -1.6$ J kg⁻¹ K⁻¹ at $\mu_0 \Delta H = 1$ T [551]. Recently, Karpenkov et al. [552] found a relatively linear correlation between the unit cell volume and transition temperatures for these rapidly quenched systems, which follows the trend reported for interstitially doped Y₂Fe₁₇A_x (A=C, N) compounds. Moreover, their MCE seems to decrease with increasing ratio of the lattice parameter, c/a.

Wang et al. [553] reported YFe_{9.5}Mo_{2.5} alloy undergoes a SOPT, showing a $\Delta S_M = -1.09$ J kg⁻¹ K⁻¹ at 275 K and 5 T and crystallizes in tetragonal ThMn₁₂-type structure (space group I4/mmm).

5.1.1.7. *Molecular magnets*

In addition to intermetallic compounds and alloys, some other complex systems have been also studied, such as organic molecules [554-557], Gd-complexes [558-562], or rare earth containing polymers [563-566]. In the case of molecular clusters [567-587], there is a mixture of rare-earth free and rare earth containing compositions, but we prefer to combine their description in a single section. These systems are generally proposed as refrigerants suitable for very low temperatures [588] to be an alternative to the use of the scarce helium-3 [589]. Evangelisti et al. [589, 590] reviewed the magnetothermal properties of molecular magnets, including MCE. Optimization of these systems should require large spins and small anisotropies. Initially, systems based on Fe₈ and Mn₁₂ clusters (with S=10) [591] were studied but they exhibit a high

anisotropy. Nearly anisotropic Cr₇Cd system ($T_C = 2$ K) shows modest MCE due to its low $S=3/2$ [592]. Promising system should be based on Fe₁₄ clusters with $S=25$ but limited to very low temperatures ($T_C=1.87$ K) [593]. MCE of these systems are competitive with, or even better than, those observed for Laves phases suitable below 10 K. Possibilities to extend to room temperature the thermal range of applicability of molecule based systems were opened after studies on Prussian blue analogues CsNi^{II}[Cr^{III}(CN)₆]·(H₂O) (FM ordering below ~90 K) and Cr₃^{II}[Cr^{III}(CN)₆]₂·12(H₂O) (FiM ordering below ~230 K) with $|\Delta S_M|=6.6$ and $0.93 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively at 7 T [594].

5.1.1.8. *Other rare earth containing crystalline compounds*

This section will focus on those intermetallic compounds containing RE (but not belonging to any of the families described above) reported with MCE properties, and in particular, crystalline intermetallics and alloys. There is a plethora of RE-containing compounds besides the already-described families, which magnetic transitions have deserved the attention of the magnetocaloric scientific community. In order to establish a classification criteria, they will be described following a descending sequence of their RE content (at.%).

A peculiar case is the high entropy alloy only composed of five rare earth elements:

Gd₂₀Dy₂₀Er₂₀Ho₂₀Tb₂₀ [595]. The alloy forms a single crystalline phase with HCP structure. For a smaller number of elements, multiple phases are formed, suggesting that phase formation is related to entropy, rather than atomic size, enthalpy, or electronegativity. The quinary alloy has a relatively large magnetic entropy change ($8.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 186 K for an applied field of 5T) and a large refrigerant capacity ($\sim 630 \text{ J}$

kg^{-1} for that same field) due to the broader phase transition emerging from the strong chemical disorder.

Compounds with 75 at. % of RE

Gd_3TM (TM=Co, Ni [596], Rh [597] and Ru [598]), Tb_3Co [599], Dy_3Co [600], Ho_3Co [601] and Er_3Co [602] crystallize in the orthorhombic cementite-like Fe_3C -type structure. The Néel temperatures are 128 and 99 K for Gd_3Co and Gd_3Ni , with $\Delta S_M \sim -11$ and $-5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$ respectively. In the case of Gd_3Rh , its MCE peaks at 112 K ($\Delta S_M = -9.2 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$) close to the ordering temperature of, presumably, a canted AFM system [597]. For Gd_3Ru compound, its GMCE is reported with $-\Delta S_M = 30 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}} = 5.7 \text{ K}$ (5 T) at around 59 K and $-\Delta S_M$ always above $5 \text{ J kg}^{-1}\text{K}^{-1}$ in the temperature range from 40 to 90 K [598]. In the case of Tb_3Co compound, reports on its magnetic behavior differ: although two transition temperatures from FM to AFM state at 72 K and from AFM to PM state at 82 K were previously reported for Tb_3Co , Arrott plots indicate a SOPT from FM to PM state at 82 K [599] ($\Delta S_M = -18 \text{ J kg}^{-1}\text{K}^{-1}$ at 84 K and $\mu_0\Delta H=5 \text{ T}$). Dy_3Co shows two successive transformations: an AFM-AFM transition at 29 K and an AFM-PM transition at 44 K. Moreover, a field induced metamagnetic transformation occurs from AFM to FM below 44 K and its MCE changes from inverse to normal near the critical field. Its $-\Delta S_M = 13.9 \text{ J kg}^{-1}\text{K}^{-1}$ at around 44 K and $\text{RC}_{\text{FWHM}} \sim 600 \text{ J/kg}$ at 5 T are ascribed to the metamagnetic transition [600]. Rapid quenching preserves Fe_3C structure but $-\Delta S_M$ reduces to $6.5 \text{ J kg}^{-1}\text{K}^{-1}$ and RC_{FWHM} to 364 J/kg (5 T) [603]. Ho_3Co also shows two successive magnetic transitions: from PM to AFM system at 21 K ($\Delta S_M = -14.5 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T) and a spin reorientation at lower temperature of 8.5 K [601]. Er_3Co undergoes a FM transition at 14 K, with $-\Delta S_M = 17 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T [602], though a higher value of $\sim 24.5 \text{ J kg}^{-1}\text{K}^{-1}$ was reported by other

authors [604]. Figure 19 shows the influence of different types of RE in RECo_3 compounds on their ΔS_M and transition temperatures.

Compounds with 71 at. % of RE

Recent results on the MCE of compounds with the stoichiometry RE_5A_2 , leading to this atomic fraction of RE, can be found for RE_5Pd_2 (RE=Dy [605] and Ho [606]) crystallizing in Dy_5Pd_2 -type structure (space group $\text{Fd}3\text{m}$). In the case of Dy-containing compound, an inverse GMCE is observed at 3.5 K ($\Delta S_M=29 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$) and a large direct MCE at 60 K ($\Delta S_M=-8.3 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$) [605]. In the case of Ho-containing compound, a maximum $-\Delta S_M=15 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$ is observed at around 40 K. Both $|\Delta S_M|$ and the paramagnetic temperature increase with increasing Pd content in off-stoichiometric samples [606]. In partially substituted Ni for Pd samples, suppression of the MCE response resulted [607].

Compounds with 70 at. % of RE

The influence of compositional modification on the MCE of Gd_7Pd_3 alloys have been studied [608], wherein MCE decreases with decreasing crystal size [609]. They crystallize in hexagonal Th_7Fe_3 -type structure (space group $\text{P6}_3\text{mc}$). Parent Gd_7Pd_3 compound [610] orders ferromagnetically above room temperature, 323 K, with $\Delta S_M=-6.5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$. The substitution of 1/3 of Ni for Pd retains the structure and shifts T_C to 310 K, whereas its $|\Delta S_M|$ reduces to $4.8 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) [608]. Slightly higher values of $|\Delta S_M|$ were reported by Oboz et al. [611] but with a larger reduction in $T_C=298 \text{ K}$. In that study, $\Delta S_M \sim -7 \text{ J kg}^{-1}\text{K}^{-1}$ is reported at the maximum field of 7 T. These results have been recently confirmed by Talik et al. [612] and they also mention that the peak temperature of MCE decreases to 240 K for 2/3 Ni substitution for Pd. While ΔS_M is almost independent of Ni content, the FWHM significantly increases with increasing Ni

content as there is a clear second maximum in $-\Delta S_M$ for Ni-containing compounds at ~ 140 K. Partial Y substitution for Gd in $Gd_3Y_4Pd_3$ compound decreases Curie temperatures, maintaining its crystalline structure [613, 614], whereas $\Delta S_M \sim -3 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 197 K [613]. More recently, the previous study was extended to single crystals of $Gd_{7-x}Y_xPd_3$ ($x=0-6$) series [615] and reported that T_C decreases close to linearity as Y content increases, and $T_C=0$ K when full Y substitution for Pd was extrapolated. Analogously, $|\Delta S_M|$ linearly decreases with Y content from ~ 7 to $\sim 3 \text{ J kg}^{-1} \text{ K}^{-1}$ (7 T) as x increases from 0 to 6.

Gd_7Rh_3 also crystallizes in Th_7Fe_3 -type structure and exhibits a Néel temperature at 140 K with $\Delta S_M = -5.7 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H = 8$ T [616]. Amorphous Gd_7Ru_3 reported by Kumar et al. [617] shows two magnetic transitions at 58 and 34 K and a maximum $\Delta T_{ad} = 5$ K at 5 T.

Compounds with 67 at. % of RE

The RE_2In series with RE=Gd [618], Tb [619], Dy [620], Ho [621] and Er [622] crystallize in hexagonal Ni_2In -type structure (space group $P6_3/mmc$). Dy- and Er-containing compounds present SOPT from FM to PM state at 130 and 46 K and an associated MCE characterized by $\Delta S_M = -9.2$ and $-16 \text{ J kg}^{-1} \text{ K}^{-1}$, respectively, at $\mu_0\Delta H = 5$ T.

The other compounds exhibit two transformation temperatures: for RE=Ho, its spin reorientation occurs at 32 K and FM to PM at 85 K (SOPT), while for RE=Gd and Tb, their AFM to FM transformations occur at 105 and 45 K respectively, and a FM to PM (SOPT) at 190 and 165 K respectively. The MCE parameters at T_C are $\Delta S_M = -4.5$

$\text{J kg}^{-1} \text{ K}^{-1}$ at 4 T for Gd intermetallic and -6.6 and $-11 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H = 5$ T for Tb and Ho intermetallics, respectively (see Figure 20). 20 at. % substitution of Al, Ga, Sn or Pb for In was studied by Tence et al. [623]. They found the Al and Ga substitution only

affects significantly the FM-AFM transition at low temperatures. However, PM-FM transition shifts to higher temperatures after substitution of Sn or Pb. This substitution also affects the transition at lower temperatures. In the case of Al substitutions, no significant alteration to ΔS_M is observed as compared to the parent compound but Pb substitution reduces it to $-3.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 4 T.

Gd_2Al crystallizes in the Co_2Si -type structure (Pnma space group). Co additions in Gd_2AlCo_x ($x \leq 0.6$) enhance MCE and also forms Gd_3Al_2 minor phase ($\text{P4}_2\text{nm}$ space group) for $x \leq 0.2$ and $\text{Gd}_{57.5}\text{Co}_{20}\text{Al}_{22.5}$ phase (Pbam space group) with higher Co content. Its $-\Delta S_M$, at around 50 K, increases from 6.5 to $7.9 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T as x increases from 0 to 0.2 [624].

The MCE of several RE_6CoTe_2 compounds (RE from Gd to Er) [625] exhibiting Fe_2P -type structure (space group P-62m) has also been reported. All the studied phases show two transition temperatures: high temperature transitions at 220, 174, 125, 60 and 30 K, and low temperature transitions at 180, 52, 26, 22 and 14 K for Gd, Tb, Dy, Ho and Er respectively. For the Gd-containing compound, $\Delta S_M = -6.5$ and $-4.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 180 and 220 K respectively. Also crystallizing in Fe_2P -type structure, Gd_6FeBi_2 [626] shows a Curie temperature at ~ 350 K with $\Delta S_M = -4.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T.

A large reversible MCE is reported for Er_4NiCd compound ($\Delta S_M = -18.3 \text{ J kg}^{-1} \text{ K}^{-1}$ for 5 T without hysteresis loss) [627], which crystallizes in Gd_4RhIn -type structure (space group F-43m) and antiferromagnetically orders at 5.9 K. For magnetic fields larger than 0.4 T, a metamagnetic transition occurs along with a Curie temperature at 7.5 K, indicating a rather unstable AFM ground state in this system. Eu_4PdMg shows a table-like MCE with $\Delta S_M = -5.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T from 20 K to 160 K, implying a large RC [628]. Also crystallizing in this structure, Tm_4AMg (A=Pd, Pt) compounds show a FM-

PM transition at very low temperatures (6.0 and 4.5 K with $-\Delta S_M=18.0$ and $16.5 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T for Pd and Pt respectively). Many other compounds of this family are also discussed by Li et al. [629] in a recent review.

MCE was also studied for $\text{Ho}_{6-x}\text{Er}_x\text{MnBi}_2$ crystallizing in Zr_6CoAs_2 -type structure, decreasing transition temperatures from 201 to 98 K as x increases from 0 to 6 [630]. A SOPT is suggested with $|\Delta S_M|$ increasing from 5.0 to $10.6 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T as x increases from 0 to 6. Crystallizing in the orthorhombic $\text{Ho}_6\text{Co}_2\text{Ga}$ structure (space group Immm), $\text{Dy}_6\text{Co}_{2.5}\text{Sn}_{0.5}$ compound exhibits a metamagnetic transition below 42 K and presents an inverse MCE at ~ 13 K and a direct MCE at ~ 52 K [631].

GdScSi and GdScGe crystallize in CeScSi -type structure (space group I4/mmm) and show Curie temperatures above room temperature (around 350 K), with $\Delta S_M(2 \text{ T})=-1.34$ and $-1.45 \text{ J kg}^{-1}\text{K}^{-1}$ respectively [632].

Compounds with 63 at. % of RE

Gd_5CoSi_2 compound crystallizes in tetragonal structure, deriving from Cr_5B_3 -type structure (space group I4/mcm), and is FM at temperatures below 168 K with $\Delta S_M \sim 9 \text{ J kg}^{-1}\text{K}^{-1}$ for $\mu_0\Delta H=4.6 \text{ T}$ [633]. This structure is also observed in Nd_5Si_3 [634] and Pr_5Si_3 [635] compounds, which undergo SOPT at 112 and 47 K with $\Delta S_M=-5.0$ and $-11.6 \text{ J kg}^{-1}\text{K}^{-1}$, respectively, at $\mu_0\Delta H=5 \text{ T}$. Derived from this structure, Mo_5SiB_2 -type structure (I4/mcm space group) is found for $\text{RE}_5\text{Ir}_2\text{X}$ ($\text{RE}=\text{Gd}, \text{Tb}, \text{Dy}$ and Ho ; $\text{X}=\text{Sn}, \text{Pb}, \text{Sb}$ and Bi) [636]. In general, the transition temperatures decrease with increasing atomic number of the RE element. For Gd-containing compounds, $\Delta S_M \sim 7 \text{ J kg}^{-1}\text{K}^{-1}$ is observed at ~ 160 K and 5 T for Sn and Pb containing alloys but for Sb and Bi-containing alloys, they exhibit $\Delta S_M \sim 9 \text{ J kg}^{-1}\text{K}^{-1}$ at ~ 125 K and 5 T. Yuan et al. [637] studied $\text{RE}_5\text{Ga}_{3-x}\text{TM}_x$ ($\text{RE}=\text{Ho}, \text{Er}$; $\text{TM}=\text{Fe}, \text{Co}$), wherein its crystal structure changes

from Mn_5Si_3 -type to Cr_5B_3 -type as TM content increases. They reported that $\text{Ho}_5\text{Ga}_{2.6}\text{Co}_{0.4}$ alloy (with Cr_5B_3 -type structure) exhibits a $\Delta S_M \sim -12.7 \text{ J kg}^{-1}\text{K}^{-1}$ at 32.5 K and 5 T.

The binary Tb_5Si_3 compound, crystallizing in Mn_5Si_3 -type structure (space group $\text{P6}_3/\text{mcm}$), undergoes a magnetic field induced transition and shows unusual positive magnetoresistance at very low temperatures [638]. Normal MCE is observed at the PM state (above 69 K) with a sign reversal of MCE in the ordered state. Their MCE results also indicate the existence of an inverse metamagnetism. Also crystallizing in the Mn_5Si_3 -type phase, Pr_5Ge_3 compound exhibits AFM ordering at low temperature (18 K) but with competing FM interactions close to the transition temperature of 22.5 K with $\Delta S_M = -5.8 \text{ J kg}^{-1}\text{K}^{-1}$ at 7 T [639]. Nd_5Ge_3 compound shows a maximum in $-\Delta S(T)$ of $\sim 5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\sim 50 \text{ K}$ and 7 T, which shifts to lower temperatures after ball milling [640]. The crystalline structure in $\text{RE}_5\text{A}_x\text{Sb}_{3-x}$ ($\text{A}=\text{Si}, \text{Ge}$) changes to Yb_5Sb_3 -type structure (space group Pnma) [641], showing multiple magnetic transitions for $\text{RE}=\text{Pr}$ and Nd , while only a Pauli paramagnetic behavior for $\text{RE}=\text{La}$.

MCE of some compounds crystallizing in the Yb_5Sb_3 -type structure has also been studied. Among them, Dy_5CoBi_2 exhibits a $\Delta S_M = -6.2 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}} = 2.2 \text{ K}$ at 38 K and 5 T [642], $\text{Gd}_5\text{Ni}_{0.96}\text{Sb}_{2.04}$ shows $\Delta S_M = -7.7 \text{ J kg}^{-1}\text{K}^{-1}$ at 112 K at 5 T and $\text{Gd}_5\text{Ni}_{0.71}\text{Bi}_{2.29}$ presents a $\Delta S_M = -5.6 \text{ J kg}^{-1}\text{K}^{-1}$ at 160 K at 5 T [643].

$(\text{Gd}_{12-x}\text{Tb}_x)\text{Co}_7$ ($0 \leq x \leq 8$) [644] crystallizes in monoclinic $\text{Ho}_{12}\text{Co}_7$ -type structure (space group $\text{P2}_1/\text{c}$). When Tb content increases from 0 to 8 at.%, its Curie temperature decreases from 161 to 119 K and $|\Delta S_M|$ (SOPT) from 8.8 to $7.1 \text{ J kg}^{-1}\text{K}^{-1}$ ($\mu_0\Delta H = 5 \text{ T}$) respectively. For $x=0$, its presence of a spin reorientation at 123 K yields a broader MCE peak with $\text{RC}_{\text{FWHM}} \sim 650 \text{ J/kg}$. Among the $\text{Ho}_{12-x}\text{Er}_x\text{Co}_7$ series, composite samples

with $\text{Ho}_{12}\text{Co}_7$ and $\text{Ho}_6\text{Er}_6\text{Co}_7$ optimize the width of the MCE response with $|\Delta S_M|=16.7 \text{ J kg}^{-1}\text{K}^{-1}$ at 37.9 K and 5 T [645]. Figure 21 shows the dependence of transition temperatures and $|\Delta S_M|$ on the atomic number of RE in these compounds (data taken from [645]). As the atomic number of RE increases, Curie temperature decreases while $|\Delta S_M|$ increases.

Compounds with 60 at. % of RE

$(\text{Gd}_{1-x}\text{X}_x)_3\text{Al}_2$ series ($\text{X}=\text{Y}$, $0 \leq x \leq 0.3$ [646]; $\text{X}=\text{Dy}$, $0 \leq x \leq 0.5$ [647]) crystallizes in tetragonal Zr_3Al_2 -type (space group $\text{P4}_2\text{nm}$). As X content increases, their lattice parameter and transition temperature decrease: from 284 K to 195 K when Y increases from 0 to 0.3 at.%, and to 215 K when Dy increases from 0 to 0.5 at.%. Their $|\Delta S_M|$ ascribed to SOPT also decreases with increasing x for Y-containing compound (from 2.3 to $1.8 \text{ J kg}^{-1}\text{K}^{-1}$, measured at $\mu_0\Delta H=1 \text{ T}$) but remains unchanged for Dy-containing compound. Also crystallizing in this structure (but with minor amounts of DyAl and DyAl_3 impurity phases) Dy_3Al_2 shows a PM-FM SOPT at 94 K and a spin reorientation at 69 K ($|\Delta S_M|=12.1$ and $5.2 \text{ J kg}^{-1}\text{K}^{-1}$ respectively, at 5 T) [648]. Two transitions were also observed for Tb_3Al_2 compound: $T_C=190 \text{ K}$ and a possible spin reorientation at 86 K [649]. It exhibits a $|\Delta S_M|>12 \text{ J kg}^{-1}\text{K}^{-1}$ at 190 K and 5 T with $\text{RC}_{\text{FWHM}}>600 \text{ J/kg}$.

RE_3Ni_2 compounds ($\text{RE}=\text{Ho}$ and Er) crystallize in a rhomb-centered rhombohedral structure with space group R-3 and experience two successive transitions: SOPT from PM to FM state (at 36 K for Ho- and 17 K for Er-containing compounds) and spin reorientation transition (at 10 K for Ho- and $\sim 12 \text{ K}$ for Er-containing compounds).

When $\text{RE}=\text{Ho}$, its $|\Delta S_M|(5 \text{ T})=21.7 \text{ J kg}^{-1}\text{K}^{-1}$ while $19.5 \text{ J kg}^{-1}\text{K}^{-1}$ for $\text{RE}=\text{Er}$ [650].

$\text{RE}_{15}\text{Si}_9\text{C}$ compounds ($\text{RE} = \text{Gd}, \text{Tb}$ and Dy) crystallize in $\text{La}_{15}\text{Ge}_9\text{Fe}$ -type structure ($\text{P6}_3\text{mc}$ space group) that derives from Mn_5Si_3 -type structure. They undergo a SOPT at

170, 130 and 70 K with $-\Delta S_M=7.0, 11.8$ and $12.9 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad}=5.06, 6.64$ and 5.52 K at 7 T for RE=Gd, Tb and Dy, respectively [651]. The scaling law of the magnetic entropy change with applied field is also applicable in these systems.

Compounds with 57 at. % of RE

The $\text{Gd}_4(\text{Bi}_x\text{Sb}_{1-x})_3$ series is FM, with T_C increasing from 266 K ($x=0$) to 332 K ($x=1$), and crystallizes in cubic anti- Th_3P_4 -type structure (space group I-43d). For $x=0$, $-\Delta S_M=6.8 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=5$ T and decreases with increasing x [652]. Monte Carlo simulations of Gd_4Sb_3 compound are in good agreement with their experimental data [653].

Gd_4Co_3 [654] and $(\text{Gd}_{0.9}\text{Dy}_{0.1})_4\text{Co}_3$ [655] ferrimagnets were assumed to crystallize in Ho_4Co_3 -type structure (space group $\text{P6}_3/\text{m}$). The ordering transitions are at 220 K and 145 K for Dy-free and Dy-containing compounds respectively, with an enhanced ΔS_M value for the latter compound (-5.74 and $-8.47 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively, at $\mu_0\Delta H=5$ T). For Dy-free compound, the presence of a second magnetic transition at 163 K was initially ascribed to spin reorientation, resulting in a very broad MCE curve and $\text{RC}_{\text{FWHM}} \sim 800 \text{ J/kg}$, which is much higher than that ($\sim 500 \text{ J/kg}$) for Dy-containing compound. However, Tence et al. [656] claim that Gd_4Co_3 compound does not exist but a mixture of $\text{Gd}_{12}\text{Co}_7$ and $\text{Gd}_4\text{Co}_{-3.23}$ is observed for this Gd/Co ratio. The latter $\text{Gd}_4\text{Co}_{-3.23}$ phase, which exhibits the $\text{P6}_3/\text{m}$ space group, should be responsible for the high temperature transformation, whereas the FM transition of the $\text{Gd}_{12}\text{Co}_7$ impurity phase could be responsible for the low temperature transformation, which was previously ascribed to a spin-reorientation. A maximal $\Delta T_{ad}=4.4$ K at 4.5 T is reported for the mixture.

Compounds with 55 at. % of RE

This atomic fraction of RE corresponds to $\text{Gd}_5\text{Si}_2\text{Ge}_2$ family described earlier.

Hexagonal $\text{Ce}_6\text{Ni}_2\text{Si}_3$ -type structure (space group $\text{P6}_3/\text{m}$) is exhibited by $\text{RE}_6\text{Co}_2\text{Si}_3$ compounds, whose Curie temperatures of 84.5 (RE=Nd), 187 (RE=Tb) [657] and 295 K (RE=Gd) [658] are reported. The MCE parameters associated with their SOPT are - $\Delta S_{\text{M}}=5.3$ (RE=Nd), 6.8 (RE= Tb) [657] and $6.3 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T, for (RE=Gd), with a broad $\Delta S(\text{T})$ peak (FWHM >100 K and $\text{RC}_{\text{FWHM}}\sim 700 \text{ J/kg}$) for the latter [658]. For $\text{RE}_6\text{Co}_{1.67}\text{Si}_3$, minor changes in the stoichiometry retain its $\text{Ce}_6\text{Ni}_2\text{Si}_3$ -type structure and SOPT character. Their MCE parameters at $\mu_0\Delta\text{H}=5 \text{ T}$ are: for RE=Pr, $-\Delta S_{\text{M}}=6.9 \text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}=262 \text{ J/kg}$ at $T_{\text{C}}=48 \text{ K}$ [659]; for RE=Tb, $-\Delta S_{\text{M}}=7 \text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}\sim 560 \text{ J/kg}$ at $T_{\text{C}}=186 \text{ K}$ [660]; for RE=Gd, $-\Delta S_{\text{M}}=5.2 \text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}\sim 520 \text{ J/kg}$ at $T_{\text{C}}=298 \text{ K}$ [661]. Tb substitution for Dy in $\text{RE}_6\text{Co}_{1.67}\text{Si}_3$ series leads to decrease in T_{C} and $-\Delta S_{\text{M}}$ although RC increases [662]. When RE=Tb and Nd in $\text{RE}_6\text{Co}_{1.67}\text{Si}_3$, its magnetization behavior is found to be sensitive to the type of measurement protocol due to the strong magnetostructural coupling and/or magnetic frustration [663]. Partial substitution of Ge for Si in $\text{Gd}_6\text{Co}_{1.67}\text{Si}_{2.5}\text{Ge}_{0.5}$ yields a $\Delta S_{\text{M}}=-5.9 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at $T_{\text{C}}=296 \text{ K}$, with a broad MCE peak (FWHM $\sim 100 \text{ K}$) [664]. $\text{Gd}_6\text{Ni}_{1.67}\text{Si}_3$ also crystallizes in $\text{Ce}_6\text{Ni}_2\text{Si}_3$ -type structure but has higher Curie temperature of $\sim 310 \text{ K}$ [665]. Its MCE parameters are $\Delta S_{\text{M}}=-5.72 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}}\sim 6 \text{ K}$ at $\mu_0\Delta\text{H}=4.8 \text{ T}$. However, to obtain relatively single phased systems will further require annealing. $\text{Gd}_6\text{Ni}_2\text{Si}_3$ and $\text{Gd}_6\text{Co}_2\text{Si}_3$ were also studied by Pathak et al. [666] showing slightly different MCE values, wherein $-\Delta S_{\text{M}}(5 \text{ T})=4.2$ and $5.3 \text{ J kg}^{-1}\text{K}^{-1}$ at $T_{\text{C}}=319$ and 298 K respectively. In addition, $\text{Pr}_6\text{Co}_2\text{Si}_3$ and $\text{La}_6\text{Ni}_2\text{Si}_3$, also crystallizing in this $\text{Ce}_6\text{Ni}_2\text{Si}_3$ -type structure, display MCE parameters of: $-\Delta S_{\text{M}}(5 \text{ T})=6.1$ and $0.1 \text{ J kg}^{-1}\text{K}^{-1}$ at $T_{\text{C}}=50$ and 7 K respectively. In the presence of a corrosive media (water flow), $\text{Gd}_6\text{Co}_{1.67}\text{Si}_3$ develops a passivated oxide layer ($\text{Gd}_2\text{O}_3+\text{SiO}_2$) without affecting MCE [667]. Closely related to this structure, $\text{Gd}_6\text{Co}_{4.85}$ was reported as the only phase

for Gd_4Co_3 stoichiometry [656, 668] and it exhibits a single ordering temperature at 219 K with $\Delta S_M \sim 2.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (2 T) and $\Delta T_{\text{ad}} = 4.4 \text{ K}$. Its transition occurring at 163 K is ascribed to impurities of the $\text{Gd}_{12}\text{Co}_7$ compound described above in the section corresponding to 63 at. % RE containing compounds.

Compounds with 52 at. % of RE

$\text{Dy}_{11}\text{Si}_4\text{In}_6$ crystallizes in $\text{Sm}_{11}\text{Ge}_4\text{In}_6$ -type structure (space group $I4/mmm$), with a $|\Delta S_M|(9 \text{ T}) = 16.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C = 52 \text{ K}$ (reduced to $\sim 9 \text{ J kg}^{-1} \text{ K}^{-1}$ for 5 T) [669].

All $\text{RE}_{11}\text{Ge}_8\text{In}_2$ (RE=Gd-Tm) phases, with space group $I4/mmm$, experience a FM ordering at 84, 96, 76, 26, 24 and 10 K for Gd, Tb, Dy, Ho, Er and Tm respectively. In addition, their unit cell volume decreases with increasing atomic number of RE. For RE from Gd to Ho, their spin reorientation transitions occur below Curie temperatures. Their $-\Delta S_M$ (5 T) obtained around their SOPT are: 4.4 (RE=Gd), 6.3 (RE=Tb) and 10.6 $\text{J kg}^{-1} \text{ K}^{-1}$ (RE=Tm), which is the largest among the series [670].

Compounds with 50 at. % of RE

Mononitrides REN compounds (RE=Gd, Tb, Dy, Ho and Er) [671-673] exhibit SOPT at cryogenic temperatures between 7.5 (for ErN) [672] and 61.2 K (for GdN) [674].

Tunable transition temperatures of the nitrides of binary systems were also studied [674-678] as shown in Figure 22. The $|\Delta S_M| \sim 30 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) and $\Delta T_{\text{ad}} = 8\text{-}10 \text{ K}$ of ErN and HoN make these systems promising and competitive with the Laves phases for applications of hydrogen liquefying and other refrigeration demands at low temperatures. Moreover, rare earth nitrides are hydrogen inert systems with high thermal conductivity. They crystallize in a NaCl-type ($\text{Fm}3m$ space group) structure and have higher RE packing densities than those of their corresponding hexagonal closed packed structures. Both their lattice parameters and transition temperatures linearly

decrease with increasing atomic mass of the RE element. Recent studies on nanopowders of REN (RE=Tb, Dy and Ho [679]) [680] produced by plasma arc discharge, report that their high MCE values are retained at slightly lower PM-FM SOPT temperatures (35.7, 19.9 and 14.2 K) than those produced via carbothermic reduction [671, 675]. The presence of Ho_2O_3 impurities in HoN samples shifts the temperature of maximum MCE response to <5 K, which is below the explored range [679].

EuO [681] also crystallizes in NaCl-type structure with a SOPT from FM to PM state at 67 K and exhibits high values of $|\Delta S_M| \sim 18 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{\text{ad}} \sim 7 \text{ K}$ at $\mu_0 \Delta H = 5 \text{ T}$. FM semiconductor EuS with this structure orders ferromagnetically at $T_C = 18 \text{ K}$ (SOPT) and has a $|\Delta S_M| = 37 \text{ J kg}^{-1} \text{ K}^{-1}$, $RC_{\text{FWHM}} = 782 \text{ J/kg}$ and $\Delta T_{\text{ad}} = 10.4 \text{ K}$ at $\mu_0 \Delta H = 5 \text{ T}$ [682].

Also crystallizing in NaCl structure, DySb was reported as a promising material for low temperature refrigeration due to its small hysteresis and large values of ΔS_M [683].

DySb is an Ising antiferromagnet with Néel temperature $\sim 9 \text{ K}$. The observed change in the sign of MCE is ascribed to a field induced metamagnetic FOPT from PM to FM state. An enhanced MCE is reported in their single crystals, which show highly anisotropic behavior [684].

CsCl-type structures (space group $\text{Pm}\bar{3}\text{m}$) are found for TbZn [685], HoZn [686] and ErZn [686] compounds, wherein their MCE is affected by spontaneous and field induced spin reorientation transitions. Theoretical studies show a table-like behavior for TbZn and HoZn between their Curie (23 K) and reorientation transitions that are dependent on the field direction. GdZn, also crystallizing in CsCl-type structure, exhibits good MCE response, with $\Delta S_M = -3.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 2 \text{ T}$. Composites of Gd plus GdZn phases were prepared in a single production step and proposed to enhance

oxidation resistance and RC with table-like MCE response with $\Delta T_{ad} \sim 3.5$ K [687].

Other REZn (RE=Tm, Ho) compounds were reviewed by Li et al. [629] along with other stoichiometries of alloys containing metals with low boiling points. HoPd suffers a SOPT from PM to FM with $\Delta S_M \sim 20$ J kg⁻¹ K⁻¹ at 10 K and 5 T [688].

GdGa compound, crystallizing in orthorhombic CrB-type structure (space group Cmc21) exhibits a Curie transition at 183 K and a spin reorientation at ~ 100 K, with $\Delta S_M = -4.81$ J kg⁻¹ K⁻¹ for $\mu_0 \Delta H = 5$ T and $RC_{FWHM} = 576$ J/kg [689]. Zheng et al. [690] studied the Gd_xEr_{1-x}Ga series, which compounds undergo a FM to PM transition followed by AFM to FM transition as temperature decreases. When x increases from 0 to 1, the low temperature transition increases from 15 to 66 K and the high temperature transition from 30 to 182 K. Differences are observed concerning low transition temperature for x=1 but the high temperature one is in agreement with the results of Zhang et al. [689]. Table-like ΔS_M curves were observed for compounds with $0.1 \leq x \leq 0.4$. In addition, both PrGa and HoGa compounds crystallize in CrB-type structure. PrGa undergoes a FM to AFM transition at 27 K and a AFM to PM transition at 37 K, leading to an almost constant $-\Delta S_M \sim 10$ J kg⁻¹ K⁻¹ and $\Delta T_{ad} \sim 2.7$ K values from 29.5 to 37.5 K [691]. In the case of HoGa, a spin reorientation occurs at ~ 20 K and a SOPT from FM to PM state is found at 69 K, leading to a broad range of high $|\Delta S_M|$ values (7.8 J kg⁻¹ K⁻¹ at spin reorientation and 17.1 J kg⁻¹ K⁻¹ at the Curie transition at 5 T) [692]. SmNi also crystallizes in CrB-type structure and shows a $|\Delta S_M| \sim 2$ J kg⁻¹ K⁻¹ and $\Delta T_{ad} \sim 1.2$ K (5 T) at around 40 K [693]. ErSi also crystallizes in CrB-type structure and has a Néel temperature of 12.5 K, below which a field induced metamagnetic FOPT from AFM to FM state occurs with $\Delta S_M = -14.8$ J kg⁻¹ K⁻¹ (5 T) [694]. GdNi and its hydride GdNiH_{3.2} also crystallize in CrB-type structure. Hydrogenation shifts the maximum of $\Delta S_M(T)$

curve from 75 to 10 K, without significantly affecting the maximum $|\Delta S_M| \sim 14 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) [695].

GdCu, crystallizing in FeB-type structure (Pnma), shows an enhancement of magnetic moment compared to that of pure Gd but their MCE values are modest ($\Delta S_M = -0.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T) and found at low temperatures in this AFM intermetallic with Néel temperature at 37 K [696]. Also crystallizing in this type of structure, CeSi compound is AFM with lower Néel temperature of 6.1 K. Its high MCE ($\Delta S_M = -13.7 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T without hysteresis loss) occurs ascribed to metamagnetic transition to FM under 0.4 T at 2 K [697].

HoAl undergoes an AFM to FM transition at 13 K and a FM to PM one at 20 K whereas ErAl changes from AFM to PM at 9 K, exhibiting $\Delta S_M(5 \text{ T}) = -22.5$ and $16.4 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively. Both compounds crystallize in orthorhombic DyAl-type structure (Pbcm space group), which is also present for other REAl intermetallics (e.g. the AFM compounds with RE=Gd, Tb and Dy, with Néel temperatures of 42, 72 and 20 K, respectively) [698].

The RE_3CoGe_2 series crystallizes in La_3NiGe_2 -type structure (space group Pnma). For RE=Gd, the system undergoes a FiM transition at 150 K with a $\Delta S_M = -4.9 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) [699].

Compounds with 43 at. % of RE

Eu_3O_4 [700], crystallizing in orthorhombic CaFe_2O_4 -type structure (Pnam space group), shows a metamagnetic transition from AFM to FM state at 5 K with high values of $\Delta S_M = -13.6 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) and $\Delta T_{\text{ad}} \sim 7 \text{ K}$. Partial substitution of Eu for other lanthanide, such as Ho or Dy, preserves the structure and leads to promising MCE parameters at

temperatures below 10 K: $\Delta S_M(8\text{ T}) = -30$ and $-25\text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad} = 13$ and 16 K for EuHo_2O_4 and EuDy_2O_4 respectively [701].

RE_3NiSi_3 compounds (RE=Gd, Tb, Dy) crystallizing in Y_3NiSi_3 -type structure (space group Immm), which is derived from Ta_3B_4 -type, exhibit modest MCE [702].

Compounds with 40 at. % of RE

$\text{Gd}_2\text{Ni}_x\text{Cu}_{2-x}\text{Mg}$ series crystallizes in the tetragonal Mo_2FeB_2 -type structure (space group P4/mbm) and the Ni substitution for Cu significantly affects their magnetic properties and shows a linear decrease of the cell volume with Ni content. For $x=0.5$, a temperature induced order to order transformation (SOPT) from FM to AFM phase occurs, while for $x=1$, a FOPT metamagnetic transition with critical field of 0.8 T is found at 5 K . Reversible MCE is reported for both compounds: $-\Delta S_M(5\text{ T}) = 9.5$ ($x=0.5$) and $11.4\text{ J kg}^{-1}\text{K}^{-1}$ ($x=1$) at about 65 K [703].

Gd_2CoSi_2 and Gd_2CoGe_2 compounds crystallize in Sc_2CoSi_2 -type structure (C2/m space group). They show $-\Delta S_M(5\text{ T}) = 3.0$ and $5.1\text{ J kg}^{-1}\text{K}^{-1}$ at Curie transition of 213 and 143 K for the compound with Si and Ge respectively [704]. In $\text{Gd}_2\text{Co}_{3-x}\text{Si}_x$ compounds, the La_2Ni_3 -type structure (Cmca space group) is stabilized by partial substitution of Si for Co. Their T_C can be tuned from 280 to 338 K ($0.50 > x > 0.29$) and shows at 2 T a $\Delta S_M \sim 1.5\text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad} \sim 1.3\text{ K}$ at $\sim 300\text{ K}$ [705]. RE_2TMSi_2 series (RE=Gd-Er and TM=Fe, Co) exhibits several transitions with several maxima of the magnetic entropy change (e.g. for RE=Er and TM=Co: $|\Delta S_M|(5\text{ T}) = 13.6$ and $8.4\text{ J kg}^{-1}\text{K}^{-1}$ at 32 and 12 K respectively) [706].

$\text{RE}_2\text{Co}_2\text{Al}$ series crystallizes in W_2CoB_2 -type structure (space group Immm). For RE=Tb and Dy compounds form $2:2:1$ and $1:1:1$ phases and show broad $\Delta S_M = -6.4\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) for RE=Tb (due to the Curie transitions of TbCoAl and $\text{Tb}_2\text{Co}_2\text{Al}$ at 74 K

and 102 K respectively), while for RE=Dy, its $\Delta S_M = -10.6 \text{ J kg}^{-1} \text{ K}^{-1}$ (due to the T_C of $\text{Dy}_2\text{Co}_2\text{Al}$ phase at 53 K, with negligible effect from DyCoAl phase with $T_C = 37 \text{ K}$) [707]. Also crystallizing in W_2CoB_2 -type structure, $\text{Dy}_2\text{Ni}_2\text{Sn}$ compound is AFM below 43 K and its spin canting occurs below 28 K. Both transitions are FOPT as evidenced by clear thermal hysteresis and their ΔS_M changes from inverse ($\sim 2 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 15 K) to direct MCE ($\sim 4 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 45 K) [708].

The tetragonal $\text{Mo}_2\text{B}_2\text{Fe}$ -type structure ($P4/\text{mbm}$ space group) is found in $\text{RE}_2\text{TM}_2\text{X}$ compounds (X=Mg, Cd, Sn or In). $\text{RE}_2\text{Cu}_2\text{In}$ shows PM-FM SOPT at 49.5, 30, 39.4 and 31.6 K and their $\Delta S_M(7 \text{ T}) = -16.5, 21.9, 18.4$ and $18.2 \text{ J kg}^{-1} \text{ K}^{-1}$ for RE=Dy [709], Ho [710], Er and Tm [711] respectively. Their composite materials were proposed to exhibit enhanced table-like MCE response [711].

Compounds with 33 at. % of RE

The MCE of RE-containing compounds crystallizing in AB_2 stoichiometry, but different to the already-described Laves phases, have been also studied. The UGa_2 compound crystallizes in hexagonal AlB_2 -type structure (space group $P6/\text{mmm}$) and minor partial substitution of Al for Ga enhances its maximum magnetic entropy change ($\Delta S_M(2 \text{ T}) \sim 2 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_C = 125 \text{ K}$) [712], which is not so significant for other elemental substitutions for Ga or pressure application [713]. The partial substitutions of Dy or Er for U gradually broaden and shift to lower temperatures the thermal range of the well defined transition observed for UGa_2 compound [714]. Also crystallizing in AlB_2 -type structure, TbB_2 shows high $|\Delta S_M|$ and ΔT_{ad} values ($5.2 \text{ J kg}^{-1} \text{ K}^{-1}$ –measured- and 14.7 K –estimated- at 2 T) at around a SOPT at 144 K [715]. The isostructural DyB_2 compound shows $\Delta S_M = -17 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) ascribed to two successive transitions: spin-reorientation at 20 K and a FM to PM transition at 50 K [716]. Anisotropic MCE has been recently

reported for RGa_2 (RE=Er and Ho) monocrystals [717]. Also crystallizing in AlB_2 -type structure, RE_2CuSi_3 (RE=Pr and Gd) compounds experience two transitions: for Pr_2CuSi_3 and Gd_2CuSi_3 , a PM to FM SOPT at 12 and 26 K respectively, and then a FM-spin glass transition at 6 and 7.5 K respectively [718]. RE_2PdSi_3 are isostructural to those compounds, with MCE reported for RE=Ho [719] and Gd, Dy and Er [719]: $|\Delta S_M|(5\text{ T}) = 27.5, 11.8, 16.6$ and $22\text{ J kg}^{-1}\text{K}^{-1}$ at transition temperatures of 10, 26, 10 and 9.5 K respectively. RE_2CoGa_3 (RE=Gd, Dy, Ho) compounds, crystallizing in RE_2RhSi_3 -type structure (space group $\text{P6}_3/\text{mmc}$ or P-62c [720]) derived from AlB_2 -type structure, experience a SOPT from PM to FM at 50, 17 and 10 K respectively, displaying - $\Delta S_M(5\text{ T}) = 12.6, 10.8$ and $13.8\text{ J kg}^{-1}\text{K}^{-1}$ for Gd, Dy and Ho respectively [721].

TbAgAl crystallizes in orthorhombic CeCu_2 -type structure (space group Imma) and exhibits a complex magnetic behavior and $\Delta S_M(5\text{ T}) = -6.2\text{ J kg}^{-1}\text{K}^{-1}$ at 59 K [722]. RECuSi compounds (RE=Ho [723] and Dy [724]) crystallize in Ni_2In -type structure, with a field induced metamagnetic FOPT from AFM to FM state. For RE=Ho, $\Delta S_M(5\text{ T}) = -33.1\text{ J kg}^{-1}\text{K}^{-1}$ at $T_N = 7\text{ K}$ [723] and $-24.0\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at $T_N = 10\text{ K}$ for RE=Dy [724]. TbH_8 hydrides ($\delta \sim 2$) crystallize in fcc CaF_2 -type structure (space group $\text{Fm}3\text{m}$) and show two FOPT at low temperatures around 10-15 K with a $|\Delta S_M| \sim 10\text{ J kg}^{-1}\text{K}^{-1}$ (9 T) [725, 726]. Recently, Zhang et al. reviewed different RETMSi compounds [727], which are also summarized in Figure 23.

The MCE properties of several intermetallic ternary compounds with stoichiometries 1:1:1 are recently reported in the literature. Concerning their crystal structure, we can find orthorhombic TiNiSi -type (space group Pnma), hexagonal ZrNiAl -type (space group P-62m) and tetragonal CeFeSi -type (space group P4/nmm) structures for these materials. GdNiGa and GdNiIn compounds [728] order ferromagnetically at 30.5 and

93.5 K, with $\Delta S_M(5\text{ T}) = -20.3$ and $-6.8\text{ J kg}^{-1}\text{K}^{-1}$ respectively. The former crystallizes in TiNiSi-type structure while the latter in ZrNiAl-type. Similar ternary Gd-X-Y alloys (X=Mn, Fe, Ni, Ru, Pd; Y=Al, In, Ge) have been studied [729, 730] and reported with $\Delta S_M(5\text{ T}) = -10.6$ (T=57 K), -10.4 (T=46 K), -4.6 (T=90 K), -2.8 (T=200 K), $-0.7\text{ J kg}^{-1}\text{K}^{-1}$ (T=274 K) and $-6\text{ J kg}^{-1}\text{K}^{-1}$ (T=70 K) for GdNiAl, GdPdAl, GdPdIn, GdFeAl, GdMnAl and GdRuGe, respectively. Later, some of the coauthors [731] reported corrected structures of MgZn₂ and MgCu₂-type for GdFeAl and GdMnAl respectively. (Gd_{0.54}Er_{0.46})NiAl, crystallizing in ZrNiAl-type structure was earlier proposed by Takeya et al. [732] as a suitable MCE material for Ericsson cycles due to its flat $\Delta S(T)$ curve between 15 and 40 K ($\Delta S_M(5.3\text{ T}) \sim 13\text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad} \sim 5\text{ K}$). RENiIn series (RE from Gd to Er) was studied by Zhang et al. [733] and they exhibit ZrNiAl-type structure. While Gd- and Er-containing RENiIn compounds show a single Curie transition (at 98 and 9 K respectively), Tb-, Dy- and Ho-containing compounds show two successive magnetic phase transitions (T_C at 71, 30 and 20 K and a transition from AFM to FM at 12, 14 and 7 K respectively). Their $-\Delta S_M(5\text{ T})$ are 7.1, 5.3, 10.4, 21.7 and $15.1\text{ J kg}^{-1}\text{K}^{-1}$ for RE=Gd, Tb, Dy, Ho or Er in RENiIn compounds, respectively, at their corresponding T_C . Moreover, for RE=Tb and Dy, a field induced metamagnetic transition occurs from AFM to FM states. Partial Cu substitution for Ni in HoNi_{1-x}Cu_xAl series changes the ordering nature of the ground state from mixed AFM/FM ($x \leq 0.1$) to AFM ($0.2 \leq x \leq 0.7$) and to FM ($x \geq 0.8$). In addition, a non-monotonous variation of MCE is observed along the series [734]. DyCuAl also crystallizes in ZrNiAl-type structure [735] with $\Delta S_M(5\text{ T}) = -10.9\text{ J kg}^{-1}\text{K}^{-1}$ at 28 K (SOPT). Likewise TbCuAl [736], HoCuAl [737] and ErCuAl [737] undergo SOPT at 52, 12 and 7 K with $\Delta S_M = -14.4$, -23.9 and $-22.9\text{ J kg}^{-1}\text{K}^{-1}$ respectively. The influence of grain size on the MCE of DyCuAl have been studied and shows that a reduction of

crystal size, from 90 nm to 38 nm, reduces T_C and also leads to a ~14 % decrease in $|\Delta S_M|$ but ~11 % enhancement in RC [738]. TbPdAl [739], also crystallizing in this structure, is a weak AFM structure and undergoes a field induced metamagnetic transition at 43 K from AFM to FM with a lower transition temperature at 22 K. Between the transition temperatures, 1/3 of the Tb magnetic moments are highly frustrated. The MCE of Tb_{1-x}Y_xNiAl series shows no significant variation after partial substitution of Y for Tb ($\Delta S_M(2\text{ T}) \sim 4\text{ J kg}^{-1}\text{K}^{-1}$ at ~45 K) [740]. However for TbNiAl_{1-y}In_y series, their ordering temperatures increase after complete substitution of In for Al and leads to a 50 % decrease of $|\Delta S_M|$ [740]. Large anisotropic MCE effect has been observed for TbNiAl monocrystals [741]. UNiGa compound also crystallizes in ZrNiAl-type structure and undergoes two transformations: from PM to FiM at 38.4 K (normal MCE) and from FiM to AFM at 34.5 K (inverse MCE) at low fields [742]. However, at higher fields, its inverse MCE weakens ($\Delta S_M = -3.6\text{ J kg}^{-1}\text{K}^{-1}$ at 37 K for 5 T) due to field induced metamagnetic phase transition.

As Cu content increases in TmNi_{1-x}Cu_xAl, its magnetic order transition reduces to below 2 K for $x \geq 0.3$. Large reversible MCE in these AFM systems are reported around ~4 K, whereby $-\Delta S_M(2\text{ T})$ increases from 5.5 ($x=0$) to 17.2 $\text{J kg}^{-1}\text{K}^{-1}$ ($x=1$) [743].

REPdSn crystallizes in TiNiSi-type or Fe₂P-type structure depending on the type of RE, being both structures possible for RE=Ho [744]. Some compounds are AFM with a transition temperature ranging from 7.5 for RE=Ce to 23.5 K for RE=Tb. However, other compounds (RE=Pr, Tm) are PM at least down to liquid helium temperature. For RE=Sm, a Néel temperature of 14 K is found. On the other hand, REPdIn compounds crystallize in ZrNiAl-type structure. For RE=Gd in REPdIn, the compound is FM with $T_C=102\text{ K}$ [744]. SmPd(Sn,In) compounds exhibit modest MCE ($|\Delta S_M| < 0.1\text{ J kg}^{-1}\text{K}^{-1}$ at

1 T) but MCE enhances one order of magnitude in $\text{Sm}_{1-x}\text{Gd}_x\text{PdIn}$ ($x=0.5$ and 1) with transitions at 60 and 83 K, whereby $-\Delta S_M(7\text{ T})=5.6$ and $6.2\text{ J kg}^{-1}\text{K}^{-1}$ respectively [744].

Also crystallizing in TiNiSi-type structure, REPtGa ($\text{RE}=\text{Dy}$, Ho and Er) orders antiferromagnetically at 15, 6.3 and 5 K ($|\Delta S_M|\sim 6, 9$ and $8\text{ J kg}^{-1}\text{K}^{-1}$, and $\text{RC}_{\text{FWHM}}=130, 180$ and 160 J/kg at 5 T) respectively [745].

Crystallizing in CeFeSi-type structure, HoTiGe and ErTiGe [746] undergo a field induced magnetic transition from AFM to PM state around their Néel temperatures (88 and 41 K respectively). While magnetic moments of Ho atoms are located along the c-axis, magnetic moments of Er atoms prefer to be perpendicularly aligned to c-axis. Its transformation is accompanied with hysteresis losses but no crystallographic changes are observed. Two peaks in their $\Delta S(T)$ curves are found at 20 and 90 K (-4.2 and $-2.5\text{ J kg}^{-1}\text{K}^{-1}$ at 5 T). Fe additions to HoTiGe do not eliminate the hysteresis, unlike for $\text{Gd}_5\text{Si}_2\text{Ge}_2$ compound, but they enhance ΔS_M , in particular to temperatures below 15 K [747]. ErTiSi also crystallizes in CeFeSi-type structure and undergoes a field induced metamagnetic transition from AFM to FM state below Néel temperature ($\Delta S_M(5\text{ T})=-8.9\text{ J kg}^{-1}\text{K}^{-1}$ at 46 K without hysteresis loss) [748]. HoFeSi compound (CeFeSi-type structure) shows two successive magnetic phase transitions (from PM to FM at 29 K and from FM to AFM or FiM at 20 K), leading to sequential inverse and normal MCE with $|\Delta S_M|=7.1$ and $5.6\text{ J kg}^{-1}\text{K}^{-1}$ at the first and second transition respectively [749]. Multicomponent Gd-Ni-Al alloys ($\text{Gd}_{28}\text{Ni}_{24}\text{Al}_{48}$ and $\text{Gd}_{33}\text{Ni}_{13}\text{Al}_{54}$) with FM phases, such as GdNiAl_2 , GdNiAl and $\text{Gd}(\text{Al,Ni})_2$, exhibit ordering close to 40 K and show $|\Delta S_M|(5\text{ T})=14.5\text{ J kg}^{-1}\text{K}^{-1}$ [750]. $\text{GdFe}_{1-x}\text{Co}_x\text{Si}$ crystallizing in CeFeSi-type structure, was recently studied by Włodarczyk et al. [751] and showed a minimum temperature for its phase transition at 70 K when $x=0.4$ while for Co-free compound a maximum at 220 K is found (wherein $|\Delta S_M|=10\text{ J kg}^{-1}\text{K}^{-1}$ at 5 T)

Gupta et al. [752] have recently reviewed 1:1:1 stoichiometry concerning, among other properties, their MCE response. Figure 24 uses the data from [752] distinguishing among the crystal structures and the type of ordered state (FM or AFM).

Compounds with 25 at. % of RE

Crystallizing in PuNi_3 -type structure (space group $R\bar{3}m$), $\text{Gd}(\text{Ni}_{1-x}\text{Fe}_x)_3$ series [753] shows a wide range of Curie temperatures: when x increases from $x=0$ (with spin reorientation at ~ 60 K) to 0.4, T_C increases from 115 K to 802 K and for $x=1$, T_C reduces to 725 K. Compensation temperatures of these FiM systems increase from 575 K ($x=0.4$) to 604 K ($x=1$). For $x \leq 0.2$, $|\Delta S_M|(1 \text{ T})$ (SOPT) decreases from $1.53 \text{ J kg}^{-1} \text{ K}^{-1}$ ($x=0$) to $0.28 \text{ J kg}^{-1} \text{ K}^{-1}$ ($x=0.2$). For $\text{Gd}(\text{Ni}_{1-x}\text{Co}_x)_3$ compounds ($x \leq 0.6$) [754] crystallizing in PuNi_3 , their Curie temperatures range from 115 to 342 K for up to $x=0.6$ ($T_C=615$ K for GdCo_3). However, $|\Delta S_M|$ decreases to $0.27 \text{ J kg}^{-1} \text{ K}^{-1}$ (1 T) for $x=0.6$.

Compounds of $\text{Tb}_{1-x}\text{Y}_x\text{CoC}_2$ series, with $\text{Amm}2$ space group, undergo a SOPT from PM to FM state and their T_C decreases from 28 to 18 K as x increases from 0 to 0.4. For $x=0.4$, $|\Delta S_M|=9.35 \text{ J kg}^{-1} \text{ K}^{-1}$ at 20 K and 5 T [755].

The orthorhombic CeNiSi_2 -type structure (Cmcm space group) [134] is found in many RETMX_2 compounds. It is related to the ThCr_2Si_2 -type structure, and polyhedra enclosing a TM atom form layers (perpendicular to the longest axis) interconnected by Si-Si bonds in both structures. With the same Cmcm space group, the crystal structure Al_2CuMg is found for FM GdNiAl_2 , which exhibits a SOPT at 28 K and $-\Delta S_M=16.0 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) [756]. The MCE of RECu_xGe_2 compounds ($\text{RE}=\text{Ho}$ or Er) crystallizing in CeNiSi_2 -type structure has been recently reported [757] with $-\Delta S_M(5 \text{ T}) \sim 6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 10 K ($\text{RE}=\text{Ho}$ and $x=0.33$) and $10.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5.5 K ($\text{RE}=\text{Er}$ and $x=0.25$). Another compound that crystallizes in this structure is TbNiGe_2 , which is AFM below 39 K and

exhibits inverse MCE [758]. REAgSn₂ compounds (RE=Ho and Er) crystallize in a cubic phase (space group Pm3m) and also exhibit AFM ordering but with even lower MCE responses [759].

Recently, Jiang et al. [760] proposed YbPt₂Sn as a metallic alternative to paramagnetic salts for demagnetization cooling from 2 to 0.2 K. This compound crystallizes in hexagonal ZrPt₂Al type structure (P6₃/mmc space group). Substitution of In for Sn yields similar magnetic properties but the structure changes to a cubic L2₁ Heusler type structure [761].

Compounds with 22 at. % of RE

Dy₂Ni₇ and Ho₂Ni₇ compounds, crystallizing in the rhombohedral Gd₂Co₇-type structure (space group R-3m), exhibit two successive transformations from PM to FM state (81 and 65 K respectively) and a spin reorientation transition (35 and 25 K respectively). Both transitions being SOPT [762] show $\Delta S_M(5\text{ T}) = -7.3$ and $-12.5\text{ J kg}^{-1}\text{K}^{-1}$ at ~40 and ~25 K respectively. Hexagonal Nd₂Co₇ crystallizing in Ce₂Ni₇-type structure has a Curie temperature of 613 K but spin reorientation transitions appear around room temperature. Anisotropic MCE was observed for their monocrystals and a maximum $\Delta T_{ad}(1\text{ T}) = 0.4\text{ K}$ at 235 K and -0.44 K at 250 K is obtained when rotating the field perpendicular and parallel to the c-axis respectively [763].

RE₂Cu₂O₅ (RE=Tb-Lu and Y) cuprates are formed with RE atomic size smaller than Gd. They ordered AFM below 17.7, 9.5, 12.3, 25, 16.7 and 17.3 K from Tb to Lu and 10.5 K for RE=Y. In the case of Dy- and Ho-containing compounds, $|\Delta S_M|$ values are $\sim 12\text{ J kg}^{-1}\text{K}^{-1}$ at 7 T [764].

Compounds with 21 at. % of RE

Lemoine et al. [765] have discussed the potential interest of $\text{RE}_6\text{Mn}_{23}$ intermetallics (RE=Y, Nd, Sm, Gd, Tb, Dy, Ho, Er, Tm and Lu) for magnetic refrigeration. They crystallize in cubic $\text{Th}_6\text{Mn}_{23}$ -type structure (space group Fm-3m). At high temperatures, Mn sublattice orders ferrimagnetically (at 505, 441, 456, 489, 457, 447, 445, 430, 411 and 398 K for RE=Y, Nd, Sm, Gd, Tb, Dy, Ho, Er, Tm and Lu respectively), whereas RE sublattice orders below 100 K. At this temperature, the $|\Delta S_M|(5\text{ T})$ are 0.34, 0.41, 0.36, 1.7, 2.5, 3.2, 5.2, 4.3, 5.2 and 0.46 $\text{J kg}^{-1}\text{K}^{-1}$ respectively. In $\text{Gd}_6(\text{Mn}_{1-x}\text{Fe}_x)_{23}$, partial substitution of Fe for Mn decreases T_C from 489 to 176 K and increases $-\Delta S_M(5\text{ T})$ from 1.67 to 2.59 $\text{J kg}^{-1}\text{K}^{-1}$ as x increases from 0 to 0.2 [766]. Partial substitution of Co for Mn decreases T_C to 118 K for $x=0.3$ [767] and leads to wide but moderate MCE peaks in $\text{Gd}_6(\text{Mn}_{1-x}\text{Co}_x)_{23}$.

Compounds with 20 at. % of RE

This atomic fraction of RE corresponds to the RETM_2X_2 compounds, which crystallize in ThCr_2Si_2 structure described above.

The MCE in RE-containing titanate family has also received attention, such as HoTiO_3 whose $\Delta S_M=-11.6\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) ascribed to a SOPT occurring at 53 K [768], and $\text{Er}_2\text{Ti}_2\text{O}_7$ whose MCE peaks below 10 K [769]. However, attention has been particularly focused on EuTiO_3 system. Alho et al. [770] have studied the anisotropic magnetocaloric effect of this AFM system and no significant differences were reported between single crystals and polycrystalline samples, which exhibits huge $|\Delta S_M|=49\text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad}=21\text{ K}$ at 5.5 K and 7 T [771]. $\text{Eu}_{1-x}\text{Ba}_x\text{TiO}_3$ series ($0.1 \leq x \leq 0.9$) shows large $-\Delta S_M$ below 10 K (40 and 6.6 $\text{J kg}^{-1}\text{K}^{-1}$ at 5 T for $x=0.1$ and 0.9 respectively) [772]. Even larger values were obtained in partial substitution of Nb for Ti: $|\Delta S_M|=51.3$

$\text{J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}}=22\text{ K}$ at $\sim 10\text{ K}$ and 9 T [773]. Partial Cr substitution for Ti (below 10 %) leads to a reduction of the MCE response without affecting the temperature of the maximum [774].

Theoretical calculations using Monte Carlo simulations and mean field model have been applied to the AFM perovskite-type EuZrO_3 compound [775] whose Néel temperature is about 4 K . A normal MCE of $\Delta T_{\text{ad}}=15\text{-}20\text{ K}$ at 2 T was calculated and an inverse MCE was predicted below 3.6 K at $\mu_0\Delta H=1\text{ T}$. In the vanadate HoVO_3 , several types of transformations occur: on cooling, SOPT from Pbnm orthorhombic to Pbn1 monoclinic at 188 K , a Néel transition of the V sublattice at 110 K , a FOPT from monoclinic to orthorhombic at 40 K (accompanied by a reordering of V moments), a FM ordering of Ho moments at 15 K and a AFM ordering of the Ho moments at 4 K [776]. This yields a large $-\Delta S_{\text{M}}\sim 15\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at $\sim 15\text{ K}$, with inverse MCE below 10 K for lower fields. In giant anisotropic REFeO_3 compounds, MCE can be even larger due to the rotation of the field from b to c axis [777]: $-\Delta S_{\text{M}}(5\text{ T})=17, 9$ and $20\text{ J kg}^{-1}\text{K}^{-1}$ at $9, 17$ and 7.5 K for Tb and Tm [778], and Dy [779], respectively. The MCE peak broadens with increasing fields. Ferroelectric $\text{DyFe}_{0.5}\text{Cr}_{0.5}\text{O}_3$ exhibits large MCE below 10 K [780]. Low temperature MCE was also studied for chromites such as $\text{Dy}_{0.7}\text{RE}_{0.3}\text{CrO}_3$ (RE=Y, Er and Ho) [781] and Gd_3CrO_3 , which exhibits large $-\Delta S_{\text{M}}>30\text{ J kg}^{-1}\text{K}^{-1}$ (4.4 T) below 5 K [782]. Ho ferro-chromites show values around $7\text{ J kg}^{-1}\text{K}^{-1}$ (7 T) but at a higher temperature of 20 K [783].

Compounds with 17 at. % of RE

NdCo_5 is a FM intermetallic that crystallizes in hexagonal CaCu_5 -type structure (space group $\text{P6}/\text{mmm}$). There is a competition between Co (favoring c-axis orientation) and Nd (favoring planar orientation) sublattices in the magnetocrystalline anisotropy. For

temperatures below 242 K, its magnetization lies in the basal plane [784] while at temperatures above 283 K, c-axis becomes the easy axis. Therefore, two spin reorientation transitions occur and a giant rotating MCE ($\Delta T_{ad}=1.6$ K at 1.3 T) has been reported at room temperature [785]. Partial substitutions of Al for Co or Dy for Nd in $Nd_{1-x}Dy_xCo_4Al$ ($x=0, 0.1$), shift their transition temperatures to higher values and MCE close to room temperature. For $x=0$, transitions occur at 299 and 311 K and their $\Delta S_M=0.25$ (inverse MCE) and $-0.6 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=1$ T respectively. For $x=0.1$, their $\Delta S_M(1 \text{ T})=0.15 \text{ J kg}^{-1}\text{K}^{-1}$ (maximum inverse MCE at $\mu_0\Delta H=0.7$ T) and $-0.6 \text{ J kg}^{-1}\text{K}^{-1}$ at transitions of 305 and 317 K respectively [786]. Other elements (not Al) have been partially substituted for Co, such as Ga in $Dy_{1-x}Nd_xCo_4Ga$ [787], which also preserves $CaCu_5$ -type structure [787]. Their Curie temperatures decrease from 498 to 410 K and spin-reorientation temperatures from 403 to 303 K as x increases from 0 to 1. The maximum entropy change are $-0.91 \text{ J kg}^{-1}\text{K}^{-1}$ (1.5 T) at 301 K for the spin-reorientation process. The $CaCu_5$ -type compound $SmNi_5$ presents a PM-FM SOPT at 29 K with $|\Delta S_M|=5.4 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) [788]. In their work, Nouri et al. collected results from Coroian, who studied other $RENi_5$ compounds with $|\Delta S_M|=0.6, 8.5, 11.3, 11.5, 15.7, 13.1$ and $16.3 \text{ J kg}^{-1}\text{K}^{-1}$ at 14, 8, 32, 23, 5, 12 and 9 K for $RE=Pr, Nd, Gd, Tb, Ho, Dy$ and Er respectively [789]. This structure is also shown by $PrNi_{5-x}Co_x$ ($1.95 \leq x \leq 3$) series [790], which exhibit several transition temperatures: spin reorientations at ~ 150 K and FM to PM transition between ~ 60 K ($x=1.95$) and ~ 537 K ($x=3$) ($T_C \sim 900$ K for $PrCo_5$). Partial substitution of Nd in $Pr_{1-x}Nd_xCo_4Al$ compounds shifts the two spin reorientation transition temperatures from 170 and 186 K to room temperature: 276 and 294 K respectively. Its MCE changes from normal to inverse as x increases from 0 to 0.8 while the magnetic entropy change ascribed to the low temperature transition is enhanced compared to that at the high temperature one. Their broad MCE peaks yield

$RC_{FWHM} \sim 30 \text{ J/kg}$ at 1 T [791]. $|\Delta S_M|$ remains below $0.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at the Curie temperature for the studied series but that associated to the spin reorientation is enhanced for $x \sim 2.4$ ($\Delta S_M = -1.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 1 \text{ T}$). Direct MCE measurements of HoCo_5 single crystals show large influence of magnetic anisotropy on the sign and magnitude of MCE at temperatures close to a spin reorientation (SOPT) at $\sim 180 \text{ K}$ [792]. A second SOPT, also ascribed to spin reorientation, occurs at a lower temperature of 50 K . Between these two temperatures, a conical phase exists. NdNi_4Si with SOPT occurring at very low temperature ($T_C = 8 \text{ K}$) shows $|\Delta S_M| \sim 7.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T [793]. Theoretical studies of DyNi_4Si [794], crystallizing in the YNi_4Si -type orthorhombic structure (space group $Cmmm$) derived from CaCu_5 -type structure, predict a FM ordering at 19 K and $|\Delta S_M|(14 \text{ T}) = 15.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 22 K , which is in good agreement with experimental values extended to other $\text{RE} = \text{Ce, Gd, Tb and Dy}$ [795]. SmNi_3TMSi compounds ($\text{TM} = \text{Mn-Cu}$) [796, 797] show FM ordering at $125, 190, 46, 17$ and 12 K for $\text{TM} = \text{Mn, Fe, Co, Ni and Cu}$ respectively, with modest ΔS_M values around the metamagnetic-like transition: the larger being $-2.6 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 65 K for Fe-containing compound [796].

Also belonging to CaCu_5 -type crystals, RENi_3TMSi compounds ($\text{RE} = \text{Gd and Tb}$; $\text{TM} = \text{Mn, Fe, Co and Cu}$) were studied by Morozkin et al. [798]. Maximum transition temperature was observed for $\text{TM} = \text{Fe}$ compound but its MCE is moderate. The largest $|\Delta S_M|(5 \text{ T}) = 12.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 25 K was found for $\text{TM} = \text{Ni}$ compound.

RECo_3B_2 compounds crystallize in hexagonal CeCo_3B_2 -type structure ($P6/mmm$ space group), which is derived from CaCu_5 -type structure [799]. These compounds are PM at room temperature and at temperatures of $150\text{-}160 \text{ K}$. Their Co-Co sublattice undergoes a magnetic ordering transition, while further cooling leads to ordering of the RE-RE

sublattice at 54, 47, 28 and 22 K for Gd, Sm, Tb and Dy respectively. Their MCE studied at the low temperature SOPT yields: $\Delta S_M = -5.0$, -4.9 and $-7.4 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 2.8$, 4.0 and 6.4 K at 2 T for Gd- [800], Tb- [801] and Dy-compounds [799], respectively. Tolinski et al. [801] reported that the Dy containing compound exhibits $\Delta S_M = -17.5 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 14 \text{ K}$ at 9 T. Other borides crystallizing with the same space group, exhibit Curie temperatures closer to room temperature: e.g. CeCo_4B [802], a FiM compound which orders at 291 K (SOPT), shows $\Delta S_M = -0.36 \text{ J kg}^{-1} \text{ K}^{-1}$ (1 T); and $\text{Ce}_2\text{GdCo}_{11}\text{B}_4$, also FiM with $T_C = 301 \text{ K}$ (SOPT) displays $\Delta S_M = -0.31 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) [803].

The effect of pressure and field on the MCE of YbInCu_4 , which crystallizes in AuBe_5 -type structure (space group $F-43m$), is reported with a maximum $|\Delta S_M| = 8.7 \text{ J kg}^{-1} \text{ K}^{-1}$ (7 T) at 6.6 kbar at 26 K near its Kondo temperature [804].

Midya et al. [805] studied the MCE of RECrO_4 (RE=Gd and Ho) compounds, which crystallize in ZrSiO_4 -type structure (space group $I4/amd$) and show $|\Delta S_M|(7 \text{ T}) = 28 \text{ J kg}^{-1} \text{ K}^{-1}$ at 25 K for RE=Gd while $29 \text{ J kg}^{-1} \text{ K}^{-1}$ at 20 K for RE=Ho. Dong et al. studied RECrO_4 (RE=Gd and Er), which shows a PM-FM SOPT with $-\Delta S_M(5 \text{ T}) = 22.8$ and $20.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 22 and 15 K respectively [806]. Their composite materials were proposed for refrigeration in the hydrogen liquefying temperature range. Recently, Palacios et al. [85] studied the MCE of GdCrO_4 attributed to the weak Gd-Cr interaction through which Cr^{5+} (FM ordered below 21.3 K) polarize the Gd^{+3} (ordered at 4.8 K). Midya et al. [807] also studied the MCE of DyVO_4 , which is an AFM compound with Néel temperature at 3.5 K, and exhibits $|\Delta S_M| = 26 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 17 \text{ K}$ at about 15 K and 8 T ascribed to a metamagnetic transition.

Compounds with 15 at. % of RE

Gadolinium gallium garnets (GGG), (e.g. $\text{Gd}_3\text{Ga}_5\text{O}_{12}$) have been proposed for very low magnetic refrigeration, i.e. <20 K, but its temperature limit can be further increased by partial Fe substitution for Ga [808]. The MCE of these Fe substituted GGG agrees with the calculations for superparamagnetic materials and there is a close relationship between large MCE and strong geometrical magnetic frustration in these systems [809]. The MCE enhancement can be as large as one order of magnitude with respect to ideal paramagnets. However, when Gd substitutes for Dy or Ho, the MCE of Fe modified GGG is not improved although Dy and Ho have a larger orbital contribution to its magnetic moment [810]. The study of bulk and nanostructured gadolinium iron garnet shows a shift of transition temperature to lower values in nanostructured samples compared to bulk ones and a non-monotonous dependence of MCE with crystal size. For bulk samples, their MCE originated from their intrinsic magnetic frustration, whereas for nanostructured samples, surface spin disorder and particle blocking effects appear as additional factors [811]. Theoretical calculations for MCE of FiM iron garnets with Y and Dy have been performed in a wide temperature range from ~ 0 to 600 K. The prediction of an inverse MCE associated to the compensation temperature (~ 225 K) is in good agreement with experimental data [812].

A completely different family of samples is that of clathrates [813]. This term derives from the Latin “clathratus”, which means “furnished with a lattice”. Therefore, in this type of compounds, usually polyatomic, one component forms a cage structure that contains the other, which is normally a large metal atom. Clathrates are known for their outstanding thermoelectric properties. If the metal ion is Eu^{2+} , it has a large magnetic moment, making it interesting for the study of thermomagnetic properties. Actually, the combination of thermoelectric and magnetocaloric response in a single material can

allow for new refrigeration technologies that combines the virtues of both thermoelectric and magnetocaloric refrigeration. Large MCE has been reported for $\text{Eu}_8\text{Ga}_{16}\text{Ge}_{30}$ clathrate compounds (and with partial Sr substitution for Eu) [814-817], which crystallize in two different complex structures: type I with space group Pm-3n and type VIII, space group I-43m . In these structures, Ga and Ge atoms form a network, which encloses Eu (and Sr) atoms. FM transitions occur at very low temperatures (13 K) and ascribed $\Delta S_M(3 \text{ T}) = -6$ and $-9.3 \text{ J kg}^{-1}\text{K}^{-1}$ are observed for types I and VIII [814]. Composite formed with clathrate and EuO , is found to exhibit a table-like MCE, which enhances RC [818]. Biswas et al. [819] reported that an enhanced MCE response of $\text{Eu}_8\text{Ga}_{16}\text{Ge}_{30}$ clathrates at lower temperatures is obtained by milling, which results in an almost constant $|\Delta S_M| \sim 5 \text{ J kg}^{-1}\text{K}^{-1}$ below 30 K. Other cage structures, whose MCE has deserved attention, are Wells-Dawson type $[\text{Ni}_6\text{Gd}_6\text{P}_6]$, with higher RE content ($|\Delta S_M| > 25 \text{ J kg}^{-1}\text{K}^{-1}$ at 3 K and 7 T) [820].

Slightly higher RE content is found in $\text{Gd}_3\text{Co}_{4+x}\text{Al}_{12-x}$ series, which crystallizes in $\text{Gd}_3\text{Ru}_4\text{Al}_{12}$ -type structure (space group $\text{P6}_3/\text{mmc}$) with a FM ordering below 40 K with $-\Delta S_M = 3.28 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}} = 1.5 \text{ K}$ at 2 T [821].

Compounds with 14 at. % of RE

The reference lanthanum manganite compound previously described, $\text{La}_{0.7}\text{Ca}_{0.3}\text{MnO}_3$ presents this RE fraction. Besides this widely studied family, studies on GdCu_6 , which orders antiferromagnetically at 16 K are also available [822]. This compound changes from inverse to normal MCE at an AFM-PM transition temperature, showing maximum $|\Delta S_M|(5 \text{ T})$ as $\sim 0.5 \text{ J kg}^{-1}\text{K}^{-1}$ (at $\sim 15 \text{ K}$) and $\sim 0.6 \text{ J kg}^{-1}\text{K}^{-1}$ (at $\sim 20 \text{ K}$) for inverse and normal MCE respectively.

Belonging to the RETM_4X_2 series ($\text{X}=\text{Si, Ge, P, As}$), which crystallize in ZrFe_4Si_2 -type phase, ErCo_4Ge_2 undergoes a PM-AFM SOPT at 2.6 K with $|\Delta S_M|=11.7 \text{ J kg}^{-1}\text{K}^{-1}$ at 2 T [823].

Compounds with 12 at. % of RE

$\text{Er}_2\text{Fe}_{14}\text{B}$ compound crystallizes in a tetragonal structure with $\text{P4}_2/\text{mmn}$ space group and has been proposed as a refrigerant close to room temperature due to its spin reorientation transition at 322.8 K and $|\Delta S_M|=0.735 \text{ J kg}^{-1}\text{K}^{-1}$ at 0.4 T [824]. The sign of the MCE is dependent on the direction of the applied magnetic field (along the a- or c-axis) [825].

Compounds with 11 at. % of RE

Besides preliminary results on intermetallics with RE_2X_{17} stoichiometry by Jin et al. [826], recent studies have focused on those that crystallize in two different structures: rhombohedral $\text{Th}_2\text{Zn}_{17}$ -type (R-3m) and hexagonal $\text{Th}_2\text{Ni}_{17}$ -type ($\text{P6}_3/\text{mmc}$). Nanometer sized ball milled powders of $\text{Pr}_2\text{Fe}_{17}$ preserving $\text{Th}_2\text{Zn}_{17}$ -type structure exhibit lower $|\Delta S_M|$ values ($4.5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$) than bulk counterparts [550] ($6.3 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H=5 \text{ T}$) although RC of the former is enhanced due to the broadening of $\Delta S_M(T)$ curves and increasing of T_C from ~ 290 to $\sim 300 \text{ K}$ [827]. Some α -Fe impurities were found, which can be avoided for Pr content above 12 at. %. MCE was optimized in single phase $\text{Pr}_{13}\text{Fe}_{87}$ alloy [828]. Partial substitution of Al for Fe in $\text{Pr}_2\text{Fe}_{17}$ compound, where Al preferentially occupies 6c positions, leads to a slight increase of the unit cell volume and T_C . ΔS_M decreases with Al content from 5.5 to $4.2 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) for $x=0$ and 1 respectively [829]. Partial substitution of Mn for Fe in $\text{Pr}_2\text{Fe}_{17-x}\text{Mn}_x$ yields a non-monotonic change of the PM-FM SOPT temperature (283, 294 and 285 K for $x=0, 1$ and 2 respectively) and ΔS_M ($5.66, 5.77$ and $4.31 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T for $x=0, 1$ and 2

respectively) [830]. Bulk $\text{Nd}_2\text{Fe}_{17}$ exhibits a single T_C at 339 K and upon ball milled, its $\text{Th}_2\text{Zn}_{17}$ -type structure is preserved while its MCE broadens ($-\Delta S_M(1.5 \text{ T})=2.7 \text{ J kg}^{-1}\text{K}^{-1}$ for bulk and $\sim 2 \text{ J kg}^{-1}\text{K}^{-1}$ for ball milled powder) [831]. Two-phase amorphous-nanocrystalline NdPrFe_{17} ribbons exhibit two close SOPT at 303 and 320 K, yielding an enhancement of refrigerant capacity [33]. Pseudo-binary $\text{Ce}_{2-x}\text{RE}_x\text{Fe}_{17}$ compounds, with $\text{RE}=\text{Y}$ ($x=1.2$), Pr ($x=1.5$) and Dy ($x=1.15$), also crystallize in $\text{Th}_2\text{Zn}_{17}$ -type structure and have been studied with the aim of tuning SOPT temperatures (253, 264 and 273 K for Y, Pr and Dy compounds respectively) [832]. $|\Delta S_M|$ and RC values decrease in the order of $\text{Pr} > \text{Y} > \text{Dy}$ (5.3, 4.3 and $3.3 \text{ J kg}^{-1}\text{K}^{-1}$ respectively, at 5 T). $\text{Ce}_2\text{Fe}_{17-x}\text{Mn}_x$ ($x \leq 2$) series also crystallizes in $\text{Th}_2\text{Zn}_{17}$ -type structure [833] and exhibits two magnetic transitions for $0 \leq x \leq 0.35$ and $1.3 \leq x \leq 2$: from FM to AFM state (decreasing from 94.4 to 22 K as x increases from 0 to 0.35 and increasing from 9 to 80 K as x increases from 1.3 to 2) and from AFM to PM state. However, for $0.5 \leq x \leq 1$, a single Néel transition is observed. Ascribed to the AFM-PM transformation (decreasing from 209 to 168 K as x increases from 0.35 to 2), $\Delta S_M \sim 3 \text{ J kg}^{-1}\text{K}^{-1}$ is obtained for low values of x , and for $x=2$, 50 % decrease in the absolute values is observed. A distribution of Curie temperatures in mechanically alloyed $\text{RE}_2\text{Fe}_{17}$ compounds ($\text{RE}=\text{Pr}$ and Nd) broadens the temperature range of the MCE [834].

Hexagonal $\text{Th}_2\text{Ni}_{17}$ -type structure is found for $\text{Tb}_2\text{Fe}_{17}$ for which Mössbauer experiments show seven different Fe sites [835]. This compound exhibits a PM-FM SOPT at 412 K with $-\Delta S_M=2.19 \text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}=88 \text{ J/kg}$ at 1.5 T. Also crystallizing in $\text{Th}_2\text{Ni}_{17}$ -type structure, $\text{Er}_2\text{Ni}_{17}$ is FiM below 150 K, with a compensation temperature at 63 K at $\mu_0 H=0.02 \text{ T}$, which shifts to lower temperatures for higher fields. ΔS_M at the FiM-PM transition is about $-0.7 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T, with an inverse MCE around 60 K [836]. The $\text{Lu}_2\text{Fe}_{17-x}\text{Mn}_x$ ($0 \leq x \leq 2$) series also crystallizes in

$\text{Th}_2\text{Ni}_{17}$ -type structure (some α -Fe impurities were found) [837]. $\text{Lu}_2\text{Fe}_{17-x}\text{Mn}_x$ series with $x \leq 0.5$ exhibits a Néel temperature at ~ 278 K but transforms to FM state at a lower temperature (from 135 to 249 K as Mn content increases from 0 to 0.5). Alloys with $0.7 \leq x \leq 2$ are FM, with $T_C = 287$ K for $x = 0.7$. Their maximum $|\Delta S_M| = 3.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 H = 5$ T is achieved at $x = 0.7$, where FOPT and SOPT merge.

Kuchin et al. [838] recently reviewed the MCE response of several $\text{RE}_2(\text{Fe}, \text{Mn})_{17}$ (RE = Ce, Tm and Lu), hydrogenated Ce compounds, and off-stoichiometric RE = Tm compounds. Despite their different nature of magnetic transitions, $|\Delta S_M| \sim 3 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 H = 5$ T at around 300 K. In the case of hydrogenated Ce compounds, the magnetic transition increases with H content.

Compounds with 10 at. % of RE or lower $\text{NdBaMn}_2\text{O}_6$ exhibits a PM-FM SOPT at 290 K and a FM-AFM FOPT at 210 K. It presents normal and inverse MCE with $\Delta S_M = -2.5$ and $1.4 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively [839].

New monoclinic GdNi_8Si_3 structure (space group C2/c) related to the tetragonal BaCd_{11} -type (space group $I4_1/\text{amd}$) was found for RENi_8Si_3 compounds (RE = Gd and Tb) [840]. These compounds order at ~ 3 K with $|\Delta S_M|(14 \text{ T}) = 19.8$ and $12.1 \text{ J kg}^{-1} \text{ K}^{-1}$, at 4 and 12 K for Gd and Tb compounds respectively.

Recently, it was shown that single crystals of $\text{KEr}(\text{MoO}_4)_2$ shows a MCE by simply rotating the crystal within the ab plane in a constant magnetic field ($|\Delta S_M| = 13 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 H = 5$ T), which is quantitatively similar to that obtained after a 5 T field increase ($|\Delta S_M| = 14 \text{ J kg}^{-1} \text{ K}^{-1}$) [841].

HoB_{12} is a frustrated incommensurate amplitude-modulated AFM crystallizing in UB_{12} -type structure (Fm-3m space group). In the AFM region, a small inverse MCE is

observed and above the Néel temperature (7.4 K), ΔS_M scales close to the predicted behavior for an ideal paramagnet [842]. Crystallizing in ThMn₁₂-type structure (space group I4/mmm), REFe₅Al₁₇ compounds (RE=Gd and Dy) show several changes in the sign of the MCE, which is also observed in the magnetoresistance data [843]. Tetragonal phase with space group P4/nmm is observed for Gd_{3-x}Y_xCo₂₉Ge₄B₁₀ compounds, whereby the Gd and Co atoms order ferrimagnetically. For Y-free compound $T_C=212$ K, with a compensation point at 128 K and a moderate MCE at T_C of $\Delta S_M=-0.5$ J kg⁻¹K⁻¹ is observed at 5 T. Partial Y substitution for Gd does not affect T_C but decreases the compensation temperature [844].

With a RE content about 7 at. %, La(FeSi)₁₃ family have been already described in previous sections.

FM RuSr₂GdCu₂O₈, a superconductor below 46 K, shows a GMCE at very low temperatures of ~4 K, characterized by $|\Delta S_M|=15.5$ J kg⁻¹K⁻¹ and $\Delta T_{ad}=14$ K at 7 T [845]. At higher temperatures of ~145 K, other maximum is found with $|\Delta S_M|\sim 0.9$ J kg⁻¹K⁻¹.

Some boride compounds with transitions temperatures above 150 K have been studied. Metastable NdFe₁₂B₆ compound can be obtained from devitrification of amorphous alloys (besides ~20 % of α -Fe phase) and exhibits an abrupt change in magnetization around its Curie temperature of ~220 K with $\Delta S_M=-8.4$ J kg⁻¹K⁻¹ and $\Delta T_{ad}=3.2$ K at $\mu_0\Delta H=1$ T [846]. This intermetallic crystallizes in hexagonal SrNi₁₂B₆-type (space group R-3m) with a high magnetic moment of 19.7 μ_B per formula unit. Mössbauer spectrometry results suggest a FOPT from FM to PM state responsible for the high MCE parameters [847]. The metastability of this compound can be overcome by alloying with the isostructural NdCo₁₂B₆ compound, which orders at 170 K. However,

the first order character of the transition is lost, leading to moderate ΔS_M values ($-1.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 1 \text{ T}$), which decrease with increasing Fe content. Nevertheless, refrigerant capacities are comparable to Co-free compound as the $\Delta S(T)$ curves become broader. Therefore, $\text{Nd}(\text{Co}_{1-x}\text{Fe}_x)_{12}\text{B}_6$ ($x=0, 0.05$ and 0.1) can be obtained in the stable $\text{SrNi}_{12}\text{B}_6$ -type structure by conventional casting and annealing methods with $\Delta S_M \sim -1.4, -1.2$ and $-0.9 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1 T and $175, 165$ and 155 K for $x=0, 0.05$ and 0.1 , respectively [848]. Similarly, $\text{PrFe}_{12}\text{B}_6$ exhibits a FOPT at 200 K from FM to PM state accompanied by a lattice contraction, which leads to high MCE values ($-11.7 \text{ J kg}^{-1} \text{ K}^{-1}$ for $\mu_0 \Delta H = 2 \text{ T}$) [849].

As a summary of this section, Figure 25 shows the plot of $|\Delta S_M|$ vs peak temperature for the described RE-containing compounds. In agreement with the description followed in this section, the different compounds are grouped according to their atomic fraction of RE. It can be observed that the behavior of the best values reported is very similar to those of Laves phases and ThCr_2Si_2 -type structures.

5.1.2. Rare earth free crystalline materials

MCE has been studied for the three pure FM transition elements (Fe, Co and Ni) and their experimental results have a fairly good agreement using mean field approach in a Hamiltonian from Hubbard-like model describing itinerant electron systems [850].

Moreover, MCE studies ascribed to the superconductor transition of Nb [851-853] and

In [854] can be also found to explore the parameters of the mixed state in these

superconductors. However, majority of transition elements compounds are based on

Mn. The main reason for considering Mn-containing compounds in the study of MCE is the large magnetic moment of the Mn atoms, which is the highest value among

transition metals (up to $4\mu_B$ in some intermetallics).

A previous review focused on Mn-containing systems was published by Brück et al. [445] including Heusler alloys, $\text{Mn}(\text{Fe}, \text{Co}, \text{Ni})(\text{Si}, \text{Ge})$, $(\text{Mn}(\text{Fe}, \text{Co}))_5(\text{Ge}(\text{Si}, \text{Sb}))_3$, $(\text{Fe}, \text{Mn})_3\text{C}$, $\text{La}(\text{MnFe})_2\text{Ge}_2$ and arsenic-containing alloys. FM lanthanum manganites, which constitute a big family among Mn-containing alloys were previously reviewed by Phan and Yu in 2007 [474] and in the present review they were already described in the section of RE-containing compounds. Curie temperatures and maximum entropy change at $\mu_0\Delta H=1$ T (when literature data were only available for a different field, data were rescaled to 1 T using mean field parameters for those systems with SOPT) are plotted in Figure 26 as a function of the magnetic moment per 3d atom.

In the following the main families of the RE-free compounds will be described.

5.1.2.1. Heusler alloys

Heusler alloys are named after the German scientist Friedrich Heusler, who in the early years of twentieth century discovered ferromagnetism in intermetallic compounds constituted by non-magnetic elements (Cu_2MnAl in 1903) [855]. Stoichiometric formula of these alloys is X_2YZ and they crystallize in L_{21} cubic structure (space group $\text{Fm}\bar{3}\text{m}$). The crystallographic positions of the different elements are distinguished as shown in Figure 27. X is a 3d element, generally Ni, although Fe, Co and Pt can be also found. Y atoms are responsible for the magnetism of the system (exchange interactions between Y and X atoms are also considered) and generally correspond to Mn (with a magnetic moment of $\sim 4 \mu_B$). Finally, several elements can be ascribed to Z atoms, being Ga, In, Sn and Sb the most interesting from the point of view of MCE. Recent reviews on the MCE of these systems can be found in Planes et al. [856, 857], Buchelnikov et al. [858], Umetsu et al. [859] and Yu et al. [860]. Besides MCE, superelasticity, shape memory effect and magnetoresistance are found among the interesting properties exhibited by these systems. Generally, the different MCE studies of Heusler alloys are

focused on FOPT martensitic transformations. However, many papers also supplied the MCE data corresponding to the SOPT austenitic Curie transition (see Table S6 of the Supplementary Material) and, recently Singh et al. [861] have reviewed the viability of this transition for MCE.

The feature leading to GMCE in these alloys is the existence of a structural transition from the high temperature austenitic phase to the low temperature martensitic phase, which for low Z concentration corresponds to the $L1_0$ tetragonal phase (space group $P4/mmm$) as shown in Figure 28. In order to characterize this transition, four temperatures are considered. On cooling the sample, martensitic start temperature, M_s , corresponds to the temperature at which martensitic phase starts to form, while martensitic finish temperature, M_f , corresponds to the end of the transformation process. Analogously, on heating the sample from the martensitic phase, austenitic start temperature, A_s , corresponds to the onset of austenitic phase formation, while austenitic finish temperature, A_f , corresponds to the end of this process. As $M_s < A_f$ and $M_f < A_s$, a thermal hysteretic behavior of up to few tens of Kelvin is observed. However, martensitic transformation is quite complex and premartensitic transformations are reported to occur in some cases. Phase diagrams show the presence of several modulated phases at higher Z content as 5M (or 10M) and 7M (or 14M), where M refers to the monoclinic structure formed due to distortion. Thermal hysteresis could be narrowed when premartensitic (or intermartensitic) transition becomes close to basic martensite-austenite one [862]. These phase diagrams are generally represented as a function of the average number of electrons in the outer shell per atom, e/a (including 3d atoms for transition elements). Martensitic phase diagram, including intermartensitic transformation (from modulated to non-modulated phase) were recently studied by Dutta et al. [863] using an ab initio approach.

In off-stoichiometric $\text{Ni}_{2+x}\text{Mn}_{1-x}\text{Ga}$ alloys, Curie temperature and structural transition coincide in the compositional range $0.18 < x < 0.27$ [858] (corresponding to $7.635 < e/a < 7.7025$) and a FOPT magnetostructural transition occurs characterized by an evident thermal hysteresis. For $x < 0.18$, martensitic transformation occurs at temperatures lower than its Curie temperature. For $x > 0.27$, martensitic transformation occurs at temperatures higher than Curie temperature. For $x > 0.36$, ($e/a > 7.77$) secondary phases appear. Theoretical studies on the disorder in Mn and Ni sites effect on the magnetic moment of Mn-Ni-Ga alloys have been recently reported by D'Souza et al. [864]. In ternary alloys, transition temperature increases with e/a ratio (see line 1 in Figure 28). However, if a transition element like Fe, Co or Cu is added, transition temperature decreases although e/a increases (see lines 2 and 3 in Figure 28). On the other hand, partial substitution of Sn for Ga has no significant change in e/a but it has a clear effect on transition temperature (see line 4 in Figure 28).

First work on MCE of Heusler alloys appeared in 2000 and corresponds to Hu et al. [865]. They observed an inverse MCE at moderate fields ($\Delta S_M \sim +4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 0.9 T) ascribed to martensitic transformation (thermal hysteresis 10 K). Inverse MCE occurs as both phases are FM and the high temperature phase (austenite) exhibits a higher magnetization than that of low temperature one (martensite) at low fields due to the strong uniaxial magnetocrystalline anisotropy of martensitic phase [866]. However, for fields above 1 T, inverse MCE changes to direct MCE [867, 868]. In the compositional region for which magnetostructural transformation occurs, GMCE is reported [869]. Zhou et al. [870, 871] systematically studied the evolution of structural and magnetic transitions as well as MCE in off-stoichiometric ternary Ni_2MnGa alloys. Ni-rich alloys exhibit a lower Curie temperature and closer to room temperature than those of Mn-rich ones. Recently, Singh et al. [872] using neutron diffraction on Ni_2MnGa and

$\text{Ni}_{1.75}\text{Mn}_{1.25}\text{Ga}$ systems show that the AFM interaction between the Mn atoms occupying inequivalent sites is the origin of inverse magnetocaloric effect. Figure 29 shows the field dependence of the ΔS_M values for two examples: Ni_2MnGa and $\text{Ni}_2\text{MnGa}_{0.95}\text{Sn}_{0.05}$ adapted from [873].

Soon after the discovery of GMCE in Ni-Mn-Ga alloys, Marcos et al. [874, 875] observed that MCE is directly related to magnetostructural coupling at the length scale of both magnetic and martensitic domains. This contribution is the dominant for those systems where Curie temperature and martensitic transition temperature are well separated ($e/a \sim 7.5$) and yields a negative GMCE, whereas for alloys in which both temperatures become close ($e/a \sim 7.7$), microscopic spin-phonon coupling becomes dominant and yields a positive GMCE. Pareti et al. [876] found a clear enhancement in ΔS_M (from -5 to -20 $\text{J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 1.6$ T) after increasing the Ni/Mn ratio to merge structural and magnetic transitions. For such alloy, Aliev et al. [877] used a statistical model of phase coexistence during martensitic transformation to describe the MCE of $\text{Ni}_{2.19}\text{Mn}_{0.81}\text{Ga}$ alloy. From a value of $\Delta T_{ad} = 1.2$ K at $\mu_0\Delta H = 2.6$ T and $T \sim 340$ K, they estimated a $|\Delta S_M| \sim 4.5 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 1.5$ T, which is much lower than the value reported by Pareti et al. Moreover, the structural entropy change linearly increases from 9.1 to 15.6 $\text{J kg}^{-1}\text{K}^{-1}$ as e/a increases from 7.48 to 7.66. Pasquale et al. [878] also found a strong discrepancy between very large $\Delta S_M = -86 \text{ J kg}^{-1}\text{K}^{-1}$ estimated from Maxwell relation and $\Delta T_{ad} = 2.2$ K at 5 T. Analysis using Clausius-Clapeyron equation reduces $|\Delta S_M|$ to $\sim 25 \text{ J kg}^{-1}\text{K}^{-1}$ at 4 T. On the other hand, Duan et al. [879] considered both magnetic and lattice contributions of $\text{Ni}_{2.15}\text{Mn}_{0.85-x}\text{Cu}_x\text{Ga}$ ($x = 0.05$ and 0.07) to its entropy change and compared the theoretical and experimental results, concluding that the main contribution to entropy change is magnetic. However, direct measurements of ΔT_{ad} (~ 1 K for $\text{Ni}_{2.19}\text{Mn}_{0.81}\text{Ga}$ at 2 T) showed to be one order of magnitude smaller than

calculated values, which is ascribed to intrinsic factors of FOPT (exchange integrals of low and high temperature phases, thermal hysteresis and phase coexistence) although the rate of martensitic transformation is assumed to be fast enough to follow the field sweeps used [880]. In fact, the applicability of Maxwell relation to FOPT (implying phase coexistence) leads to artifacts affected by the integration method [881]. Zhang et al. [882] pointed that the use of Maxwell relation is correct for magnetic field values below the critical field. Khovailo et al. [880] also showed an irreversible character of ΔT_{ad} upon applying a magnetic field at a constant temperature when reached by cooling from $T > A_f$. The first application of the field can induce the formation of FM martensitic phase (contributing as an extra to the temperature change registered) but consequent removal of the magnetic field cannot induce the formation of austenitic phase (as martensitic phase is already stable) and thus a second cycle will show a reduced ΔT_{ad} as it occurs for $Ni_{2.19}Mn_{0.81}Ga$. The history dependence of MCE on the field and thermal sequence followed has been also pointed by Sasso et al. [883], whose conclusions agree with those of Khovaylo et al., assigning differences between field applied and field removed curves to the magnetostructural hysteresis. Porcary et al. [884] established the convergence of direct and indirect methods in the study of Ni(Co)-Mn-Ga alloy. In fact, there are many objections to using Heusler alloys as magnetic refrigerants due to the irreversible character of the magnetic field induced martensitic transformation and the large hysteresis found [885]. Hence, reducing hysteresis is the main goal, nowadays, prosecuted by researchers working in the MCE of Heusler.

Substitution of 1 at. % of Fe for Ni in $Ni_{55.5}Mn_{20}Ga_{24.5}$ increases Curie temperature and decreases martensitic transformation temperature [886]. Inverse MCE is observed at 0.5 T, changing to direct MCE at higher field changes. Partial substitutions of Co or Cu for Mn in stoichiometric Ni_2MnGa alloy [887, 888] tune martensitic and Curie

temperatures to coincide for $\text{Ni}_2\text{Mn}_{0.7}\text{Co}_{0.3}\text{Ga}$ and $\text{Ni}_2\text{Mn}_{0.75}\text{Cu}_{0.25}\text{Ga}$. The latter alloy exhibits a GMCE of $\Delta S_M(2\text{ T}) = -17.6\text{ J kg}^{-1}\text{K}^{-1}$ at 315 K and further studies showed that magnetostructural transition shifts to higher temperatures with increasing Mn deficiency in off-stoichiometric alloys while a nearly constant value of ΔS_M is observed [889]. As Cu substitution for Mn increases, magnetic transition decreases, delocalization of magnetism of Mn atoms increases and the chemical bond between Ni-Ga is reinforced, which increases the martensitic transition temperature [890]. Co substitution for Cu in $\text{Ni}_2\text{Mn}_{0.75}\text{Cu}_{0.25-x}\text{Co}_x\text{Ga}$ ($0 \leq x \leq 0.05$) [891] progressively shifts FOPT to higher temperatures, maintaining GMCE ($\Delta S_M \sim -50\text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 5\text{ T}$). Magnetostructural transition temperature of $\text{Ni}_2\text{Mn}_{0.75}\text{Cu}_{0.25}\text{Ga}$ alloy could be tuned (preserving similar values of MCE) to lower temperatures in $\text{Ni}_2\text{Mn}_{0.71}\text{Cu}_{0.27}\text{Fe}_{0.02}\text{Ga}$ ($\sim 299\text{ K}$) and to higher temperatures in $\text{Ni}_2\text{Mn}_{0.70}\text{Cu}_{0.30}\text{Ga}_{0.95}\text{Ge}_{0.05}$ ($\sim 316\text{ K}$) [892]. Partial substitution of Co for Mn in $\text{Ni}_{2.15}\text{Mn}_{0.85-x}\text{Co}_x\text{Ga}$ shows magnetostructural transition in the range $0.04 \leq x \leq 0.06$ [893]. As Co content increases, the transition temperature increases while $|\Delta S_M|$ decreases from a maximum of $14.2\text{ J kg}^{-1}\text{K}^{-1}$ obtained for $x=0.04$ to $7\text{ J kg}^{-1}\text{K}^{-1}$ for $x=0.06$ at $\mu_0\Delta H = 1.9\text{ T}$. When Co is substituted for Ni in Mn-rich/Ga-poor alloys, Curie temperature of the austenitic phase as well as the magnetization of this phase increase. This fact, along with a decrease of the Curie transition of the martensitic phase, leads to an inverse MCE [894]. Pathak et al. [895] studied the $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{32-y}\text{Fe}_y\text{Ga}_{18}$ alloy series ($x=0$ and 8, and $y=0, 1$ and 1.5) and found a maximum value of $\Delta S_M = +31\text{ J kg}^{-1}\text{K}^{-1}$ for $x=8$ and $y=2$ alloy with a hysteresis loss of 40 J/kg .

Partial substitution of Bi for Ga in $\text{Ni}_2\text{MnGa}_{1-x}\text{Bi}_x$ ($x \leq 0.05$) [896] leads to an increase of the martensitic transformation temperature and a decrease of the Curie temperature of the austenitic phase, without merging both temperatures in the explored compositional range. However, as it occurs for stoichiometric ternary alloy, MCE changes from

inverse to normal with field while it decreases for Bi-containing alloys. Partial substitution of Sn for Ga in $\text{Ni}_{50}\text{Mn}_{25}\text{Ga}_{25-x}\text{Sn}_x$ alloys yields a decrease of martensitic temperature of 90 K as x increases from 0 to 2 at. % [897]. This decrease is out of the general trend found for increasing e/a above 7.5 (see Figure 28). Moreover, 10 % substitution of Sn for Ga in Ni_2MnGa alloy suppresses the martensitic transformation [873]. Partial substitution of Al for Ga has been reported without deleterious effect on MCE and can significantly reduce the cost of the alloy [898].

Addition of Gd in $\text{Ni}_{56}\text{Mn}_{18.8}\text{Ga}_{24.5}\text{Gd}_{0.7}$ [899] yields the formation of two phases, being the new one an intergranular phase rich in Gd. The intergranular phase was presumed to enhance the mechanical properties of these Heusler alloys. The enhanced mechanical behavior of Heusler alloys was also the reason to propose Ni-Fe-Ga systems [900].

As V content increases from 0 to 4 at. % in $\text{Ni}_{54}\text{Mn}_{21-x}\text{V}_x\text{Ga}_{25}$, both martensitic and Curie temperatures decrease, with a higher rate of the former temperature, which leads to a separation of these two transitions. A 27 % decrease in $|\Delta S_M|$ at the Curie temperature is observed as x increases from 0 to 4 [901].

Pressure effect on Ni-Mn-Ga alloys differs between Ni-rich and Mn-rich alloys [902]. In both cases, structural and magnetic transition temperatures increase with applied pressure. However, while the rate of increasing the transition temperatures is similar for both structural and magnetic transitions for the Ni-rich alloy (5.8 and 4.7 K/GPa, respectively), the structural transition temperature of Mn-rich alloy increases about three times faster than that of the magnetic one (17.0 and 6 K/GPa respectively), leading to a progressive overlapping of both transitions and thus enhancing MCE. Mandal et al. [903] also reported a large decrease of the magnetic hysteresis in samples submitted to 8 kbar of hydrostatic pressure. Comparison between the magnetocaloric and elastocaloric

properties in Ni-Mn-Ga(Co) alloy series showed a similar relative cooling power after applying 0.8 T or 10 MPa [904].

In 2005, Krenke et al. [905] reported inverse GMCE in $\text{Ni}_{50}\text{Mn}_{37}\text{Sn}_{13}$ and, in 2006, Han et al. [906] reported inverse GMCE also for In-containing Heusler alloys. Later, in 2007, Khan et al. [907] and Du et al. [908] reported similar behavior for $\text{Ni}_{50}\text{Mn}_{37}\text{Sb}_{13}$. $\text{Ni}_2\text{Mn}_{1+x}\text{Z}_{1-x}$ alloys (Z=In, Sn, Sb) close to stoichiometric composition ($x < 0.3$) shows a cubic L_{21} austenitic phase and magnetic transition is typical FM-PM one preserving this austenitic phase. However, martensitic transformation is observed for alloys with a larger deviation from 2:1:1 stoichiometry. Unlike for Ga-containing Heusler alloys, AFM coupling between Mn atoms in these alloys is enhanced in the martensitic phase, leading to a metamagnetic transition from AFM to FM, which can be tailored to be a metamagnetostructural transition from AFM martensitic to FM austenitic phase. The presence of both inverse and normal MCE has been suggested as a mechanism of improved refrigeration cycle [909].

Values of the different temperatures as a function of e/a ratio are shown in Figure 30 for several ternary and quaternary Ni-Mn-Sn systems.

Recent studies of the phase diagram of Ni-Mn-Sn Heusler alloys have been reported by Cakir et al. [910], including their intermartensitic transformations. As e/a increases in ternary Ni-Mn-Sn alloys, transition temperature linearly increases (~ 80 K per 0.1 increase in e/a) [911-913] (e.g. see lines corresponding to solid squares in Figure 30).

However, after partial substitution of transition elements, the general rule based on e/a is no longer fulfilled. In $\text{Ni}_{2+x}\text{Mn}_{1.4-x}\text{Sn}_{0.6}$ Heusler alloys, excess Ni has no monotonic effect on ΔS_M ascribed to the inverse MCE with maximum value at $x=0.12$. However, RC increases continuously [914].

Studies have been done in tin-containing quaternary Heusler alloys with Co [915-921] or Fe [922-924]. The substitution of Co for Ni reduces the magnetostructural transformation temperature and increases the magnetic moment, and thus the Curie temperature of the austenitic phase. Co addition also slightly reduces thermal hysteresis [915, 918] and martensitic transformation temperature but Curie temperature of the austenitic phase is reported to reduce [919, 925], to slightly increase [926] or to be unaffected [927] depending on the alloy composition. Results on partial substitution of Fe for Mn differ between a continuous decrease of the martensitic transformation temperature (14 at.% of Sn) [922] and an increase of martensitic transformation temperature for low Fe substitution (~11 at. % Sn alloys) [923]. In both cases, Curie temperature of the austenitic phase increases and thermal hysteresis becomes broader. Ghosh et al. [924] studied Fe substitution for Ni or Mn for $e/a < 8$ and, in general, martensitic temperature increases with e/a . However, Curie temperature of both phases increases with Fe content, independently whether the substitution is for Mn or Ni.

Cr addition also leads to a decrease of the martensitic transformation but without significant effects on the Curie temperature of the austenitic phase [928]. Nb substitution for Ni yields a decrease of the martensitic transformation temperature and $|\Delta S_M|$, whereas Curie temperature of martensitic phase remains constant and that of the austenitic phase slightly decreases [929].

Substitution of non TM elements for Sn has also been explored, such as Sb [930], In [1040], Al [931, 932], Ge [933, 934] and Si [935]. In the case of Al and Ge, lattice contraction is responsible for the increase of the martensitic transformation temperature, which is in agreement with the observed increase of this temperature after applying hydrostatic pressure [936]. However, the addition of interstitial boron yields a lattice expansion and also shifts the martensitic transformation temperatures to higher values

[937]. For $\text{Ni}_{48}\text{Mn}_{39.5}\text{Sn}_{12.5-x}\text{Al}_x$ ($x=0, 1, 2, 3$) melt spun ribbons, martensitic transition temperature increases with Al content ($\sim 250, 275, 290$ and 295 K for $x=0, 1, 2$ and 3 respectively). While T_C of the austenite phase slightly decreases from 300 K to 296 K but that of the martensite phase is more affected by Al substitution ($229, 215, 205$ and 193 K respectively). Their MCE was studied above 200 K, presenting an inverse MCE at their structural transitions from PM martensite to FM austenite ($\Delta S_M(2\text{ T})=+7.8, 5.7, 4.8$ and $2.8\text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}(2\text{ T})=56, 40, 32$ and 18 J/kg) and a normal MCE at the austenite Curie transition ($-\Delta S_M(2\text{ T})=1.8, 1.7, 1.5$ and $0.9\text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}}(2\text{ T})=92, 53, 31$ and 15 J/kg) [938]. Regarding Si substitution for Sn in $\text{Ni}_{48}\text{Mn}_{39}\text{Sn}_{13-x}\text{Si}_x$ Heusler alloys, a non-monotonic effect of pressure on MCE is reported although martensitic temperature increases with pressure [935]. Minor substitutions of Nb for Ni in $\text{Ni}_{45-x}\text{Nb}_x\text{Co}_5\text{Mn}_{40}\text{Sn}_{10}$ ($x=1$ and 1.5) help to tune its martensitic temperature down to room temperature (68 K per at. % of Nb) and also reduce the Curie temperature of the austenite phase while retaining large values of magnetic entropy and low hysteresis [939].

Figure 31 shows the transition temperatures of different Ni-Mn-In alloys as a function of e/a ratio. Their martensitic transformation temperature increases with e/a ratio, whereas slightly altering the Curie temperature of the austenitic phase.

Stern-Taulats et al. [940] proposed a relationship to correlate the caloric parameters and the temperature difference between the martensitic transformation temperature and austenitic Curie temperature in Ni-Mn-In with differing findings for barocaloric and magnetocaloric responses. Figure 32 shows $\Delta S_M(5\text{ T})$ of FOPT as a function of the relative temperature difference between the austenitic T_C and M_S (absolute values for the SOPTs are also shown) using the data shown in Table S6 of the Supplementary Material. As Stern-Taulats proposed, ΔS_M improves as the relative temperature

difference reduces. The use of different experimental data from the literature could explain the dispersion of the data.

Different quaternary In-containing Heusler alloys have been recently studied by Dubenko et al. including additions of Si, B, Cu, Fe, Ag and Al [941]. Replacing Co for Ni decreases e/a ratio and transition temperature, which is a trend similar to that of ternary systems. For $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{In}_{15}$ alloys, martensitic transition is not observed at temperatures above 77 K for $x > 2$ [942]. In addition, the FOPT of this alloy series is close to the SOPT and significant changes in the MCE can be observed [943]. For Cr substitution in $\text{Ni}_{55}\text{Mn}_{33-x}\text{Cr}_x\text{In}_{17}$ alloys [944], martensitic transformation does not follow a monotonous trend with the average e/a ratio due to the formation of precipitates affecting the Heusler phase composition. If the e/a ratio of this phase is considered, M_s increases with e/a . In the case of 2 at.% Cu additions in $\text{Ni}_{50}\text{Mn}_{34}\text{In}_{16}$, transition temperature shifted from 240 to 295 K [945]. The effects of the strength of magnetic exchange on MCE of this compound using Monte Carlo simulations has been studied by Sokolovskiy et al. [946]. Regarding substitutions for In, although Si and Sb increase e/a ratio, they lead to a decrease of transition temperature [947, 948]. Pathak et al. [949] reported an enhancement of refrigerant capacity in $\text{Ni}_{50}\text{Mn}_{35}\text{In}_{15}$ alloy by substituting 1 at. % of Si, Ge or Al for In. In terms of ΔS_M , it enhances for Si- and Ge-containing samples as compared to that of the ternary alloy but not for Al-containing series, with a similar compositional dependence of the inverse MCE at FOPT and the normal MCE at the Curie temperature of the austenitic phase. $\text{Ni}_{50}\text{Mn}_{35}(\text{In}_{1-x}\text{Cr}_x)_{15}$ series ($0 \leq x \leq 0.1$) were studied by Quetz et al. [950], presenting a large $-\Delta S_M = 11 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T at around 380 K for $x = 0.02$. For $\text{Ni}_{50}\text{Mn}_{35}\text{In}_{15}$, the substitution of small amount of B for In [951] yields large shifts of martensitic Curie and transition temperatures. Dubenko et al. [952] reported that for 1 at. % B, FOPT occurs at 320 K due to the

transformation from FM martensitic phase to PM austenite and its direct characterization indicates a $\Delta T_{ad}=1.5$ K for $\mu_0\Delta H=1.8$ T. However, Pandey et al. [951] assign this transition to a FM-PM austenite transition, being the martensitic transformation around 240 K and the Curie transition of the martensite phase at ~ 200 K. Phase diagram at low fields shows that the effect of B is not monotonous and while T_C of austenite is almost insensitive to B content, for around 1 at. % of B substitution, martensitic transformation and Curie temperatures present a minimum and a maximum respectively [951]. For $Ni_{48}Mn_{39}In_{13-x}Ge_x$ ($x=1-3$), partial Ge substitution for In changes the crystalline structure (e.g. $L2_1$ for $x\leq 2$ and modulated martensite for $x=3$). When x increases from 1 to 3 at. %, its martensitic transformation increases from 220 K to 320 K. For $x\leq 2$, positive $\Delta S_M(5\text{ T})\sim 30\text{ J kg}^{-1}\text{K}^{-1}$ occurs at 210 and 255 K while negative $\Delta S_M(5\text{ T})\sim -6\text{ J kg}^{-1}\text{K}^{-1}$ at 280 and 275 K, whereas for $x=3$ alloy, it shows a weak inverse MCE of $\Delta S_M=4.8\text{ J kg}^{-1}\text{K}^{-1}$ at 320 K [953]. In addition, the martensitic transition temperature increases while the Curie temperature of the austenitic phase decreases with increasing Ge content. The atomic disorder in $Ni_{1.81}Co_{0.22}Mn_{1.46}In_{0.51}$ enhances ΔS_M but sharpens the MCE peak, resulting in an invariant refrigerant capacity [954].

Interstitial H additions to these Heusler alloys shift the martensitic transformation to lower temperatures [955]. This result agrees with the increase of the martensitic transformation after pressure application [956]. However, Pandey et al. [957] found in $Ni_{50}Mn_{35}In_{14.9}Ag_{0.1}$, instead of a decrease of the transformation temperature, a splitting of the MCE peak into two broader peaks after applying 6.6 kbar.

The initial reported MCE results of $Ni_{50}Mn_{37}Sb_{13}$ alloy differ between $\Delta S_M(5\text{ T})\sim 9$ [908] and $\sim 18\text{ J kg}^{-1}\text{K}^{-1}$ [907]. They are analyzed using Maxwell relation and the samples were prepared in similar ways (arc-melting followed by annealing at 850 or

900 °C for one day) although Khan's samples were slowly cooled to room temperature while Du's samples were water-quenched. No comment about presence of impurities appears in any of these two papers. Martensitic transition temperature increases with e/a while Sb/Mn ratio changes [907, 908] (e.g. trend line of 50 at. % Ni in Figure 33), indicating a progressive increase as Sb content is reduced. However, when changing the Ni/Mn ratio, martensitic temperature slightly increases with e/a [958] (e.g. trend line of 12 at. % Sb in Figure 33). The Curie temperature of the austenitic phase is relatively composition independent.

Partial Co substitution for Ni progressively decreases martensitic transition temperature but not significantly affecting the magnetic transitions of martensite and austenite phases [959, 960]. Good MCE values at low fields (1 T) were obtained although thermal hysteresis increases after substitution [959]. For Co-containing Ni-Mn-Sb alloys, the effects of partial substitutions of Si and Ga [961] and Ge [962] for Sb have been also studied. For $\text{Ni}_{49-x}\text{Cu}_x\text{Mn}_{39}\text{Sb}_{12}$, it shows an increasing trend of martensitic transition temperatures with e/a ratio but for partial substitution of Cu for Ni (assuming the typical e/a of 11 electrons per atom in Cu for the calculations), an inverse trend is observed, wherein M_s decreases from 340 to 257 K as x increases from 0 to 2 [963]. Likewise for $\text{Ni}_{50}\text{Mn}_{38-x}\text{Fe}_x\text{Sb}_{12}$ alloys, a decrease of M_s with e/a is also observed [964].

The addition of interstitial boron in Sb-containing Heusler alloys increases the Curie temperature of the austenitic phase and decreases the martensitic transformation (from AFM martensite to FM austenite) [965], which gives a 30 % enhancement in ΔS_M . As it is generally found, these results agree with those derived from pressure application, wherein M_s shifts to higher temperatures with increasing pressure [966].

Other Heusler alloy families have been a subject of study due to its MCE. Results on Ni-Mn-Ge were reported for temperatures around their SOPT [967] wherein T_C decreases with increasing Ni/Mn ratio. Mn-free systems can be also found in the literature, such as Ni-Fe-Ga [900, 968-970], Ni-(Fe,Co)-Ga [971] and Co-Ni-Ga [972]. Ni-Fe-Ga was proposed as a more appropriate MCE material than Ni-Mn-Ga due to its better ductility. Its field dependence of MCE effect is similar to that of Ni-Mn-Ga and at low fields, it changes from inverse MCE to normal MCE. Addition of Co shifts transition temperature to higher values and helps to match Curie temperatures of martensite and austenite phases with the martensitic transformation. Moderate MCE values were reported: $\Delta S_M = -2.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 360 K at 5 T [971]. On the other hand, $\text{Co}_{50}\text{Ni}_{22}\text{Ga}_{28}$ alloy was reported with higher MCE values: $\Delta S_M(5 \text{ T}) = -10.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 313.5 K, ascribed to the martensitic transformation between FM martensite to FM austenite [972]. However, its sharp $\Delta S_M(T)$ curve ($\text{FWHM} < 2 \text{ K}$) and the large thermal hysteresis of the martensitic transformation ($\sim 20 \text{ K}$) must be also considered. Vivas et al. [973] studied the effects of the number of valence electrons on the MCE of half-metal $\text{Fe}_2\text{MnSi}(\text{Ga})$ alloys and found a phenomenological linear relationship between ΔS_M and this parameter.

As a summary, Figure 34 shows the ΔS_M values at 5 T as a function of the transition temperatures of various Heusler alloys. For Ni_2MnGa , the magnetic moment of Mn atoms is $4 \mu_B$ while that of Ni atoms is below $0.3 \mu_B$ [871]. Accounting only the magnetic moment of Mn, a magnetic moment per formula unit of $\sim 4 \mu_B$ for Ni_2MnGa can be considered, while that of $\text{Ni}_2\text{Mn}_{1.4}\text{X}_{0.6}$ ($\text{X} = \text{In, Sn, Sb}$) should be considered $\sim 5.6 \mu_B$. The latter yields a maximum magnetic entropy change, $\Delta S_M = R \cdot \ln(2J+1)$, between 10 and $12 \text{ J mol}^{-1} \text{ K}^{-1}$, which should correspond to $40\text{-}45 \text{ J kg}^{-1} \text{ K}^{-1}$. However, several values exceeding this theoretical limit (see Table S6 of the Supplementary Material)

were reported but they were obtained using Maxwell relation without employing an appropriate measurement protocol to avoid artifacts. Amaral and Amaral [974] have described how artificially high values of ΔS_M resulted from applying Maxwell relation to non-equilibrium states and when a mixture of phases exists in the analyzed range. Other authors also reported overestimated ΔS_M values from using Maxwell relation in comparison with those calculated using Clausius-Clapeyron equation [858, 881, 975]. However, Bourgault et al. found similar ΔS_M values computed from both equations [976] while Fabbri et al. [894] obtained even lower $|\Delta S_M|$ values for $\text{Ni}_{50-y}\text{Co}_y\text{Mn}_{30+x}\text{Ga}_{20-x}$ ($5 \leq y \leq 9$, $0 \leq x \leq 2$) with Maxwell relation than those obtained from Clausius-Clapeyron equation. Zhang et al. [882] pointed that Maxwell relation is valid for magnetic fields below the critical field. Nevertheless, although the presence of significant magnetoelastic interactions would explain these extraordinarily large ΔS_M values [977], the distinctive shapes of these $\Delta S_M(T)$ curves are with sharp peaks of $\text{FWHM} < 3$ K (a general feature in the present cases), which reduces the refrigerant capacity of the system. On the other hand, Caballero-Flores et al. [978] analyzing Ni-Mn-Ga samples, explained that smaller ΔT_{ad} values of samples with FOPT when compared to those of SOPT (assuming them with same magnetic entropy change), are due to their latent heat.

In fact, purely magnetic contribution is even neglected in some models: e.g. that of Recarte et al. [979] which describes the transformation in Ni-Mn-In-Co alloy and the influence of the field dependence of the martensitic temperatures on the measurements of magnetic entropy change. They assumed two constant entropy values for each phase: martensite and austenite. Therefore, the maximum achievable value of the magnetic entropy change would correspond to the structural one. For a field H , when heating the alloy, its martensitic transformation occurs between $A_s(H)$ and $A_f(H)$ temperatures and

Recarte et al. assume a linear change of the entropy between the values of the martensite and the austenite phases within the temperature interval of A_s and A_f . With increasing H , the transition temperatures would ideally shift to lower values (stabilization of the austenitic phase) and ΔS would correspond to the difference between the entropy at certain field H_1 and that at zero field. Therefore, if $A_f(H_1) < A_s(0)$, the maximum ΔS_M allowed ($\Delta S_M = S(\text{austenite}) - S(\text{martensite})$) could be obtained in the thermal range $A_f(H_1) < T < A_s(0)$. In this range, austenite phase is stable at H_1 but upon removing the field, martensite phase becomes the only stable phase and a complete transformation is induced. This is observed from the plateau in its $\Delta S(T)$ curve and ΔS_M is expected to be field-independent for fields above H_2 , which corresponds to $A_f(H_2) = A_s(0)$. For lower fields, there is no thermal range at which a complete transition can be induced and the achieved ΔS_M is only a fraction of the structural entropy change. Borgault et al. [1980], studying $\text{Ni}_{45.5}\text{Co}_{4.5}\text{Mn}_{37}\text{In}_{13}$ single crystals, extended this approximation using field dependence calorimetric data, by considering two new factors: decrease of entropy of the FM austenitic phase as field increases and a smeared transformation instead of an abrupt FOPT.

Recently, it has been proposed to unify both driving factors of the martensitic transition (magnetic field, H , and temperature, T) as a single effective temperature (T^*) to describe its hysteresis. This T^* is correlated to the actual temperature and the applied magnetic field, expressed as $T^* = T + a(T) \cdot H$, where the temperature dependent parameter a is related to the difference of the magnetization between martensitic and austenitic phases and to the entropy change during the martensitic transformation [1981]. This study assumes an equivalent effect of both driving factors: temperature and field. However, Diestel et al. [1982], studying thin films of Ni-Mn-Ga-Co, pointed a different character

of field and temperature on the nucleation and growth mechanisms due to the vector and scalar character of those two magnitudes, respectively.

Gottschall et al. [72] reported that nucleation is a slower phenomenon than growth process, being hindered for fast field sweep (limitation for cycle cooling operation beyond 100 Hz). For a similar alloy composition, Stonaha et al. [983] characterized the microstructures using inelastic neutron scattering and found an unexpectedly large increase in the phonon entropy when heating at temperatures ranging from below to above the Curie temperature.

The total entropy change ascribed to the martensitic transformation is found to exponentially decay with the difference (Δ) between the austenitic Curie and the martensitic transformation temperatures: from ~ 45 to $5 \text{ J kg}^{-1} \text{ K}^{-1}$ as Δ increases from 0 to 150 K [984]. Ab initio studies on the MCE of Heusler alloys can be found in [946, 985-990] and other thermodynamic models are developed to describe the GMCE of Heusler alloys [991].

As FOPT materials, Heusler alloys exhibit both thermal and magnetic hystereses, which are undesirable for applications. In general, there is no clear dependence of thermal hysteresis with MCE results, as observed from Figure 35, which shows the difference between final austenite temperature and start martensite temperature for several Heusler compositions (collected from Table S6 of the Supplementary Material). However, in $\text{Ni}_2(\text{Mn,Fe})\text{Ga}$ alloy series, thermal hysteresis increases with Fe content [992]. In addition, Hu et al. [993] found thermal hysteresis below 2 K for $\text{Ni}_{51}\text{Mn}_{49-x}\text{In}_x$ ($x=15.6$, 16 and 16.2), with ΔS_M up to $33 \text{ J kg}^{-1} \text{ K}^{-1}$. For $x=16$ and 16.2, the rate of transformation temperature reduces between 3 and 4 K/T, while for $x=15.6$, its transition temperature is not significantly affected by field. Similarly, low thermal hysteresis are reported for

Ni₅₅Mn_{19.6}Ga_{25.4} melt-spun ribbons, with also low hysteresis losses [994] and Ni₄₃Mn₄₂Co₄Sn₁₁ ribbons [18]. Thermal hysteresis can be drastically reduced if the geometric compatibility between martensite and austenite phases is achieved [995]. This can be summarized in two conditions: no volume change during transformation and the direct compatibility between the austenitic phase and a single variant of martensite phase. Following these ideas, Srivastava et al. [918] studied Ni₄₅Co₅Mn₄₀Sn₁₀ alloy and reported a thermal hysteresis of 6 K. For Ni_{45.7}Mn_{36.6}In_{13.5}Co_{4.2} alloy, Gottschall et al. [996] reported a high $\Delta T_{ad}(2\text{ T}) = -8\text{ K}$ but it is irreversible while the reversible value is -3 K. It was also found that minor loops lead to reduced hysteresis because the remnant untransformed phase upon incomplete transformation avoids the slow nucleation phenomenon. Recently, Franco et al. [87] proposed the FORC analysis of these minor loops as a new method to further analyze the magnetostructural transformation.

Metallic foams also have been proposed to reduce the thermal hysteresis of Ni-Mn-Ga alloys, for which, additionally, an enhancement of MCE is observed [997]. However, the behavior of Ni₄₃Co₇Mn₃₉Sn₁₁ foams is almost identical to that of their as-atomized powder [998].

Virgin effects, i.e. irreversibility between the first curve obtained and the following ones, have been observed by different authors [880, 999-1006] and are ascribed to the field-induced arrest of the austenitic phase. This indicates the important role of the measurement protocol [39, 1007] affecting the magnitude and sign of ΔS_M [54] or differences in the ΔS_M values derived from heating and cooling experiments [1008].

Therefore, in order to erase the thermal history of the sample in the case of a PM martensitic phase changing to a FM austenitic one, the sample must be cooled down at zero field below M_f before heating to the selected temperature to perform the isothermal field change measurement to really check the change from a starting pure martensite

[883]. In some compositions, austenitic phase can be also arrested at high fields [1009, 1010]. Khovaylo et al. [1011] and Booth et al. [1012], performing repetitive cycles, show a decrease of MCE ascribed to the presence of a remnant fraction of untransformed phase. Dincer et al. [926] studied the effect of irreversibility on MCE by comparing continuous and non-continuous heating experiments of (Ni,Cu)-Mn-Sn alloys. The influence of long thermal cycling on martensitic transformation was reported by Kokorin et al. [1013] with an enhanced field shift of the transformation temperatures (from 17 to 10 K/T) during 1000 cycles at magnetic fields of 1.1 T.

On the other hand, energy losses due to magnetic hysteresis are found ascribed to irreversibility of the magnetization curves at temperatures around the magnetostructural transformation. In order to consider the different data available ascribed to different field changes, a relative hysteresis loss with respect to RC_{FWHM} (both magnitudes at the same magnetic field change) have been calculated. This calculated percentage when presented as functions of e/a , ΔS_M (5 T) and the FWHM of the $\Delta S_M(T)$ peak respectively, shows that hysteresis losses roughly decrease with increasing FWHM (the sharper the MCE peak, the higher the hysteresis losses) but increases with increasing $|\Delta S_M|$ (see Figure 36).

In order to enhance the sample quality of Heusler alloys, it is worth noticing that martensitic transformation is highly dependent on quenching [1014, 1015] and postquenching annealing [1016-1020], whereby both affect the atomic ordering. This is particularly important for these samples which are highly sensitive to tiny compositional changes [943] and thus to heterogeneities in the atomic distribution. In addition, their MCE is also affected by the texture in the samples [1021-1025], particularly during stress assisted thermal cycles [1026].

In 2008, melt-spinning technique was proposed by Hernando et al. [1027] as a way to produce Heusler alloys with enhanced homogeneity, reduced annealing time and lowered amount of undesirable impurity phases. Ga-containing Heusler melt-spun ribbons exhibit a diminished thermal hysteresis of $\sim 2\text{K}$ [994] and also reduced hysteresis losses ($\sim 2\text{ J/kg}$) [1028]. However, for Ni-Mn(Co)-In, a lower thermal hysteresis for bulk samples than that for ribbons was reported [1029]. Recently, Zhang et al. reported a large enhancement of both ΔS_M ($\sim 40\%$) and ΔT_{ad} ($\sim 15\%$) for $\text{Ni}_{43}\text{Mn}_{46}\text{Sn}_{11}$ melt spun ribbons compared to the bulk counterpart [1030]. In general, rapid quenching techniques enable a better homogeneity, smaller crystal size (which stabilizes the cubic austenite) and the presence of disorder. Post-annealing of ribbons leads to crystal growth, compositional homogenization and ordering, generally increasing the martensitic and Curie temperatures. Other recent studies on MCE of melt-spun Heusler alloys can be found in the literature [927, 942, 984, 1028, 1031-1041].

Ball milling was also used to produce Heusler alloys or to induce mechanical stresses to the alloys [1038]. Varzaneh et al. produced $\text{Ni}_{47}\text{Mn}_{40}\text{Sn}_{13}$ alloy via mechanical alloying of pure powders, and reported results comparable to those obtained by arc-melting or rapid quenching techniques [1042]. Ni-Mn-Ga powders produced by spark erosion [1043] were reported with $|\Delta S_M| \sim 2\text{ J kg}^{-1}\text{K}^{-1}$ (2 T) about 363 K, which is in agreement with the Curie transition of austenitic phase although its martensitic transformation was detected at 313 K from calorimetry. Singh et al. show that martensitic phase becomes stabilized in crushed powders due to the induced residual stresses [1044].

MCE in Heusler systems of reduced dimensions has been recently reviewed by Khovaylo et al. [1045]. In general, their MCE are reduced compared to those of their bulk counterparts. Different factors are considered as possible ones to explain this: spin

canting, presence of a distribution of transition temperatures and suppression of the magnetic moment at interfaces. The fabrication of $\text{Ni}_{50.95}\text{Mn}_{25.45}\text{Ga}_{23.6}$ as microwires [1046] stabilizes the tetragonal martensitic phase, with an easy axis along the wire direction and a Curie temperature of 315 K. Their MCE corresponds to a SOPT, with $\Delta S_M = -0.7 \text{ J kg}^{-1} \text{ K}^{-1}$ (3 T) at around room temperature. With Cu additions in $\text{Ni}_{48.8}\text{Mn}_{26.7}\text{Ga}_{20.8}\text{Cu}_{3.7}$ microwires, larger $\Delta S_M = -8.3 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) was obtained, for which the martensitic and Curie transitions overlap at 359 K and FM martensite directly transforms to PM austenite [1047].

Ni-Mn-Ga thin films with coincidence of martensitic and magnetic transitions enable the implementation in very small devices ($\Delta S_M = -8.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 6 T and 346 K) [882, 1003, 1048, 1049]. The transformations in thin film samples are affected by film thickness, whereby in Ni-Mn-In, it is ascribed to increasing grain sizes with film thickness [1050]: while samples with film thickness smaller than 90 nm do not show martensitic transformation, a FOPT is observed for thicknesses above 153 nm. Similarly, this occurs for Ni-Mn-Sn thin films on various types of substrates, which retains an untransformed layer of austenite [1051]. This layer is reduced when better compatibility between the martensite and the austenite is attained, which also lowers hysteresis. For Ni-Mn-Sb thin films with thickness of 800 nm, the partial substitution of Al for Sb increases both martensitic transformation temperature and the maximum magnetic entropy change, as reported by Akkera et al. [1052]. However, Barman et al. [1053], although agree that MCE enhances with increasing Al content, found that the martensitic temperature decreases with increasing Al content. It is worth noticing that, for Barman's samples, grain size decreases as Al content increases and no martensitic transformation is reported for samples with grain size below 100 nm. Dutta et al. [1054] recently studied the effect of strains on the MCE of Ni-Mn-Ga/Ni-Mn-Sn bilayers,

which exhibit combined direct and inverse MCE. GMCE was found for $\text{Ni}_{41}\text{Co}_{10.4}\text{Mn}_{34.8}\text{Al}_{13.8}$ films, with $\Delta S_M = 17.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T, ascribed to the transformation from 14M modulated weakly magnetic martensite to FM austenite, which is detected at $\sim 340 \text{ K}$ on cooling and ~ 380 on heating [1055]. Combination of magnetocaloric Heusler thin films with ferroelectric thin films have been recently proposed for multicaloric applications [1056].

5.1.2.2. *Mn(Fe,Co,Ni)(Si,Ge)*

Similar to Heusler alloys, MnCoGe compound undergoes a martensitic transformation from orthorhombic TiNiSi-type (space group Pnma) to hexagonal Ni_2In -type structure (space group $\text{P6}_3/\text{mmc}$) at 650 K. The lattice parameters of the two phases are correlated as: $a_{\text{TiNiSi}} = c_{\text{Ni}_2\text{In}}$, $b_{\text{TiNiSi}} = a_{\text{Ni}_2\text{In}}$ and $c_{\text{TiNiSi}} = \sqrt{3}a_{\text{Ni}_2\text{In}}$, leading to the volume correlation: $V_{\text{TiNiSi}} = 2V_{\text{Ni}_2\text{In}}$. Moreover, it shows a FOPT at 334 K, which implies a reduced unit cell volume of the orthorhombic volume cell [1057]. The sharp $M(T)$ decrease ascribed to this FOPT in off-stoichiometric $\text{MnCo}_{0.95}\text{Ge}_{1.14}$ compound yields interesting ΔS_M values ($-6.4 \text{ J kg}^{-1} \text{ K}^{-1}$ for 1 T at 331 K) as reported by Fang et al. [1057]. For $\text{Mn}_{0.975}\text{Co}_{0.975}\text{Ge}_{1.05}$, its Curie temperature decreases with excess Ge and depends on sample preparation while $|\Delta S_M|$ is not significantly affected ($1 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1.5 T) [1058]. The ab-initio calculations for off-stoichiometric $\text{MnCo}_{1-x}\text{Ge}$ alloys predict that, for $x > 0.08$, its TiNiSi structure changes to Ni_2In [1059] with a large volume change (3.9 %). Experimental validation of Fang et al. [1060] reported for $x < 0.06$ mainly TiNiSi structure whereas Ni_2In -type dominates at higher Ge concentration. They also reported ΔS_M about $-2 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H = 1 \text{ T}$ for these compositions which Curie temperatures can be tailored around room temperature. For TM-deficient $\text{Mn}_{1-x}\text{Co}_{1-x}\text{Ge}$ series, magnetostructural transition is achieved in the range $0.015 < x < 0.025$ [1061]. For Mn-

poor $\text{Mn}_{1-x}\text{CoGe}$ alloys ($x \leq 0.05$), magnetostructural transition from PM Ni_2In phase to FM TiNiSi phase can be induced and its $-\Delta S_M > 25 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 290 K for $x=0.035$ [1062]. On the other hand, the relative Mn enrichment over Co in $\text{Mn}_{1+x}\text{Co}_{1-x}\text{Ge}$ alloys yields a progressive decrease of transition temperatures with x , giving a maximum $-\Delta S_M = 2.8 \text{ J kg}^{-1} \text{ K}^{-1}$ (1.2 T) at 331 K [1063]. Similarly, FOPT was observed in Mn-enriched $\text{MnCo}_{0.95}\text{Ge}_{0.97}$ (with $\Delta T_{\text{ad}} \sim 1.5 \text{ K}$ at 282 K and 1.9 T) although SOPT is reported for $\text{MnCo}_{0.95}\text{Ge}_{0.95}$ (with $\Delta T_{\text{ad}} \sim 1.5 \text{ K}$ at $\sim 330 \text{ K}$ and 1.9 T) and $\text{MnCo}_{0.95}\text{Ge}$ compounds (with $\Delta T_{\text{ad}} \sim 0.9 \text{ K}$ at $\sim 285 \text{ K}$ and 1.9 T) [1064]. Large thermal and magnetic hystereses were found for $\text{MnCo}_{0.95}\text{Ge}_{0.97}$, indicating the alloy behavior is highly dependent on its thermal history.

Compositional tailoring of the structural change between Ni_2In and TiNiSi type phases has been reported. Previously in 2006, Lin et al. [1065] studied the $\text{MnFe}_{1-x}\text{Co}_x\text{Ge}$ series, whereby the hexagonal phase characteristic of MnFeGe extends up to $x < 0.8$ while the orthorhombic phase characteristic of MnCoGe extends down to $x > 0.85$, and reported a continuous increase of T_C from ~ 150 to $\sim 350 \text{ K}$, regardless of its crystalline structure. For $x=0.8$, a maximum absolute magnetic entropy change of $9 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at $T_C = 289 \text{ K}$ is observed, for which the magnetic moment is maximum ($4.69 \mu_B/\text{f.u.}$). However, traces of Ni_2In -type impurities were found [445]. For $\text{CoMnGe}_{0.945}\text{Ga}_{0.055}$, a FOPT occurs at 305 K, giving a GMCE of $-\Delta S_M(5 \text{ T}) = 29.6 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{\text{ad}}(2 \text{ T}) = 6.5 \text{ K}$ [1066]. For $\text{MnCo}_{1-x}\text{In}_x\text{Ge}$ compounds, the partial substitution of Co for In can yield overlapping between martensitic transformation and Curie temperatures, which results in a magnetostructural coupling with a large ΔS_M for $x=0.015$ [1067].

MnNiGe is a single spiral AFM with a Néel temperature of 346 K, which crystallizes in TiNiSi -type structure at room temperature and transforms into Ni_2In -type structure

(with volume change -1.6 %) at 470 K. Off-stoichiometry in $\text{Mn}_{1.9-x}\text{Ni}_x\text{Ge}$ compounds [1068] reduces their structural transition temperatures below the Curie temperature of the Ni_2In type phase and produces two MCE peaks: an inverse magnetocaloric peak (for $x=0.85$, $\Delta S_M=+5$ (1 T) and $+27 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at about 135 K) ascribed to a FOPT metamagnetic structural transformation from AFM TiNiSi to FM Ni_2In structure, and a direct magnetocaloric peak ($\Delta S_M=-2$ (1 T) and $-6.5 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at 205 K) ascribed to the SOPT Curie transition of the Ni_2In -type phase. Annealing of polycrystalline $\text{MnNiGe}_{1.05}$ melt-spun ribbons yields a sharpening of MCE peak with an enhanced $|\Delta S_M|$ and higher transition temperatures [1069]. For $\text{MnNi}_x\text{Co}_{1-x}\text{Ge}_{1.05}$ alloys with excess Ge, an inverse MCE below 200 K ($\Delta S=+1.9 \text{ J kg}^{-1}\text{K}^{-1}$ at 1 T) ascribed to the transition from AFM TiNiSi -type to FM Ni_2In -type structure is observed [1070]. For $x\sim 0.05$ in $\text{MnNi}_{1-x}\text{Co}_x\text{Ge}_{1.05}$ series, structural transition from TiNiSi to Ni_2In -type phases and magnetic transition from PM to FM Ni_2In phase are obtained [1070]. $\text{MnNi}_{1-x}\text{Co}_x\text{Ge}$ alloys with higher Co substitution ($x=0.38$ and 0.40) [1071], exhibit both inverse and normal MCE, in a relative small thermal span, around the structural and magnetic transformations respectively. When Fe partially substitutes for Ni, the structural transformation temperatures decrease whereas the Curie temperatures are less significantly affected. The MCE curves of $\text{MnNi}_{0.75}\text{Fe}_{0.25}\text{Ge}$ show overlapped inverse and direct curves, with maximum extremes of $+1.5 \text{ J kg}^{-1}\text{K}^{-1}$ (1 T) at 175 K and $-1.2 \text{ J kg}^{-1}\text{K}^{-1}$ (1 T) at 210 K [1072]. Magnetostructural coupling between martensite transformation and PM to FM states in Fe substituted MnNiGe alloys is discussed [1073]. When Fe substitutes for Mn or Ni in MnNiGe alloys, structural transition decreases below the magnetic transition of the low temperature phase and converts AFM (Fe-free parent compound) to FM, yielding a FM martensite to PM austenite transition upon heating the sample and GMCE ($\text{MnNi}_{0.77}\text{Fe}_{0.23}\text{Ge}$: $|\Delta S_M|(5 \text{ T})\sim 18$

$\text{J kg}^{-1}\text{K}^{-1}$ at ~ 265 K; $\text{Mn}_{0.82}\text{Fe}_{0.18}\text{NiGe}$: $|\Delta S_M|(5\text{ T}) \sim 30\text{ J kg}^{-1}\text{K}^{-1}$ at 205 K). For $\text{Mn}_{0.93}\text{Cr}_{0.07}\text{NiGe}_{1.05}$, a broad $-\Delta S_M$ peak of $\sim 7\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at around ~ 350 K is found, whereas an inverse MCE is observed at $\mu_0 H = 1$ T [950]. In addition, a larger $-\Delta S_M$ value with sharper peak is observed at 365 K.

Partial substitution of Co for Al in $\text{MnCo}_{1-x}\text{Al}_x\text{Ge}$ [1074] yields a crystal structure of hexagonal Ni_2In -type for $x > 0.03$. Its Curie temperature can be tuned between 286 and 347 K by changing the Co/Al ratio and the maximum magnetic entropy change achieved in the studied series is $1.52\text{ J kg}^{-1}\text{K}^{-1}$ (1.5 T) at 310 K for $x = 0.08$. When Al is substituted in $\text{MnNiGe}_{1-x}\text{Al}_x$ compounds, a FOPT magnetostructural transition at ~ 190 K is observed for $x \sim 0.09$ with $|\Delta S_M| = 17.6\text{ J kg}^{-1}\text{K}^{-1}$ at 5 T [1075]. $\text{MnCo}_{1-x}\text{V}_x\text{Ge}$ alloys crystallize in Ni_2Ti -type structure for $x > 0.02$ and are reported that increasing V content leads to a reduction of Curie temperature (e.g. for $x = 0.06$, $|\Delta S_M|(1.5\text{ T}) = 3.9\text{ J kg}^{-1}\text{K}^{-1}$ at 265 K) [1076]. However, when Mn instead of Co substitutes for V in $\text{Mn}_{1-x}\text{V}_x\text{CoGe}$, thermal hysteresis decreases yielding large MCE at room temperature (for $\text{Mn}_{0.98}\text{V}_{0.02}\text{CoGe}$, $|\Delta S_M|(1.2\text{ T}) = 9.5\text{ J kg}^{-1}\text{K}^{-1}$) [1077]. For $\text{Mn}_{1-x}\text{Cr}_x\text{CoGe}$, when $x = 0.11$, structural and magnetic transitions coincide, and induce a single FOPT from FM to PM at 322 K [1078]. In addition, for $0.04 \leq x \leq 0.25$, the first order character of the transition is retained with large $-\Delta S_M$ values (e.g. $28.5\text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at 322 K for $x = 0.04$) but for $x = 0$ and $x = 0.27$, SOPTs with much lower values of $-\Delta S_M$ are observed. Both substitution of Cr and hydrostatic pressure application in these systems shift their Curie temperatures to lower values [1079], showing a correlation between the effects of chemical and external pressures in this family of alloys.

Partial substitution of Sn for Ge in $\text{CoMnGe}_{1-x}\text{Sn}_x$ (x between 0.04 and 0.055) leads to a magnetostructural transition with $|\Delta S_M| = 4.5\text{ J kg}^{-1}\text{K}^{-1}$ but accompanied by large thermal

hysteresis of ~20 K [1080]. Sandeman et al. [1081] studied the magnetocaloric effect ascribed to the metamagnetic transition of $\text{CoMnSi}_{1-x}\text{Ge}_x$ compounds ($x \leq 0.3$) with TiNiSi-type structure. Their FOPT shift to lower temperatures with increasing Ge concentration while their SOPT, ascribed to the Curie temperature ($T_C \sim 390$ K for $x=0$), tune to higher temperatures. Previous reported values for the transition temperatures are broadly dispersed, which indicates that they are highly sensitive to sample preparation. The single phase CoMnSi compound presents $\Delta S_M \sim 6 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T \sim 1.5$ K ascribed to the metamagnetic transition at $\mu_0 \Delta H = 5$ T. Zhang et al. [1082] found that the metamagnetic transition of $\text{CoMnSi}_{0.88}\text{Ge}_{0.12}$ is highly field-sensitive, leading to large inverse MCE of $\Delta S_M(2 \text{ T}) > 6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 185 K but with large magnetic and thermal hystereses. Interstitial B additions to MnCoGe yield GMCE near room temperature (e.g. for $\text{MnCoGeB}_{0.02}$, $\Delta S_M(5 \text{ T}) = -47.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T = 287$ K but with a thermal hysteresis of 14 K) [1083]. A magnetostructural FOPT is obtained after coincidence of the structural TiNiSi to Ni_2In transition with the FM to PM one. Annealing of $\text{Mn}_{1-x}\text{Cu}_x\text{CoGe}$ melt-spun ribbons leads to a magnetostructural transformation and an enhanced MCE (e.g. for $x=0.02$, $\Delta S_M(1 \text{ T}) = 7.2 \text{ J kg}^{-1} \text{ K}^{-1}$ at 286 K) [1084]. $\text{MnV}_{0.02}\text{CoGe}_{0.99}$ ribbons, with a single Ni_2In -type phase, were prepared using melt-spinning by Ma et al. [1085], presenting a SOPT at 252 K with $-\Delta S_M(1 \text{ T}) = 0.9 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\text{RC}_{\text{FWHM}}(1 \text{ T}) = 36 \text{ J/kg}$.

The behavior of this alloy family can be generally described with MCE as a function of the average number of electrons in the outer shell per atom, e/a (including 3d atoms for transition elements), which was previously used to describe Heusler alloys. This is illustrated in Figure 37 for several compounds. The maximum $|\Delta S_M|$ values are obtained for those e/a values leading to coincide the structural and magnetic transitions.

Moreover, some RE-TM-Ge compounds also exhibit TiNiSi structure with large magnetic moments above $4 \mu_B$ but, in general, with transition temperatures well below room temperature. See some examples in the section *Compounds with 33 at. % of RE* described earlier.

5.1.2.3. $(Mn, Fe, Co)_5(Ge, Si, Sb)_3$

In 2002, Songlin et al. studied the $Mn_{5-x}Fe_xSi_3$ ($0 \leq x \leq 5$) [1086] and $Mn_{5-x}Ge_{3-x}Sb_x$ ($0 \leq x \leq 0.3$) [1087] series, both crystallizing in the hexagonal Mn_5Si_3 -type structure (space group $P6_3/mcm$). For the former series, Mn_5Si_3 compound is an AFM system which experiences two FOPT (from AFM to AFM at 66 K and AFM to PM at 99 K). Inverse MCE is observed below the first transition while direct MCE is observed at higher temperatures [1088]. However, Fe_5Si_3 compound is FM up to 373 K. The maximum $|\Delta S_M|$ value as well as the maximum magnetic moments observed in the compounds of the Mn/Fe series were found for $x=4$, which undergoes a Curie transition from FM to PM ($|\Delta S_M| \sim 2 \text{ J kg}^{-1} \text{ K}^{-1}$ at 305 K and 2 T). The AFM character of the alloy ($x \leq 2$) changes to FM one ($x \geq 4$) as Fe content increases. However, some discrepancies can be found in the literature. For $Mn_2Fe_3Si_3$ compound, while Songlin et al. [1086] found FM behavior with a Curie temperature at about 175 K, Candini et al. [1089] reported an AFM behavior with Néel temperature at 150 K. This reflects the magnetic character of the system is highly dependent on the bondings between transition metals, and possibly on the Fe and Mn occupancies of the 4d and 6g sites. Recently, Wu et al. [1090] reported that B additions to Fe_4MnSi_3 increase the lattice parameters while preserving the crystal structure, leading to the assumption of B addition to occupy interstitial positions. As a result, their Curie temperature increases but not their ΔS_M . The MCE of hexagonal $MnFe_4Si_3$ single crystals, with ordering transition at ~ 300 K, shows anisotropy with $\Delta S_M \sim 2.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) for the field applied along $[100]$

direction while $1.25 \text{ J kg}^{-1} \text{ K}^{-1}$ for the field applied along $[001]$ direction [1091]. Studies on the density of phonon states were performed to correlate MCE and lattice dynamics, resulting an influence limited to long wavelength contributions [1092].

Regarding $\text{Mn}_{5-x}\text{Ge}_{3-x}\text{Sb}_x$ series, Mn_5Ge_3 alloy is FM with Curie transition at 296 K. The ascribed absolute value of $|\Delta S_M| \sim 9 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T is comparable to that of Gd [1087]. Partial substitution of Sb for Ge retains its FM order while increasing T_C about 50 K per Ge atomic substitution for Sb in the formula unit. For $x=0.3$, $\sim 40\%$ decrease in ΔS_M while $<10\%$ reduction in RC is observed since the MCE peak becomes wider upon Sb substitution. Partial Fe substitution for Mn in $(\text{Mn}_{0.9}\text{Fe}_{0.1})_5\text{Ge}_3$ leads to enhancement of both ΔS_M and RC [1093]. Unlike the influence of Sb additions on the MCE of $\text{Mn}_{5-x}\text{Ge}_{3-x}\text{Sb}_x$, Si [1094, 1095] and Ga [1096, 1097] substitutions lead to a decrease of Curie temperatures. Partial substitution of Al and Sn have been also studied [1096]. Recently, Tolinski et al. [1098] found that mechanical milling of Mn_5Ge_3 reduces its MCE ($\Delta S_M \sim 4 \text{ J kg}^{-1} \text{ K}^{-1}$, $T_C=297.4 \text{ K}$ and $\Delta T_{ad}=1.8 \text{ K}$ at 2 T).

Zhao et al. [1099] reported the increase of T_C ($\sim 250 \text{ K}$) for $\text{Mn}_{5-x}\text{Sn}_{3-x}\text{Ga}_x$ series with Ga content. Although its magnetic moment increases as compared to Ga-free alloy, their $|\Delta S_M|$ values are moderate ($<1 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=2 \text{ T}$). Similar results are obtained for ball milled $(\text{Mn}_{1-x}\text{Fe}_x)_5\text{Sn}_3$ series ($\Delta S_M=0.87 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0\Delta H=1.5 \text{ T}$), with a flat curve in a wide temperature range [1100], probably due to inhomogeneities inherent to the production process.

Some RE-containing alloys also exhibit the Mn_5Si_3 -type phase and were already discussed in the section *Compounds with 63 at. % of RE*.

5.1.2.4. *MnAs alloys*

Although Kuhrt et al. [1101] reported MCE results for MnAs(P) as early as in 1985 (with $\Delta T_{ad}=0.2$ K at 0.65 T and 300 K accompanied by thermal hysteresis of ~ 10 K), the reason for the rise in studying this family can be found in the work of Wada et al. [1102], which reported in 2001 its GMCE ascribed to a FOPT from FM to PM in MnAs alloy with ($\Delta S_M=-30$ J kg⁻¹K⁻¹ and $\Delta T_{ad}=13$ K at 5 T and 318 K). The FOPT observed in the MnAs compound is ascribed to a magnetostructural transition of FM hexagonal NiAs-type structure (space group P6₃/mmc) to PM orthorhombic MnP-type structure (space group Pnma), which could be described as a kind of soft mode transition with large reduction in the modulus of compression and stress at T_C [1103]. The relationships among the lattice parameters of these two structures can be found in [1104]. In addition, a SOPT occurs at higher temperature of 378 K, which corresponds to a reversal transition from MnP to NiAs-type phase [1102, 1105]. In general, arsenic-containing alloys attractive for MCE, can be grouped into two different families: alloys derived from MnAs compound and compounds crystallizing in Fe₂P-type structure. The former alloy family is described under this section while the latter will be discussed in the following section.

The FOPT of MnAs sharpens for monocrystalline samples and slightly shifts to lower temperatures (315.5 K), whereas its SOPT decreases to 391 K [1106]. Excess Mn in Mn_{1+δ}As compounds leads to linearly decreases of both the magnetic moment and the Curie temperature with δ as: $\mu=(3.55-2.78\delta)$ μ_B /f.u. and $T_C=(577-900\delta)$ K respectively [1105]. Unusual attenuation of surface acoustic waves has been recently observed in magnetocaloric/piezoelectric MnAs/GaAs hybrid thin films, showing a weak dependence on the film thickness [1107].

Partial Sb substitution for As in $\text{MnAs}_{1-x}\text{Sb}_x$ effectively reduces both transition temperatures (e.g. 280 K in $\text{MnAs}_{0.9}\text{Sb}_{0.1}$) and thermal hysteresis (from 6 to <1 K for $x=0$ to 0.1), although the system remains in NiAs-type structure and transformation apparently becomes a SOPT for above 10 at. % Sb [445]. For $0.125 \leq x \leq 0.25$, a good agreement between theoretical calculations and experimental measurements is found, while larger deviations are observed for $x=0.3$, whereby a prominent MCE reduction is observed [1108]. Recently, Trassinelli et al. [1109] reported that the thermal hysteresis in MnAs thin films is suppressed by the induced defects from ion bombardment, which facilitate nucleation during the FOPT process.

A relatively linear correlation of the pressure application to $\text{MnAs}_{1-x}\text{Sb}_x$ and their transition temperatures has been reported with a rate of ~ 10 K/kbar (higher rate in Sb-free samples) [1110]. Shock compaction of $\text{MnAs}_{1-x}\text{Sb}_x$ ($x=0.068$ and 0.073) decreases their MCE properties in FOPT although the outstanding MCE values ($\Delta S_M(1\text{ T})=-28$ $\text{J kg}^{-1}\text{K}^{-1}$ for $x=0.073$) are recovered after annealing at 1073 K for a week. However, Curie temperature changes compared to that of the starting powder [1102, 1111-1116]. Recently, Rocco et al. [1117] studied the influence of pressure and chemical doping (to resemble negative pressure) on magnetic phases of $\text{Mn}_{1-x}\text{Fe}_x\text{As}$ series ($x \leq 0.018$) and supplied a magnetic phase diagram of this compound under pressure conditions. They found that Fe doping only produces structural deformation and AFM orthorhombic phase coexists with FM hexagonal phase for the highest pressure studied below the critical temperature.

Besides Sb substitution for As [1102, 1111-1116], other compositional tailoring have been studied. Si substitution for As in $\text{MnAs}_{1-x}\text{Si}_x$ ($x \leq 0.09$) series is reported to reduce thermal hysteresis (almost zero for $x=0.09$), which is due to the transformation from FOPT to SOPT. Moreover, transition temperatures increases with higher Si

concentration (from 270 to 302 K for $x=0.03$ to 0.09 [1118], although these values are below that of Si-free alloy [1102]), while maintaining high $|\Delta S_M|$ values ($>10 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T and $>5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2 T). These $\text{MnAs}_{1-x}\text{Si}_x$ samples were prepared by ball milling and subsequently annealed but, unfortunately, a reference Si-free sample was not prepared in the study for comparison. Interstitial N additions to these Si-containing alloys yields a reduction in T_C and a slight MCE enhancement [1119]. Severe lattice distortions produced by addition of interstitial C in MnAsC alloys lead to a decrease of T_C with optimum $|\Delta S_M| > 22 \text{ J kg}^{-1} \text{ K}^{-1}$ for $\text{MnAsC}_{0.03}$ [1120]. Substitution of P for As reduces the hysteresis and retains high $|\Delta S_M|$ (For $\text{MnAs}_{0.994}\text{P}_{0.006}$, $|\Delta S_M| = 11.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2 T and 294.5 K with thermal hysteresis of 4 K) [1121]. Large microstrains in $\text{MnAs}_{0.97}\text{P}_{0.03}$ due to ball milling, suppress the orthorhombic-hexagonal FOPT and promotes a helimagnetic state at low temperatures. These factors lead to a decrease of hysteresis but also to their MCE [1122]. Mitsiuk et al. [1123] studied the MCE of P substituted alloys at a high field change of 14 T ($|\Delta S_M| \sim 40 \text{ J kg}^{-1} \text{ K}^{-1}$) and pointed that the structural instability is due to the anomalous expansion-compression between 350 and 250 K.

Besides As substitution, compositional tailoring by Mn substitution can be also found in the literature. Fe substitution for Mn in $\text{Mn}_{1-x}\text{Fe}_x\text{As}$ decreases the transition temperature with an enhanced thermal hysteresis from 10 to 30 K as x increases from 0 to 0.015 [1124] unlike for Fe additions in $\text{Gd}_5\text{Si}_2\text{Ge}_2$ family [28]. Balli et al. [1125] showed that the combined substitution of V and Ti for Mn, retains the high $|\Delta S_M|$ values of $\sim 30 \text{ J kg}^{-1} \text{ K}^{-1}$ while tuning T_C from 318 to 266 K (for MnAs and $\text{Mn}_{0.9}\text{Ti}_{0.05}\text{V}_{0.05}\text{As}$ respectively). No monotonous trend for $|\Delta S_M|$ is reported for $\text{Mn}_{1-x}\text{Al}_x\text{As}$, alloys with Al substituted for Mn but an evident decrease in MCE is observed for the $x=0.03$ alloy, although thermal hysteresis persists in this alloy [1126]. Cu substitution for Mn does not

significantly affects the transition temperature but increases thermal hysteresis while reducing the unit cell volume as compared to MnAs alloy ($|\Delta S_M| \sim 20\text{-}30 \text{ J kg}^{-1} \text{ K}^{-1}$) [1127]. Xu et al. [1128] grew $\text{Mn}_{1-x}\text{Co}_x\text{As}$ epitaxial films on GaAs and found a double $|\Delta S_M|$ value for 0.24 % Co. The Cr substitution for Mn in $\text{Mn}_{1-x}\text{Cr}_x\text{As}$ reduces transition temperature (e.g. 267 K for $x=0.01$) and enhances MCE by reducing thermal hysteresis (e.g. 5 K for $x=0.01$) [1129]. For ball milled $\text{Mn}_{0.95}\text{Cr}_{0.05}\text{As}$ alloy, it exhibits two FOPT: a combined magnetostructural transition from PM MnP-type phase to FM NiAs-type phase at 234 K ($\Delta S(T) = -5.28 \text{ J kg}^{-1} \text{ K}^{-1}$, at 213 K and 5 T) and from FM NiAs-type phase to helimagnetic (H_a -type) MnP-type phase at 159 K (inverse MCE with $\Delta S(T) = 6.06 \text{ J kg}^{-1} \text{ K}^{-1}$, at 163 K and 5 T) [1130]. For $\text{Mn}_{0.92}\text{Ba}_{0.08}\text{As}$ thin films grown on Al_2O_3 (0001) substrate, its T_C increases up to 350 K and exhibits high $|\Delta S_M|$ values [1131].

Colossal MCE phenomenon (magnetic entropy change far beyond the theoretical magnetic limit) was also reported in several papers [1124, 1127, 1132, 1133] and some explanation was attempted to ascribe this colossal behavior to magnetovolume coupling [1124, 1133-1136]. However, the use of Landau theory shows that the so-called colossal MCE could be an artifact derived from the estimation of ΔS_M when not accounting for the irreversibility and the mixed-phase regime typical of FOPTs [50, 56, 974, 1137-1139]. Balli et al. [1140] also questioned the use of Maxwell relation to obtain ΔS_M data, in agreement with the work by Bratko et al. [1141]. It is worth pointing that Wada et al. [1102] already described the spikes in ΔS_M reaching $-40 \text{ J kg}^{-1} \text{ K}^{-1}$ could probably be artifacts, as observed in their results. Therefore, the maximum values obtained at such spikes reported are neglected in this section. Niemann et al. [1142] describe the inapplicability of Maxwell relation to quantify MCE, when magnetic induced reorientation occurs (e.g. variants in magnetic shape memory alloys or domain structure

in hard magnetic systems). They disagree to use Maxwell relation even when the measurement protocol proposed by Caron et al. [57] is used (to avoid artifact colossal MCE peaks) as it can yield spurious non-reversible MCE peaks.

5.1.2.5. *MnFe(P,As) alloys*

The compounds belonging to the other As-containing alloy family with interesting MCE properties crystallize in the hexagonal Fe_2P -type structure (space group P-62m). A recent review on these materials can be found in [1143]. Tegos et al. [1144] reported in 2002 the GMCE for $\text{MnFeP}_{0.45}\text{As}_{0.55}$, for which a magnetostructural FOPT occurs ($\Delta S_M = -14.5$ and $-18 \text{ J kg}^{-1} \text{K}^{-1}$ at 2 and 5 T respectively, and with thermal hysteresis below 1 K at about 300 K). Recently, direct ΔT_{ad} measurements of ~ 2.5 K were reported under millisecond pulse field by Cugini et al. [44]. In stoichiometric compounds, Mn and Fe occupy specific sites (3g and 3f respectively), whereas P and As are statistically distributed over 2c and 1b sites. The unit cell volume decreases with field and a metamagnetic transition from PM to FM is induced just above Curie temperature [1145]. Ab initio studies have been performed to determine the preferential occupancy of Mn to the 3g site in $(\text{Fe}_{1-x}\text{Mn}_x)_2\text{P}_{1-y}\text{Si}_y$ doped compounds [1146]. Transition temperatures can be also tuned by varying Mn/Fe ratio (e.g. transition temperature decreases with increasing Mn content in $\text{Mn}_{2-x}\text{Fe}_x\text{P}_{0.5}\text{As}_{0.5}$ series with $|\Delta S_M|(2 \text{ T}) > 15 \text{ J kg}^{-1} \text{K}^{-1}$ for $x \geq 0.8$ [1147] and AFM phase stabilizes for $x = 0.5-0.6$ as observed from Mössbauer spectroscopy [1148]) or As/P ratio (T_C can be tuned between 168 and 332 K for As/P ratios of 3/1 and 13/7 respectively [1088]) without losing its GMCE. The phase diagram of $\text{MnFeP}_{1-x}\text{As}_x$ can be found in [1149], with an AFM phase for $x < 0.35$ and a magnetic triple point FM-AFM-PM at 210 K.

Recent results from Szymczak et al. [1150] showed that, for $\text{Mn}_{2-x}\text{Fe}_x\text{P}_{0.5}\text{As}_{0.5}$, while for $x=1$, a single FOPT from PM to FM occurs at 275 K, for $x=0.7$ this is splitted into two transitions: SOPT from PM to FM at 190 K and FM-AFM at 90 K. Tailoring Mn/Fe ratio in MnFe(P,As) alloys can also diminish thermal hysteresis below 1 K but transforms FOPT into a SOPT. For $\text{Mn}_x\text{Fe}_{1.95-x}\text{As}_{0.5}\text{P}_{0.5}$ series, their Arrott plots indicate that the magnetoelastic FOPT obtained for $x<1.4$ changes to a SOPT with increasing x and, although a stable hexagonal Fe_2P phase appears for $x\leq 0.8$, a magnetostructural transformation between FM hexagonal phase and a body centered orthorhombic phase (space group Imm2) occurs for higher x values [1151]. Small changes in the crystallographic order imply significant modifications to the magnetic order in these compounds [1152], which highlight the importance of sample preparation. Particularly, samples quenched in water exhibit lower hysteresis and enhanced MCE [1153]. Tegus et al. [1154] studied TM substitution in compositions close to $\text{MnFeP}_{0.5}\text{As}_{0.5}$ for MCE optimization, whereby Cr and Co substitutions for Fe reduce their transition temperatures. Contrarily, 10 at.% Mn substitution for Fe enhances MCE while not significantly affecting the transition temperature. On the other hand, 10 at.% substitution of Fe for Mn yields an increase of transition temperatures in these Fe-enriched samples. Griбанov et al. [1147] extended the study to Mn-rich composition in $\text{Mn}_{2-x}\text{Fe}_x\text{P}_{0.5}\text{As}_{0.5}$ and found a continuous decrease of transition temperatures as x decreases and that FM phase disappears when $x=0.5$. Maximum $|\Delta S_M|$ values $>20 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) at 225 K were obtained for $x=0.9$, while reduced about one half when $x=0.6$. Recently, Balli et al. [1155] have studied the complex magnetic behavior exhibited by MnRhAs single crystals: metamagnetic AFM to FM transition at 158 K, SOPT from FM to another AFM state at 196 K and finally PM state occurs above 238 K. The former transition yields an inverse MCE of $\Delta S_M=3 \text{ J kg}^{-1}\text{K}^{-1}$ at 1 T, while the

second transition corresponds to a weaker direct MCE response ($\Delta S_M = -2.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T).

The presence of As in this promising alloy family, as well as the previously described one, should be considered an important drawback due to its inconvenient toxic nature especially for domestic appliances. Therefore, this prompted the search of substituting other elements (P, Si and Ge) for As in MnFe(P,As) alloys. Although GMCE could be preserved in As-free alloys, thermal hysteresis is generally increased.

Partial or complete substitution of Ge for As in $\text{MnFeP}_{0.5}\text{As}_{0.5-x}\text{Ge}_x$ does not change the Fe_2P -type structure but progressively increases T_C (up to 570 K in As-free sample). However, minor substitution of Ge for As converts the character of the transition to second order, implying a weakening of MCE (e.g. $\text{MnFeP}_{0.5}\text{As}_{0.44}\text{Ge}_{0.06}$, ~50% decrease in $|\Delta S_M|$ is obtained compared to that of $\text{MnFeP}_{0.5}\text{As}_{0.5}$) [1156]. In the P- and Mn-enriched series, i.e. $\text{Mn}_{1.1}\text{Fe}_{0.9}\text{P}_{0.7}\text{As}_{0.3-x}\text{Ge}_x$ ($0 \leq x \leq 0.3$), T_C increases with x from 150 to 380 K, while field-induced FOPT is lost for $x > 0.15$ [1157]. For Mn-rich $\text{Mn}_{1.1}\text{Fe}_{0.9}\text{P}_{1-x}\text{Ge}_x$ ($0.20 \leq x \leq 0.23$) [1158, 1159] and $\text{Mn}_{1.2}\text{Fe}_{0.8}\text{P}_{1-x}\text{Ge}_x$ ($x = 0.22$) series [1160], complete substitution of P and Ge for As preserves their GMCE ($|\Delta S_M| > 30 \text{ J kg}^{-1} \text{ K}^{-1}$ for 5 T). The excess Mn atoms preferentially locate at 3f sites and affect the magnetic moment, hysteresis and critical field to induce the PM to FM transition [1161]. In both cases, Curie transition temperature increases with Ge content although the unit cell volume increases only in a certain x range [1159]. Liu et al. [1162] highlight the phase fraction of the transformed phases as a relevant parameter to describe $|\Delta S_M|$ values (It is worth mentioning that coexisting phases lead to artifacts in MCE curves, as described in the previous sections). With melt spinning, the initial good MCE results of ball milled $\text{Mn}_{1.1}\text{Fe}_{0.9}\text{P}_{0.7}\text{Ge}_{0.24}$ samples are further enhanced due to the improvement of

compositional homogenization [1163]. Mössbauer results showed the coexistence of FM and PM phases in a temperature range of 50 K around T_C for $Mn_{1.1}Fe_{0.9}P_{1-x}Ge_x$ compounds, with $0.19 < x < 0.26$, [1159]. For $Mn_{2-y}Fe_yP_{0.75}Ge_{0.25}$ series, thermal hysteresis could be reduced while preserving high MCE responses, by tuning Fe/Mn and optimizing the annealing conditions [1164]. The compositional limit of FOPT is $y=0.8$ and its $\Delta S_M = -20 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2 T with negligible hysteresis. Spark plasma sintered $Mn_{1.1}Fe_{0.9}P_{0.8}Ge_{0.2}$ is reported with large $|\Delta S_M|$ values close to $50 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T ($\Delta T_{ad} = 1.2 \text{ K}$ at 1.5 T) at 253 K but accompanied by thermal hysteresis of about 15 K [1165]. In addition, higher hysteresis were reported in spark plasma sintered $MnFeP_{0.89-x}Si_xGe_{0.11}$ compounds ($0.14 \leq x \leq 0.26$) [1166]. The inhomogeneous distribution of Ge could affect the behavior of such compositions, whereby neutron diffraction studies found different Ge contents in coexisting FM and PM phases [1167]. Optimal thermal annealing treatment is reported to enhance homogenization, leading to the MCE enhancement and a decrease in thermal hysteresis [1168]. Chen et al. [1169] proposed that the crossover from SOPT to FOPT in $Mn_{1.1}Fe_{0.9}P_{1-x}Ge_x$ is related to the lattice parameter of its parent phase. They used Landau theory along with Bean-Rodbell theory to describe the transformation process in this series [1170]. They have further optimized the RC of $Mn_{1.1}Fe_{0.9}P_{1-x}Ge_x$ by preparing nanostructured $Mn_{1.1}Fe_{0.9}P_{0.79}Ge_{0.21}$, which exhibits MCE of $\Delta S_M = -26.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 281.5 K and 5 T and $RC_{FWHM} = 1138 \text{ J/kg}$ [1171]. Al doping in these systems retains the Fe_2P -type structure and MCE but affects the unit cell volume (increases with Al content) and reduces thermal hysteresis [1172]. Ru substitution in $MnFeP_{0.8}Ge_{0.2}$ and $Mn_{1.2}Fe_{0.8}P_{0.5}Si_{0.5}$ reduces thermal hysteresis from 10 K to 2-3 K, while preserving a sharp magnetic transition [1173].

Partial Si substitution for As in $MnFeP_{0.5}As_{0.5-x}Si_x$ ($0 \leq x \leq 0.3$) retains the Fe_2P -type structure and GMCE, which is ascribed to a field induced FOPT ($|\Delta S_M| = 32 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2

T and 294 K) without a monotonous trend of T_C with x [1174]. Complete substitutions of P and Si for As in $\text{MnFeP}_{1-x}\text{Si}_x$ compounds successfully lead to the formation of hexagonal Fe_2P -type structure in the compositional range $0.28 \leq x \leq 0.64$, whereas orthorhombic Co_2P -type structure is formed for smaller Si content [1175]. For $0.28 < x < 0.48$, a secondary Fe_2MnSi phase (space group $\text{Fm}\bar{3}\text{m}$) is detected, whose phase fraction increases with Si content. Large MCE values although accompanied with large thermal hysteresis, are reported for $x=0.5$ ($\Delta S_M = -30 \text{ J kg}^{-1}\text{K}^{-1}$ at $T_C = 300 \text{ K}$ and $\mu_0\Delta H = 2 \text{ T}$ and thermal hysteresis $> 20 \text{ K}$). The phase diagrams of this system show that the single-phased regions for Fe_2P and Co_2P phases are limited to $0.24 \leq x \leq 0.50$ and $x \leq 0.15$ respectively [1176]. Virgin effects due to irreversible structural changes are reported [1177]. Li et al. [1178] ascribed this virgin effect to partial chemical disordered structure (as it occurs for As containing compounds, Mn atoms preferentially occupy 3g sites [1179]). When the PM state of $\text{MnFeP}_{0.75}\text{Si}_{0.25}$ transforms to FM state, it results in the formation of an intermediate metastable AFM state, whereby AFM/FM ratio can be controlled by modifying the kinetics of atomic diffusion [1180]. Off-stoichiometric $\text{Mn}_x\text{Fe}_{1.95-x}\text{P}_{0.50}\text{Si}_{0.50}$ compounds exhibit a gradual change from SOPT to FOPT as Mn/Fe ratio increases [1151]. Recently, Bartok et al. [1181] prepared off-stoichiometric $\text{Mn}_{1.3}\text{Fe}_{0.65}\text{P}_{0.5}\text{Si}_{0.5}$ compound via different techniques to produce bulk and spark plasma sintered powdered samples. The isothermal magnetization of powdered samples shows a smoother transition than those of bulk ones, which show a staircase shape. Latent heat measurements have been recently proposed to estimate MCE, assuming proportionality between this magnitude and the magnetic entropy change [1182]. For $\text{Mn}_{1.2}\text{Fe}_{0.75-x}\text{Ni}_x\text{P}_{0.5}\text{Si}_{0.5}$ and $\text{Mn}_{1.2}\text{Fe}_{0.8-x}\text{Ru}_x\text{P}_{0.5}\text{Si}_{0.5}$, minor partial substitutions of Ru or Ni for Fe retain good MCE responses, while decreasing the transition temperatures and hysteresis [1183].

GMCE was maintained upon combined substitutions of Si and Ge for As in $\text{MnFe}(\text{P}_{0.89-x}\text{Si}_x)\text{Ge}_{0.11}$ (e.g. $|\Delta S_M|(2\text{ T})=16\text{ J kg}^{-1}\text{K}^{-1}$ at 292 K for $x=0.26$). Nevertheless, an increase of thermal hysteresis $>15\text{ K}$ is observed together with virgin curve effects [1184, 1185]. However, when Ge or Si partially substitute for P, T_C linearly increases with Ge content while it significantly varies in a non-linear dependence in Si substituted samples. T_C is suggested to be related to lattice parameters a and c : the larger a and smaller c , the higher T_C [1186]. In $\text{MnFeP}_{0.5}\text{Ge}_{0.5-x}\text{Si}_x$ compounds with Fe_2P -type structure, T_C increases from 300 to 548 K as x decreases from 0.45 to 0.1 [1187]. X-ray circular magnetic dichroism experiments (L_3 Mn edge) performed on these As-free samples evidence magnetic field dependence spectra, indicating a modification of the electronic levels of Mn above the Fermi level during the field induced FOPT [1188]. However, this result disagrees with those obtained for $\text{Mn}_{1.1}\text{Fe}_{0.9}\text{P}_{0.45}\text{As}_{0.55}$, which exhibits unaltered electronic configuration of both Fe and Mn [1189]. On the other hand, first principle studies conclude that the competing exchange interactions between the different pairs of TM atoms are responsible for the large MCE ascribed to the field induced FOPT [1190]. Tailoring Mn/Fe ratio of As-free $\text{Mn}_{2-x}\text{Fe}_x\text{P}_{0.6}\text{Ge}_{0.15}\text{Si}_{0.25}$ compounds ($0.7 \leq x \leq 1$) was explored by Wang et al. [1191], whereby, for $x=0.7$ and 0.8 compounds, the conversion of FOPT to SOPT is observed to reduce MCE and transition temperatures. The character of the transition in these compounds is analyzed by the scaling law of the entropy change [229] while the analysis from Arrott plots is not as straightforward [1192]. Addition of B to $\text{MnFeP}_{0.63}\text{Ge}_{0.12}\text{Si}_{0.25}\text{B}_x$ ($0 \leq x \leq 0.03$) produces no significant effects on MCE but a continuous increase in T_C (from 300 to 347 K) along with a decrease in thermal hysteresis (from 20 to 6 K) [1193]. However, Guillou et al. [1194] show that B substitution for Si or P changes FOPT to SOPT. They [1195] analyzed the characteristics to improve MCE in FOPT materials indicating that although

latent heat should be sufficiently high to ensure a large ΔS_M , it must be limited to avoid a reduction in ΔT_{ad} . Hence, they proposed minor additions of B in $MnFe(P,Si)$ compounds and found a highly reversible $\Delta T_{ad}=2.8$ K and $\Delta S_M=-9.8$ J kg⁻¹K⁻¹ at 1 T and 280 K, with minimum hysteresis for $MnFe_{0.95}P_{0.595}B_{0.075}Si_{0.33}$, which poises this alloy family very promising and competitive to hydrogenated $La(FeSi)_{13}$ compounds. Both of these systems exhibit common advantages, such as not using critical and toxic elements, but Guillou et al. [1195] claim the former exhibits better mechanical stability. Recent X-ray absorption experiments show large changes in the K-edge of phosphorous in $MnFe_{0.95}P_{0.66-x}Si_{0.34}B_x$, which carries a small magnetic moment in the FM state [1196]. Recent results report a similar evolution of the magnetic moments of Mn and Fe through the transition, which contradicts theoretical predictions of quenched Fe moments [1197]. The cyclability [1198] and mechanical stability [1199] of these $MnFe(P,Si,B)$ systems are also investigated. Enhancement of mechanical stability upon B doping and TM substitution was also recently reported by Roy et al [1200]. In addition, mechanical stability can be also enhanced by interstitial N [1201].

The application of pressure affects the transition temperatures differently for different compositions of this alloy family. Moreover, the dependence is not related to the volume change but to the anisotropic evolution of the lattice parameters [1202]. The magnetostructural character of the transition implies an important role of the interactions [1079]. Transition temperature and its thermal hysteresis is thus related with the microstructural parameters of the Fe_2P . Figure 38 shows these two parameters as a function of both the unit cell volume and the ratio between the lattice parameters of the hexagonal phase. Finally, regarding this family of alloys, the presence of impurity phases, such as Mn_2O_3 and Fe_2MnSi should be taken into account when comparing different data.

With a similar composition to that of the previous compounds, $(\text{Co}_{0.35}\text{Mn}_{0.65})_2\text{P}$ crystallizes in Co_2P -type structure and exhibits a FOPT (AFM to FM) at 240 K and a SOPT (FM to PM) at 320 K. Ascribed to this SOPT transition, $\Delta S_M = -2.3 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T [1203]. In addition, Fe and Ni partial substitution to this compound have been recently studied [1204]. The effects of B addition and varying Mn/Co ratio on the MCE of MnCoP can be found in Ref. [1205].

5.1.2.6. *Mn antiperovskites*

In 2003, two different groups independently reported the MCE characterization of the cubic antiperovskite Mn_3GaC compound (space group $\text{Pm}\bar{3}\text{m}$) [1206, 1207], whose cubic cell consists of a formula unit with a lattice parameter of 3.98 Å as shown in Figure 39. Assuming the Ga atom is located at the corner of the cell, the three Mn atoms sit at the center of the faces while C atom is positioned at the center of the cubic cell.

These compounds are also attractive for their giant magnetoresistance among other properties [1208]. Upon heating, Mn_3GaC undergoes a FOPT from AFM to a canted FM, without symmetry change but with a volume change of -0.46 % at about 160 K, followed by a SOPT at 164 K, with an ascribed inverse GMCE. A SOPT from FM to PM phase is observed at 250 K [1209]. Yu et al. [1206] reported a small thermal hysteresis of 2 K for the FOPT wherein $\Delta S_M = 13 \text{ J kg}^{-1} \text{ K}^{-1}$, which is almost field-independent (from 1 to 5 T) except that its peak broadens and shifts to lower temperatures (i.e. relatively constant ΔS_M within a temperature range of about 30 K at 5 T). In agreement with these results, Tohei et al. [1207] estimated from entropy measurements $\Delta S_M = 15 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{\text{ad}}(2 \text{ T}) = -5.4 \text{ K}$ for Mn_3GaC at ~160 K, which were later confirmed by direct MCE measurements [1210]. The X-ray magnetic circular dichroism studies, performed at the excitation energy range from 2p to 3d Mn levels, show that electronic changes near the Fermi level occur at the FOPT of this compound

[1211]. Although ΔS_M spikes were reported [1212], they were probably due to artifacts as observed for other compositions. Cakir and Acet [1213] studied the reversibility of MCE by direct measurements observing only a 10 % decrease in ΔT_{ad} from the virgin curve to the subsequent application/removing magnetic field cycles (2.8 K change is measured at 3 T). In Mn antiperovskite compound, the field sweep rate of 700 mT/s is shown to preserve its GMCE but the FOPT is incomplete when using a fast rate of 1000 T/s [1214].

The MCE of $Mn_{3+x}Ga_{1-x}C$ series at the Curie transition was also studied, which reported that this transition shifts to higher temperatures from 250 to 323.5 K when x increases from 0 to 0.08 [1215]. In addition, $|\Delta S_M|$ decreases from ~ 2.5 to $1 \text{ J kg}^{-1} \text{ K}^{-1}$ (for a field of 4.5 T) for increasing x from 0 to 0.08, while RC_{FWHM} (in J/cm^3 units) increases up to $x=0.07$. For Mn_3GaC_x ($x=0.8, 1.0$ and 1.05), an increase in C content is concluded to convert the system from a FM metal ($x=0.8$) into an AFM semiconductor ($x=1.05$) [1216].

With partial Co substitution for Mn, its AFM to FM transition shifts to lower temperatures while MCE parameters are preserved, presenting a wide thermal range of 60 to 160 K [1217]. Partial Ni substitution for Mn in $Mn_{3-x}Ni_xGaC$ weakens AFM interaction, which then favors the magnetic field induced transition to the canted FM state, the AFM phase being suppressed for $x=0.1$ [1218]. As x increases, the transition temperatures of AFM to canted FM as well as those from canted FM to FM phase shift to lower temperatures, whereas Curie temperature increases. In addition, its absolute ΔS_M value decreases as x increases. Carbon deficiency in $Mn_3GaC_{1-\delta}$ enhances the FM character at low temperatures leading to the absence of AFM phase [1219]. The MCE

observed in this case is typically SOPT with $|\Delta S_M| = 3.7 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at a $T_C = 295 \text{ K}$ [1212].

For $\text{Mn}_3\text{GaC}_{0.85}\text{N}_{0.15}$, its FOPT is displaced to higher temperatures of 180 K and with almost negligible hysteresis by the partial substitution of N for C, which enhances its AFM character ($\Delta S_M = +8 \text{ J kg}^{-1} \text{ K}^{-1}$ at 4 T and $\Delta T_{ad} = 2.8 \text{ K}$ at 5 T) [1220]. FM domains above FOPT are recently found absent based on polarization studies [1221].

Antiperovskite $\text{Mn}_3\text{Cu}_{0.89}\text{N}_{0.96}$ shows a $\Delta S_M = 13.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at 150 K, with $\text{RC}_{\text{FWHM}} = 39 \text{ J/kg}$ ascribed to a magnetostructural change from tetragonal to cubic [19].

Partial substitution of Al for Ga in $\text{Mn}_3\text{Ga}_{1-x}\text{Al}_x\text{C}$ ($0 \leq x \leq 0.15$) produces an increase in T_C from 250 to 312 K. Although saturation magnetization increases with Al content, $|\Delta S_M|$ lowers while the MCE peak broadens [1222]. Ga-free Mn_3AlC compound shows a SOPT at 287 K with the following MCE parameters at 4 T: $\Delta S_M = -3 \text{ J kg}^{-1} \text{ K}^{-1}$, $\text{RC}_{\text{FWHM}} = 300 \text{ J/kg}$ and $\Delta T_{ad} = 1.5 \text{ K}$ at 285 K [1223].

Partial substitution of Ge for Ga in $\text{Mn}_3\text{Ga}_{1-x}\text{Ge}_x\text{C}$ ($x = 0, 0.1$ and 0.2) [1224] increases the FOPT temperature but reduces the Curie temperature. When $>71 \text{ at.}\%$ Sn partially substitutes for Ga, the alloys display conventional MCE ($\Delta S_M(7 \text{ T}) \sim -2.5$ and $-8 \text{ J kg}^{-1} \text{ K}^{-1}$ at 200 and 280 K for 71 and 100 at.% Sn substitution respectively). However, for Sn substitution below 55 %, inverse MCE is preserved although the magnitude of the MCE decreases with Sn content: for 0, 20, 41 and 55 % of Sn substitution for Ga, $\Delta S_M(7 \text{ T}) \sim 15, 11, 8$ and $4 \text{ J kg}^{-1} \text{ K}^{-1}$, at $\sim 165, 140, 145$ and 150 K respectively [1225]. In comparison to Mn_3GaC compound (6.25 g/cm^3), Mn_3SnC compound, which exhibits a higher density (7.78 g/cm^3), experiences a FOPT close to room temperature ($|\Delta S_M| = 17 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T and 279 K) from FiM to PM state. In addition, a thermal hysteresis of 2.5 K and a reduced unit cell volume (-0.1%), smaller than that of Mn_3GaC compound,

were observed [1226]. Abrupt changes in the Seebeck and Hall coefficients near T_C , indicate a modification of the density of states near the Fermi level. In off-stoichiometric $Mn_{3+x}Sn_{1-x}C$, the Curie temperature initially decreases with x for $x \leq 0.1$, for which the transition changes from FOPT to SOPT. Then, Curie temperature increases with x . Their MCE continuously decreases with increasing x [1227].

In $Mn_{3-x}Fe_xSnC$ series, partial substitution of Fe for Mn decreases both Curie temperature (well correlated to the lattice parameter) and $|\Delta S_M|$. However, FWHM increases with x , being the optimum RC_{FWHM} for $x=0.05$ [1228]. In addition, the effects of partial Co substitution for Sn on MCE of $Mn_3Sn_{1-x}Co_xC$ has been also reported [1229].

Antiperovskite Mn_3SbN crystallizes in a tetragonal structure at room temperature (P4/mmm space group) and shows a FOPT from FiM to PM phase, which is associated with a structural change from tetragonal to cubic phase. MCE shows a sharp peak around this transformation temperature (FWHM ~ 3 K and $|\Delta S_M| \sim 7 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T) [1230]. Similarly, a weak FOPT from PM-FiM accompanied by a cubic to tetragonal structural transformation is found in $CuNMn_3$ compound at 141 K, with a $|\Delta S_M| = 3.49 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2 T [1231].

5.1.2.7. *FeRh alloys*

FeRh deserves particularly attention as it was the first intermetallic reported with GMCE close to room temperature in early 90s of the last century with a maximum $-\Delta T_{ad}$ as high as 13 K (2 T) at 305 K for quenched $Fe_{49}Rh_{51}$ [1232]. However, this value reduces to only 4 K and the temperature of the peak shifted to 325 K upon annealing [1233].

The $\text{Fe}_{100-x}\text{Rh}_x$ compound crystallizes in an ordered α' CsCl-type structure in a wide compositional range [1234]. The formation of its AFM phase occurs for equiatomic composition (from ~49 % to ~56 %, which is also known as the α'' phase). From α'' phase, a FOPT from AFM to FM state (α' phase) can be induced with an associated 1 vol.% increase [1235] (about 350 K for stoichiometric FeRh). The FM phase undergoes a typical SOPT with Curie temperature at 675 K for FeRh [367]. Ascribed to the FOPT, $\text{Fe}_{48}\text{Rh}_{52}$ shows an inverse GMCE of $\Delta S_M = +12 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at $T \sim 300 \text{ K}$ with $\text{RC}_{\text{FWHM}} \sim 500 \text{ J/kg}$ [1235]. However, this is accompanied with a thermal hysteresis about 10 K and significant virgin curves effects [1233]. However, Annazarov et al. proposed the development of heat pumps using stress-induced AFM-FM transition (orders of 1 GPa) [1236, 1237] or magnetic field-induced (orders of 2 T) [1238]. Manekar and Roy [1235] described a refrigeration protocol for FeRh to achieved a reversible cycle: prior to increasing the adiabatic field, an initial field increasing and decreasing cycle at hot temperature should be performed isothermally [1184]. Recent results of subjecting $\text{Fe}_{49}\text{Rh}_{51}$ alloy to two-step cycling of the field (maximum 1.9 T) show $\Delta T_{\text{ad}} = 9.2 \text{ K}$ prior to a large $\Delta T_{\text{ad}} = 6.2 \text{ K}$ in the final step due to thermal hysteresis [1239]. Stern-Taulats et al. [1240] further investigated the cyclic dependence of MCE by increasing to ten cycles and found a good MCE reproducibility. Moreover, they also show the giant barocaloric effect in this system [1241]. Liu et al. [1242] have recently proposed a method to overcome the irreversibility of these systems by using a combined magnetic-electric stimulus, and attained 96% reduction of losses.

Theoretical studies based on Monte Carlo simulations [1243, 1244] and density functional theory calculations [1245, 1246] highlight that the magnetic moment of Rh atoms plays an important role to stabilize FM phase ($\sim 1 \mu_B$ in FM phase and null in AFM phase). The main contribution to the entropy change is of electronic nature

[1247]: a reduction of the carrier density by one order of magnitude in the AFM state compared to FM state, observed from Hall-effect measurements [1248]. Zverev et al. [98] have recently reported both theoretical and experimental investigations on the influence of defects and inhomogeneities on the MCE of $\text{Fe}_{50.4}\text{Rh}_{49.6}$.

MCE studies of pseudobinary Fe-Rh alloys with small additions of other metals, such as Pd [1249] and Ni [1250], show a reduction of transition temperatures compared to the binary alloy. On the other hand, Pt [1251, 1252] and Ir [1251] substitutions shift the transition temperatures to higher values, however, to our knowledge, no MCE studies have been performed on these alloys.

In the case of $\text{FeRh}_{1-x}\text{Pd}_x$ alloy [1249], the transition temperature decreases from 325 to 154 K and ΔS_M decreases from ~ 14 to $6 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 2 \text{ T}$, whereas thermal hysteresis increases from 10 to 43 K for $x=0$ to 0.08. Although ΔS_M decreases with increasing x , for $x=0.08$, its full width at half maximum of the MCE peak is about double of that of $x=0$. Moreover, although ΔS_M almost saturates for $\Delta H \geq 2 \text{ T}$, the MCE peak broadens linearly with field, from FWHM $\sim 10 \text{ K}$ to 80 K when the field increases from 1 to 7 T. For $\text{FeRh}_{0.45}\text{Pd}_{0.55}$, 1 % volume change is predicted [1253]. $\text{Fe}(\text{Rh}_{1-x}\text{A}_x)$ ($\text{A}=\text{Cu}, \text{Pd}$) and $(\text{Fe}_{1-x}\text{Ni}_x)\text{Rh}$ ($0 < x < 0.06$), which were produced via arc-melting and subsequent annealing, present increased RC in the ternary alloys (as compared to that of binary alloys) although ΔS_M decreases [1250]. When slight amounts of Ni partially substitutes for Fe in Fe-Ni-Rh, over 50% increase in MCE, in terms of RC, was already obtained for $\text{Fe}_{0.975}\text{Ni}_{0.025}\text{Rh}$. Although, ΔS_M decreases to $9 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T and 260 K, the full width at half maximum reaches 80 K [1254].

For $\text{Fe}_{0.45}\text{Mn}_{0.55}\text{Rh}$, a large volume change of 3.6 % in the AFM (CuAu-I type) to FM (CsCl-type) transition with a magnetic moment per formula unit of $4.5 \mu_B$ is predicted [1253].

5.1.2.8. *Fe₂P alloys*

Fe_2P orders ferromagnetically below 214 K and is the prototypical phase for the homonymous structure type (space group P-62m) found in MnFe(P,As) family with GMCE [1255]. Fe is located in two different sites with different magnetic moments: $2.2 \mu_B$ is found for Fe atoms in 3g pyramidal site (five P neighbors) and $0.6 \mu_B$ for Fe atoms in 3f tetrahedral sites (four P neighbors). The small negative moment of P yields a total magnetic moment per formula unit of $2.7 \mu_B$ in Fe_2P . The FM to PM transition in Fe_2P is a magnetoelastic FOPT, which implies a small volume shrinkage of 0.04 % despite preserving the hexagonal symmetry. Both the elastic and magnetic transitions are displaced to higher temperatures with the application of an external magnetic field. Hudl et al. [1256] recently described the H-T phase diagrams of this system considering the influence of the direction of the applied field.

Regarding MCE properties, pure Fe_2P shows a $-\Delta S_M = 2 \text{ J kg}^{-1} \text{ K}^{-1}$ at about 220 K and $\mu_0 \Delta H = 1.3 \text{ T}$ [1257], which also agrees with recent results reported for polycrystalline ($-\Delta S_M > 4 \text{ J kg}^{-1} \text{ K}^{-1}$ for 5 T) and monocrystalline samples ($-\Delta S_M$ obtained are much larger when subjected to field applied perpendicular to the c-axis) [1255]. Theoretical studies based on Landau theory and a magnetovolume coupling term agree with experimental data [1258].

Partial substitution of Ru for Fe in $\text{Fe}_{1.85}\text{Ru}_{0.15}\text{P}$ [1259] increases transition temperatures up to 240 K, although $|\Delta S_M|$ decreases to $1.2 \text{ J kg}^{-1} \text{ K}^{-1}$ ($\mu_0 \Delta H = 1.3 \text{ T}$) and its crystal

symmetry changes from hexagonal to orthorhombic (Co₂P-type structure). However, for larger Rh content in Fe_{1.75}Rh_{0.25}P [1259], the hexagonal symmetry is retained while increasing transition temperature up to 315 K. For this compound, $\Delta S_M = -1.2 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\mu_0 \Delta H = 1.3 \text{ T}$.

In (Fe_{1-x}Ni_x)₂P series, Ni atoms are only located at 3f sites [1260] and for x=0.025, Curie temperature increases up to 313 K. However, for x=0.1 no significant increase of T_C is observed. The maximum - ΔS_M among the series is found for x=0.025 ($2.6 \text{ J kg}^{-1} \text{ K}^{-1}$ at 3 T) [1261]. In addition, its transition temperature is pressure-independent and 0.3 % increase in the lattice volume is observed during FM to PM transition [1260].

Delczeg-Czirjak et al. [1262] using first principle studies, found that B, Si and As dopants in Fe₂P lead to an increase in T_C. The hexagonal phase is preserved in Fe₂P_{1-x}M_x for x<0.15 (M=B), x≤0.25 (M=Si) and x≤0.65 (M=As). In addition, the Monte Carlo simulations of FeCrAs compound crystallizing in Fe₂P-type structure, show table-like $\Delta S_M(T)$ curves [1263].

5.1.2.9. *Fe-transition metals compounds*

Fe nanocrystals embedded in Cu matrix have been reported with small MCE values at the blocking temperature and the Curie temperature of its interfacial FeCu phase [1264].

Fe_xPt_{100-x} (x=78-88) alloys exhibit a structural transition from γ phase to α phase induced by cooling or by applying a magnetic field near the transition [1265]. A GMCE of $|\Delta S_M| > 30 \text{ J kg}^{-1} \text{ K}^{-1}$ (5 T) at ~250 K reported for x=79 is ascribed to a huge negative thermal expansion occurring at the transition (26.2%). The thermal hysteresis observed in the magnetization curve of this system, which is hundreds of Kelvins, is a clear drawback. The formation of Fe₃Pt phase for x<78 leads to disappearance of this thermal hysteresis.

Recarte et al. [1266] studied the martensitic transformations occurring in $\text{Fe}_{70}\text{Pd}_{30}$ alloys and found a reversible transformation occurs at 290 K upon cooling (from face centered cubic to face centered tetragonal) and an irreversible transformation at 216 K (from face centered tetragonal to body centered tetragonal). They used MCE to study the magnetoelastic coupling of the alloys. With increasing magnetic field, MCE changes from inverse to direct MCE. At the transition temperature of the reversible martensitic transformation, the inverse MCE shows a maximum value of $\Delta S_M \sim -0.015 \text{ J/kg}$ at 0.1 T. For $\text{Fe}_{68.4}\text{Pd}_{30.6}\text{Mn}$ alloy ($|\Delta S_M| = 1.2 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\sim 290 \text{ K}$ and 6 T), partial substitution of 1 at. % Mn for Fe inhibits the irreversible martensitic transformation found in $\text{Fe}_{70}\text{Pd}_{30}$ alloys at 216 K, although it appears again for 2.5 at % of Mn substitution for Fe [1267]. Prida et al. [1268] studied the SOPT occurring at 550 K for melt-spun $\text{Fe}_{73.2}\text{Pd}_{26.8}$, which results in a $\Delta S_M = -1.04 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1.5 T. This SOPT is well described using the scaling law of field dependence of the MCE proposed by Franco et al. [76].

Ipus et al. [1269] produced $\gamma\text{-FeNi}$ phase in $(\text{Fe}_{70}\text{Ni}_{30})_{89}\text{Zr}_7\text{B}_4$ alloy after ball milling and subsequent annealing at 973 K. Its MCE parameters ascribed to the SOPT from FM to PM state, are $\Delta S_M = 0.182 \text{ J kg}^{-1} \text{ K}^{-1}$ (0.5 T) with a broad peak at 342 K. This phase was also obtained upon milling FeNi and its T_C can be tuned by controlling oxidation [1270]. Mo addition in milled $(\text{Fe}_{0.7}\text{Ni}_{0.3})_{100-x}\text{Mo}_x$ ($x=3, 4$) also stabilizes the fcc $\gamma\text{-FeNi}$ phase and shows $\text{RC}_{\text{FWHM}} = 440$ and 432 J/kg (5 T) for $x=3$ and 4 respectively [1271]. The 8 at.% Mn addition in ball milled $(\text{Fe}_{0.7}\text{Ni}_{0.3})_{92}\text{Mn}_8$ reduces T_C from 443 (Mn-free alloy) to $\sim 375 \text{ K}$ (α phase) and further to $\sim 335 \text{ K}$ (γ phase which stabilizes after annealing) [1272]. Broad MCE peaks of $\Delta S_M \sim -1.5 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\text{RC}_{\text{FWHM}} \sim 500 \text{ J/kg}$ for 5 T are observed. Addition of B to milled FeNi samples in $(\text{Fe}_{0.7}\text{Ni}_{0.3})_{89}\text{B}_{11}$ yields a progressive increase of γ phase fractions with increasing milling time [1273]. Its $\Delta S_M \sim -2 \text{ J kg}^{-1} \text{ K}^{-1}$ and large $\text{RC}_{\text{FWHM}} \sim 640 \text{ J/kg}$ at 5 T are calculated. In addition, the multiple

phase coexistence can further enhance RC_{FWHM} [1274]. Austenite $\gamma\text{-Fe}_{49}\text{Ni}_{29}\text{Cr}_{22}$ shows a PM-FM SOPT at 125.5 K with a $\Delta S_M = -1.8 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T [1275].

High entropy FeCoCr_xNi ($0.85 \leq x \leq 1.15$) alloys were also studied [1276]. T_C increases from ~50 K for $x=1.15$ to ~200 K for $x=0.85$. In addition, cold rolling broadens the MCE peak, showing a $\Delta S_M = -0.35 \text{ J kg}^{-1} \text{ K}^{-1}$ at 2 T. Zhong et al. [1277] studied the MCE of $\text{Fe}_{50}\text{Mn}_{15-x}\text{Co}_x\text{Ni}_{35}$ alloys produced by arc-melting. When x increases from 0 to 5, T_C increases from 237 to 342 K. A relatively constant $\Delta S_M \sim -2.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T is obtained for the studied series. NiFeCoCrPd_x ($x \leq 0.5$) series, with a tunable SOPT from 100 K to above room temperature, shows $|\Delta S_M| < 1 \text{ J kg}^{-1} \text{ K}^{-1}$ at 5 T [1278].

Tan et al. [1279] have recently prepared AlFe_2B_2 compound, which crystallizes in a orthorhombic lattice (space group Cmmm) by forming slabs of Fe_2B_2 chains separated by Al layers. They pointed several advantages for this composition: only earth-abundant elements are used and good MCE properties ($\Delta S_M = -7.7 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} \sim 3 \text{ K}$ at 307 K and $RC_{FWHM} = 162 \text{ J/kg}$ at 5 T). Further studies on this systems can be found in the recent literature [1280, 1281]. The PM-FM transition at 282 K disappears after Mn or Cr completely substitute for Fe [1282] despite retaining the structure. Partial Mn substitution for Fe leads to a reduction of the maximum magnetic entropy change and its temperature peak [1283].

5.1.2.10. Other rare earth free crystalline compounds.

Recently, Kaeswurm et al. [1284] studied the MCE of cementite at its $T_C = 475 \text{ K}$, which exhibits a $\Delta S_M = -3.07 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 1.76 \text{ K}$ at 2 T. Partial substitution of Mn for Fe in Fe_3C cementite [445] allows for tunable T_C below and above room temperature and also preserves the orthorhombic Fe_3C structure while its MCE remains low: $\Delta S_M = -1.39$

$\text{J kg}^{-1}\text{K}^{-1}$ (5 T) at 305 K for $(\text{Fe}_{0.9}\text{Mn}_{0.1})_3\text{C}$. Mn-free $(\text{FeCr})_3\text{C}$ cementite-type was also reported with a $\Delta S_M = -0.9 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 1 \text{ T}$ [1285].

Zhang et al. [1286] explored the complete compositional range in the Heusler $\text{Fe}_2\text{MnSi}_{1-x}\text{Ge}_x$ series. The crystallization of the amorphous phase produced by milling leads to the formation of a metastable D0_3 phase (space group $\text{Fm}\bar{3}\text{m}$) for $x \leq 0.5$, tuning the Curie temperature closer to room temperature (for $x=0$ and $x=0.5$, $T_C = 243$ and 260 K respectively) than that of D0_{19} phase (space group $\text{P6}_3/\text{mmc}$; for $x=0.6$ and 1 , $T_C = 355$ and $>400 \text{ K}$ respectively). ΔS_M value of $-1.5 \text{ J kg}^{-1}\text{K}^{-1}$ at 5 T at the corresponding Curie temperature is reported for $\text{Fe}_2\text{MnSi}_{0.5}\text{Ge}_{0.5}$ alloy.

Mn_3Sn_2 compound crystallizes in Ni_3Sn_2 -type structure (space group Pnma) below 753 K . Its anomalous MCE is due to the presence of two successive magnetic transitions: at 262 K (from PM to FM) and at 227 K (from FM to FM), assumed of second order character. Hence, its $\Delta S_M(T)$ curves exhibit two maxima centered at the two transition temperatures and of similar magnitude ($\Delta S_M \sim -3.3 \text{ J kg}^{-1}\text{K}^{-1}$ at $\mu_0\Delta H = 5 \text{ T}$), leading to a broad curve and thus $\text{RC}_{\text{FWHM}} > 300 \text{ J/kg}$ [1287]. Recently, the same group has reported a third anomaly at 197 K , ascribed to the formation of a weak AFM interaction in this compound [1288].

$\text{Mn}_{1.9}\text{Co}_{0.1}\text{Sb}$ compound crystallizing in tetragonal Cu_2Sb -type structure, with P4/nmm space group, derives from the simple FiM Mn_2Sb . In Co added intermetallic, FOPT from AFM to FiM occurs and shows a $\Delta S_M = 5 \text{ J kg}^{-1}\text{K}^{-1}$ (1 T) at 165 K [1289]. On the other hand, CoMnSb crystallizes in $\text{F}\bar{4}3\text{m}$ space group and shows a high $|\Delta S_M| = 2.06 \text{ J kg}^{-1}\text{K}^{-1}$ (0.9 T) at a SOPT with $T_C = 471 \text{ K}$ [1290]. Annealing at higher temperatures leads to broadening of the MCE peak at the expense of decreasing $|\Delta S_M|$ [1291]. Partial substitution of Nb for Mn in Co-Nb-Mn-Sb slightly lowers T_C but more significantly

reduces MCE ($\Delta T_C \sim 10$ K with a 75 % decrease in $|\Delta S_M|$ for $\text{CoNb}_{0.6}\text{Mn}_{0.4}\text{Sb}$) [1292].

$\text{Mn}_{2-x}\text{Cr}_x\text{Sb}_{1-y}\text{In}_y$ compounds show a AFM-FiM FOPT at 306 K with a $|\Delta S_M| \sim 5 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad} = 1.75$ K at 5 T, for a composition of $x=0.13$ and $y=0.05$ [1293]. Partial substitution of In for Sb is studied in the aim of suppressing the MnSb impurity phase.

Mn_5PB_2 crystallizes in a tetragonal structure, with space group $I4/mcm$, and experiences a SOPT from FM to PM at 302 K with a $\Delta S_M \sim 5 \text{ J kg}^{-1}\text{K}^{-1}$ (5 T) in the presence of some Mn_2P impurities [1294].

$(\text{Mn}_{0.83}\text{Fe}_{0.17})_{3.25}\text{Ge}$ compound is reported with an inverse MCE and a high ΔS_M of $+11.6 \text{ J kg}^{-1}\text{K}^{-1}$ (7 T) at 93 K, which is attributed to its field induced metamagnetic FOPT from collinear AFM to triangular AFM phase [1295]. This hexagonal compound undergoes a Néel transition to the PM state at 240 K.

The MCE of spinel compounds has been also studied. Non-magnetic substitution in Mn/Zn ferrites has been shown to be an effective way to tune the peak temperature of MCE without significant losses in MCE [1296]. $\text{Ni}_{1-x}\text{Zn}_x\text{Fe}_2\text{O}_4$ ferrite shows SOPT [1297], with a continuous decrease of T_C from 845 to 302 K as x increases from 0 to 0.7. At 2.5 T, the maximum $|\Delta S_M| = 1.39 \text{ J kg}^{-1}\text{K}^{-1}$ is observed for $x=0.3$ ($T_C=664$ K) while the maximum $\text{RC}_{FWHM} = 161 \text{ J/kg}$ is attained for $x=0.5$ ($T_C=481$ K). Ball milling of ZnFe_2O_4 yields flat $\Delta S_M(T)$ curves ($\sim 0.25 \text{ J kg}^{-1}\text{K}^{-1}$ at 3 T from 100 to 300 K with $\text{RC}_{FWHM} > 75 \text{ J/kg}$) [1298]. However, deposited nanosized CoFe_2O_4 prepared by sol-gel technique, exhibits an inverse MCE of $\Delta S_M = +0.25 \text{ J kg}^{-1}\text{K}^{-1}$ at 210 K and 1.3 T [1299].

Spinel vanadate MnV_2O_4 becomes FiM at 57 K (SOPT) although a structural phase transition occurs in the vicinity, which is at 53 K. Its GMCE of $|\Delta S_M| = 24 \text{ J kg}^{-1}\text{K}^{-1}$ (4 T) and $\Delta T_{ad} \sim 3$ K (2 T) is ascribed to the combined contributions of spin entropy change plus the orbital entropy change induced in V^{3+} ions [1300]. However, it is worth noting

that Maxwell analysis is used for this MCE determination and the maximum is sharp. Small substitution of Zn for V yields to a better decoupling of the two transitions [1301]. With 2.5 % substitution of Al for V, smaller $\Delta S_M = -8.2 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 2.6 \text{ K}$ at $T_C = 59.6 \text{ K}$ (4 T) are obtained but with enhanced $RC_{FWHM} = 177 \text{ J/kg}$ as compared to Al-free compound [1302]. The critical exponents for Al-containing compound is consistent with mean field theory unlike for MnV_2O_4 compound [1303]. Partial Fe substitution for Mn in $\text{Mn}_{1-x}\text{Fe}_x\text{V}_2\text{O}_4$ broadens the MCE peak and aids RC enhancement [1304]. Sulfospinel CdCr_2S_4 with SOPT at a Curie temperature of 87 K, shows $\Delta S_M = -7 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\Delta T_{ad} = 2.6 \text{ K}$ at 5 T [1305] due to spin-lattice coupling, as proved by electron paramagnetic resonance experiments [1306]. Minor partial substitutions of Cu or Fe for Cd retain the second order character of the transition, whereas T_C increases to 119 K for $\text{Cd}_{0.7}\text{Fe}_{0.3}\text{Cr}_2\text{S}_4$. However, $|\Delta S_M|$ is reduced to $\sim 5 \text{ J kg}^{-1} \text{ K}^{-1}$ [1307]. For CuCr_2Se_4 [1308] and CuCr_2Te_4 [1309] spinels, with $T_C = 430$ and 330 K and $\Delta S_M = -0.7$ and $-0.6 \text{ J kg}^{-1} \text{ K}^{-1}$ (1 T) respectively, scaling laws of ΔS_M and RC_{FWHM} are reported to be applicable and show critical exponents typical for a 3D-Heisenberg model. Lower T_C values are found for CdCr_2Se_4 ($\sim 130 \text{ K}$) and HgCr_2Se_4 (113 K) alloys [1310], which both exhibit the same $\Delta S_M = -1.5 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1.5 T. Mean field exponent was proposed for the field dependence of the MCE for these systems.

There are also some works reported on the MCE of CrO_2 . Zhang et al. [1311] observed $\Delta S_M \sim -5 \text{ J kg}^{-1} \text{ K}^{-1}$ (1.5 T) at 390 K for 400 nm sized CrO_2 particles but the response decreases for smaller particles. Smaller ΔS_M values with a broader T_C distribution were observed for CrO_2 films on TiO_2 substrate, leading to higher RC [1312]. Recently, partial substitution of O for F has been shown to effectively reduce the Curie temperature to room temperature [1313].

Zhong et al. [1314] studied several RE-free double perovskite systems, such as $\text{Ba}_2\text{Fe}_{1+x}\text{Mo}_{1-x}\text{O}_6$ [1315, 1316], $\text{Sr}_2\text{FeMo}_{1-x}\text{W}_x\text{O}_6$ [1315], followed by other authors studying $\text{Ba}_2\text{CrMo}_{1-x}\text{W}_x\text{O}_6$ [541] and $\text{Ba}_2\text{Fe}_{1-x}\text{Cr}_x\text{MoO}_6$ [1317]. In $\text{Ba}_2\text{Fe}_{1+x}\text{Mo}_{1-x}\text{O}_6$ series, two contrary effects yield an almost unchanged T_C : large superexchange Fe-O-Fe interactions for $x>0$ and reduced number of itinerant electrons (3 and 5 electrons per Fe and Mo respectively) [1315]. Although $|\Delta S_M|$ reduces from 1.54 to 1.2 $\text{J kg}^{-1}\text{K}^{-1}$ (1 T) at ~ 335 K as x increases from 0 to 0.3, the peak becomes broader (FWHM from 20 to 30 K). Partial substitution of W for Mo reduces the Curie temperature to 285 K and, with 50 at.% W substitution, slight enhancement in MCE is attained: $|\Delta S_M|$ increases from 1.25 to 1.6 $\text{J kg}^{-1}\text{K}^{-1}$ (1 T) [541]. T_C can also be tuned to room temperature with partial Cr substitution in $\text{Ba}_2\text{Fe}_{1-x}\text{Cr}_x\text{MoO}_6$ cubic perovskite series. However, its MCE decreases with increasing Cr content: for $x=0$ and 1, $|\Delta S_M|(1\text{ T})=2.08$ and $0.65\text{ J kg}^{-1}\text{K}^{-1}$ at 340 and 310 K respectively [1317]. The MCE studies of $\text{Li}_{4-x}\text{Ti}_{5-2x}\text{Fe}_{3x}\text{O}_{12}$ system shows that the transition temperature of its SOPT is highly compositional dependent on x . However, the system shows moderate MCE of $|\Delta S_M|\sim 1\text{ J kg}^{-1}\text{K}^{-1}$ at 5 T [1318].

High entropy alloys composed of transition metal elements of the family FeCoNiCuX , with $X=\text{Mn, Ag, Pt, and Mo}$ have been proposed as a way of tuning transition temperature of the alloys with the hope of enhancing their magnetocaloric response for near-room-temperature applications [1319]. The distribution of exchange interactions intrinsic to these materials provoke a broad magnetic entropy change peak. However, in the current state of the art, the enhancement of the refrigerant capacity comes with the price of a small magnetic entropy change. This is in contrast with the rare earth high entropy alloys presented in section 5.1.1.8, where an increased refrigerant capacity was achieved while keeping a large magnetic entropy change peak [595].

Recently, $\text{Cs}_2\text{NaAl}_{1-x}\text{Cr}_x\text{F}_6$ elpasolites (space group R-3m) were proposed for very low temperature cooling as they exhibit large MCE at 3 K: e.g. for $x=0.62$, $|\Delta S_M| \sim 11 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{\text{ad}}=8 \text{ K}$ at 5 T [1320].

A more exotic type of magnetocaloric material is graphene, in which oscillatory behavior of the magnetic entropy change and adiabatic temperature change as a function of the applied magnetic field has been predicted [1321]. This oscillating behavior is intrinsic to diamagnetic materials [1322] and it emerges from the crossing of the Landau levels through the Fermi energy. In graphene, the value of the magnetocaloric effect is three orders of magnitude larger than for other two dimensional non-relativistic diamagnetic materials. This oscillating behavior has been the basis for a magnetocaloric cycle with six stages [1323], which is being proposed for low temperature magnetic refrigeration.

5.2. Amorphous materials

Among the advantages of amorphous materials as compared to their crystalline counterparts, compositional tailoring, low eddy current losses, low hysteresis losses and corrosion resistance can be listed. Amorphous alloys with attractive MCE undergo SOPT at their magnetic ordering temperatures. They are classified into two families: RE-based alloys (mainly Gd-based) and TM-based alloys (mainly Fe-based). Figure 40 shows the various data of these alloys found in the literature and estimated at $\mu_0\Delta H=1 \text{ T}$ using the exponential laws for the field dependence of ΔS_M and RC_{FWHM} [80] (using mean field values for the critical exponents of $2/3$ and $4/3$ respectively [80]). The MCE of crystalline Gd, which is the reference material for MCE and particularly for SOPT materials, is also included.

Both alloy families exhibit dissimilar behavior RE-based amorphous alloys show transition temperatures extending from cryogenic temperatures to close to room temperature but not beyond that of crystalline Gd. On the other hand, the transition temperatures of TM-based alloys extended from above 100 K to clearly above room temperature. In general, with increasing peak temperatures, $|\Delta S_M|$ decreases for RE-based alloys, whereas relatively increases for TM-based alloys. For amorphous alloys with similar transition temperatures to that of Gd, $|\Delta S_M|$ is clearly lower than that of Gd but it is not the case when comparing RC_{FWHM} of these systems. This is due to the highly disordered structure of amorphous systems, which smears out the magnetic transition and consequently, broadens the MCE peak.

The partially amorphous RE-based and TM-based samples also show dissimilar behaviors. In the case of Gd-based amorphous alloys, the crystalline phases that appear as products of primary crystallization (or as crystalline products of rapid quenching) exhibit transition temperatures close to those of their amorphous systems (i.e. hexagonal Gd crystals). Nevertheless, the formation of α -Fe crystallites with a Curie temperature above 1000 K in Fe-based alloys are detrimental to room temperature MCE ascribed to the magnetic transition of the amorphous phase. Although the microstructures of both systems are similar (i.e. crystallites embedded in an amorphous matrix), the difference in their behaviors is due to the differences in the transition temperatures of both phases formed within the composites [1324, 1325].

5.2.1. Rare earth based amorphous materials

Besides the review of Gschneidner and Pecharsky [11], reviews on different properties of bulk metallic glasses, including their MCE can be found in [1326, 1327]. Binary alloys of Gd can show ΔS_M values similar to that of hexagonal Gd but with reduced T_C

(see Figure 41). On the other hand, these alloys exhibit broader transitions, resulting in enhanced RC compared to that of pure Gd.

Figure 42 shows the Curie temperature and maximum magnetic entropy change at 5 T as a function of the average atomic number of the RE for different amorphous alloys. It is clear that the maximum transition temperatures are obtained for Gd-containing amorphous alloys. This temperature is generally well below that of hexagonal Gd, however, amorphous GdCo [1328-1330], GdCoB [1330], GdCoMn [1331], GdCoSi [1329], GdFe [1332], GdFeAl [1333, 1334] and GdFeAlSi [1335] can show T_C above 200 K. Moreover, MCE peak temperatures of partially crystalline GdY and GdZn [115] can also exceed 200 K. Amorphous $Gd_{65}Mn_{35-x}Si_x$ ($x=5$ and 10) ribbons show $-\Delta S_M=4.6$ and $4.7 \text{ J kg}^{-1}\text{K}^{-1}$ and $RC_{FWHM}=625$ and 660 J/kg at $\sim 230 \text{ K}$ (5 T) for $x=5$ and 10 respectively. Nanocrystallization sharpens the peak and shifts it to higher temperatures without affecting ΔS_M [1336]. On the other hand, amorphous/nanocrystalline $GdAl_2$ milled powder exhibits a table-like peak from 30 to 165 K, with enhancement of MCE below 30 K due to the presence of superparamagnetic particles [1337]. Although the higher temperatures are among these RE=Gd systems (see Figure 42), partially amorphous $Nd_{45}Fe_{30}Co_{15}Al_{10}$ alloys have been recently shown to have a peak entropy change (moderate $\Delta S_M \sim -0.25 \text{ J kg}^{-1}\text{K}^{-1}$ for 1 T) at high temperature (430-445 K) [1338]. Previously, the maximum peak temperature for RE-based amorphous alloy was reported by Foldeaki et al. [1332] as $T_C=288 \text{ K}$ for $Gd_{70}Fe_{30}$ amorphous ribbons, obtained by rapid quenching, which exhibit $\Delta S_M=-1.50 \text{ J kg}^{-1}\text{K}^{-1}$ at 1 T. Recently, Zhang et al. systematically studied the MCE of heavy RE-based metallic glasses in the series of $(RE^{(1)}_{1-x}RE^{(2)}_x)_{55}Al_{27.5}Co_{17.5}$, whereby $RE^{(i)}=Gd, Er$ or Ho , and $x=0, 0.3, 0.5, 0.7$ and 1 [1339]. Their T_C can be tuned between 8 and 93 K and a relative linear correlation is observed between the absolute value of ΔS_M and $T_C^{-2/3}$. Higher Curie temperature is

reported for $\text{Gd}_{55}\text{Al}_{20}\text{Co}_{20}\text{Ni}_5$ bulk metallic glass, with $-\Delta S_M = 9.8 \text{ J kg}^{-1}\text{K}^{-1}$ and $\Delta T_{ad} = 4.74 \text{ K}$ at 5 T and 105 K [1340]. Recently, Liu et al. [1341] reported table-like MCE response of amorphous $\text{Gd}_{50}\text{Co}_{45}\text{Fe}_5$ close to room temperature ($\Delta S_M = 3.8 \text{ J kg}^{-1}\text{K}^{-1}$ at 289 K and 5 T).

Amorphous samples can be produced by different routes, typically by fast cooling (e.g. melt-spinning or suction casting) or by mechanical methods (mechanical grinding or alloying). Amorphous GdNiAl produced by ball milling [1342] exhibits a 25 % smaller $|\Delta S_M|$ than that observed for rapidly quenched samples [1343] with no significant change in the FWHM $\sim 55 \text{ K}$.

The MCE of amorphous micro-fibers and microwires were studied for $\text{Gd}_{55}\text{Al}_{25}\text{Co}_{20}$ ($\Delta S_M = -9.7 \text{ J kg}^{-1}\text{K}^{-1}$ and $\text{RC}_{\text{FWHM}} = 870 \text{ J/kg}$ at 5 T and 100 K [1344]), $\text{Gd}_{68}\text{Ni}_{32}$ ($\Delta S_M = -4.5 \text{ J kg}^{-1}\text{K}^{-1}$ at 122 K and 3 T) and $\text{Gd}_{53}\text{Al}_{24}\text{Co}_{20}\text{Zr}_3$ ($\Delta S_M = -7 \text{ J kg}^{-1}\text{K}^{-1}$ at 97 K and 3 T) [1345].

The group of von Ranke performed different theoretical studies to describe the MCE of amorphous systems: e.g. $\text{Er}_{80}\text{Au}_{20}$ [1346] (Handrich-Kaneyoshi model is applied to this case), REAg and GdCuAl systems [1347], and other binary Dy-TM compounds [1348].

5.2.2. Transition metal based amorphous materials

To reduce materials cost, much interest is focused on RE-free compositions, which necessarily led to MCE investigations of TM-based amorphous alloys. These materials were previously studied due to their soft magnetic properties, which are related to absent magnetocrystalline anisotropy. Therefore, a large number of works on characterizing the compositional dependence of the magnetic properties of these systems can be found in the literature [1349]. Unlike RE-based amorphous alloys, TM-based amorphous systems exhibit a general increase of $|\Delta S_M|$ as transition temperature

increases (as shown in Figure 40). A review on the MCE of TM-based amorphous alloys can be found in [1350] for rapidly quenched systems and in [1351] for mechanically alloyed ones.

Maeda et al. in 1983 [1352] early studied the MCE of pseudobinary $(\text{Fe}_{1-x}\text{Ni}_x)_{90}\text{Zr}_{10}$ and $(\text{Fe}_{0.95}\text{M}_{0.05})\text{Zr}_{10}$ amorphous series and found that T_C increases with x up to 0.05 as well as with increasing atomic number of the M element ($\text{M}=\text{Al}, \text{Si}, \text{Ga}, \text{Ge}$ and Sn). Almost constant $|\Delta S_M|$ values with less than 1 % change is reported ($\sim 1.4 \text{ J kg}^{-1} \text{ K}^{-1}$ at 1.4 T assuming a density of 7.7 g/cm^3) among the different studied alloys. Maeda et al. also mentioned that the MCE behavior of these amorphous systems departs from the predictions of molecular field theory, which was later confirmed by Franco et al. [76].

The vast majority of studied TM-based compositions are Fe-based alloys and only a few examples on Co-based [1353, 1354] and Pd-based [1355] alloys (the latter with significantly smaller MCE response) can be found in the literature. Among Fe-based alloys, those derived from $\text{Fe}_{90}\text{Zr}_{10}$ eutectic composition, or ternary and quaternary variations around eutectic composition $\text{Fe}_{80}\text{B}_{20}$ (including Cr, Mo, Nb or Mn) exhibit T_C close to room temperature. Figure 43 shows the compositional dependence (Fe content) of ΔS_M and T_C for some Fe-Zr-M amorphous alloys. The Fe-based amorphous alloy exhibiting the largest MCE reported to date belongs to this Fe-Zr-M family with 4 at % B and some partial substitution of Co and Ni for Fe. The results for some compositions of this series are also included in Figure 43. The MCE of $\text{Fe}_{84-x}\text{Ni}_x\text{Zr}_6\text{B}_{10}$ ($0 \leq x \leq 6$)

Nanoperm-type alloys is studied using the master-curve analysis, and also shows that Ni additions increase their T_C and MCE ($-\Delta S_M = 1.52 \text{ J kg}^{-1} \text{ K}^{-1}$ at $\sim 480 \text{ K}$ for 1.4 T) [1356].

Ball milled and mechanically alloyed TM-based amorphous alloys are also studied in addition to the earlier-discussed ultra-rapid quenched systems [432, 433, 1351, 1357-

1362]. The ribbon samples produced by the latter technique allows the assumption of a zero demagnetizing factor when the applied field direction is contained in the plane of the ribbon. However, for powder samples obtained from milling, a demagnetizing factor of 1/3 is a better approximation. In fact, if the demagnetizing factor is not considered, ΔS_M values are only slightly affected (~5 %) but their field dependence completely departs from the predicted behavior, requiring a demagnetizing factor of 1/3 to properly account for the field dependence of this magnitude [1362]. Moreover, a broader Curie distribution in milled samples is observed due to compositional heterogeneities or strain distribution [1363].

Ball milling of amorphous $\text{Fe}_{75}\text{Nb}_{10}\text{B}_{15}$ melt-spun ribbons enlarges the metal-metal distance and thus enhances magnetization but $|\Delta S_M|$ is reduced, possibly due to a broader distribution of Curie temperatures after mechanical treatment [1361].

Compaction of these ball milled powders leads to different effects depending on the composition [1364], due to a general B enrichment of the amorphous phase upon annealing. In fact, B incorporation by milling is not trivial but this element remains as inclusions and is not completely incorporated to the matrix [1358]. This leads to clear differences between the MCE of B-containing compositions obtained via rapid quenching or ball milling [1362]. Besides B inclusions, remnant α -Fe type nanocrystals are generally found (though not clearly detected by some techniques: e. g. X-ray diffraction). Procedures to extract the information of the amorphous phase have been developed based on the power dependence of MCE with magnetic field [1365].

In general, for Fe-based amorphous alloys and unlike Gd-based amorphous ones, the presence of crystalline phases (due to quenched in crystallites or nucleated ones after partial crystallization) leads to a deterioration of MCE, as the Curie transitions of the

crystalline phases are far beyond that of the amorphous phase [1365-1368]. Only few exceptions appear due to the formation of intermetallics with lower T_C [1360].

The MCE of $\text{Fe}_{49.7}\text{Cr}_{18}\text{Mn}_{1.9}\text{Mo}_{7.4}\text{W}_{1.6}\text{B}_{15.2}\text{C}_{3.8}\text{Si}_{2.4}$ gas atomized powders shows that amorphicity enhances with the reduction of particle size [1369]. $(\text{Fe}_{93}\text{Zr}_7)\text{C}_{0.11}$ alloys, which are prepared from carbon implantation of amorphous $\text{Fe}_{93}\text{Zr}_7$ (produced by magnetron sputtering) shows a MCE enhancement: $-\Delta S_M$ increases from 0.66 to 1.01 J $\text{kg}^{-1}\text{K}^{-1}$ and RC_{FWHM} from 85 to 156 J/kg at 1.5 T as well as the Curie temperature increases from 160 to 311 K after C implantation [1370].

The tunable T_C by fine compositional modification of amorphous alloys favors the development of composite systems with table-like MCE responses [1325, 1371-1373].

Furthermore, several works of Fe-based amorphous alloys with very high MCE values (using the terms of GMCE and even colossal MCE) can be found in the literature [1374-1378]. However, these results are also self-contradictory for the same alloy compositions in works reported by some of the coauthors (e.g. [1379]). Recent results reported conventional MCE behavior for these compositions and thus only these results are included in Table S7 of the Supplementary Material [1380].

To finish this section it should be helpful to position the different families in their way to be used in the main application of materials with MCE: magnetic refrigeration. As it will be shown in Section 8, first devices for low temperature applications used paramagnetic salts. Progressively GGG was also used. Nowadays, in addition to these two types of materials, Laves phases are also employed in the prototypes developed for the temperature range of hydrogen liquefaction, becoming the benchmark materials against which the new developed materials (such as molecular magnets or RE nitrides) are compared. The exponential decay of the magnetic entropy change observed in Laves

phases as transition temperature increases (regardless of the order of the transition) is the typical behavior found for most of the RE-based systems, either crystalline or amorphous.

Concerning room temperature applications, the Gd and its alloys are by far, the most profusely used in the proposed devices. However, La(Fe,Si)_{13} -based systems (including hydrides) are becoming a usual material. Few refrigerators prototypes also use Lanthanum-manganites and $\text{Gd}_5(\text{Si,Ge})_4$. The use of MnFePSi alloys in the prototype presented by Haier, Astronautics Corporation of America and BASF at the CES-Las Vegas in 2015 [10] opens a bright technological future for this kind of materials. The absence of Heusler alloys in the refrigeration devices should be ascribed to problems such as hysteresis and mechanical stability, leading to irreversibilities. These problems are nowadays under active research to look for solutions.

6. Enhancing the magnetocaloric effect through material processing techniques

Typically, the optimization of MCE is centered on the search for new alloys with enhanced performance and in particular to those exhibiting magnetism-structure coupling in FOPT, which gives rise to GMCE properties. Hence, a higher magnetic moment without modifying the phase transition is highly desired and even preferable if combined with a transition temperature near room temperature, which are usually not all easily attained in real cases. Moreover, material cost can play a significant role in the tradeoff between performance and cost, especially when the required elements of the strategic materials could only be extracted from few certain locations in the world, risking the ultimate success of any new technology subjected to the changes in

geopolitical situation. Therefore, materials engineering approach (i.e. materials processing) could complement the former tradeoffs with a larger certainty of its success. For example, already-established compounds used as building blocks for composites, nanostructures, graded materials etc., could provide promising results in part due to the existing deep knowledge of the material's constituent phases. In this section, the materials processing methods currently used to enhance MCE will be discussed while many earlier publications on the material selection for MCE and their related physical properties were reported and some mainly focused on the optimization of LaFeSi-related alloys, a type of GMCE material [2, 12, 18, 19, 1381, 1382].

6.1. Multiphase materials and composites

An ideal material for Ericsson thermodynamic magnetic refrigeration cycles should exhibit a table-like MCE (i.e. a constant ΔS_M or ΔT_{ad} in the refrigeration temperature range), which could be achieved by either incorporating different phases or combining different tunable magnetic phase transitions in the same material [818, 1325, 1383, 1384]. These methods could, in turn, optimize RC because the combined phases of tunable transition temperatures would broaden the MCE peak (provided that the magnitude of the peak value is not compensated). In particular, a theoretical ~90% RC enhancement in biphasic materials compared to their starting pure phases was reported when appropriate transition temperatures of phases and proportion of constituents were selected [1324]. Its main phase constituent should exhibit the largest transition temperature to maximize RC enhancement in the composite in comparison to that of the main phase. These results from the numerical calculations agree with the experimental measurements of a layered composite, which indicates MCE for moderate magnetic fields could be enhanced using this approach to design magnetic refrigerant materials that use the Ericsson cycle [1325]. Furthermore, multilayered composites were also

used for the hydrogen liquefaction as well as active magnetic regenerators (AMR) [1325, 1385, 1386]. In addition, the optimization of the shape of the peak (to obtain a table-like *MCE* and to broaden the peak to enhance *RC*) is best obtained explicitly at applicable magnetic fields used in refrigerator devices, as the optimal phase proportion can change with field. A magnetocaloric nanocomposite of $\text{La}_{0.8}\text{Ca}_{0.2}\text{MnO}_3$ (LCMO)/ $\text{La}_{0.8}\text{K}_{0.2}\text{MnO}_3$ (LKMO), with a mass fraction of 1:1, has been reported to display a table-like $\Delta S_M(T)$ curve with slightly lowered ΔS_M^{peak} and an *RC* enhancement of 33%-35% compared to those of the components [533]. Very recently, a biphasic Gd+GdZn composite, developed in a single fabrication step by induction melting, was reported with an enhanced *RC* in comparison to single-phased Gd (11%) and GdZn (45%) respectively (see Figure 44) [687]. GdZn, whose Curie temperature lies near room temperature and near that of Gd, could co-exist with the latter as a composite via varying the Gd:Zn ratio according to its phase diagram. The earlier theoretical studies also reveal the tunable *MCE* from the variant Gd+GdZn phase ratio, which enables them as highly effective *MCE* composite regenerator materials for a flexible choice of thermodynamic cycle [116].

6.2. Influence of fabrication techniques

6.2.1. Rapid solidification

Rapid solidification has been a popular choice to prepare many magnetocaloric alloys (either with FOPT or SOPT) since it is a well-known method to fabricate amorphous/nanocrystalline/polycrystalline materials with excellent composition homogeneity, controlled thickness catered for applications, good mechanical properties etc. For example, rapid solidification process in melt spinning or splat quenching uses a cooling rate of 10^4 – 10^7 Ks^{-1} , enabling one to easily obtain various microstructures, such

as amorphous/polycrystalline states or a more complex microstructure of nanocrystallites in amorphous matrix. Another example: The resulted refined microstructures from higher cooling rates in rapid solidification help to shorten the duration for heat treatment, which is typically required for homogenization [1382]. The La(Fe,Si)_{13} ribbons or flakes prepared by melt spinning only required a heat treatment duration of a few hours and resulted in a uniform microstructure and homogenous element distribution, in which the NaZn_{13} -type 1:13 phase in La–Fe–Si-based alloys (the origin of the GMCE in these materials) is not readily produced as a bulk material using conventional melting [1387]. Moreover, a bulk sample fabricated by conventional methods of 1:13 phase will entail a typical homogenization process at high temperatures of 1100 K for at least one week followed by quenching [323]. Another advantage includes the kilogram-scale production of high performance magnetocaloric materials by strip casting [335]. It had also been reported that micro-sized spherical particles and wires, capable of good heat transfer under high operating frequency, were produced by rapid solidification [336, 1388].

6.2.2. Powder metallurgy

Despite the above-mentioned advantages of rapidly solidified flakes and ribbons, they usually cannot be directly utilized in AMR and will need extra steps to form specific geometries, such as grinding them into fine powders as precursors for powder metallurgy. In addition, powder metallurgy processing has become an attractive fabrication method for magnetocaloric materials when Vacuumschmelze GmbH produced blocks of sintered La–Fe–Si-based alloys with this technique and furthermore it could be conveniently accompanied by solid-state sintering when preparing Fe_2P -type materials (another type of GMCE material) [1389]. In addition, mechanical alloying

could address the challenges to melt toxic or volatile elements, such as P in Mn–Fe–P–(Ge, Si)–based materials.

6.2.3. Liquid reactive sintering and Thermal Decomposition and Recombination process

AMR refrigerants are typically required as thin plates of thickness well below 0.5 mm. However, machining the phase with a Curie temperature close to room temperature causes the formation of micro-cracks in the samples due to the local temperature gradients around the cutting tool [1390] and the large magnetovolume effect. In fact, microstructural characterization of a fully sintered La(Fe,Si,Co)_{13} alloy reveals segregation of La-rich phase along the grain boundaries and the large thermal expansion of NaZn_{13} -type phase when above its T_C led to the formation of micro-cracks [1382]. This could be overcome by the thermally induced decomposition and recombination process (TDR), whereby decomposing the $\text{La(Fe, Co, Si)}_{13}$ phase into a lamellar structure of α -Fe and 1:1:1 phases, both of highly enhanced mechanical properties, allows easier machining [1382, 1390]. The full recovery of $\text{La(Fe, Co, Si)}_{13}$ phase can be attained by annealing the cut decomposed thin plates at around 1323 K.

6.3. Nanostructures

Nanostructured magnetocaloric materials generally includes nanocrystalline materials, core-shell structures as well as nanoparticles, and they could be suitable for magnetic refrigeration of microfluidic reactors, electronic chips or other devices of small dimensions [18]. In addition, it has been demonstrated that nanostructuring can tune not only the quantitative response of a material but also its qualitative behavior, which can be completely dissimilar from that of its bulk counterpart [1391]. The magnetocaloric response of nanostructured materials will not only depend on the characteristics of the

bulk material (the composition of the phase) but also highly dependent on other factors such as nanocrystallite size (and size distribution), nanoparticle concentration, interactions between particles, finite size scaling, anisotropy etc. Models accounting for these effects have been proposed [1391-1394]. For example, the magnetization of nanostructured magnetic materials with single-domain crystallites could rapidly vary (due to the dynamical freezing near blocking temperature, T_B) with a much larger magnetic response due to superparamagnetism [1394] and the authors also reported broadened ΔS_M extended over a wider temperature range near T_B in the nanostructured system.

6.3.1. Nanocrystalline materials

Nanocrystalline materials could be prepared by melt spinning using a lower cooling rate (lesser than that used for fabricating fully amorphous ribbons as mentioned earlier) and/or partial devitrification of an amorphous precursor. Nanocrystalline $\text{Nd}_2\text{Fe}_{17}$ and $\text{Pr}_2\text{Fe}_{17}$ powders milled for 10 h show ΔS_M^{peak} extended over a wider temperature span as compared to their bulk counterparts, which can aid to improve RC [1395]. Another example: Nanocrystalline FeSiBCuNb-type alloys exhibit superparamagnetic transition above the T_C of the residual amorphous phase (when the interactions between particles vanish) and this enables the tunable transition temperatures to even above room temperature, which is more useful than low temperature applications of superparamagnetic nanoparticles with enlarged magnetic moment (these are better magnetic refrigerants than paramagnetic materials) [1396-1398]. Moreover, superparamagnetic clusters could group together the spins, enabling the magnetic moments to be more easily aligned than those in paramagnetic systems [1399]. As a result, the entropy of the spins is more easily modified by the magnetic fields for certain ranges of the magnetic field, temperature and cluster size. However, in the case of

superparamagnetic particles embedded in a residual amorphous matrix, the presence of Fe-rich nanocrystals in the amorphous matrix was later found to reduce MCE and RC due to the large separation between the transition temperatures of the different phases [1324, 1325, 1366, 1400]. Very recently, $\text{Gd}_{(50+5x)}\text{Al}_{(30-5x)}\text{Co}_{20}$ microwires exhibiting amorphous/nanocrystalline phases produced by melt extraction, were reported to display larger RC values than those of GdAlCo -based alloys in the forms of bulk metallic glass or melt-spun ribbons [1401]. The authors attributed the enhanced RC values to the magnetic coupling between the amorphous and nanocrystalline phases (20 vol.% of ~ 10 nm nanocrystallites as observed by TEM).

6.3.2. Magnetocaloric nanoparticles

In the case of magnetocaloric nanoparticles, they could be prepared using various chemical synthesis techniques, mechanical milling [1269, 1402], ion-exchange technique, spray pyrolysis [1403] and even on a large scale using plasma arc discharge method [680]. The typical chemical synthesis methods reported include sol-gel [517, 1404, 1405], sol-gel based Pechini [511, 1406], citric sol-gel [484, 1407], glycine nitrate process [1403] and glycine-assisted combustion method [1408].

The reduced particle size of magnetic nanoparticles may not usually aid in large MCE but it could extend ΔS_M^{peak} over a larger temperature span [1409], which could sometimes increase RC . However, for the case of superparamagnetic nanoparticles, the effect of the particle size on MCE is reasonably evident. For example, GdNi_5 superparamagnetic nanoparticles exhibit larger $-\Delta S_M$ than that of $\text{GdNi}_5/\text{Gd}_2\text{O}_3$ nanocapsules ($\sim 4\times$ larger diameter), despite both fabricated in the same process using the arc-discharge technique [1410]. This is attributed to the superparamagnetic nanoparticles having a smaller magnetic anisotropy energy barrier due to their smaller

particle size than the nanocapsules, which could impede the reversal of magnetic moment vector of nanoparticles during field variation. In addition, the transformation of antiferromagnetic behavior (in $\text{La}_{0.48}\text{Ca}_{0.52}\text{MnO}_3$ bulk state) to ferromagnetic via reducing the particle size dimension into nanometer scale has been recently reported [517]. The $\text{La}_{0.48}\text{Ca}_{0.52}\text{MnO}_3$ nanoparticles, synthesized by the conventional sol-gel method, show that their MCE and relative cooling power (*RCP*) systematically increase with smaller particle sizes. The authors also reported the *RCP* of a particle size sample of 45 nm is about 400% larger than that of the bulk counterpart. Another recent study on the MCE of $\text{La}_{0.7}\text{Ca}_{0.3}\text{MnO}_3$ nanoparticles prepared by reactive milling technique shows that larger grain sizes could decrease T_C and increase ΔS_M^{peak} [1411]. The authors report that the weakening of exchange interaction at the surfaces of smaller particles resulted in the decrease in saturation magnetization.

However, nanostructuring can also cause a decrease in MCE, in particular to materials that present a FOPT when in bulk form, due to the loss of the first-order character of the phase transition [1412]. A recent example would be the MCE report of $\text{Sm}_{0.6}\text{Sr}_{0.4}\text{MnO}_3$ nanotube and nanoparticles prepared by a sol-gel modified technique (Pechini) [511]. The nanostructured samples display broader $\Delta S_M(T)$ curves as compared to the bulk counterpart due to their extended widths of temperature range. However, their ΔS_M^{peak} were compensated due to the suppression of FOPT by nanostructuring. The authors reported that the nanotube sample (constituted of smaller nanoparticle size: 25 nm) exhibits an undesired M-shape $\Delta S_M(T)$ profile due to a uprising superparamagnetic behavior, which also indicates a threshold of nanostructuring, below which the advantage to make the transition wider is no longer valid.

Exceptional from the above, the effect associated with surface spin disorder in core-shell nanoparticles had been reported with a significant MCE [1413] despite being a second order phase transition, which could be studied using the techniques developed for Curie transitions [1414]. Another example is the system of isolated Gd_2O_3 nanoparticles embedded in the pores of a mesoporous silica matrix [1415], which recently reports extraordinarily large values of magnetic entropy change in the cryogenic temperature range, with the added advantage that the nanoparticles are not free standing but embedded in a matrix, which facilitates handling and possible technological applications.

One interesting point about magnetocaloric nanoparticles is that they extended the range of applications for magnetocaloric materials beyond magnetic refrigeration. Wang et al. developed magnetocaloric stimuli-responsive hydrogels of Fe_3O_4 colloidal nanocrystal clusters in poly(*N*-isopropylacrylamide) matrix loaded with dyed oil [1416]. Using MCE, they demonstrated controlled release of the incorporated hydrophobic molecules for each reversible shrinking-swelling cycle under AC magnetic field. The authors also suggested MCE offers advantages of remote control and precise selection, giving rise to further potential for practical engineering applications, like sensors, displays or controlled release for targeted medical therapy.

6.3.3. Nanostructured films (or thin films)

The materials' properties typically differ when the microstructure of the material alters, likewise when the microstructural dimensions reduced to nano-scale. Thus, nanostructured or thin films would exhibit different magnetocaloric properties than their bulk counterparts. As an example, the microstructure of the thin films of ferromagnetic shape memory alloys (which is dependent on alloy composition, film thickness, substrate temperature and post-deposition annealing) can strongly influence their

characteristic phase transformation temperatures. This was experimentally observed in the magnetron-sputtered Ni-Mn-In, whereby the phase transformation temperatures were tuned to lower temperatures with decreasing film thickness [1050]. The authors also reported observing the completion of martensitic phase transformation in the thick film, in which its larger grain size reduced grain boundary density. In addition, there is another study on epitaxial Ni-Mn-Sn thin films on MgO (001) substrate fabricated by magnetron sputtering, which reports that the film thickness significantly affects the fraction of martensitic transformation, the transformation temperatures and the temperature range of transformation [1051]. The authors ascribed the former to the deferred transformation near the interface due to the rigid substrate, resulting in a residual layer of austenite at low temperatures, while the lower transformation temperatures and broadened martensitic transformation are presumably due to size effects. The films displayed an inverse MCE with a martensitic transition around 260 K, with comparable MCE values as those of bulk counterparts [1417].

Desautels et al. recently report that the formation of a FeCu interfacial phase in the nanostructured thin film of Cu matrix/Fe nanocrystallites, could control magnetic ordering temperature and MCE [1264]. The authors prepared the Fe/Cu thin films using a reactive ion beam deposition method, alternating the deposition of Fe (30s or 60s) then Cu (60s) for five repeats (ten layers total). The intermix between the Fe nanocrystallites and deposited Cu was stimulated resulting in the formation of an interfacial FeCu phase. A larger *RCP* was observed for the sample with shorter deposit time of Fe, whereby the authors attributed to the overlap of the temperature ranges of the dynamical freezing (T_B) of Fe crystallite magnetizations and the spin fluctuations of FeCu alloy near T_C .

In addition, multilayers formed by layers with different composition (which exhibit a distribution of T_C) can improve the magnetic field responsiveness of magnetocaloric materials in the range of fields suitable for technological applications [123, 1418]. In these cases, ΔS_M^{peak} correlates linearly to H in a broad temperature range close to the transition temperature, which is an enhancement with respect to the $H^{0.7}$ behavior typical of bulk materials.

6.3.4. Self-assembled nanostructures

Self-organized ferromagnetic nanowires embedded into highly ordered nanoporous anodic alumina templates have been found exhibiting a coexistence of conventional MCE together with inverse MCE (their sign and magnitude are controlled by the magnitude of the field) upon perpendicularly magnetized to their easy axes of magnetization [1391]. These results validate the numerical calculations that aim to establish the concept of tailoring magnetocaloric response via controlling the sample's microstructure. In contrast to previous results in which combined normal and inverse MCE was reported in other bulk magnetocaloric compounds with several different magnetic transitions [1419], the field-controlled combination of direct and inverse MCE arises due to the superparamagnetic transition of the nanowires and their peculiar anisotropy. This can lead to new possibilities to develop completely new applications of the materials.

Another example demonstrating that self-assembled nanostructures are promising for magnetic cooling near room temperature at low magnetic fields is illustrated in the enhanced magnetic phase transition temperature and MCE in the framework of interfacial coupling in self-assembled $\text{Mn}_3\text{O}_4\text{-La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ nanocomposites [532]. The authors used a dual-target pulsed-laser deposition (PLD) technique to fabricate the

growth of self-assembled nanostructures with varying amounts of Mn_3O_4 in the form of nanocrystals embedded in the $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ matrix. The authors reported an enhanced ΔS_M^{peak} (the highest being similar to 130%) in the nanocomposite doped with low Mn_3O_4 content in comparison to that of pure $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ thin film.

6.4. Special treatment conditions

The typical examples of special treatment conditions reported to contribute to enhanced MCE are hydrogenation and annealing, while porous/metal foam structuring improves the resistance to fatigue-cycling.

6.4.1. Hydrogenation

Hydrogenation is typically attained from conventionally annealing $\text{La}(\text{Fe-Si})_{13}$ -type compound in H-atmosphere at high pressures of 5 MPa, reactive milling in the presence of H_2 , which could avoid the need of high pressures from the former technique, and electrolytic hydriding [223, 264, 265, 279, 281, 532]. The FOPT and itinerant-electron metamagnetic (IEM) transitions (which contributes to the GMCE characteristic of $\text{La}(\text{Fe-Si})_{13}$ alloy) were maintained with hydrogenation while significantly increasing T_C from ~195 K to near RT, which are desirable as RT magnetocaloric refrigerant. The effects of hydrogen absorption on the MCE of $\text{La}(\text{Fe}_{1-x}\text{Si}_x)_{13}\text{H}_y$ alloys were only significant at magnetic fields of 5 T, where $\Delta T_{ad}^{\text{peak}} = 12.6\text{K}$ for $y = 1.5$, which is about 50% larger than that for $y = 0$ ($\Delta T_{ad}^{\text{peak}} = 8.6\text{K}$) [265]. Otherwise, it is typically reported that hydrogenation treatment applied to this family of alloys increased T_C to RT without significantly altering MCE, even with full hydrogenation in reactive milling prior to a dehydrogenation process [330]. Partial dehydrogenation is a less optimal way to tune T_C . In the case of partially dehydrogenated $\text{LaFe}_{11.6}\text{Si}_{1.4}\text{H}_y$ alloys, two coexisting phases of different H concentration form prior to turning into one phase of single H content via

thermal cycling but the two phases reinstate again upon storage near the T_C of the single phase [330]. Two distinct ΔS_M^{peak} were also reported in partially hydrided $\text{LaFe}_{11.6}\text{Si}_{1.4}\text{H}_y$ compounds prepared by electrochemical hydriding [281].

The hydrogenation of $\text{LaFe}_{11.57}\text{Si}_{1.43}$ into $\text{LaFe}_{11.57}\text{Si}_{1.43}\text{H}_{2.27}$ compound was also reported to display dramatic heating effect with time when subjected to a constant AC magnetic field and reached the ideal therapeutic range for hyperthermia treatment in less than a minute [1420]. It demonstrated unique self-regulated induction heating effect near first-order T_C since its temperature was controlled at 320 K (within the therapeutic T_{range}) due to its abrupt reduction of spontaneous magnetization at the T_C of its FOPT. The authors also found that the $\text{LaFe}_{11.57}\text{Si}_{1.43}\text{H}_{2.27}$ compound exhibits high estimated values of specific adsorption rate (SAR), which are more than one order magnitude larger than conventional Fe oxide-based materials observed under the same conditions. Moreover, its self-regulated heating effect within therapeutic temperature range with excellent SAR value made it stood out exceptional from the conventional superparamagnetic or ferromagnetic materials of relatively high SAR values.

6.4.2. Annealing

In the case of annealing method, grain size typically increases upon annealing and this is also reported in post-annealed $\text{Ni}_{41.98}\text{Mn}_{42.97}\text{V}_{2.08}\text{Sn}_{12.97}$ melt-spun ribbons [1421]. In addition, a decrease in lattice constants, enhanced exchange bias effect and increased martensitic transformation temperatures are observed. Large ΔS_M^{peak} of $41.6 \text{ J kg}^{-1}\text{K}^{-1}$ ($\Delta H=30 \text{ T}$) near the martensitic transformation temperature was reported. On the other hand, annealing is an essential step in preparing NaZn_{13} -type $\text{La}(\text{Fe},\text{Si})_{13}$ -based compounds because the 1:13 phase is difficult to be directly attained. The typical annealing process requires high temperatures for more than one week and followed by

rapid quenching. Recently, an optimized annealing profile (1323 K for 24 h) has been studied for non-stoichiometric $\text{La}(\text{Fe}, \text{Co}, \text{Si})_{13}$ composition, i.e. $\text{La}_{1.7}\text{Fe}_{11}\text{Co}_{0.8}\text{Si}_{1.2}$, and this alloy displayed a large ΔS_M^{peak} of $7.5 \text{ J kg}^{-1} \text{ K}^{-1}$ ($\Delta H=2 \text{ T}$) at 271 K and a reversible magnetic phase transition with zero hysteresis [322]. The authors also reported a complex phase combination, which consisted of $\text{La}(\text{Fe}, \text{Co}, \text{Si})_{13}$, La_5Si_3 , $\text{La}(\text{Fe}, \text{Co})\text{Si}$, α -Fe and La oxide phases, as compared to the stoichiometric $\text{La}(\text{Fe}, \text{Co}, \text{Si})_{13}$ alloy.

Recently, it has been reported that increasing annealing temperatures can effectively tune MCE of $\text{La}_{0.7}\text{Ba}_{0.3}\text{MnO}_3/x\text{TiO}_2$ system ($x = 0.04, 0.06$), in which up to 24 % increase of *RCP* could be observed by increasing the annealing temperature from 600 °C to 800 °C [527]. The authors attributed the enhanced *RCP* to the TiO_2 distribution in the doped composites, which alters the resistance of the grain boundaries affecting magnetization pinning.

6.4.3. Porous/metal foam structuring

In addition to the two above-mentioned techniques, porous/metal foam structures have been considered for enhancing the resistance to fatigue-cycling, showing potential to improve the lifetime of the material. Lyubina et al. report that by introducing porosity, the internal constraints imposed by grain boundaries could be removed, resulting in a desirable decrease of hysteresis in the field- and thermally induced magnetic phase transitions [227]. In addition, a distinct enhancement in mechanical stability during cycling was attained, leading to a fully reversible MCE despite having a lower $\Delta T_{ad}^{\text{peak}}$ than that of the bulk alloy. The authors suggested that the ideal porous materials to sharpen the transition and enhance MCE should comprise of single-crystalline particles with a uniform size distribution, based on their observations of the evolution of para- to ferromagnetic transition from magneto-optic Kerr microscopy. They also added that the

magnetic and mechanical properties could be further improved through the fabrication of foams [227] and by embedding magnetic coolant particles in an elastic matrix [227].

7. Fundamental studies in magnetism using magnetocaloric characterization

Nowadays, a large majority of MCE research papers focus on the characterization of materials' properties, and the design and evaluation of possible magnetic refrigeration devices. However, in early times of this field, magnetocaloric characterization was considered a useful tool to enhance the knowledge that we had about phase transitions [1422, 1423]. This fundamental aspect of magnetocaloric research was never completely forgotten but regained recent attention with the emergence of new methods for determining critical exponents or the order of phase transitions.

7.1. Critical phenomena

It has been demonstrated that the magnetocaloric magnitudes of materials with a SOPT, in the proximity of the critical region, follow power laws of the field of the type $magnitude \propto H^{exponent}$, with exponents that are related to the critical exponents of the material [80, 1424] (Table 1). Therefore, it is possible to extract the values of these critical exponents by studying the field dependence of MCE. For simple cases, like a purely SOPT material wherein a single magnetic phase exists, the critical exponents obtained from the conventional Kouvel-Fisher method [1425] and those attained from the field dependence of MCE show a good agreement [1424]. However, for more complex cases, the conventional methods are not applicable, e.g. when there are several phase transitions coexisting in the same material. Nevertheless, it has been

demonstrated that the scaling of MCE is capable of providing accurate values of the critical exponents even for these more complicated situations [1426].

Table 1: Exponents controlling the field dependence of different magnitudes related to MCE ($magnitude \propto H^{\text{exponent}}$).

Magnitude	Exponent
$T_{pk} - T_C$ (not mean field)	$1/\Delta$
$T_{pk} - T_C$ (mean field)	0
$\Delta S_M (T = T_c)$	$1 + 1/\delta(1 - 1/\beta) = (1 - \alpha)/\Delta$
ΔS_M^{pk}	$1 + 1/\delta(1 - 1/\beta) = (1 - \alpha)/\Delta$
RC_{Area} or RC_{FWHM}	$1 + 1/\delta$

For SOPT materials, a phenomenological construction of a universal curve was proposed [76]. This implied the possible identification of curve shape onto which all rescaled experimental data of that studied material should collapse without the prior knowledge of its critical exponents or EOS. The practical applications of this phenomenological universal curve ranges from the extrapolation of experimental data for fields and temperatures less commonly accessible in laboratories [1427] to the prediction of MCE responses of different alloys from series of materials with similar values of critical exponents, provided that a limited number of data points of those other alloys are known [1427]. The method is also applicable to addressing the noise in the measurements, which also allows the enhancement of data resolution recorded for low field [80]. In brief, the construction of the phenomenological universal curve involves:

1) normalizing each magnetic entropy change curve (each calculated for a certain value of the field) with respect to their own peak, 2) identifying the field dependent reference temperatures (T_r) that provide a specific value of the normalized curve (which typically ranges between 0.5 and 0.7), and 3) rescaling the temperature axes of these curves as

$$\theta = (T - T_c) / (T_r - T_c).$$

The phenomenological construction was later theoretically justified by using scaling relations [229] and extended to the adiabatic temperature change of magnetocaloric materials [33]. It is important to note that both the universal scaling of MCE and the determination of critical exponents using MCE are restricted to the critical region of the material, i.e. for temperatures that are relatively close to the transition temperature and fields that are not too large [1428]. A detailed investigation on the applicability limits of scaling in magnetocaloric materials [1429] shows that the usual experimental conditions of fields up to 10 T are still appropriate for this type of studies. This is supported by numerous papers in which scaling is observed in many different materials by different authors [78, 500, 1278, 1309, 1430-1437]. It is also worth noticing that the demagnetizing field experienced by experimental samples, as well as the various coexisting phases in materials or the distribution of Curie temperature of the phases due to sample inhomogeneity might alter the field dependence of magnetocaloric magnitudes [55, 1363, 1438-1440].

7.2. Analysis of the order of phase transitions

While there were previous methods, based on only magnetic measurements, to determine the order of a thermomagnetic phase transition like the Banerjee criterion [1441], there were cases reporting that the reliability of this criterion was not appropriate as it predicted a SOPT behavior while calorimetric measurements indicated the transition was of the first order type [147]. On the other hand, the use of the universal curve enabled the demonstration, by using purely thermomagnetic data, that DyCo₂ undergoes a SOPT, in agreement with its calorimetric data [148]. The basis of procedure is that SOPT should collapse onto a universal curve when the magnitudes are properly rescaled, while this is not valid for FOPT. The validity of the phenomenological universal curve method, in comparison to the Banerjee criterion, is

due to the latter imposing a specific EOS for the material to study the nature of its phase transition, while the method proposed by Franco et al. [80] does not rely on any EOS.

Recent theoretical studies indicate that the scaling of MCE is valid, in the proximity of the critical region, for both SOPT materials and for those at the tricritical point [82].

Besides the frequent topics of MCE research (like the search for new magnetocaloric materials and, more recently, the determination of critical exponents via the scaling of MCE), there are also similar research fields with minimal magnetocaloric characterization, which could be because either these materials are less appropriate as potential refrigerants or that their main properties were not directly related to thermomagnetic properties. One of such examples is the field of skyrmions. It has recently been shown that the population of skyrmions in MnSi as a function of field can be determined from the analysis of the scaling of MCE [1442]. This compound exhibits helimagnetic order and an additional skyrmion phase for a range of magnetic field and temperature. The deviations from universal scaling allowed the authors to determine that the maximum concentration of skyrmion vortices took place for a temperature of 28.5 K and a field of 1.6 kOe.

8. Refrigerator devices

Magnetic refrigerators based on MCE are basically composed by three elements: the magnetic field source, the magnetocaloric material (the regenerator is formed by this magnetocaloric material along with the bed that contains it) and the heat exchange system.

In 2015, Kitanovski et al. [22] published an excellent book addressing the main engineering aspects and problems of magnetic refrigeration, including a deep description of the magnetic sources, heat transfer fluids and the corresponding heat

transfer mechanisms, mainly for room temperature magnetocaloric applications. In this section, we briefly describe the main parameters to be considered in magnetic refrigerators as well as a comprehensive list of devices based on the magnetocaloric effect for a broad temperature range, including cryogenic temperatures.

There are several ways of performing magnetization/demagnetization of the MCE materials and that usually define the type of magnetic refrigerator: static, when the magnetic field intensity varies with time (e.g. powered by an AC current source); rotary, when the magnetic field source (or the regenerator) moves circularly around the regenerator (or the magnetic source); and reciprocating, when the magnetic field source or the MCE material performs a linear motion to create a variable magnetic field on the material. Different types of magnets are utilized for the magnetic field source, namely: superconducting magnets for low temperature applications while arrays of permanent magnets are preferred nowadays for room temperature devices.

Regarding the MCE materials used in the devices, gadolinium gallium garnet, $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, has been extensively used in the development of magnetic refrigerators working at cryogenic temperatures, with a more recent appearance of Laves phases, while Gd metal has been considered the reference material for those operating close to room temperature. In the recent years, room temperature devices also frequently tend to include $\text{La}(\text{Fe},\text{Si})_{13}$ type materials.

Heat is continuously stored and released in the heat exchangers, with a heat transfer fluid to transport the heat between exchangers. He gas is commonly used as heat transfer fluid to reach low temperatures for He or H_2 liquefaction or even lower temperatures, while water, with some antifreeze additives, is generally used in the development of room temperature prototypes. Liquid nitrogen can be found for

applications at intermediate temperatures (O_2 or air liquefaction) or as an additional step for cryogenic temperatures.

In this section, we will categorize devices into two groups, depending if they are developed for cryogenic or room temperature applications. Prototypes will be presented in a chronological order.

8.1. Low temperature devices

In 1953, Collins et al. built a reciprocating adiabatic demagnetization refrigerator (ADR), whereby its magnetic material composed of two separate paramagnetic salt samples (Iron-Ammonium-Alum). The upper sample was initially magnetized by a magnetic field of 0.185 T, which was supplied by a permanent magnet. Next, this cold sample was moved out of the magnetic field and then kept in thermal contact with the lower sample until a thermal equilibrium between both samples was attained, and subsequently the cycle was started again. The authors reported a minimum temperature of 0.73 K for this device [1443].

In 1976, Rosenblum et al. presented a device based on that reported by Collins in 1953. In the new design, a magnetic field of 0.35 T was supplied by an alternate superconducting solenoid in its center gap, where a cylinder filled with $CeMn_{1.5}(NO_3)_6 \cdot 12H_2O$ and Au powders were placed. Experimental results showed that this device attained a minimum temperature of around 10 mK [1444].

In 1977, Pratt et al. developed a rotary refrigerator using adiabatic demagnetization cooling. $Gd_2(SO_4)_3 \cdot 8H_2O$ powders were used as the magnetic refrigerant material and were compacted to fill the rim of a wheel (wherein they were to be placed), which would be rotated through the high and low magnetic field regions generated by two

permanent magnets. For a wheel operation frequency of 0.5 Hz, a cooling power of 150 mW and a minimum temperature of 2.1 K were obtained [1445].

In 1978, Steyert presented a rotary magnetic refrigerator, which employed a superconducting magnet to supply a constant magnetic field of 5 T and a paramagnetic Gd compound as the working material, which was placed on the surface of a wheel. As the wheel rotates, part of it moves into the region where the field is applied. Using He gas as a heat transfer fluid, a cooling power of ~1 kW was measured with a working temperature of around 2 K [1446].

In 1979, Barclay presented a reciprocating magnetic refrigerator. This device, operating in a temperature range from 4.2 to 2.2 K, could only perform two cycles at a frequency of 0.016 Hz, showing a cooling power of 52 mW [1447].

A year later, Barclay developed a rotary type magnetic refrigerator operating under Carnot cycle. DyPO_4 single crystal was used as the working material, rotating inside the gap of a couple of coils, which generated a constant magnetic field of 7 T. The movement of the crystal was performed in such a way that its c-axis rotated perpendicularly to the direction of the magnetic field to perform the magnetization/demagnetization processes of the working material. He gas was used as the heat transfer fluid to transfer the heat between the reservoirs. This device was experimentally shown working within a temperature range of 20 to 4 K and generated a cooling power between 1 mW to 1W, depending on its operational frequency (e.g. 1 W of cooling power could be obtained with a frequency of about 2 Hz) [1448].

In 1981, Delpuec et al. presented a reciprocating magnetic refrigerator, in which HoPO_4 , $\text{Gd}_2(\text{SO}_4)_3$ and $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ were used as the magnetocaloric working materials. The magnetocaloric refrigerant moves in and out of a superconducting magnet, which gives

a magnetic field of 5 T. Both the working material and magnet were immersed in liquid He, which acts as a refrigerant for the magnet and as a heat transfer fluid. When HoPO_4 was used as the working material, experimental results showed that for an operational frequency of 0.3 Hz, a temperature difference of 2.1 K and a cooling power of 0.12 W were obtained, reaching a temperature of 1.98 K. In the case of $\text{Gd}_2(\text{SO}_4)_3$, a working temperature of 2.1 K with a cooling power of 0.36 W were observed, reaching a temperature of 1.67 K. For $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, a temperature of 1.8 K with a cooling power of 1.2 W were measured for an operation frequency of 0.95 Hz [1449].

In 1984, a Carnot type magnetic refrigerator was presented by Numazawa et al. In this device, $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ single crystal in a cylindrical shape was located in the center gap of a superconducting magnet, whereby the magnetization/demagnetization of the working material was due to the switching on/off the magnetic field. He gas was used as the heat transfer fluid to carry the heat between the material and the heat exchange system. The working material reached temperatures below 4.2 K, thus, it was used for liquefaction of He [1450].

In 1985, Matsumoto et al. presented a reciprocating magnetic refrigerator operating under Carnot cycle. A superconducting magnet, supplying a magnetic field of 5 T, was operated in switch on - off mode to magnetize and demagnetize the magnetocaloric material. Two pieces of $\text{DyAl}_{2.2}$ were used as the magnetocaloric material and He gas as the heat transfer fluid. Experimental results showed that this device could provide a working temperature in the range from 77 K to 20 K [1451].

In the same year, Nakagome et al. developed a magnetic refrigerator using a cylinder working material, which could be formed by $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, Er-Ni alloy, $\text{Dy}_3\text{Al}_5\text{O}_{12}$, $\text{Gd}_3\text{Al}_5\text{O}_{12}$, GdPO_4 or $\text{Dy}_2\text{Ti}_2\text{O}_7$ compounds. Its magnetic field was applied by

superconducting coils powered by AC current source, which magnetize and demagnetize the working material placed in the center gap of the coils. Using He gas as the heat transfer fluid, this device allows the liquefaction of He [1452].

In addition, a magnetic refrigerator operating under Carnot cycle was developed in 1985. In this device of Hakuraku et al., a superconducting magnet with a maximum magnetic field of 3T was operated by switching on and off with a frequency of 0.3 Hz. Using $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ as the working material, a cooling power of 0.6 W was obtained with a temperature of 1.8 K [1453].

Barclay et al. presented a rotary magnetic refrigerator working under Carnot cycle in the same year. $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ placed on the rim of a wheel rotates through the gap of superconducting Helmholtz solenoids, and He gas was used as the heat transfer fluid in the hot and cold ends of the wheel. Tests were performed for applied magnetic fields of 4 and 5.6 T over the working material rotating at a frequency of 0.08 Hz. Results showed a cooling power of ~0.9 and ~1.2 W, reaching a temperature below 4.2 K [1454].

In 1986, a rotary magnetic refrigerator operating under Carnot cycle was presented by Hakuraku et al. The working material consisted of 12 pieces of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ placed surrounding the rotor. A superconducting magnet was used to apply a magnetic field of 3 T into the bore where the working material rotates. Operating with a frequency of 0.4 Hz, a cooling power of 1.79 W was obtained, reaching a temperature of 1.8 K [1455].

In the same year, Hakuraku et al. presented a rotary magnetic refrigerator, working under Carnot cycle, wherein 8 disks of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ were placed equidistant on a cylinder. The cylinder was located inside of three superconducting Helmholtz coils, which generate a constant magnetic field of 2.94 T and rotate around the cylinder with

the working material. Operating with a frequency of 0.2 Hz a cooling power of 1.5 W was obtained, reaching a minimum temperature of 1.9 K [1456].

Researchers from Massachusetts Institute of Technology presented a Carnot type magnetic refrigerator in 1986. The regenerator was formed by a set of parallel plates of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ in a cylindrical configuration and located in the bore of the magnet system, which consisted of a superconducting solenoid operating in AC mode. He gas was used as the heat transfer fluid driving heat between the heat exchangers and the regenerator. Test performed varying the magnetic field between 1 and 4 T showed a cooling power of 0.40 W, reaching a temperature of 3.9 K [1457].

Also in 1986, Nakagome et al. developed a reciprocating magnetic refrigerator which used $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ single crystal in a cylindrical shape as the regenerator. It was moved in and out of the center gap of a superconducting magnet, which supplied a magnetic field of 4.5 T. Using He gas as the heat transfer fluid, experimental results showed a cooling power of 0.95 W under an operational frequency of 0.38 Hz, reaching a temperature below 4.2 K [1458].

In 1988, Nakagome et al. presented a rotary magnetic refrigerator which operated under a reverse Carnot cycle. Fourteen monocrystals of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, the working material, were contained in a disk that rotates in the gap of a superconducting magnet (a magnetic field gradient between 4.5 to 1.1 T was generated within the gap where the disk rotates). With an operation frequency of ~ 0.01 Hz, experimental results showed that the device developed a cooling power of 0.12 W in the working temperature range from 11.5 to 4.2 K [1459].

In 1991, Kuzmin et al. built a rotary Carnot type magnetic refrigerator. This device was formed by a superconducting solenoid and the working material in the shape of cylinder

in the center of the gap. A DyAlO_3 monocrystal was used as the working material, which rotates from its b axis to the c axis. This motion of the monocrystal enabled its magnetization and demagnetization depending on the instantaneous orientation of the easy magnetization axis. Tests performed showed that this device worked within a temperature range of 20 and 4.2 K [1460].

In 1992, Filin et al. presented a static type magnetic refrigerator operating under Carnot cycle. The device consisted of a $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ monocrystal as the working material, which was placed in the bore of pulsed superconducting magnet, and He gas as the heat transfer fluid. Experimental results showed that the device worked in the temperature range of 4.2 to 1.8 K [1461].

In the same year, DeGregoria et al. presented a reciprocating magnetic refrigerator operating under Brayton cycle. Its magnetic system consisted of a superconducting solenoid which could move in a reciprocating mode. The regenerator was formed by two beds filled with $\text{Er}_x\text{Gd}_{1-x}\text{Al}_2$ particles with a cold heat exchanger placed between them and placed in the gap of the solenoid. He or N_2 gas were used as the heat transfer fluid depending on the working temperature range of the device. Tests were performed for different magnetic fields and results showed that the device reached a minimum temperature of around 4 K which gave the maximum temperature change of around 2 K for an applied magnetic field of 7 T [1462].

In 1994, Bezaguet et al. developed a magnetic refrigerator operating in a quasi Carnot cycle between 4.5 and 1.8 K. $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ single crystal was used as the working material and its magnetization/demagnetization process was performed by a superconducting magnet which supplied a magnetic field of 3.5 T. Liquid He was used as the heat

transfer fluid. Estimations showed that this device could allow a cooling power of about 25 W [1463].

In the same year, Jeong et al. presented a Carnot type tandem magnetic refrigerator. This device consisted of two magnetic systems, each filled with $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ particles, which constitute the working material. A superconducting magnet supplied a magnetic field varying between 0.4 to 2.8 T in the bore where the working material was placed. Helium-3 gas was used as the heat transfer fluid carrying the heat between the heat exchangers through the working material. For switching the magnetic field with a frequency of 0.4 Hz, a cooling power of 12 mW per each magnetic system, reaching a temperature of 1.8 K were reported [1464].

In 1995, Zimm et al. presented a reciprocating active magnetic regenerator (AMR) refrigerator. The regenerator was composed by two beds filled with ErNi and was moved in and out of a superconducting solenoid, which supplied a constant magnetic field of 3 T. He gas, below 0.3 atm of pressure, was used as the heat transfer fluid of this device. Experimental tests showed a maximum cooling power of 1.6 W at 3.5 K [1465].

In the same year, Kashani et al. developed a Carnot type magnetic refrigerator, which used two flat pieces of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ separated by a small gap, as its working material.

When the material was isothermally magnetized by a magnetic field of 6.5 T, the gap was filled with He gas from the high temperature heat-end, and when the material was isothermally demagnetized the gap was filled with liquid He from the cold-end.

Experimental results showed a maximum cooling power of 30 mW operating from 9.6 to 1.8 K [1466, 1467].

Also in 1995, Satoh et al. developed a magnetic refrigerator, which used Pb and $\text{ErNi}_{0.9}\text{Co}_{0.1}$ as the working materials, and obtained a cooling power of 1.5 W for a working temperature of 4.2 K [1468].

In 1996, Zimm et al. reported a design similar to that described above [1465] but using GdNi_2 particles instead of ErNi to fill both beds and a larger magnetic field of 7 T. A temperature span of 38 K was obtained [1469].

In 1996, Ohira et al. presented a magnetic refrigerator operating under Carnot cycle in which a cylindrical $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ monocrystal was used as the working material and a Gifford-McMahon refrigerator as a hot reservoir. When the working material was isothermally magnetized with an applied field of 8 T, the material was then placed in thermal contact with the hot reservoir, and when the material was isothermally demagnetized, it was taken out from the reservoir. Experimental results showed that the device developed a cooling power of 0.21 W at a temperature of 20 K, when operated at a frequency of 0.002 Hz [1470].

In 1998, Hall and Barclay presented a rotary AMR refrigerator, wherein its regenerator consisted of a ring filled with the magnetocaloric material. A superconducting magnet supplied a constant magnetic field and He gas was used as its heat transfer fluid. During the rotation of the ring, only one part of it enters the region where the field is applied. Calculations showed that the device was capable of producing efficiencies of around 60 % for a temperature span from 100 to 300 K [1471].

In 2000, Yayama et al. developed a static magnetic refrigerator operating under Brayton cycle and working along a conventional gas cooling cycle. ErNi alloy was used as its working material and a superconducting magnet powered by AC current source was used to produce a maximum magnetic field of 5 T. He gas was used as heat transfer

fluid to transport the heat between the exchanger and as gas for the conventional cycle.

The device showed an enhanced cooling power with respect to the single gas cooling apparatus, obtaining 0.36 W and reaching a temperature of 4 K. Finally the authors proposed to change its magnetic source to a mobile array of permanent magnets for room temperature devices [1472].

In 2001, Mullard Space Science Laboratory from UK developed an ADR for ultra low temperatures. This device consisted of: a paramagnetic material which was located in the bore of the magnet; the magnetic field source that can be formed by an array of permanent magnets or a superconducting magnet; a heat switch which cooled the paramagnetic material and extracted heat from the paramagnetic material before demagnetization; and a cold bath which provided the temperature for operation. The operation process of the ADR starts with the isothermal application of the magnetic field, followed by the adiabatic reduction of the magnetic field to attain a lower temperature. The next step involves the isothermal demagnetization of the material. Finally, the paramagnetic material absorbs heat to reach the conditions of the initial stage. This device realizes a double ADR, thus the process consisted of two-step cooling, in fact the first ADR acts as lower bath temperature and cools the second ADR. Tests performed with double ADR showed that the device allowed refrigeration in the temperature range from 4 to 0.01 K [1473].

In 2006, Yamiya et al. presented a magnetic refrigerator device for H_2 liquefaction operating under Carnot cycle. A superconducting magnet was used to apply a magnetic field of 6 T to a vertically moving working material (DyGaAl alloy) at a frequency of 0.71 Hz. H_2 gas was used as the heat transfer fluid and the results showed a maximum cooling power of 37.4 W attained for this device [1474].

In 2008, Numazawa et al. presented a reciprocating magnetic refrigerator for H_2 liquefaction. Its regenerator was formed by $Dy_{2.4}Gd_{0.6}Al_5O_{12}$ sheets, filling the interior of a non-magnetic cylinder, which moves in and out of the superconducting magnet (6 T) and H_2 gas was used as the heat transfer fluid. Tests performed with a frequency of ~ 0.02 Hz show a cooling power of 0.3 W for a working temperature between 77 to 20 K [1475].

In 2009, Matsumoto et al. reported a reciprocating magnetic regenerator, based on the device presented by Numazawa in 2008, but the precooling refrigerator was changed by an AMR refrigerator in the new device [1476].

In 2013 Numazawa et al. presented a reciprocating device for nitrogen liquefaction. In this device the regenerator was formed by three beds, each one filled with three magnetocaloric materials with different Curie temperatures ($Gd_5(Ga_{1-x}Fe_x)O_{12}$, $HoAl_2$ and $DyAl_2$). He gas was used as heat transfer fluid. Experimental results showed a range of working temperature from 77 to 20 K [1477].

Also in 2013, Kim et al. presented a two-step magnetic refrigerator. The regenerator consisted of two beds filled with different magnetocaloric materials. The first bed was filled with $GdNi_2$ and $Dy_{0.85}Er_{0.15}Al_2$ while the second with $Dy_{0.5}Er_{0.5}Al_2$ and $Gd_{0.4}Dy_{0.9}Al_2$ particles. Each bed was located inside a superconducting magnet operated in AC mode, which supplied a magnetic field peak of 4 T with a frequency of 0.1 Hz, each magnet. Experimental results showed that this device reached a temperature of 24 K [1478].

Additional devices working at cryogenic temperatures are collected in Table S8 of the Supplementary Material.

8.2. Devices working close to room temperature

The first person to use Gd element to build a laboratory scale device for magnetic refrigeration was Brown in 1976 [9, 1479]. This reciprocating device used 0.175 kg of Gd disks separated by fine wires, and a mixture of water and ethyl alcohol as heat transfer fluid. The magnetocaloric material moved in and out of the gap of a superconducting magnet, which allowed an alternating magnetic field of maximum value of 7 T. A maximum temperature span of 47 K was obtained after fifty cycles of operation.

In 1978, after Brown's prototype, Steyert proposed the principle of the active magnetic regeneration, AMR. He presented a rotary device, in which Gd was placed on the rim of a wheel, rotating in and out of the magnetic source. Working with a frequency of 1 Hz, this device reached a temperature span of 14 K [1446, 1480].

In 1987, Kirol and Dacus presented a design of rotary heat pump that was built using 125 coaxial disks of Gd, forming a cylinder with a mass of 0.270 kg. A magnetic field of 0.9 T produced by an array of permanent magnets of Fe-Nd-B, was used to get a working temperature span of 11 K while operating in an Ericsson type magnetic cycle [8, 1481].

Green et al. in 1990 reported the construction of a reciprocating magnetic refrigerator using beds of Gd and Tb as the AMR and nitrogen gas as the heat transfer fluid. A superconducting magnet was used to apply a maximum magnetic field of 7 T and, with very low frequency, a temperature span of around 24 K was obtained after operating for several hundreds of cycles [1482].

New progresses in this field were reported in 1997, when Zimm et al. presented a reciprocating magnetic refrigerator that used Gd spheres as the magnetocaloric material,

placed on beds. This device consisted of a superconducting magnet, which allowed for a maximum magnetic field of 5 T, and operated over 18 months without significant problems, indicating an improvement in the design as compared to earlier devices, which were only typically tested for few days. This equipment was able to produce a maximum cooling power of 600 W with a temperature span of 10 K [1483, 1484]. Almost one year later, an enhancement in the performance of the device was reported, obtaining a larger temperature span of 38 K for a maximum applied magnetic field of 5 T despite a lower cooling power of 100 W [14].

In 2000, Bohigas et al. reported the development of a rotary magnetic refrigerator, which used permanent magnets as the magnetic field source. In this design, Gd ribbons were placed on the surface of a plastic wheel, which rotated around the pole pieces of the magnet at very low frequencies (between 0.06 and 0.8 Hz) and olive oil was utilized as the heat transfer fluid. In the optimal arrangements, the permanent magnets allowed for a maximum magnetic field of 0.9 T, leading to a maximum temperature span of 5 K [1485].

In the same year, Chubu Electric and Toshiba Company presented a reciprocating magnetic refrigerator, which employed Gd spheres forming beds as the magnetocaloric material and a superconducting magnet for a magnetic field of 4 T. The maximum cooling power obtained was 100 W with a temperature span of 21 K [1486].

In 2001, a new reciprocating device was presented by Rowe and Barkley from University of Victoria. It was composed of a superconducting solenoid, which generated a maximum magnetic field of 2 T, and AMR beds of layers of Gd and $\text{Gd}_{1-x}\text{Tb}_x$ alloys in the form of grains. This device was reported to generate a maximum temperature span of 14 K [1487].

In the same year, Astronautics Corporation of America developed a new rotating magnetic refrigerator, wherein its AMR was composed of three sets of beds filled with layers of Gd, Gd and $\text{Gd}_{94}\text{Er}_6$ alloy spheres (sizes between 0.25 and 0.5 mm) and only $\text{La}(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}\text{H}_{1.0}$ particles in a wheel configuration. In this setup, the magnetocaloric material rotates through the gap of the magnetic source, which comprised of Fe-Nd-B permanent magnets yielding a maximum magnetic field of 1.5 T. Several tests using water with inhibitor as the heat transfer fluid showed a maximum temperature span of 18 and 24 K for a working frequency of 1 and 4 Hz respectively, generating a maximum refrigeration power of 50 W when only Gd layers were used. In the case of using Gd and Gd(Er) alloys, the temperature span and cooling power attained were 27 K and 25 W respectively. If the AMR consisted of solely La-Fe-Si-H intermetallics to fill the beds, lower values are expected. Finally was concluded that successive layers with different Curie temperatures were necessary to enhance the performance of the regenerator [1488-1490].

In 2002, a new superconducting regenerative magnetic refrigerator was presented by Blumenfeld et al. In this device, without moving parts, its magnetic field was supplied by an array of cryocooled coils, which allowed for a maximum magnetic field of 1.7 T. The regenerative bed was formed by Gd powder particles filling a cylinder, located inside of the coils, where the magnetic field was alternatively applied and removed. From the various tests performed, the best result showed a maximum temperature span of 15 K for cold and heat end temperatures of 293 and 308 K respectively [1491].

A group from Nanjing University in China also built a reciprocating magnetic refrigerator in the same year. The magnetocaloric material was placed in two beds, filled with Gd and Gd-Si-Ge or $\text{Gd}_5\text{Si}_{1.985}\text{Ge}_{1.985}\text{Ga}_{0.03}$ alloys in powder form, and the magnetic field was applied by an array of Halbach type magnets, which led to a

magnetic field of 1.4 T. In this refrigerator, water was used as the heat transfer fluid.

Results published show a maximum temperature span of 23 K when operating near room temperature around 298 K [8, 1492].

In 2003, Tokyo Institute of Technology in collaboration with Chubu Electric Company developed a rotary magnetic refrigerator. The magnetocaloric refrigerant composed of four types of Gd-based alloys ($\text{Gd}_{0.91}\text{Y}_{0.09}$, $\text{Gd}_{0.84}\text{Dy}_{0.16}$, $\text{Gd}_{0.87}\text{Dy}_{0.13}$ and $\text{Gd}_{0.89}\text{Dy}_{0.11}$ with Curie temperatures: 277, 278, 280 and 283 K respectively) for a total of 1 kg of material. An array of Fe-Nd-B permanent magnets rotates around the magnetocaloric beds, supplying a variant magnetic field and a maximum intensity of 0.77 T. Using water as the heat transfer fluid, each bed was filled with these alloys positioned in an ascending order of their Curie temperatures to enhance the efficiency of the magnetocaloric beds. The experimental results showed a maximum cooling power and temperature span of 60 W and 8 K respectively, with a very low operation frequency of 0.4 Hz [1493].

In the same year, a group from Grenoble in France developed a reciprocating magnetic refrigerator. This prototype composed of an array of Halbach type permanent magnets (a maximum magnetic field of 0.8 T) and 0.230 kg of Gd sheets, piled to form the bed of magnetocaloric material, through which water was allowed to pass as heat transfer fluid. They reported a maximum cooling power of 8.8 W and a temperature span of 4 K, with an operation frequency of 0.5 Hz [1494, 1495].

In 2004, groups from Universities of Victoria and Quebec, in Canada, developed a reciprocating magnetic refrigerator, which employed an array of permanent magnets as the magnetic field source for a maximum field intensity of 2 T. The regenerator consisted of two cylindrical beds filled under three different configurations: with 0.090

kg of Gd, 0.045 kg of Gd and 0.040 kg of $\text{Gd}_{0.74}\text{Tb}_{0.26}$ alloys and with 0.135 kg of Gd, $\text{Gd}_{0.74}\text{Tb}_{0.26}$ and $\text{Gd}_{0.85}\text{Er}_{0.15}$. In all cases, He gas was used as the heat transfer fluid. Experimental results reported a maximum temperature span of 16 K for the bed with Gd while 20 K for the bed of Gd and $\text{Gd}_{0.74}\text{Tb}_{0.26}$ alloy for an applied magnetic field of 2 T. For the beds composed of three materials and an applied magnetic field of 1.5 T, when the working temperatures were 301 and 299 K, the maximum temperature span of 37 and 44 K were obtained at the operational frequencies of 0.65 and 1 Hz, respectively. For an applied magnetic field of 2 T, the measured temperature spans were 46.8 and 51 K for frequencies of 0.65 and 1 Hz respectively [1496, 1497].

The University of Quebec presented a reciprocating magnetic refrigeration device, using a superconducting magnet to supply a magnetic field of 2 T and its regenerator formed by a layered bed of Gd-Tb alloy. The reported experimental results showed a maximum temperature span of 14 K [1496].

In the same year, a group from Washington State University presented another reciprocating magnetic refrigerator. This equipment comprised of a bed of Gd powder as the magnetocaloric material and an array of permanent magnets, which provided a magnetic field of 2 T. The maximum temperature span reported was 5 K [1498].

In 2005, the company Cooltech Applications in France reported the construction of a rotary magnetic refrigerator, which composed of eight beds of magnetocaloric material inserted in a thermally isolated disc. An assembly of rotating Halbach type magnets of Fe-Nd-B alloy provided a magnetic field of 1.47 T in the air gap. Each bed was composed of fifteen sheets of Gd plates, with a total mass of 0.165 kg, where the heat exchanger fluid flowed between them. The experimental results showed that the

maximum cooling capacity obtained ranged between 100 and 360 W depending on flow rate of the fluid [1499].

In the same year, scientists from Xi'an Jiaotong University in China developed a reciprocating magnetic refrigerator working at room temperature. A water cooled electromagnet was used to provide a maximum magnetic field of 2.18 T, and the AMR was built using Gd and $\text{Gd}_5\text{Si}_2\text{Ge}_2$ alloys. Experimental results performed at different temperature ranges and flowrates of the heat transfer fluid showed a maximum cooling power of 18.7 W [1500, 1501].

Also in 2005, another reciprocating magnetic refrigerator was developed at the Graduate School of Engineering of the Hokkaido University. In this device, Gd was used as working material to fill the beds and distilled water as heat transfer fluid. The magnetic field was supplied by a permanent magnet with a maximum field of 2 T. At the operation frequency of ~ 0.2 Hz, a maximum temperature span of 10 K was reported [1502].

A magnetic refrigerator was presented by Kim et al. also in 2005. In this device, its regenerator was a block of $\text{Gd}_5(\text{Si}_2\text{Ge}_2)$ with a Si wafer containing microchannels for the flow of the heat transfer fluid. An electromagnet was used to apply a variably magnetic field with maximum value of 1.5 T and perform the magnetization/demagnetization process of the magnetocaloric material. Tests showed a maximum temperature change of 7 K for a temperature of 258 K was attained [1503].

In 2006, the Chinese Academy of Science developed a prototype with two cylindrical reciprocating active regenerative beds, which were filled with a total of 1.1674 kg of Gd particles. The magnetic field was supplied by an array of two permanent magnets, yielding a maximum magnetic field of 1.5 T. Several tests were performed by varying

the operating frequency between 0.4 and 1.0 Hz, as well as the length of the piston that moved the exchange fluid (helium gas). Results show a maximum cooling power of 51.3 W and a temperature span of 18.2 K for a length of 0.11 m, while for a length of 0.20 m, values of 25.2 W and 11.0 K were reported [1504].

In the same year, Tokyo Institute of Technology and Chubu Electric, performed an update of their device that was formerly developed in 2003. In the new rotary type refrigerator, besides an increase in the size of the equipment, eddy current losses were reduced, enhancing its cooling power. The applied magnetic field was also enhanced from 0.77 T to 1.1 T by changing the pole pieces for only one permanent magnet in V form. The magnetocaloric material used for the beds was retained like in the initial design (Gd) but the exchange fluid flowed in circumferential form in the new device instead of the former axial configuration. Tests performed on this device showed a maximum cooling power of 560 W [1505, 1506].

Also in 2006, University of Applied Sciences of Western Switzerland developed a rotary magnetic refrigeration device for air conditioning technology. This equipment was presented in The World's Largest Industrial Fair at Hannover, Germany (2006), and they won the first prize of the Swiss Technology Award. It utilized air as the working fluid and a magnetic fluid with Gd particles as the magnetocaloric material to fill the bed, which was ring-shaped. This bed rotates inside the gap of a permanent magnet assembly, which provided a magnetic field of 0.8 T over half of the rotating ring, so the magnetic field was applied only on half of the bed [8, 1507].

In the same year, Baotou Research Institute of Rare Earth in China presented a reciprocating magnetic refrigerator. The AMR bed was filled with particles of Gd (0.750 kg) and $\text{LaFe}_{10.97}\text{Co}_{0.78}\text{Si}_{1.05}\text{B}_{0.2}$ (0.200 kg) alloys and its magnetic field was

supplied by an array of permanent magnets yielding a maximum field of 1.5 T. Using alkaline water solution as the heat transfer fluid, a maximum temperature span of 5 K and a cooling power of 20 W upon 45 minutes of operation were obtained [1508, 1509].

In 2007, a new rotary magnetic refrigerator was built by Astronautics Corporation of America. The device consisted of a base made of twelve static beds placed in the form of parallel disks filled with Gd as magnetocaloric material, with 0.946 kg of Gd per bed, and an array of rotating permanent magnets with a maximum applied magnetic field of 1.5 T. Tests performed, using water with inhibitors as heat transfer fluid and an operational frequency of 4 Hz, showed a maximum temperature span of 12 K with a cooling power of 220 W [1510].

Also in 2007, researchers from University of Victoria developed a rotary magnetic refrigerator to study different types of magnetic refrigerants in different configurations. This refrigerator consisted of an array of Fe-Nd-B permanent magnets: one external part of this array was static while another internal part could rotate, yielding a maximum magnetic field of 1.47 T. Water was used as the heat transfer fluid and its operational frequencies were between 0 and 4 Hz. Using cylindrical regenerators filled with Gd flakes and varying the operation frequency, the temperature span increases with frequency, reaching a maximum of 13.2 K [1511].

In 2007, Chelyabinsk State University built a rotational type magnetic refrigerator. The wheel containing Gd and NiMnGa alloys rotated with an operational frequency between 1 and 10 Hz, through the gap of the permanent magnet, which was formed by Fe-Nd-B pieces in a Fe yoke and supplied a magnetic field of 1 T. A cooling power of 40 W was predicted and presented [1512].

In the same year, Sichuan University in China designed a rotary magnetic refrigerator. Its regenerator consisted of a wheel divided in 36 sectors, wherein each was filled with 1 kg of Gd particles as the magnetocaloric material, and was allowed to rotate with a frequency between 0.1 and 0.7 Hz. Water was used as the heat transfer fluid and its magnet unit was a Halbach array of permanent magnets of Fe-Nd-B and Fe-Co-V, which supplied a maximum magnetic field of 1.5 T. This device provided a maximum temperature span of 11.5 K [1513].

In addition, Riso National Laboratory in Denmark developed a reciprocating magnetic refrigerator in the same year. An array of Gd plates was used as refrigerant, which could move into the gap of a ring-shaped Halbach type magnet (produced a maximum 1.2 T). For operation frequency below 1 Hz, a temperature span of ~9 K was predicted [1514].

INSA Institute and Cooltech Applications Company in France developed a rotary refrigerator prototype also in 2007. The device consisted of four static beds of Gd placed on a ring, water as the working fluid and a rotary array of permanent magnets, which applied a maximum field of 1.3 T [1515].

In 2008, researchers of Hokkaido University in Japan developed a reciprocating AMR equipment composed of two mobile permanent magnets, which led to a maximum magnetic field of 2 T in the middle of its gap, where the magnetocaloric material (a pack of 0.0334 kg Gd particles) was located during operation. This equipment was tested under different conditions using either air or water as the heat transfer fluid.

Results show that for an optimal operation, a larger flow rate was needed for air than that used for water [8, 1516].

Also in 2008, Bahl et al. developed a reciprocating AMR refrigerator. The regenerator consisted of a plastic cylinder, which contained parallel sheets of Gd and moved

laterally in and out of the electromagnet (magnetic field of 1.4 T). The magnet was orientated with its magnetic field parallel to the plane of the Gd sheets. Several heat transfer fluids were used and results showed that for operation frequencies between 0.06 and 0.08 Hz, temperature spans of 6.9, 6.4, 6.2 and 6.0 K were measured for distilled water and ethanol, ethylene glycol, propylene glycol and olive oil respectively [1517].

In 2009, Hirano et al. built a reciprocating AMR device with a mobile bed containing particles of magnetocaloric material. The magnetic field was produced by an array of two opposing permanent magnets, which allowed for a maximum field of 2.3 T. The authors reported a temperature span of ~2 K when La-Fe-Si alloy was used [1518].

In the same year, South China University of Technology reported a reciprocating AMR system with its regenerator formed by two rectangular beds with a separation between them filled with Gd particles as the magnetocaloric material. An array of moving Fe-Nd-B permanent magnets led to a maximum magnetic field of 1.5 T in the center of the gap, where the two beds were located, performing a double regenerator cycle. Only theoretical studies on the dependence of the temperatures of hot and cool heat exchangers on the refrigeration power had been performed, showing that its cooling power decreased as the hot heat exchanger temperature increased [1519].

Cooltech Applications in France also built a reciprocating AMR in 2009. The prototype was formed by an array of 37 parallel plates of Gd, with a spacing between them to allow the flow of the passing heat transfer fluid and a Halbach type array of three Fe-Nd-B magnets, which supplied a maximum magnetic field between 0.8 and 1.1 T in the gap. Tests were performed varying the temperature of the heat and cool exchangers and showed that with an initial temperature of ~293 K, a temperature span of 16.1 K could be obtained [1520].

In the same year, researchers of Campinas State University in Brazil built a wheel-type AMR for a rotary magnetic refrigerator. The wheel, containing six beds filled with 0.160 kg Gd each, rotated inside the gap of the magnet but only half of it was inside the field region. The magnetic unit was an electromagnet, which supplied a magnetic field in the range of 0.24 to 2.30 T. Results of this device using ethyl alcohol as the heat transfer fluid show that the maximum temperature span obtained was 11 K upon about 20 minutes of operation at a frequency of 0.4 to 0.5 Hz [1521].

In 2009, Dupuis et al. developed a new reciprocating AMR to study the influence of demagnetizing factor and packing ratio of the magnetocaloric material, operational frequency, and flow rate of the exchanger fluids on the performance of the device. Gd plates were piled into a cylinder with its axis oriented parallel to the magnetic field. A Halbach array supplying a magnetic field of 0.8 T was moved to magnetize and demagnetize the magnetocaloric material. Results from the different tests showed that a maximum temperature span of 7.8 K was obtained with a heat transfer fluid flow rate of 3 ml/s [1522].

In the same year, Korea Advanced Institute of Science and Technology built a reciprocating magnetic refrigerator. Gd particles were used to form the bed that moved with a maximum frequency of 1 Hz in and out of the magnetic field area, which was generated by sixteen segments of permanent magnets in the shape of a cylinder, yielding a magnetic field of 1.58 T in the center of the gap. A maximum temperature span of 16 K was reported when using He gas as the heat transfer fluid [1523].

Adopted from their previous prototype, RISO DTU in Denmark developed a new device in 2009 by changing the regenerator. The bed was formed by parallel plates of

$\text{La}_{0.67}\text{Ca}_{0.33-x}\text{Sr}_x\text{MnO}_3$ ($0 \leq x \leq 0.33$), whereby the size of the regenerator could be modified and the heat exchange fluid was a mixture of water and ethanol [1524].

Also in 2009, University of Applied Sciences of Western Switzerland built a reciprocating equipment to characterize different magnetocaloric materials. The magnetic field was supplied by Fe-Nd-B permanent magnets, which led to a maximum field of 2 T in the air gap. The magnetocaloric material was placed into a holder, which was displaced in and out of the magnetic field region. Pulsed air was used as a heat transfer fluid. $\text{LaFe}_{11}\text{Co}_{0.9}\text{Si}_{1.1}$ and Gd were used to test the equipment measuring the temperature change on the magnetocaloric material under adiabatic and non-adiabatic conditions and showed that the device could determine the temperature change of the tested materials for different applied magnetic fields [1525].

Furthermore, University of Genoa in Italy presented a linear reciprocating magnetic refrigerator in the same year. Its magnetic field was supplied by an array in a crossed form of ten Fe-Nd-B magnets surrounded by a soft magnetic material, which led to a field of 1.55 T in the air gap. Two regenerators, each filled with 0.400 kg of Gd particles, were attached to a plastic shuttle for moving them in and out of the magnetic field region. A mixture of water and ethanol was employed as the heat transfer fluid with a maximum operative frequency of 0.2 Hz. Preliminary results showed a maximum temperature span of around 5.0 K and a degradation of the performance with increasing frequencies [1526, 1527].

In 2009, Tura and Rowe improved their initial rotary prototype that was developed two years ago. In this new device, its regenerator was filled with 0.055 kg of Gd particles and its cold and hot end exchangers were modified to reduce dead volume. Tests

performed showed a temperature span of 29 K, which was larger than that obtained from the previous device [1528].

University of Ljubljana in Slovenia developed a rotary magnetic refrigerator in 2009. This device consisted of a regenerator cylinder with 34 beds filled with ~0.600 kg Gd plates and operated at frequencies up to 4 Hz. The cylinder, which consisted of a stationary Fe soft magnetic alloy core, rotates inside a structure of a set of four permanent magnets. With the geometry of the two magnetic circuits, a magnetic field in the range of 0.05 to 0.98 T was attained in the air gap. Theoretical studies on this device predicted a temperature span of approximately 7 K [1529].

In 2010, Trevizoli et al. developed a reciprocating device with static AMR. The magnetic field was provided by a Halbach array of Fe-Nd-B permanent magnets, which led to a maximum field of 1.65 T in the air gap. This Halbach array moved linearly to magnetize and demagnetize the AMR during the cycle. The AMR consisted of an array of twenty-eight parallel sheets of Gd (total mass of 0.1954 kg) with a space between them for the flow of the heat transfer fluid. Initial results showed a temperature span of 4.4 K and a cooling power of 3.9 W [1530].

In the following year, researchers from Korea Advanced Institute of Science and Technology presented a reciprocating AMR with its regenerator consisting of a stainless steel tube filled with 0.021 kg of Gd particles, which moved in and out of the applied field region. The magnetic field was supplied by a hollow cylindrical array of permanent magnets, Halbach array, which generated a magnetic field of 1.5 T in the center of the gap, where the generator moved. He gas was used as the heat transfer fluid. Experimental results showed a maximum temperature span of 14 K [1531].

In the same year, Balli et al. built a reciprocating magnetic refrigeration device intended for industrial application. It consisted of two static Fe-Nd-B permanent magnets, which supplied a maximum magnetic field of 1.45 T in the space between them. Two mobile regenerators were used and each one was divided into two parts, which were separated by 0.030 m. This division of the regenerator was done to compensate the forces appearing in the regenerator when it entered into the magnetic field region. Each of these parts was made of 0.200 kg of Gd parallel sheets, using a total mass of 0.800 kg of magnetocaloric material. Experimental results showed a cooling power between 80-100 W and a temperature span of 20 K for an operational frequency of 0.5 Hz [1532].

In the same year, Tura et al. also presented a rotary AMR device, which used Gd as magnetocaloric material. Its magnetic field was supplied by a cylindrical nested Halbach array. In this configuration only the inner magnets in the array rotated, varying the magnetic field in the center gap, which led to a maximum magnetic field of 1.47 T. A static bed filled with 0.110 kg of Gd particles was placed in the gap of the Halbach array and a mixture of water and glycol was employed as the heat transfer fluid. Its experimental results showed a maximum temperature span of 29 K [1533].

In addition, Technical University of Denmark presented a reciprocating AMR in 2011. Its magnetic field was supplied by an array of permanent magnets in a Halbach cylinder type, with a maximum magnetic field of 1.03 T in the gap. The regenerator comprised of a bed filled with parallel plates of magnetocaloric material and it was moved vertically in and out of the field region, using a mixture of water and 25% vehicle antifreeze liquid as the heat transfer fluid. Tests were performed using three types of magnetocaloric materials. When Gd was used, a temperature span of around 10 K was measured. In the case of La(Fe,Co,Si)_{13} compound, a temperature span of around 7 K was obtained. However, when two La(Fe,Co,Si)_{13} compounds of different Curie

temperatures were used, a temperature span of ~10 K was observed. On the other hand, tests using ceramic $\text{La}_{0.67}\text{Ca}_{0.26}\text{Sr}_{0.07}\text{Mn}_{1.05}\text{O}_3$ in the device, yielded a maximum temperature span of around 5 K [1386].

In 2012, Park et al. from Korea Advanced Institute of Science and Technology built a new reciprocating AMR refrigerator, which utilized a static Halbach array of permanent magnets to provide a maximum magnetic field of 1.4 T in its center gap. Its regenerator was formed by two beds containing 0.186 kg of Gd particles (with the cold heat exchanger placed between the beds) and moved in and out of the magnet. Tests using deionized water as the heat transfer fluid were performed, showing a maximum temperature span of 26.8 K and cooling power of 33 W [1534].

In the same year, researchers from Technical University of Denmark reported the construction of a rotary AMR device with its regenerator made of a nylon ring, which was divided into 24 separate beds. Gd particles were used as the magnetocaloric material to fill the beds and a mixture of deionized water and ethylene glycol was used as heat transfer fluid. The regenerator rotated in the gap of two cylindrical permanent magnets array. In this configuration, there were four regions with a high magnetic field of around 1.24 T and four others with low field close to zero. Experimental results with a variant fluid flow rate showed a maximum temperature span of around 20 K and a cooling power of 200 W for an operational frequency of around 2 Hz [1535].

In 2013, researchers from University of Victoria in Canada modified the magnet design of their rotary AMR device that was formerly built in 2007. The new magnet design comprised of a Halbach array of three concentric cylinders. Each of these cylinders was formed by twelve segments of permanent magnets in a configuration, whereby the internal cylinder was static, while the other two moved in opposite directions to create a

sinusoidal magnetic field with a peak value of around 1.6 T. Tests performed using 0.650 kg of Gd and a operation frequency of 0.8 Hz, yielded a maximum temperature span of 33 K with a cooling power of 50 W [1536].

In the same year, Korean Advanced Institute of Science and Technology presented a named tandem reciprocating device. Its magnetic field was provided by a Halbach configuration with six segments of Fe-Nd-B permanent magnets located within an Fe yoke, to increase the magnetic field in the center gap, giving a maximum of 1.41 T in the direction where the regenerator moves. The regenerator plates were composed of two AMR beds filled with Gd particles, which were connected to the same cold end heat exchanger. Experimental results indicated a temperature span of 20 K with a cooling power of 2 W when using water as the heat transfer fluid [1537].

Also in 2013, the Chinese Academy of Science presented a new device, which utilized a combination of AMR refrigeration and Stirling regenerative refrigeration cycles. The magnetic field was supplied by an array of two hollow Halbach cylinders, each composed of sixteen segments of permanent magnets. The Halbach cylinders were configured in a manner that one was inside the other and the inner magnet rotated, providing a variable magnetic field from 0 to 1.5 T. The regenerator, located inside the rotating magnet, was a stainless steel tube filled with Gd sheets. Two cycles were tested for this regenerator: a Stirling cycle and an AMR refrigeration cycle. Tests performed with He gas as the heat transfer fluid and at operating frequency of 1.5 Hz, showed a temperature span of 14.9 K and 7.9 K for a cooling power of 6 W and 10 W respectively [1538].

Romero-Gómez et al also reported the construction of a reciprocating AMR refrigerator in 2013. In this device, the magnetic field source was formed by a set of twenty pieces

of permanent magnets (Fe-Nd-B) placed in the extremes to close a C-shaped Fe yoke. This configuration of the magnet supplies a constant magnetic field of 1.015 T in the air gap, where the regenerator is moved. The regenerator was formed by two separated beds filled of parallel Gd sheets, piled with its plane perpendicular to the direction of the magnetic field. Distilled water was used as the heat transfer fluid and the operational frequency was varied between 0.15 and 0.25 Hz. Performed tests showed a maximum temperature span of 3.5 K and a cooling power of 3 W [1539].

In 2014, Aprea et al. from University of Salerno in Italy presented a rotary AMR equipment. Its magnet design consisted of four pieces of permanent magnets configured in two parallel semicircles, which supplied a maximum magnetic field of 1.25 T and rotated to magnetize/demagnetize the regenerators. The device utilized 1.20 kg of Gd particles distributed into 8 static beds, which were positioned at 45 degrees and fixed to an aluminium ring, and distilled water as its heat transfer fluid. Operating at a frequency of 0.5 Hz, the maximum temperature span measured was 13.5 K [1540].

In 2015, Technical University of Denmark presented a new rotary AMR prototype. In this device, its regenerator was placed in between two concentric cylindrical magnets and was formed by two concentric non-magnetic steel cylinders. Eleven beds were placed between the two cylinders in a periodic sequence of Gd, $\text{Gd}_{97.5}\text{Y}_{2.5}$, Gd_{95}Y_5 and $\text{Gd}_{90}\text{Y}_{10}$ particles. One of the two cylindrical magnets was a soft magnetic Fe-based alloy, which was located within the regenerator cylinder, while the other one surrounding the regenerator, was formed by two arrays of six blocks of Fe-Nd-B connected by Fe yokes. The last magnet could rotate creating a maximum magnetic field of ~ 1.2 T in the region where the regenerator was located. Using a mixture of demineralized water and ethylene glycol as the heat transfer fluid and at an operational

frequency between 0 and 4 Hz, a maximum temperature span of 10.2 K and a cooling power of 103 W were obtained [1541].

In 2016, Institute of Non Ferrous Metals in Poland presented a magnetic refrigerator with a reciprocating magnetic field source. This source consisted of an array of four Fe-Nd-B permanent magnets placed in a C-shape structure of a soft magnetic Fe-based alloy, which supplied 0.8 T in the gap. It was allowed to move laterally in between two regenerators. These two regenerators were formed by parallel sheets of Gd and a liquid with a specific heat of about 4000 J/kgK was employed as the heat transfer fluid. Experiments performed with an operational frequency of 0.5 Hz showed an adiabatic temperature change of 2.5 K [1542].

In the same year, researchers from Federal University of Santa Catarina in Brazil and Technical University of Denmark presented a rotary magnetic refrigerator, wherein it utilized a stationary ring shaped regenerator that was formed by eight pairs of beds, all filled with 1.7 kg of Gd particles and located surrounding the rotary magnetic source. This comprised of a two poles rotor-stator magnetic assembly formed by an array of Fe-Nd-B permanent magnet pieces, producing a maximum magnetic field of ~1.5 T. Experimental tests showed that for an operational frequency of 0.8 Hz, a cooling power of 80.4 W and a temperature span of 7.1 K was attained using a mixture of ethylene glycol and distilled water as the heat transfer fluid [1543].

Also in 2016, Abdelmessih et al. developed a magnetic cooling apparatus employing $\text{Gd}_5\text{Si}_2\text{Ge}_2$ alloy as its regenerator and an array of permanent magnets supplying a maximum field of 1 T, to magnetize/demagnetize the magnetocaloric material. Experimental results showed a maximum temperature change of 2.3 K for this device [1544].

In the same year, Benedict et al. presented a rotary AMR refrigerator. In this device, the regenerator was formed by two cylindrical beds filled with Gd particles and each bed was placed inside a Halbach array magnet. The magnet consisted of a pair of cylindrical magnets, which were formed by ten pieces of Fe-Nd-B permanent magnet, and could be independently rotated supplying a magnetic field between 0.05 and 1.5 T into the gap area where the regenerator was placed. Tests performed on this device were done with a maximum operational frequency of 2 Hz and using a mixture of water and ethylene glycol as the heat transfer fluid. Results showed a maximum temperature span of 21 K [1545].

Other devices working around room temperature for which some characteristics are available are collected in Table S8 of the Supplementary Material.

8.3. General considerations about system design

Any new technology that is being developed has to overcome limitations before reaching the market. The competition with already established systems, optimized for longer periods of time, poses several questions about the feasibility of the new technology. In the case of magnetic refrigeration, general questions arise about field application and moving parts (or fluids).

With respect to the application of magnetic field, as the magnetic entropy change and adiabatic temperature change both scale with the applied magnetic field in a non-linear way [33, 76], it is natural to consider that the larger the field the larger the performance of the device. Currently, fields of the order of ~ 2 T can be applied using permanent magnets. For larger fields, superconducting magnets would be needed, increasing the cost of use of the refrigerator. While superconductor based devices seem unacceptable for home appliances, they might find their niche of application in central cooling plants.

Variable magnetic fields constitute an intrinsic requirement of magnetic refrigeration. This can be produced by mechanical movement of the permanent magnet or rotation of a set of halbach cylinders. Sample reciprocation is also possible, although it implies more technical challenges. Alternatively, studies on anisotropic magnetocaloric effect suggest that the rotation of the magnetocaloric material inside a constant magnetic field could be a way of inducing temperature changes [1546-1548].

Heat transfer from the magnetocaloric material to the refrigerator content or to the environment would, in principle, need heat transfer fluids or, alternatively, mechanical movement of the sample to put it in contact with the thermal reservoirs [1549]. This is seen as another apparent bottleneck for developing the technology, as it is one of the intrinsic limitations for increasing the frequency of operation [70]. New approaches considering thermal diodes could notably improve heat transfer and, therefore, the coefficient of performance of the devices [1550].

8.4. Other caloric refrigeration technologies

Although the focus of this review paper is magnetocaloric materials and their application in refrigerator devices, it is important to mention that there are other technologies based on other “caloric” effects, i.e. reversible temperature changes of the material produced by external fields. Among them, the most relevant nowadays are mechanocaloric (either elastocaloric or barocaloric), electrocaloric and magnetocaloric technologies. This generalization is currently attracting a lot of attention in the research literature, with a very recent example of an special issue of Journal of Physics D: Applied Physics (volume 50, 2017), edited by Mañosa and Planes, which covers many of the current approaches to generalize the magnetic field excitation to other fields like electrical or mechanical. A relatively recent review on caloric effects was published by Moya[1551].

The application of mechanical excitation to a material might cause barocaloric and magnetocaloric effects, depending if the stress is applied in one direction or isotropically, respectively. In fact, the concept of elastocaloric effect was first described in 1805 [1552] and, over time, the concept and its thermodynamical interpretation [1553] were generalized to predict other caloric effects, like the magnetocaloric effect. For a very recent review on mechanocaloric materials the reader can consult [1554].

Electrocaloric effect consist in the reversible temperature change of a material under the application of an electric field. There is a broad variety of materials exhibiting this effect, ranging from perovskites to polymer films. An overview of this field of research can be found in [1555].

A comparison between different caloric refrigeration devices can be found in recent literature [22, 1556]. A particularly interesting approach includes SMA foils in a demonstrator that incorporates mechanical work recovery [1557], as this design is compatible with the implementation of micro-cooling devices.

Among all caloric refrigeration technologies, the most technologically developed at this moment is magnetocaloric, followed by mechanocaloric. It is expected, however, that each of these three caloric effects will find its own market niche, depending on the desired operation conditions of the device and the intrinsic limitations of the technology [1556, 1558].

9. Conclusions and future outlook

Magnetic refrigeration is an active field of research, with relevant activity both in the search of new magnetocaloric materials as well as in the application of these materials into magnetic refrigeration devices.

On the materials research side, one of the mainstream lines is the discovery of new phases with significant magnetocaloric response. At the same time, the application of materials engineering techniques to modify the microstructure of the samples, the combination of different phases to develop composites, etc. are helping researchers to understand the role of the different physical magnitudes in order to optimize the applicability of the materials, or to eliminate some of the limitations which seemed to be inherent to some of the phases.

The design of magnetic refrigeration devices has also experienced a big transition, from laboratory prototypes to the exhibition of pre-commercial systems at large industrial fairs in the recent years. The most used materials in the most recent refrigerator prototypes are Laves phases for low temperature applications and Gd (and its compounds), La(FeSi)_{13} type phases and MnFePSi type phases for room temperature applications.

As shown in the previous sections, there are other materials that exhibit large magnetocaloric response, but they are not applied in magnetic refrigerator devices yet because of their large hysteresis or slow kinetics of the transformation. These detrimental factors could eventually be ameliorated by proper processing of the samples or by a clever combination of temperature and external field excitation during the refrigeration cycle.

In our opinion, the interface between these two interconnected fields of research, materials and devices, will give the most relevant results in the near future. The optimization of magnetocaloric materials in order to enhance their dynamic response and reduce their hysteresis will allow them to be incorporated into refrigerator devices, which are intrinsically cyclic in their operation.

From a more fundamental point of view, magnetocaloric characterization has been proven to be a reliable method for the determination of the characteristics of the phase transition that originates the phenomenon. Fundamental studies of the hysteresis of the materials will allow developing models for a comprehensive description of the magnetic refrigerant of devices under normal operating conditions, improving the optimization of the design. Techniques like FORC, broadly used in the characterization of magnetic hysteresis in permanent magnets, are being introduced into magnetocaloric research to analyze the thermal hysteresis of the materials. In addition, the new techniques for determining critical exponents, the order of phase transitions and the identification of the tricritical point separating first and second order phase transitions using magnetocaloric methods are being applied in the broader scope of the general characterization of magnetic materials.

At the same time, the successful application of the magnetocaloric effect in magnetic refrigerators has established the seed for the study of other caloric phenomena (like elasto-caloric, electro-caloric, etc.) and this will probably lead to alternative technologies for refrigeration.

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Figure Captions:

Figure 1: S - T diagram representing the different Carnot, Ericsson and Brayton thermodynamic cycles used in a magnetic refrigerator.

Figure 2: Effect of the addition of some elements to hexagonal Gd on its T_C and ΔS_M . Data taken from [104, 106, 110-113, 119]. For [106], [111] and [119], values of ΔS_M have been rescaled from $\mu_0\Delta H=1$ T to 2 T using the field dependence law with an exponent $n=0.67$ [76].

Figure 3: Structure of the Laves phases presenting interesting MCE properties. Blue spheres correspond to Mg atoms.

Figure 4: Vegard law behavior for $\text{Gd}(\text{Co},\text{Mn})_2$ alloy series crystallizing in cubic Laves structure [187] and for several RENi_2 compounds crystallizing in a cubic superstructure of cubic Laves phase [192, 198-200, 202, 1559].

Figure 5: Relative enhancement of ΔS_M and RC_{FWHM} for HoCo_2 Laves phase, with a FOPT, and $\text{Ho}(\text{Co}_{0.9}\text{Ni}_{0.1})_2$ Laves phase, with a SOPT, using the data of Ref. [156]. The solid lines corresponds to the fitting of the SOPT data to power laws $\Delta S_M(H)=AH^n$ ($n=0.48\pm0.04$) and $RC_{\text{FWHM}}=BH^m$ ($m=1.12\pm0.04$), respectively. The dashed lines represent the same fitting using mean field values ($n=2/3$ and $m=4/3$).

Figure 6: Maximum magnetic entropy change as a function of the corresponding transition temperature for different Laves phases at $\mu_0\Delta H=5$ T. Solid circles correspond to FOPT and hollow circles correspond to SOPT. Data taken from [139, 151, 153, 156, 158, 169, 184, 185, 187, 188, 203-205, 215, 546, 1560-1564].

Figure 7: Relative enhancement of ΔS_M and RC_{FWHM} for $\text{LaFe}_{13-x}\text{Si}_x$ phase, with a FOPT ($x=1.2$) and a SOPT ($x=1.8$) using the data of Ref. [226]. The solid lines correspond to the fitting of the SOPT data to power laws $\Delta S_M(H)=A\cdot H^n$ ($n=0.77\pm0.03$) and $RC_{\text{FWHM}}=B\cdot H^m$ ($m=1.087\pm0.016$), respectively. The dashed lines represent the same fitting using mean field values ($n=2/3$ and $m=4/3$).

Figure 8: Relationship between ΔS_M and T_C at $\mu_0\Delta H=5$ T for several compositions of the $\text{La}(\text{Fe},\text{Si})_{13}$ family (data taken from [15, 231, 238, 260, 265, 314, 338, 1565-1570]). Lines are guides to the eye to show the trend after changing the Fe/Si or RE/La ratio (black), Co content (red) and H (green).

Figure 9: Relationship between RC_{FWHM} and T_C at $\mu_0\Delta H=5$ T for several compositions of LaFe_{13} family (data taken from [231, 260, 314, 338, 1566, 1568, 1571]). Lines are guides to the eye to show the trend upon changing the Fe/Si or RE/La ratio (black) and Co content (red).

Figure 10: Peak temperatures of MCE as a function of Ge content in $\text{Gd}_5\text{Si}_{4-x}\text{Ge}_2$ (data taken from [7, 11, 359, 365, 372, 375, 378, 380, 381, 383, 385, 391, 393, 395, 407, 420, 1572-1575]). For a more complete view of the different structural and magnetic transitions see [357].

Figure 11: The maximum ΔS_M (above) and RC_{FWHM} (below), both at $\mu_0\Delta H=5$ T, as a function of transition temperatures for several compositions of the $\text{Gd}_5(\text{Si},\text{Ge})_4$ family (data taken from [7, 11, 28, 359, 365, 372, 376-378, 383, 384, 390, 391, 393, 395, 414, 419, 433, 1576-1585]).

Figure 12: Relationship between $|\Delta S_M|$ and transition temperatures of several compounds crystallizing in ThCr_2Si_2 -type structure at $\mu_0\Delta H=5$ T (data taken from [89, 440, 446-448, 456-462, 466, 1586-1598]).

Figure 13: Relationship between ΔT_{ad} and transition temperatures of several compounds with ThCr_2Si_2 -type structure at $\mu_0\Delta H=5$ T (data taken from [89, 446, 456, 458-460, 462, 1586, 1588, 1589, 1591, 1594, 1595]).

Figure 14: Curie temperature as a function of the tolerance factor (t) for several manganite compounds with monovalent, divalent and trivalent substitution for La^{3+} (data taken from [474]).

Figure 15: Curie temperature dependence on the fraction of Mn substituted in manganites by TM element, with 67-70 at % of La and A as a divalent ion (data taken from [1599-1609]).

Figure 16: Effect of different TM partial substitutions for Mn on the Curie temperatures of (La-A)-(Mn-TM)-O₃.

Figure 17: Peak magnetic entropy change and refrigerant cooling power at 1 T as a function of the tolerance factor (data taken from [474]).

Figure 18: Relative enhancement of ΔS_M and RC_{FWHM} for $La_{1-x}Ca_xMnO_z$ phase, with a FOPT ($x=0.3$, $z=1$ [505]) and a SOPT ($x=0.33$, $z=8$ [480]). The solid lines correspond to the fitting of the SOPT data to power laws $\Delta S_M(H)=A \cdot H^n$ ($n=0.80 \pm 0.02$) and $RC_{FWHM}=B \cdot H^m$ ($m=1.14 \pm 0.03$), respectively. The dashed lines represent the same fitting using mean field values ($n=2/3$ and $m=4/3$).

Figure 19: Magnetic entropy change and transition temperature for RE₃Co compounds crystallizing in Fe₃C structure.

Figure 20: Magnetic entropy change and transition temperature for RE₂In compounds crystallizing in Ni₂In structure.

Figure 21: Magnetic entropy change and transition temperatures for different RE₁₂Co₇ compounds (solid symbols represent those alloys with PM-FM transition and hollow ones for those with FM-AFM transition). Data adapted from [645].

Figure 22: Magnetic entropy change and transition temperature for different rare earth nitrides crystallizing in NaCl structure. Data adapted from [678].

Figure 23: Transition temperatures (Néel temperatures for hollow symbols and Curie temperatures for solid ones) and maximum magnetic entropy change at 5 T for different RETMSi compounds: black, Fe; red, Co; blue, Ni; and green, Cu. Data taken from [727].

Figure 24: Magnetic entropy change as a function of the transition temperature for several RETMX compounds grouped according to their crystal structures. Hollow symbols corresponds to AFM ground states and solid symbols to FM or FiM ground states. Data taken from [752].

Figure 25: Plot of $|\Delta S_M|$ at 5 T vs. peak temperature values for the different RE-containing compounds described above as a function of the atomic fraction of rare earth in their composition. Corresponding references can be found in the main text.

Figure 26: Curie temperatures and maximum entropy change at 1 T as a function of the magnetic moment per 3d atom (see corresponding tables for references).

Figure 27: Unit cells for $L2_1$ cubic austenitic (left) and $L1_0$ tetragonal martensitic (right) phases in Ni_2MnGa alloy.

Figure 28: Temperatures of martensitic transition (T_t), premartensitic transition (T_t^*) and Curie temperatures of the austenitic phase (T_c) and martensitic phase (T_c^*) as a function of the number of external electrons per atom (e/a) for ternary Ni-Mn-Ga alloys [870, 871, 875, 878, 999, 1610-1614] and with Fe [992], Cu [888], Co [894], Cu and Co [891], Bi [896] and Sn [873, 897]. Lines are guides to the eye showing the different trends. For number in parenthesis see text.

Figure 29: Field dependence of ΔS_M for $Ni_2MnGa_{1-x}Sn_x$ ($x=0$ and 0.05) alloys, adapted from [873].

Figure 30: Temperatures of martensitic transition (T_t) and magnetic transition temperatures of the austenitic phase (T_c) and martensitic phase (T_c^*) as a function of

the number of external electrons per atom (e/a) for ternary Ni-Mn-Sn alloys [911-913, 1615, 1616] and with Fe [915, 922-924], Cu [919, 925, 926], Co [915-920, 1617], Cr [928], Nb [929], Sb [930], In [1618], Ge [933], Al [931, 932] and B [937]. Lines are guides to the eye showing the different trends described in the text.

Figure 31: Temperatures of martensitic transition (T_t) and magnetic transition temperatures of the austenitic phase (T_c) and martensitic phase (T_c^*) as a function of the number of external electrons per atom (e/a) for ternary Ni-Mn-In alloys [993, 1035, 1619-1623] and with Co [942, 976, 979, 1000, 1624], Cr [944], Si [1625] and Ga [1626]. Lines are guides to the eye showing the different trends described in the text.

Figure 32: Absolute values of the maximum adiabatic entropy changes (data taken from Table S6 of the Supplementary Material) corresponding to the martensitic FOPT and the austenite Curie SOPT as a function of the relative difference between the austenitic T_c and the start martensitic transition temperatures.

Figure 33: Temperatures of martensitic transition (T_t) and magnetic transition temperatures of the austenitic phase (T_c) and martensitic phase (T_c^*) as a function of the number of external electrons per atom (e/a) for ternary Ni-Mn-Sb alloys [907, 908, 958, 1627] and with Co [959, 960], Cu [963] and Fe [964]. Lines are guides to the eye showing the different trends.

Figure 34: ΔS_M values as a function of the peak temperatures for different Heusler alloys at 5 T. Data taken from Table S6 of the Supplementary Material.

Figure 35: Thermal hysteresis as a function of ΔS_M (5 T) and e/a for different Heusler compositions taken from Table S6 of the Supplementary Material.

Figure 36: Hysteresis losses for Ni-Mn-X (X=Ga, Sn, In, Sb) as a function of the number of electrons per atom (top), the full width at half maximum of the MCE peak (middle) and the maximum value of the magnetic entropy change (bottom). Data taken from [888, 889, 891, 907, 949, 958, 964, 994, 1034, 1621, 1624, 1628-1631]

Figure 37: Magnetic entropy change (5 T) and transition temperatures of MnCoGe [1065, 1078, 1081, 1083] and MnNiGe alloy families [1068, 1073, 1075] as a function of their outer electrons per atom (e/a). Solid symbols represent SOPTs whereas hollow symbols represent magnetostructural or metamagnetic FOPTs. Half-filled symbols correspond to a first order Curie transition.

Figure 38: Magnetic transition temperature and thermal hysteresis of FeMnP(AsSiGe) compounds as a function of the ratio between the lattice parameters a/c (left) and its unit cell volume (right). Data from [1153, 1159, 1164, 1166-1168, 1174, 1175, 1184, 1187, 1632].

Figure 39: Unit cell of Mn₃GaC compound.

Figure 40: Magnetic entropy change at peak and refrigerant capacity as a function of peak temperature for RE-based amorphous (solid black squares) and partially amorphous (solid red circles) alloys as well as for TM-based amorphous (hollow green squares) and partially amorphous (hollow blue circles) alloys. Corresponding references are collected in Table S7 of the Supplementary Material.

Figure 41: Curie temperatures and ΔS_M values (1 T) of some binary Gd-based amorphous alloys (see Table S7 of the Supplementary Material for references). Solid symbols correspond to fully amorphous alloys, half-filled symbols correspond to partially amorphous alloys and hollow symbols correspond to fully crystalline alloys.

Figure 42: Curie temperatures and ΔS_M values (5 T) of some RE-based amorphous/partially amorphous alloys as a function of average atomic number of RE (see Table S7 of the Supplementary Material for references).

Figure 43: Compositional dependence (Fe content) of the maximum absolute magnetic entropy change and transition temperature of Fe-Zr-M amorphous alloys at 1.5 T (see Table S7 of the Supplementary Material for references).

Figure 44: Table-like MCE experimental results observed in biphasic Gd+GdZn alloys with the extended temperature span as indicated by the arrows.

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Table Captions:

Table 1: Exponents controlling the field dependence of different magnitudes related to

MCE (*magnitude* $\propto H^{\text{exponent}}$).

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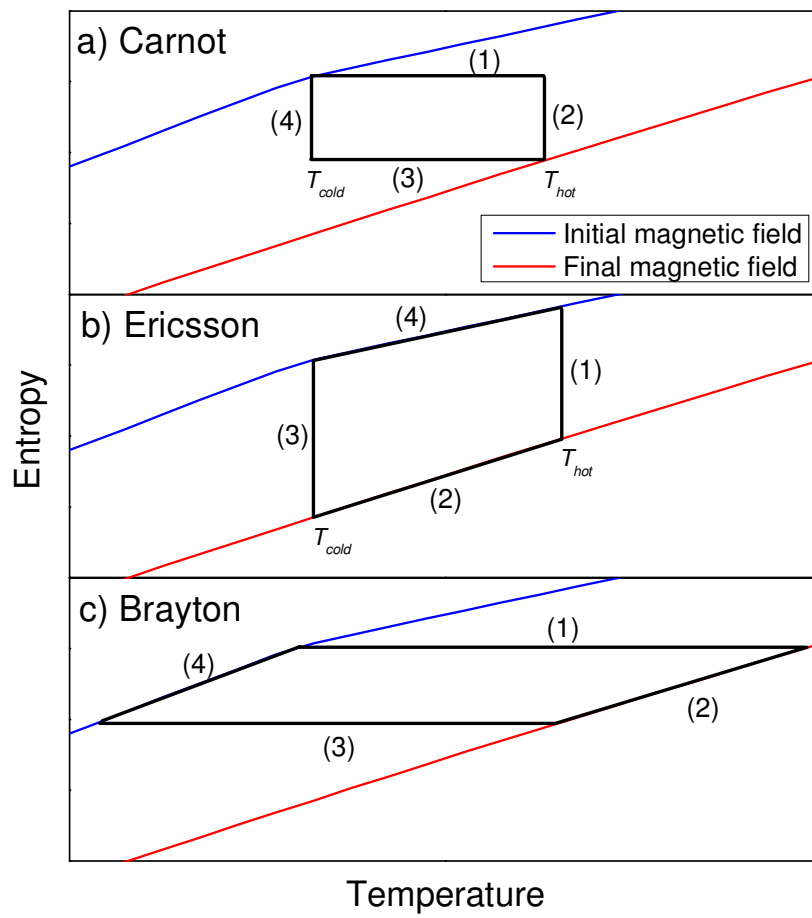


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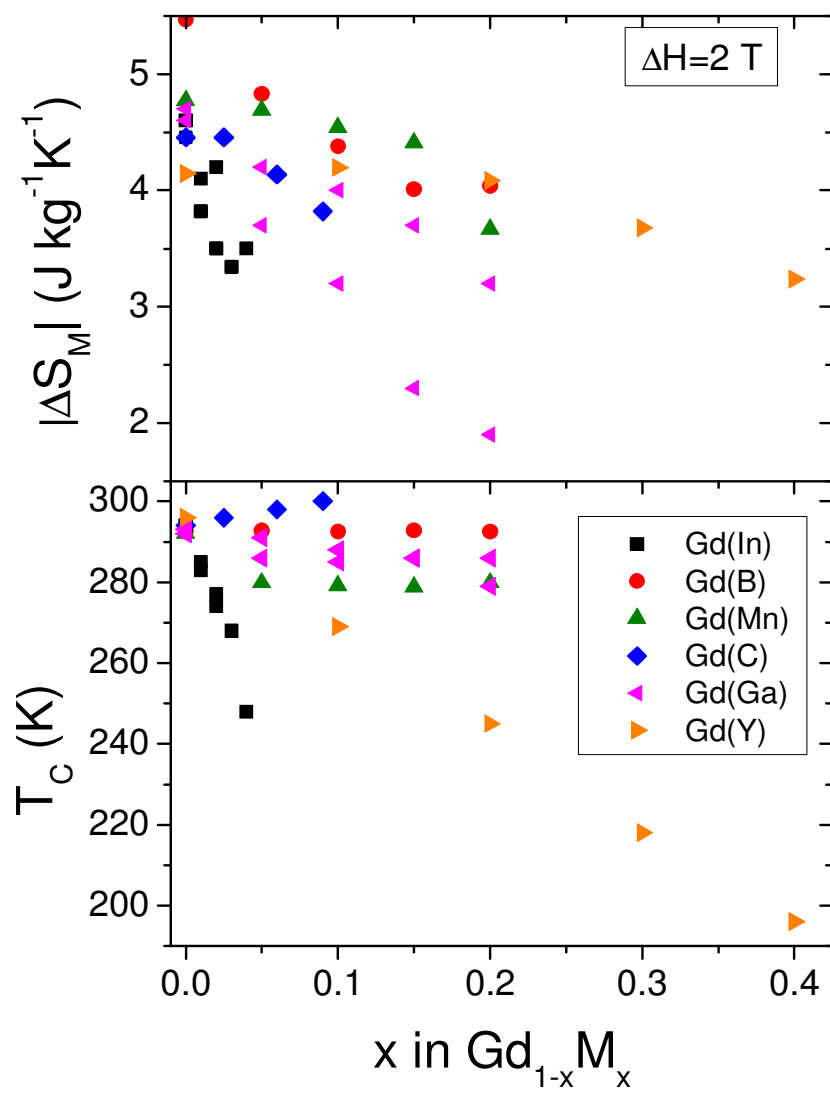


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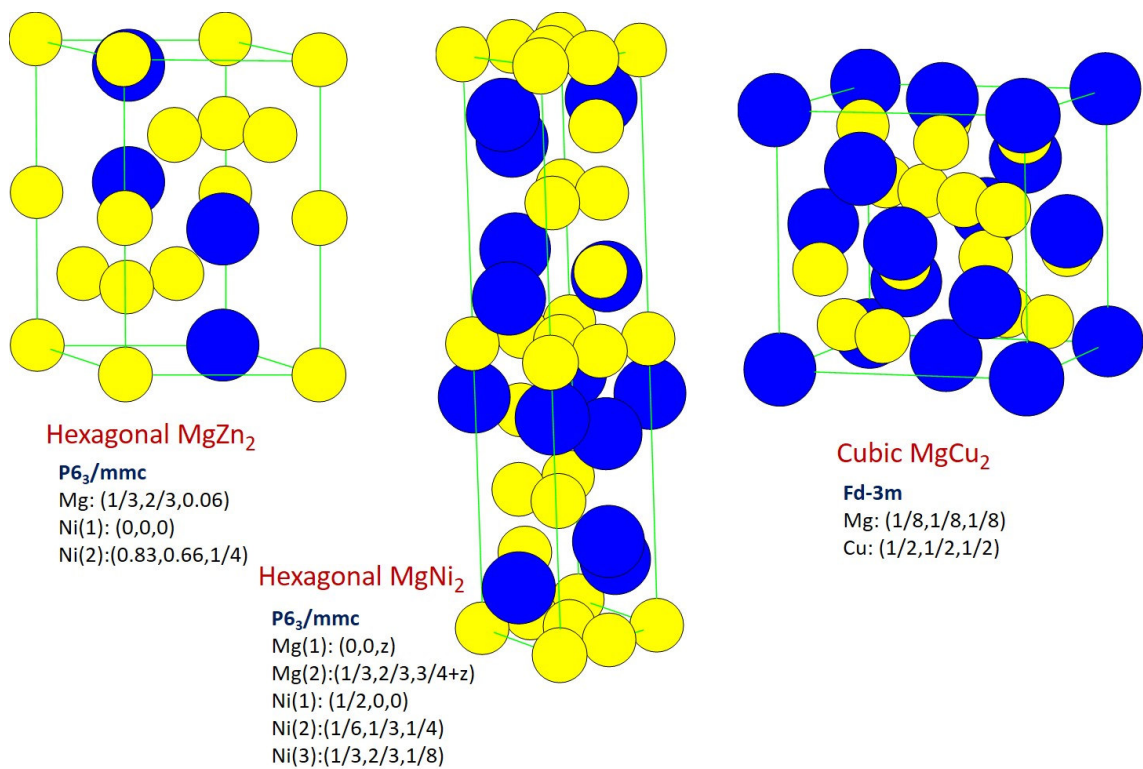


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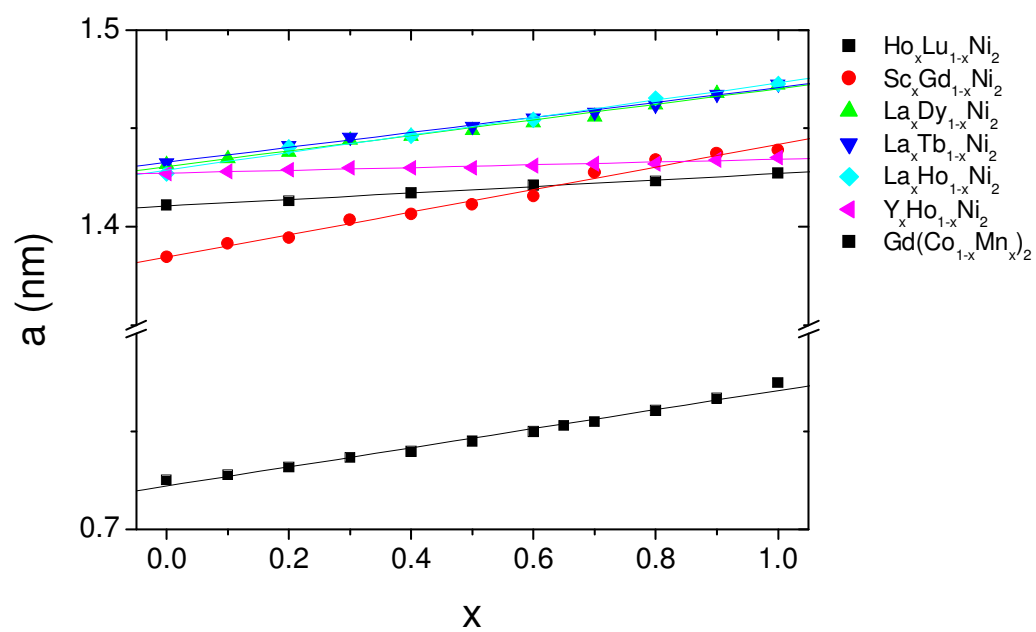


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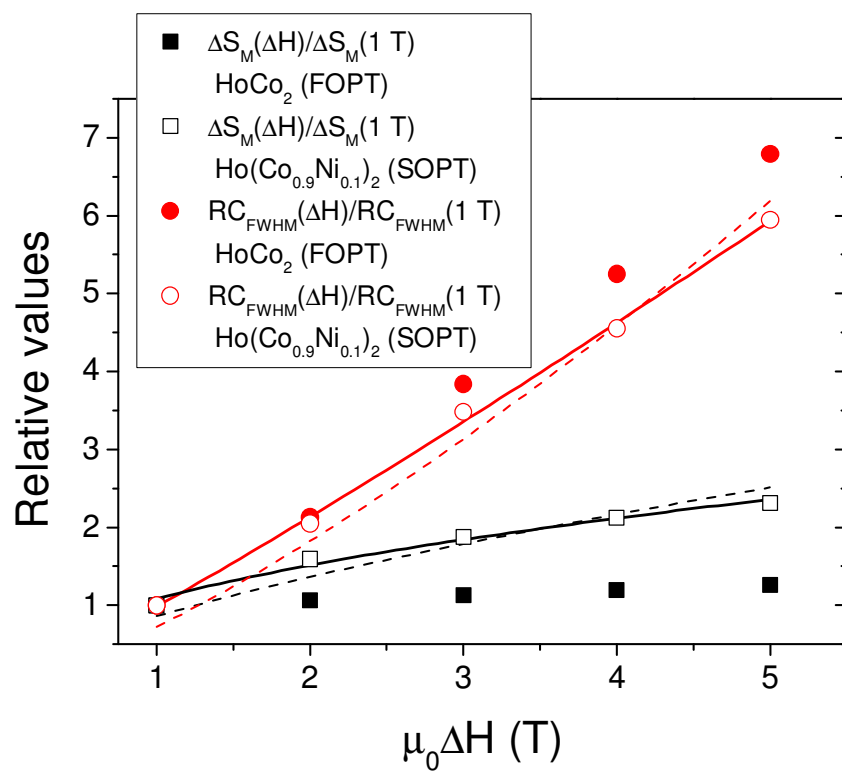


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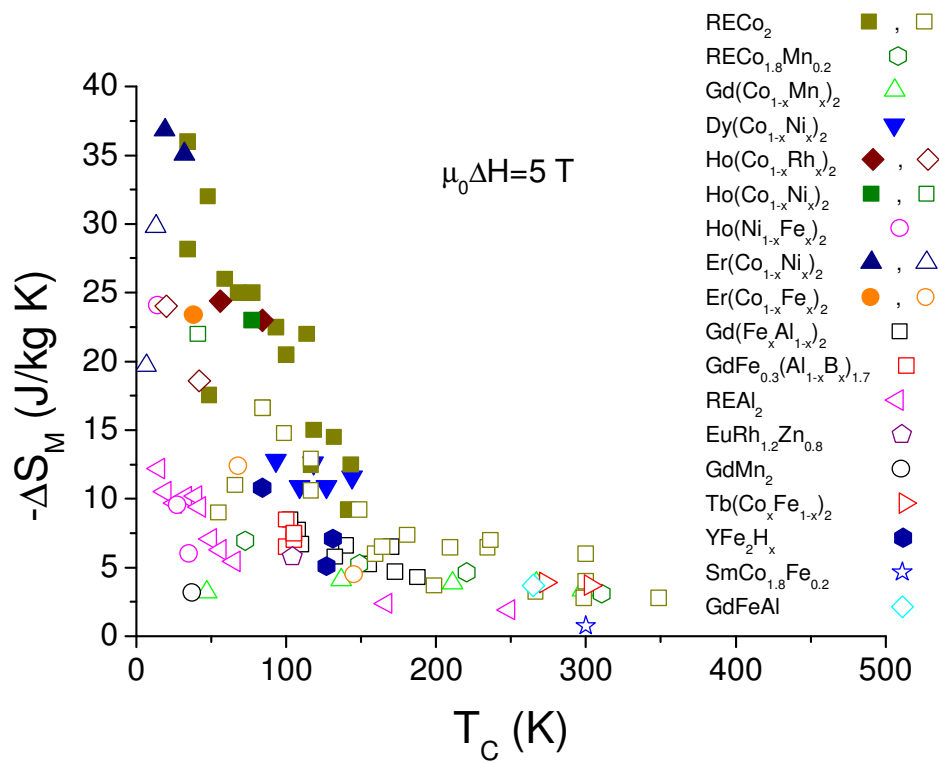


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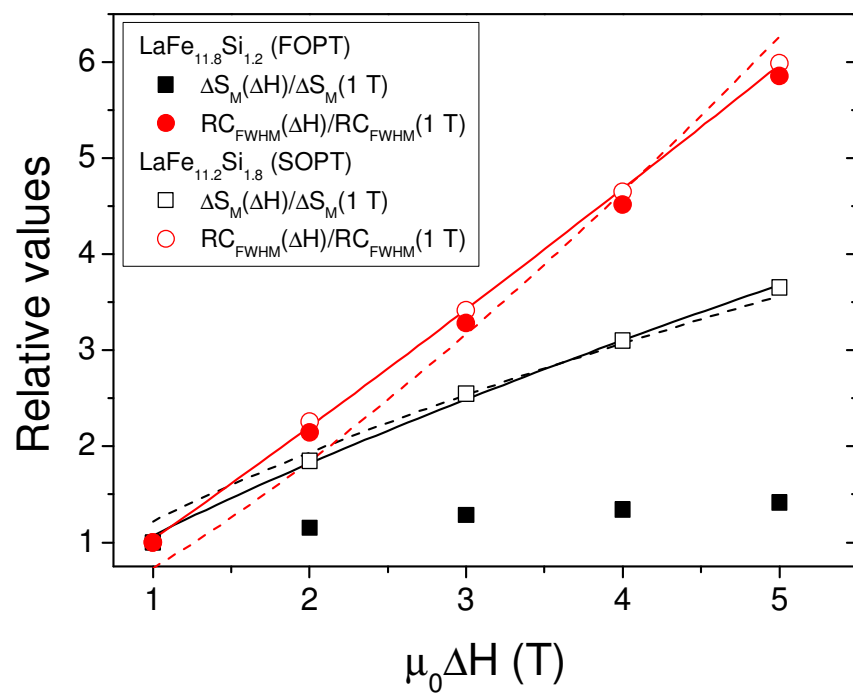


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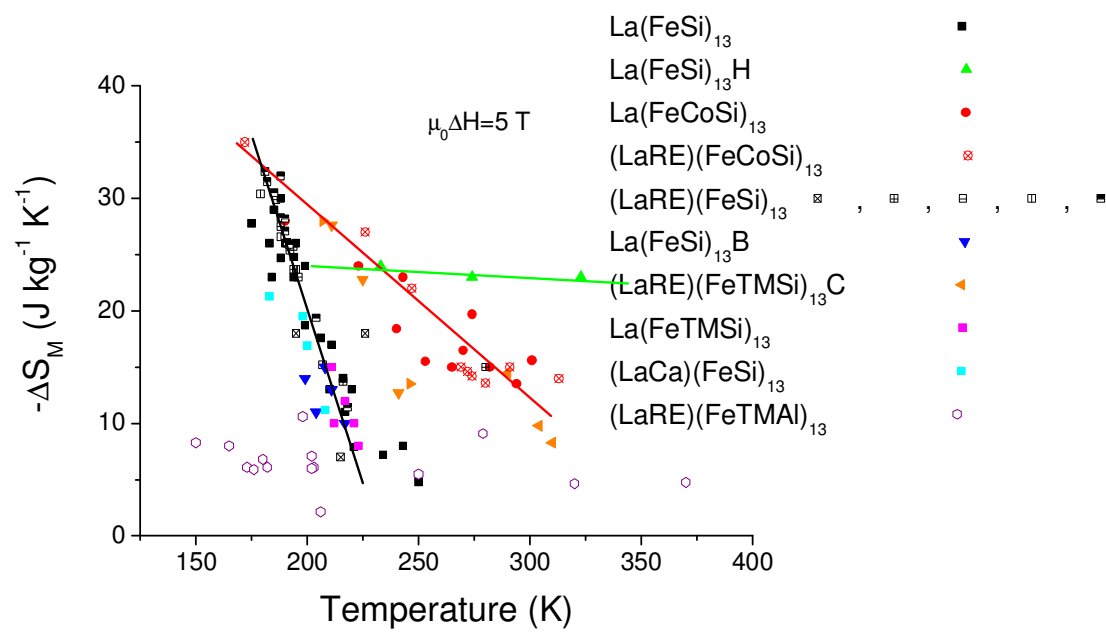


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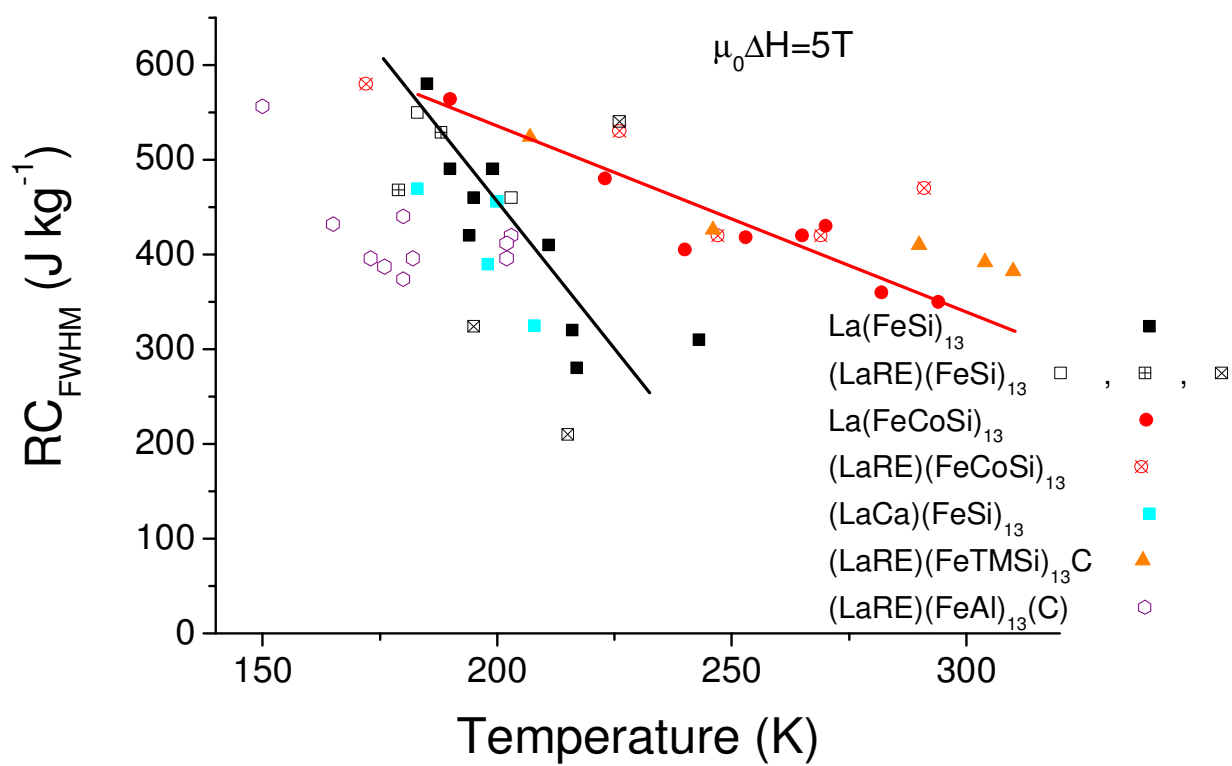


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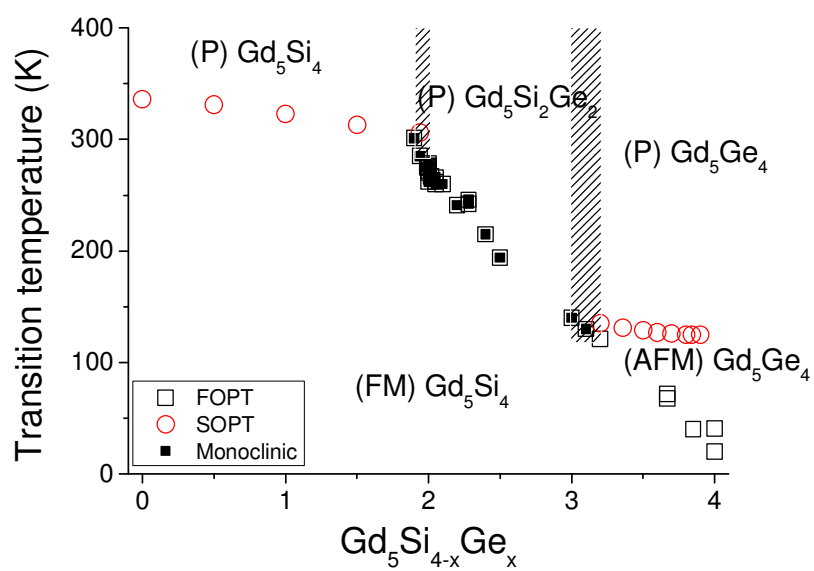


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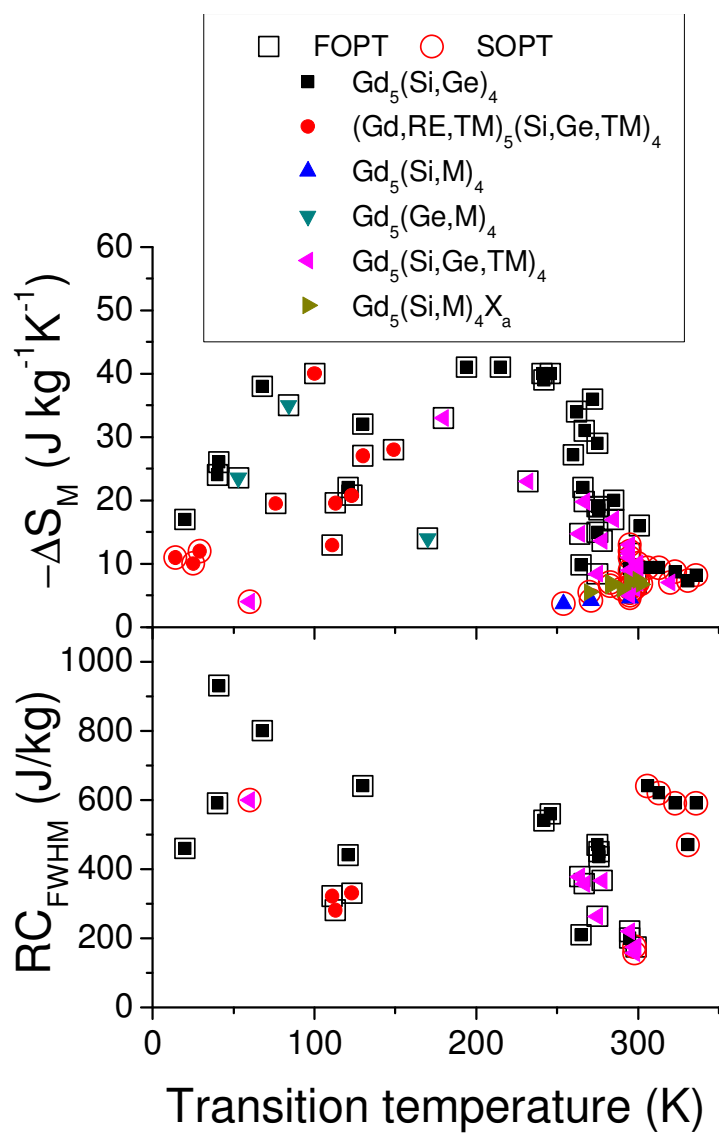


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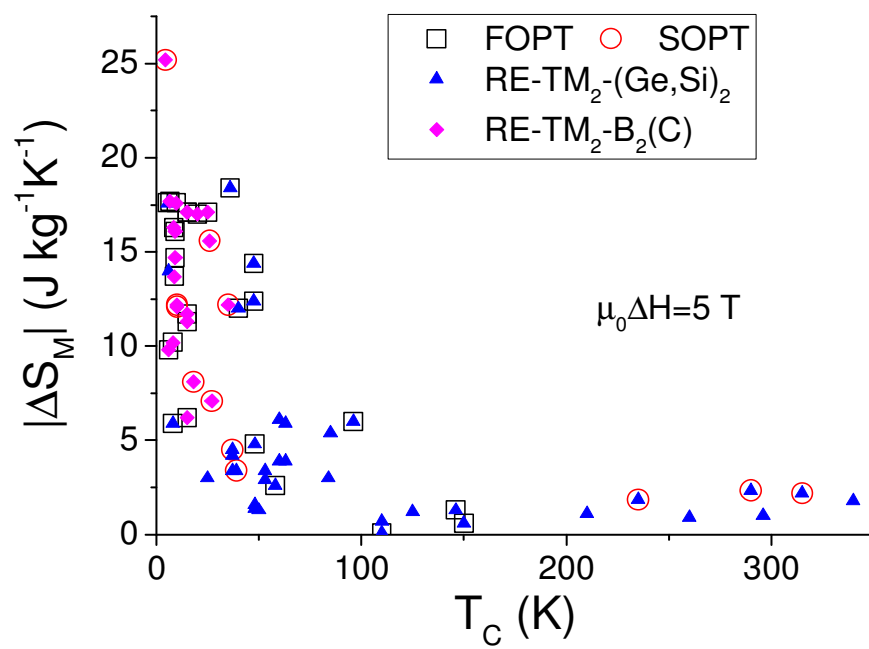


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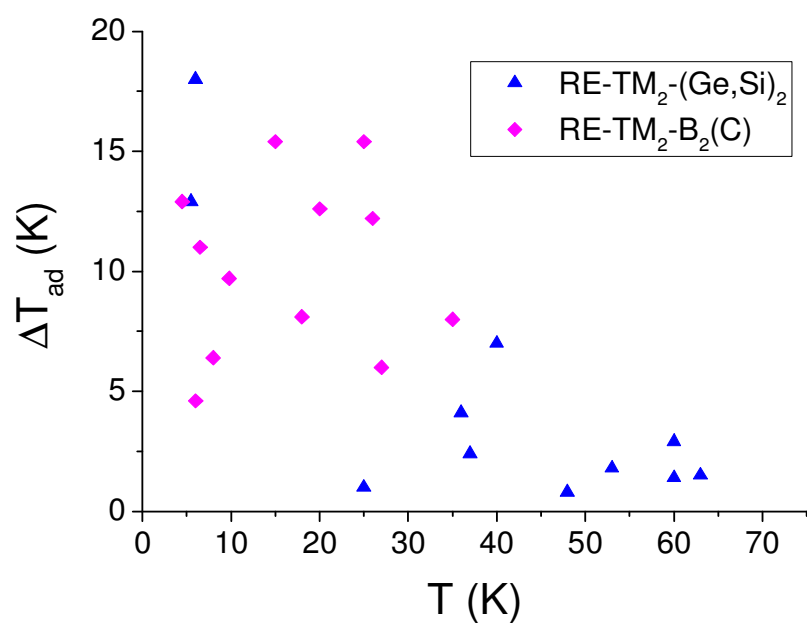


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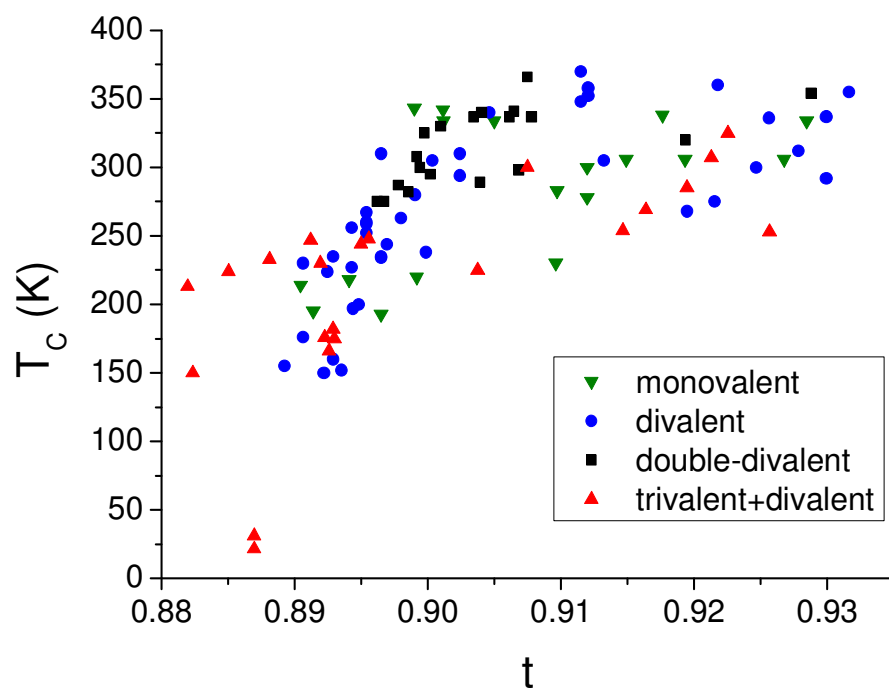


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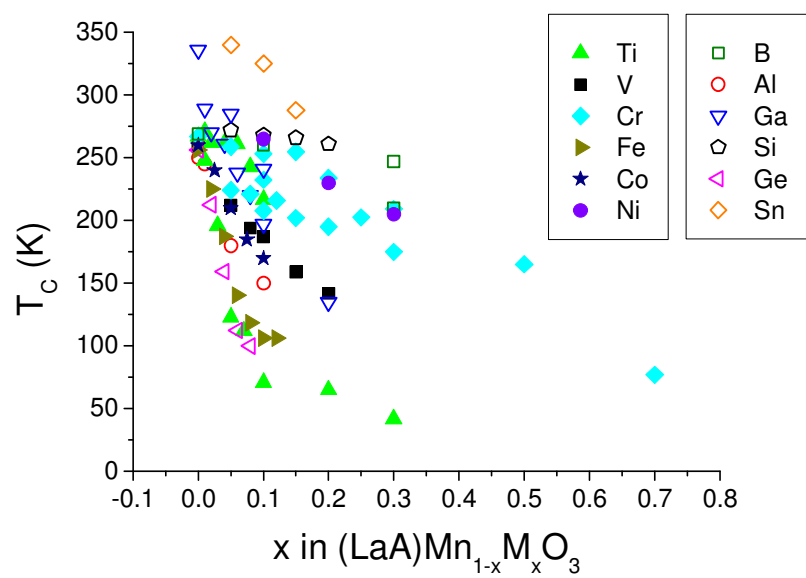


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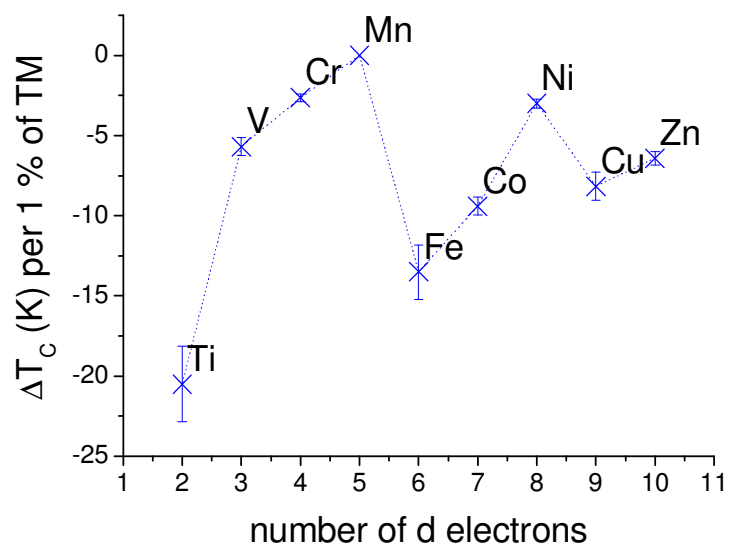


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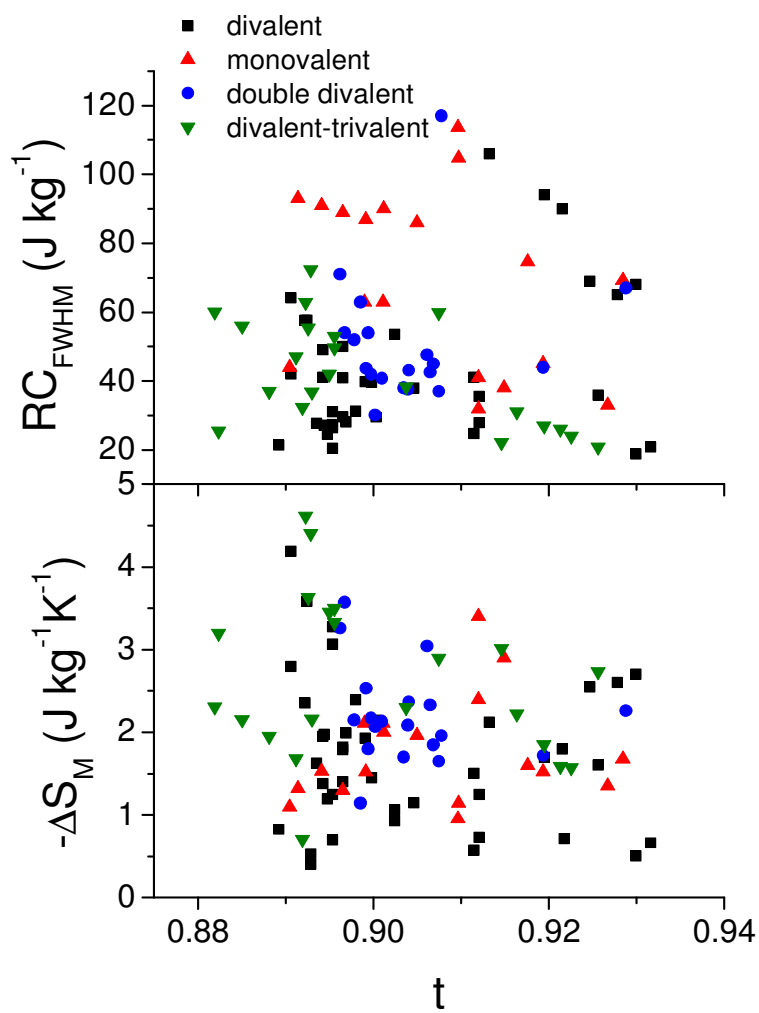


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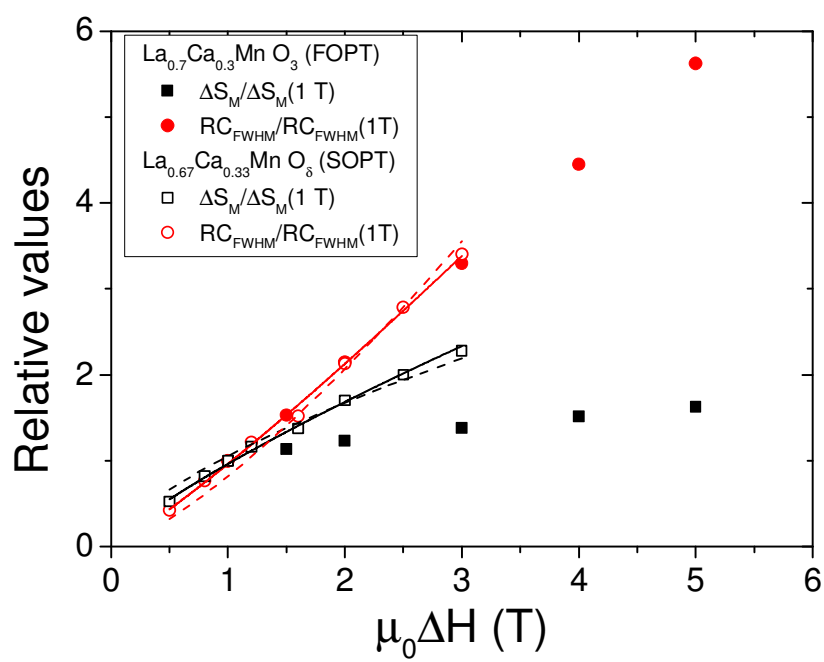


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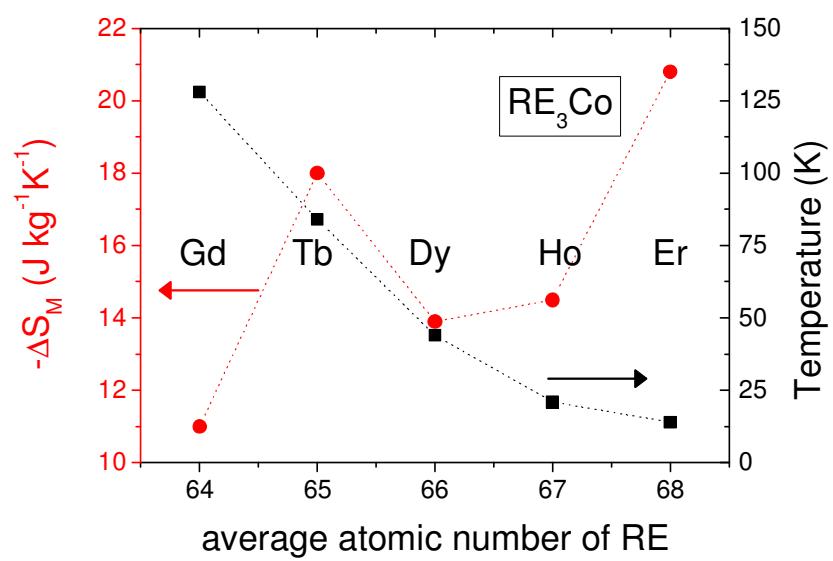


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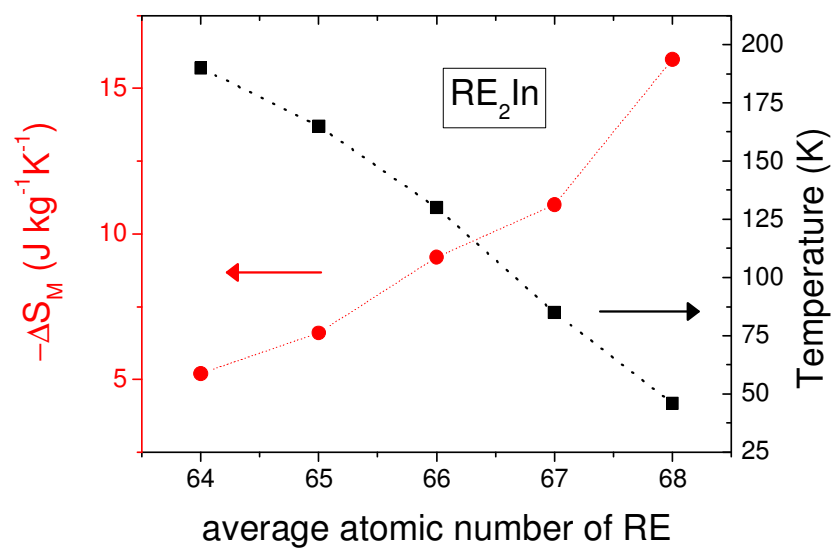


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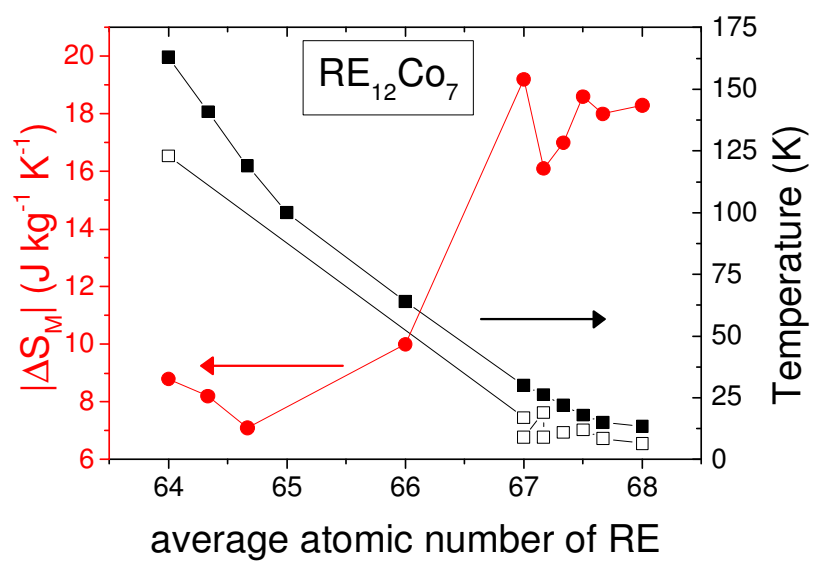


Figure 21

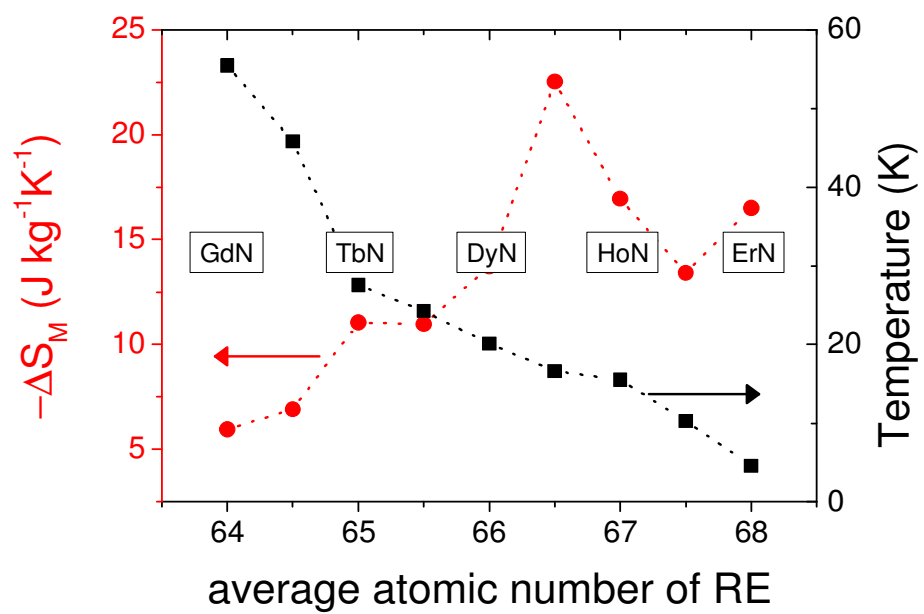


Figure 22

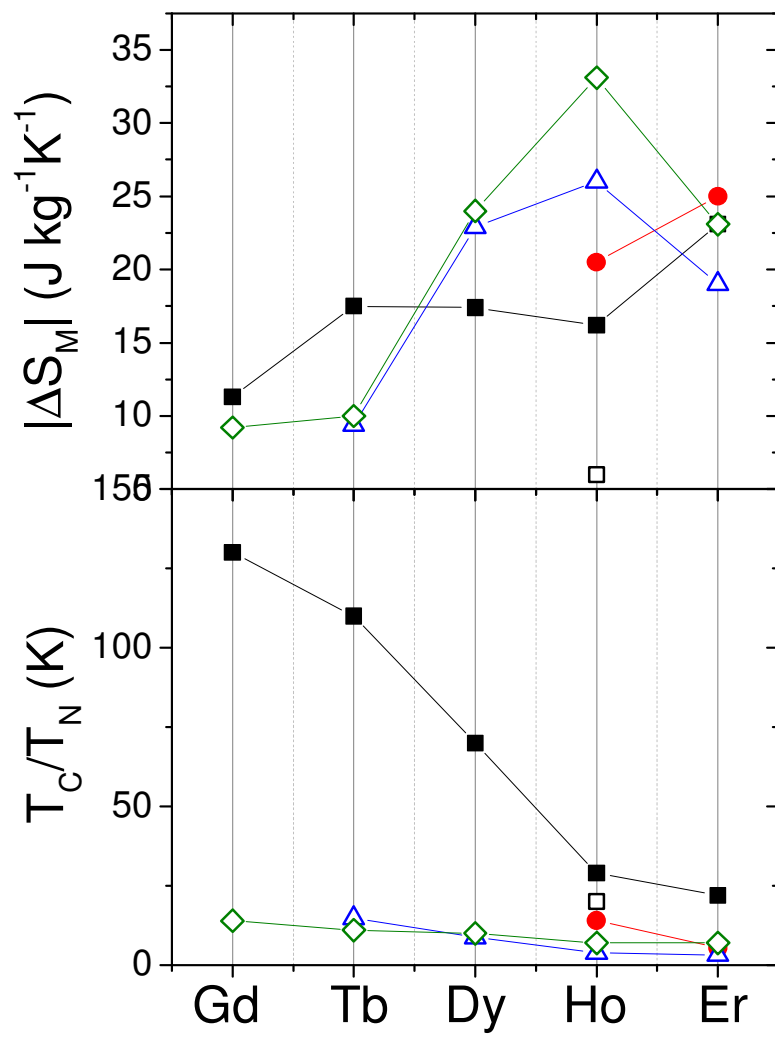


Figure 23

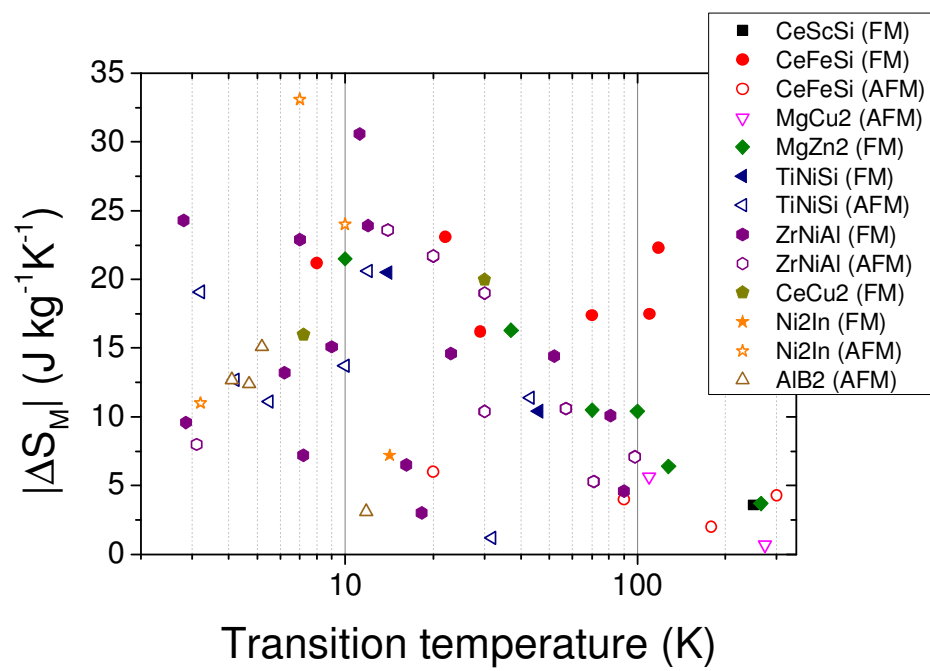


Figure 24

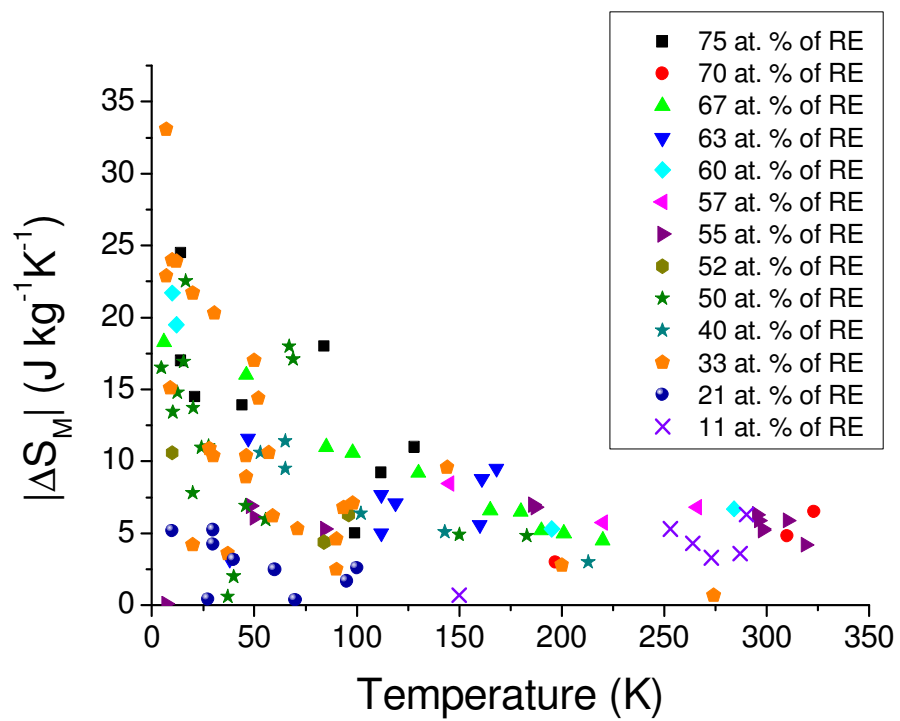


Figure 25

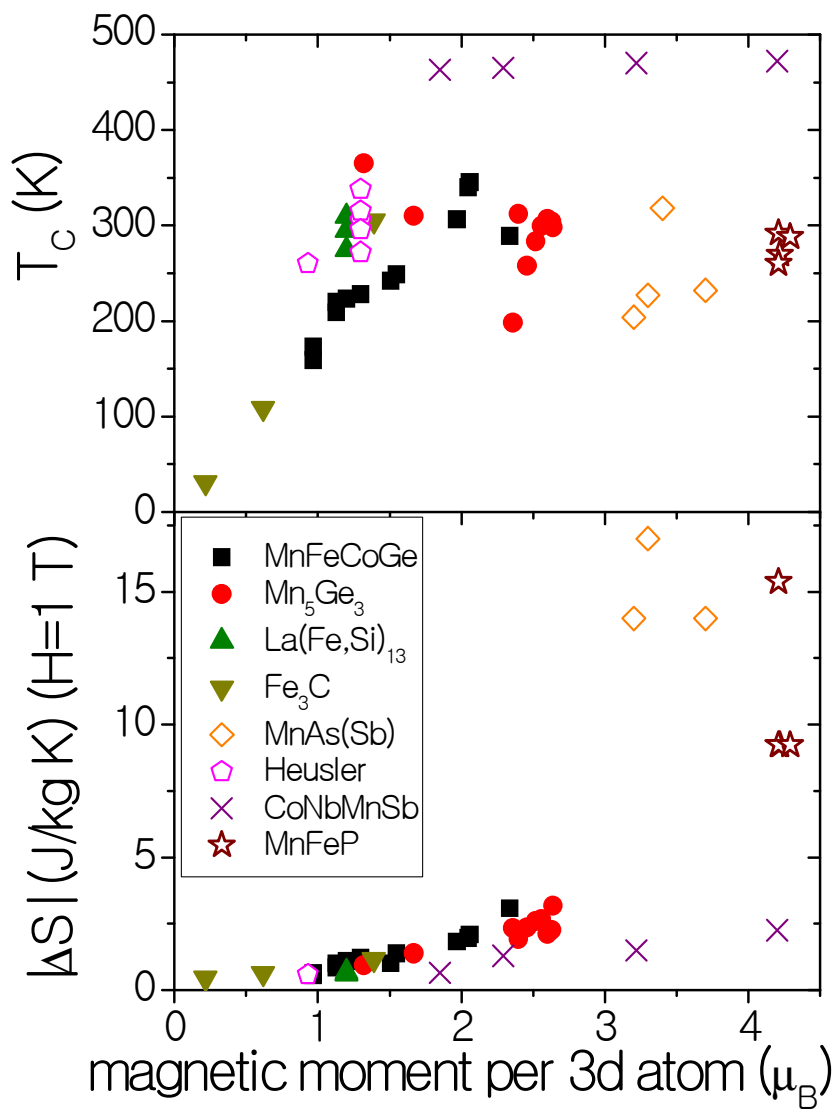


Figure 26

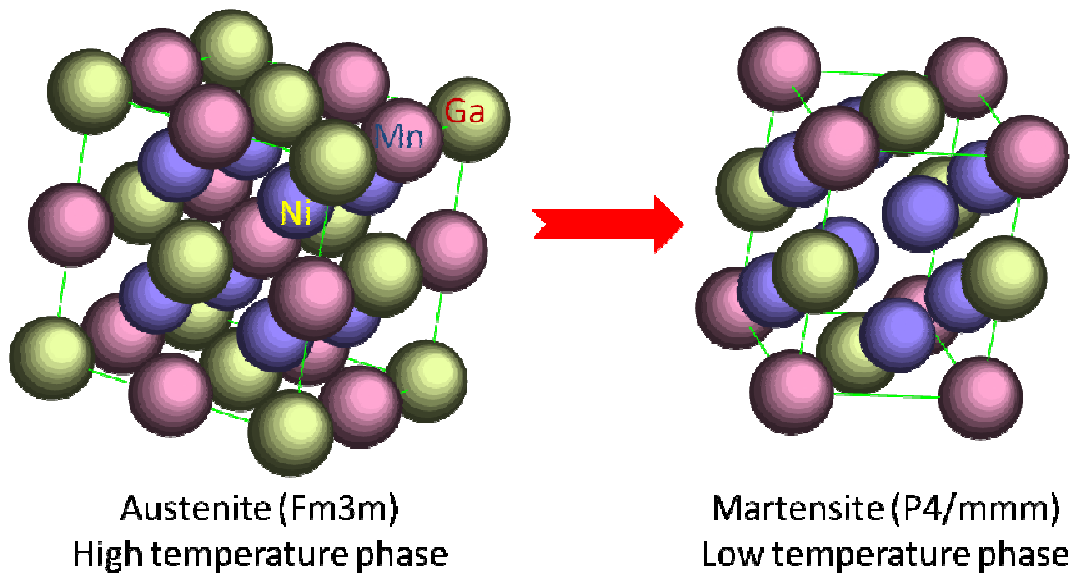


Figure 27

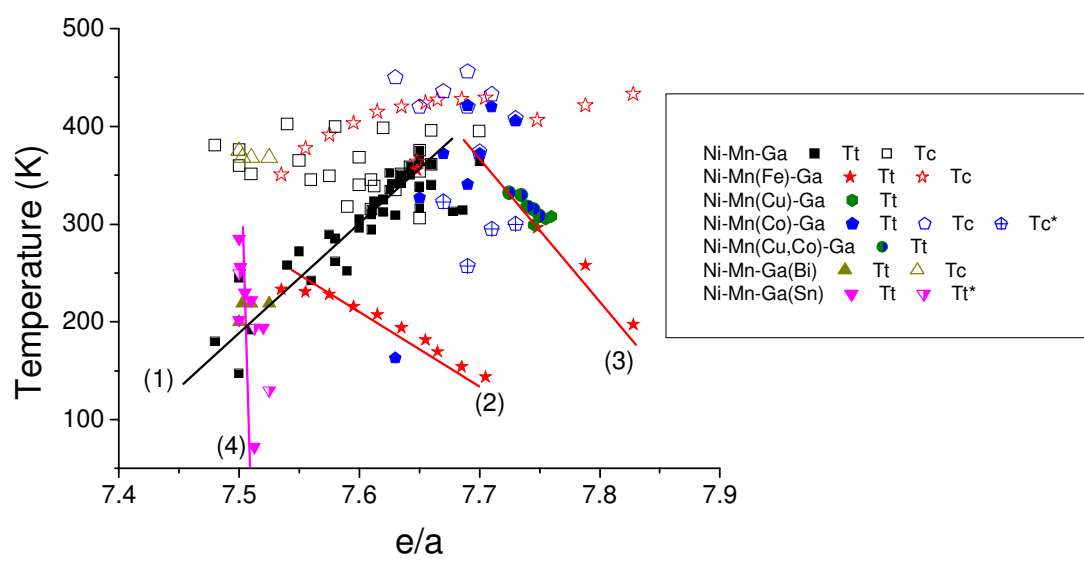


Figure 28

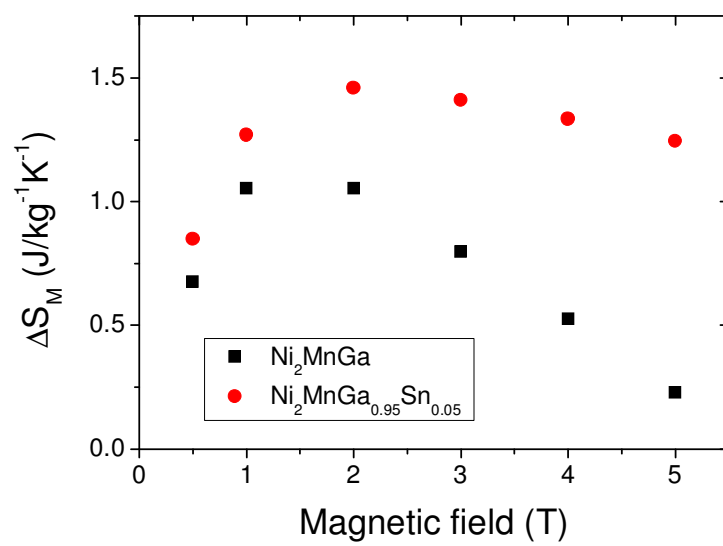


Figure 29

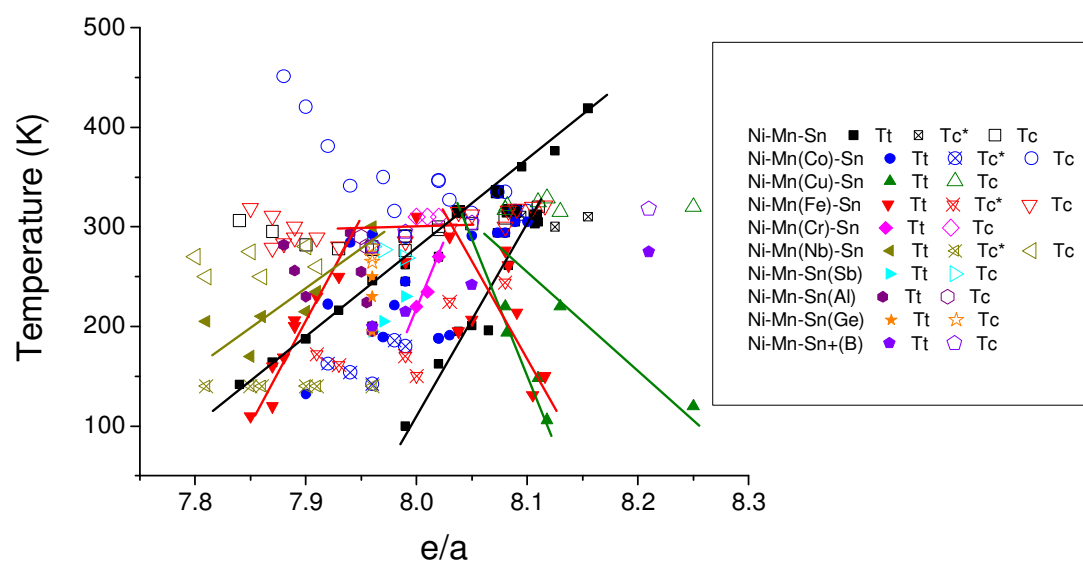


Figure 30

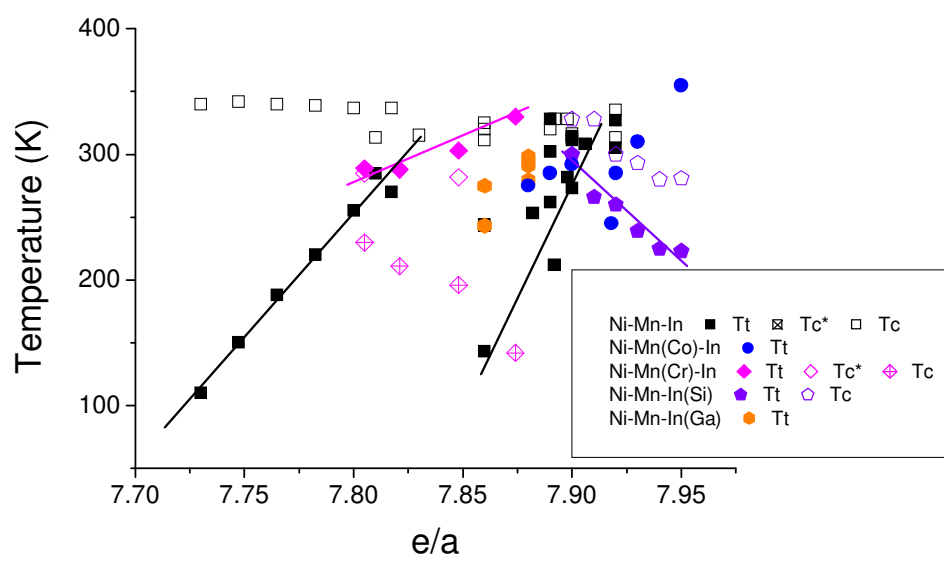


Figure 31

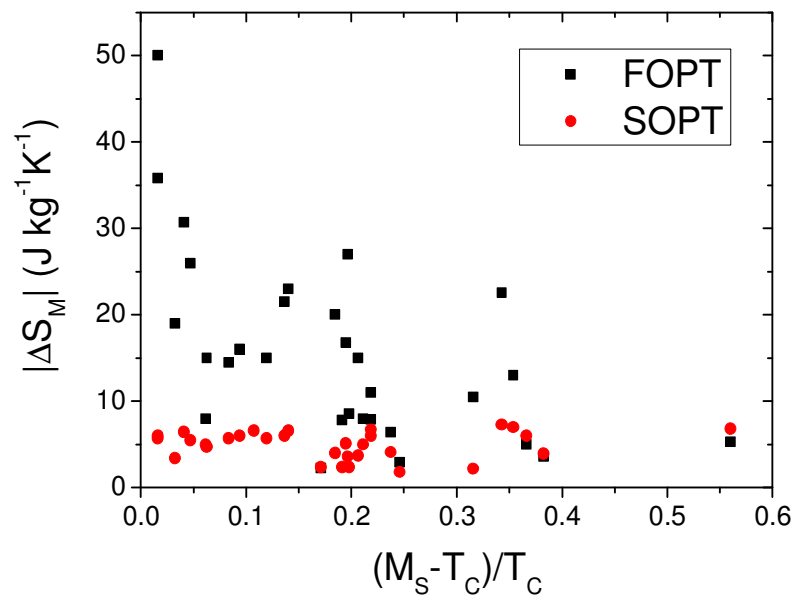


Figure 32

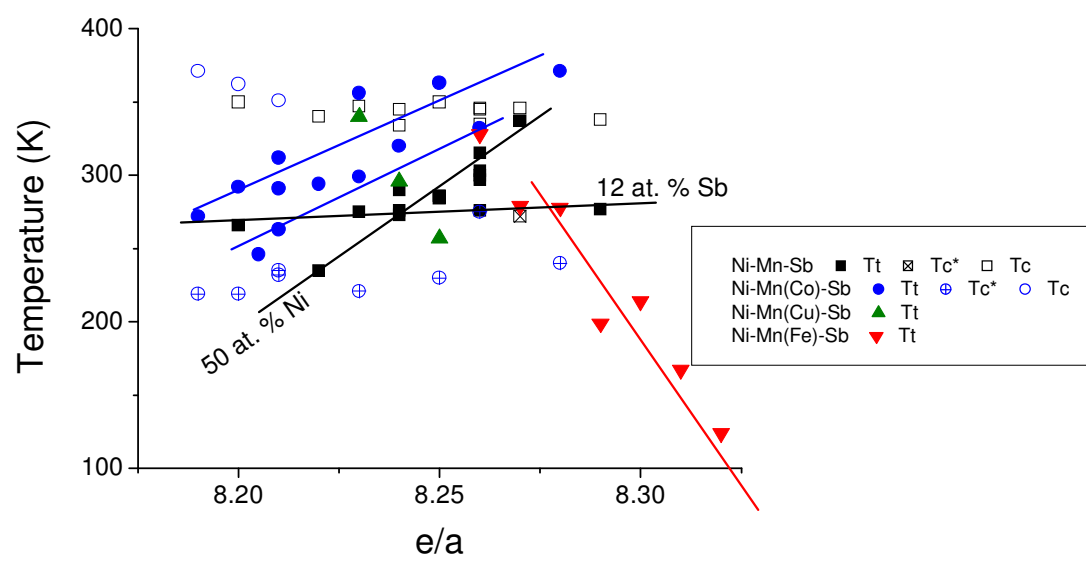


Figure 33

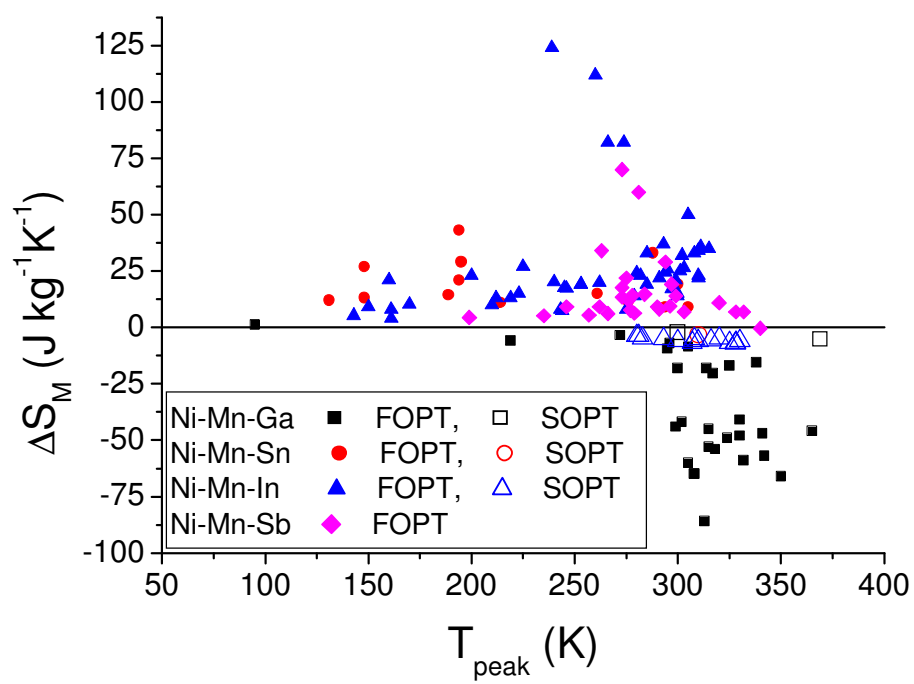


Figure 34

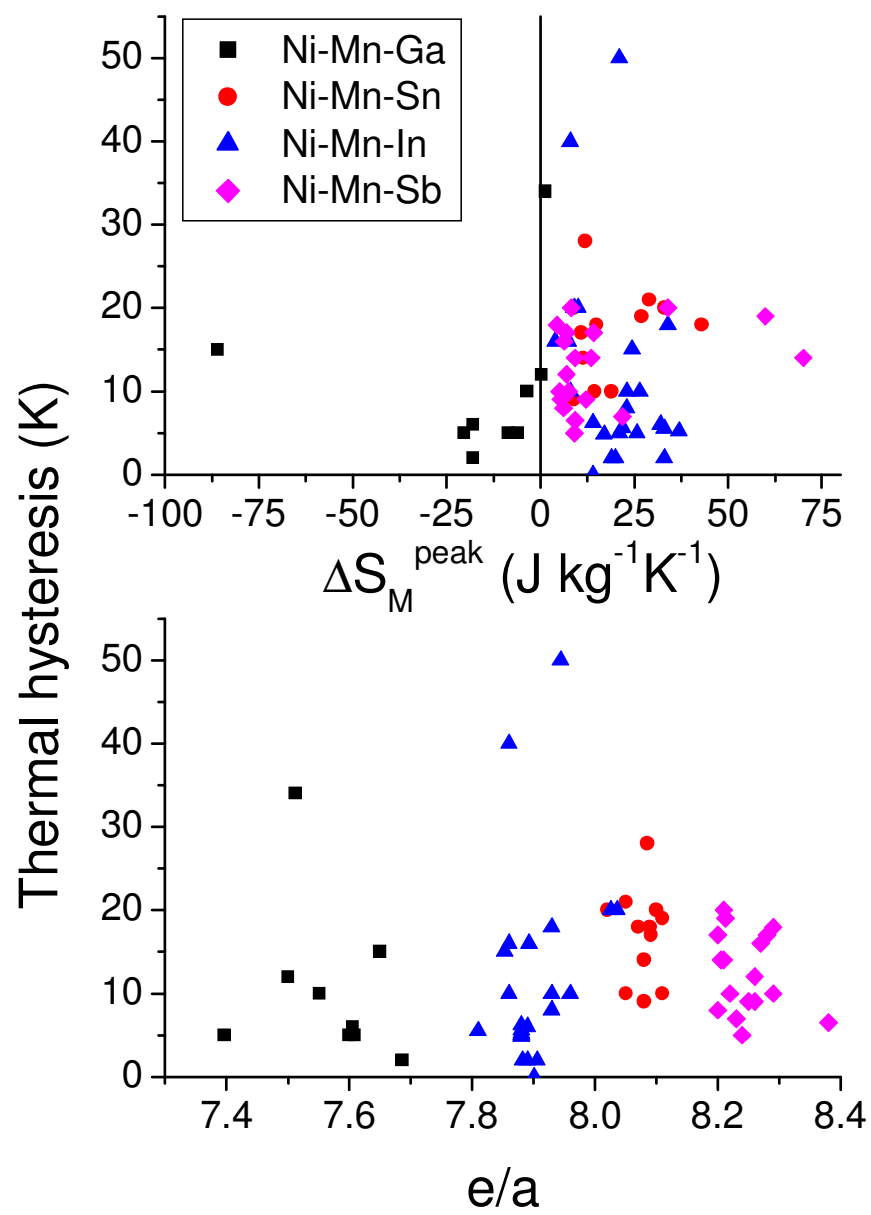


Figure 35

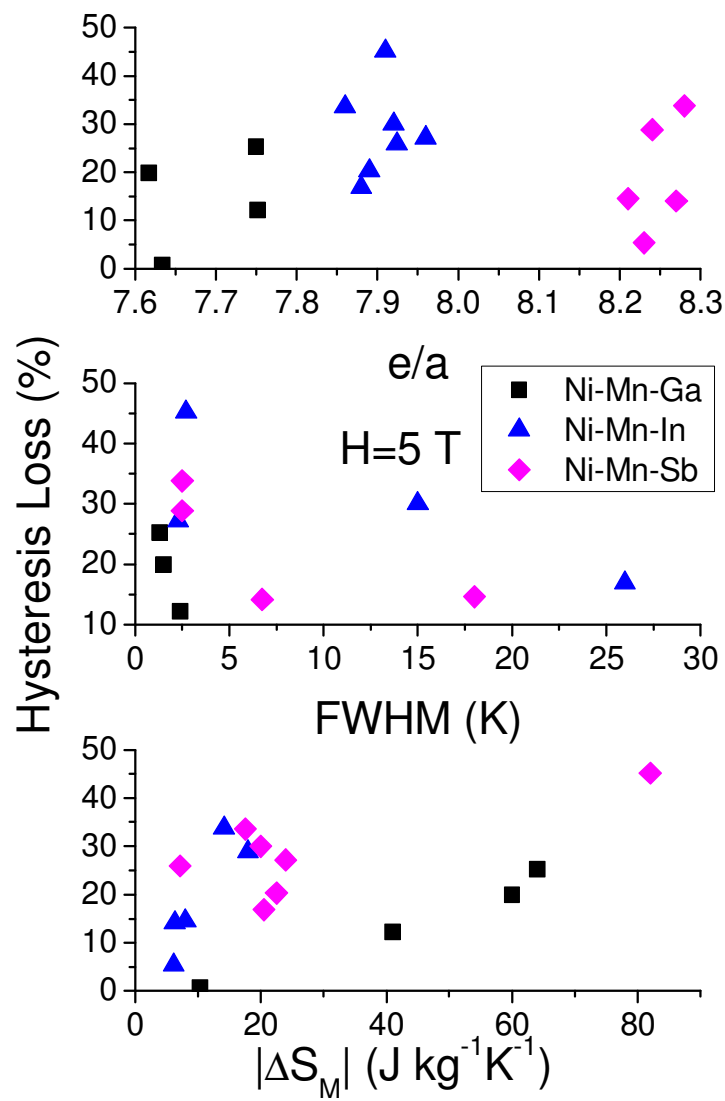


Figure 36

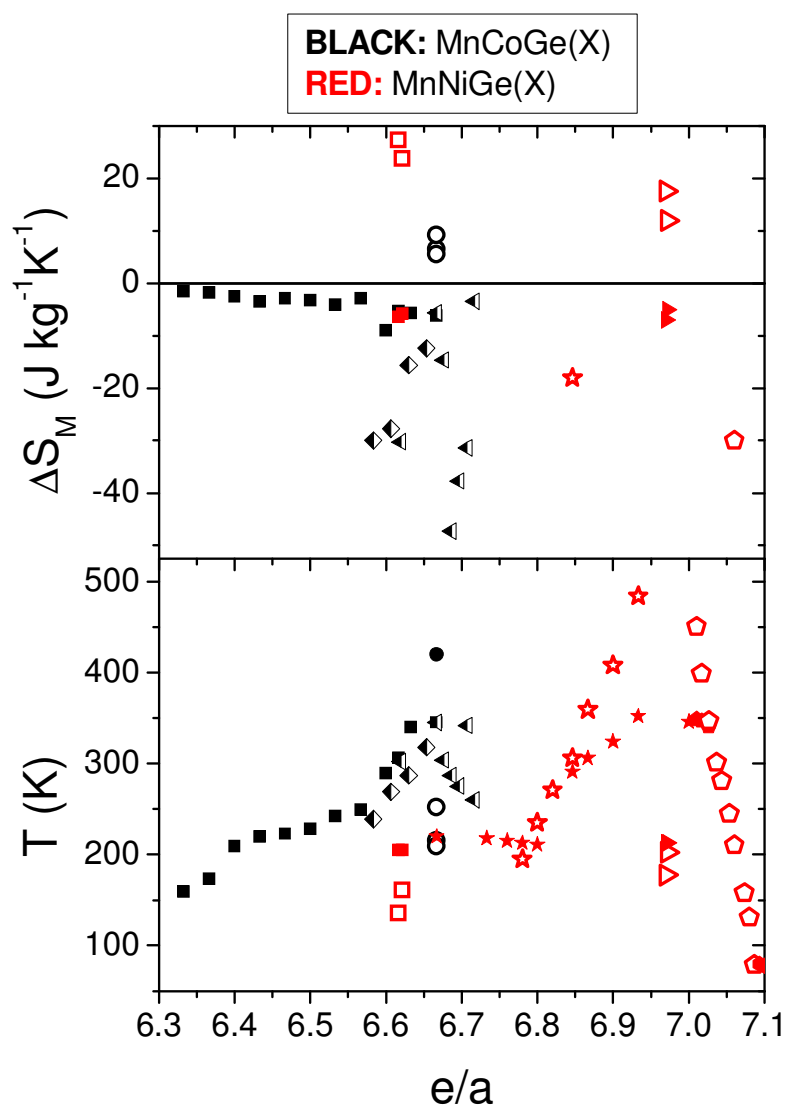


Figure 37

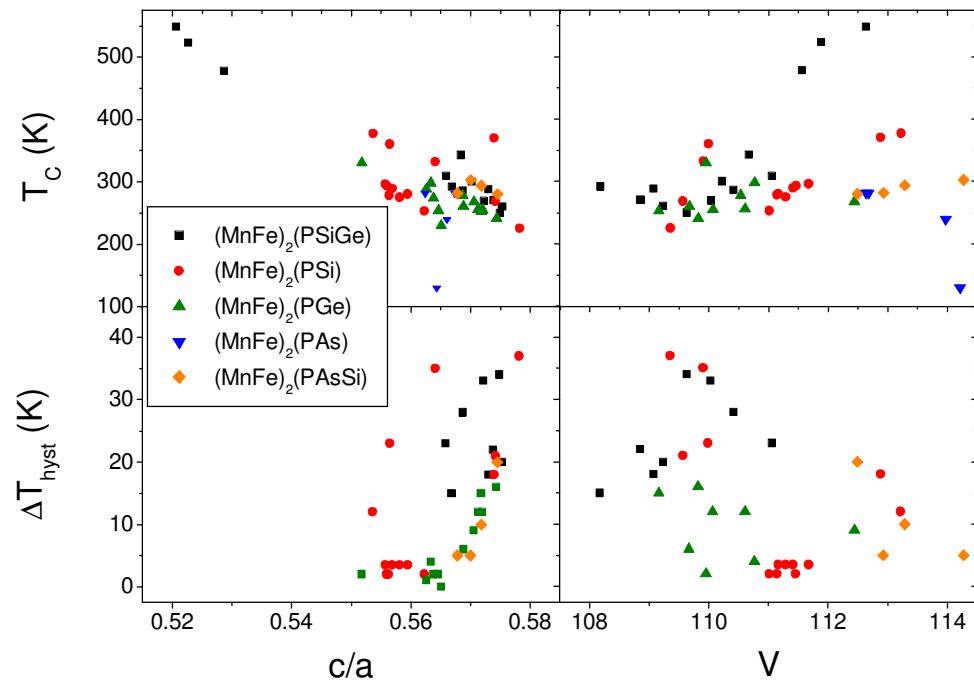


Figure 38

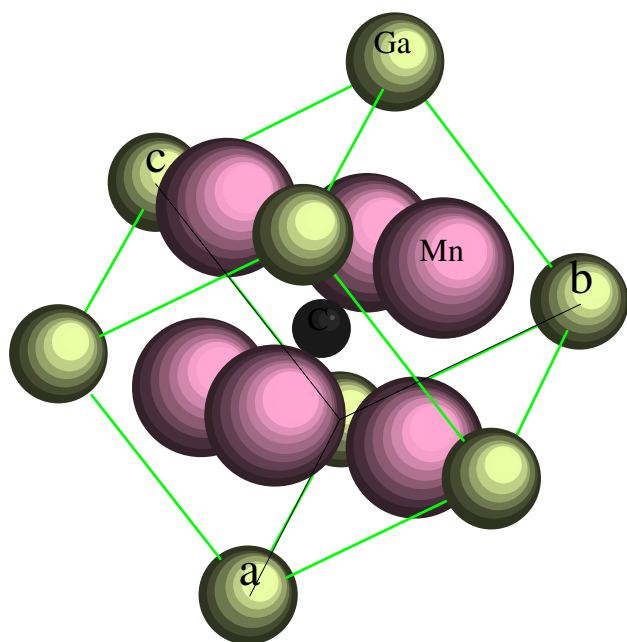


Figure 39

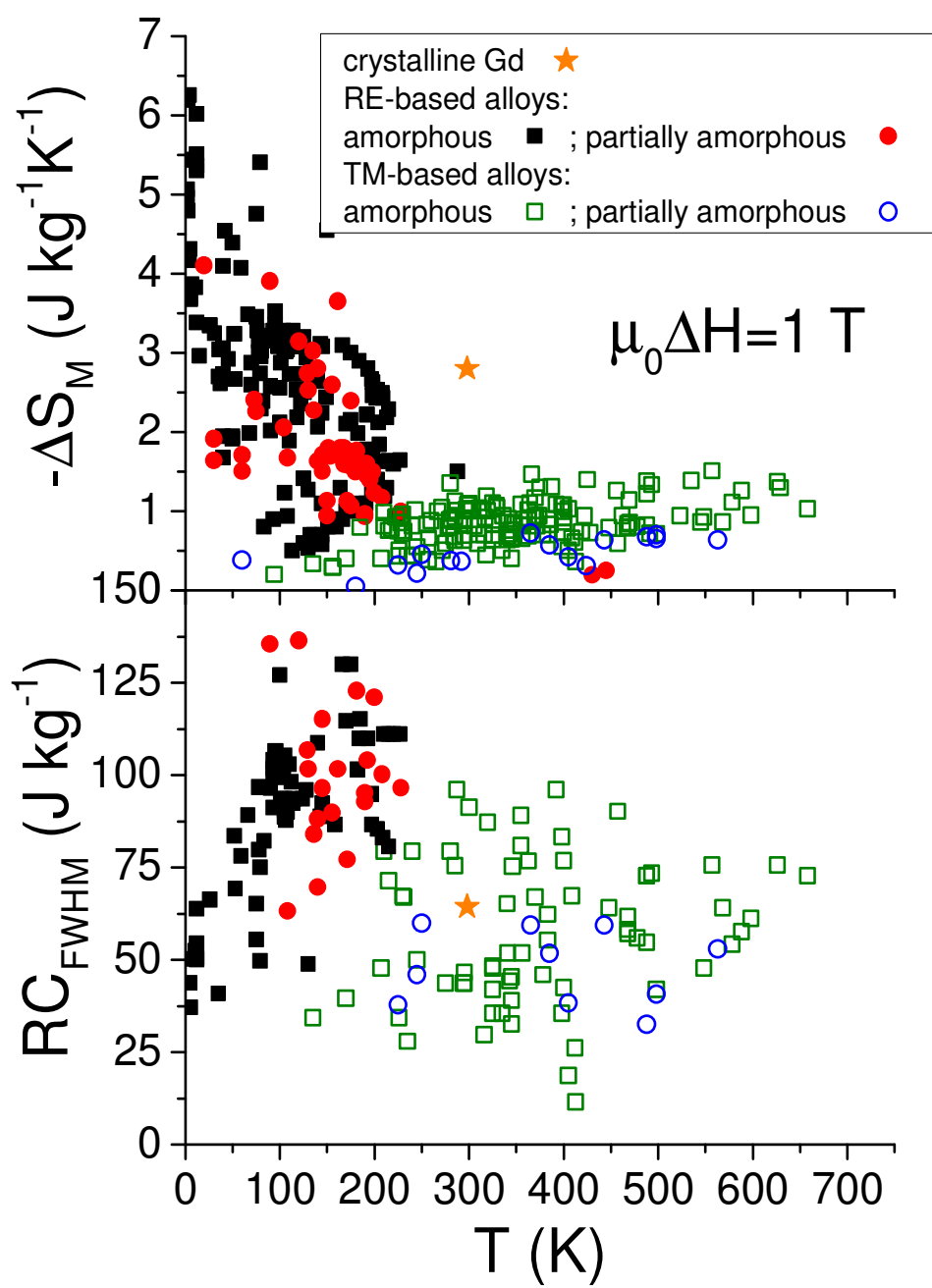


Figure 40

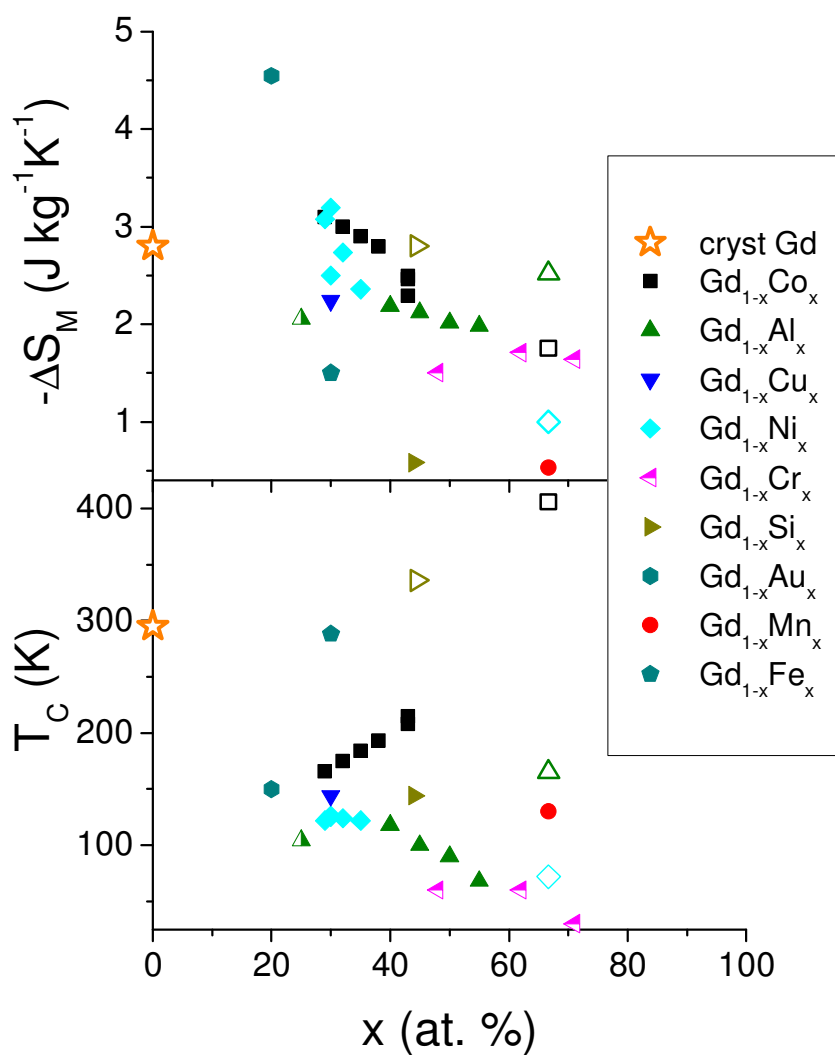


Figure 41

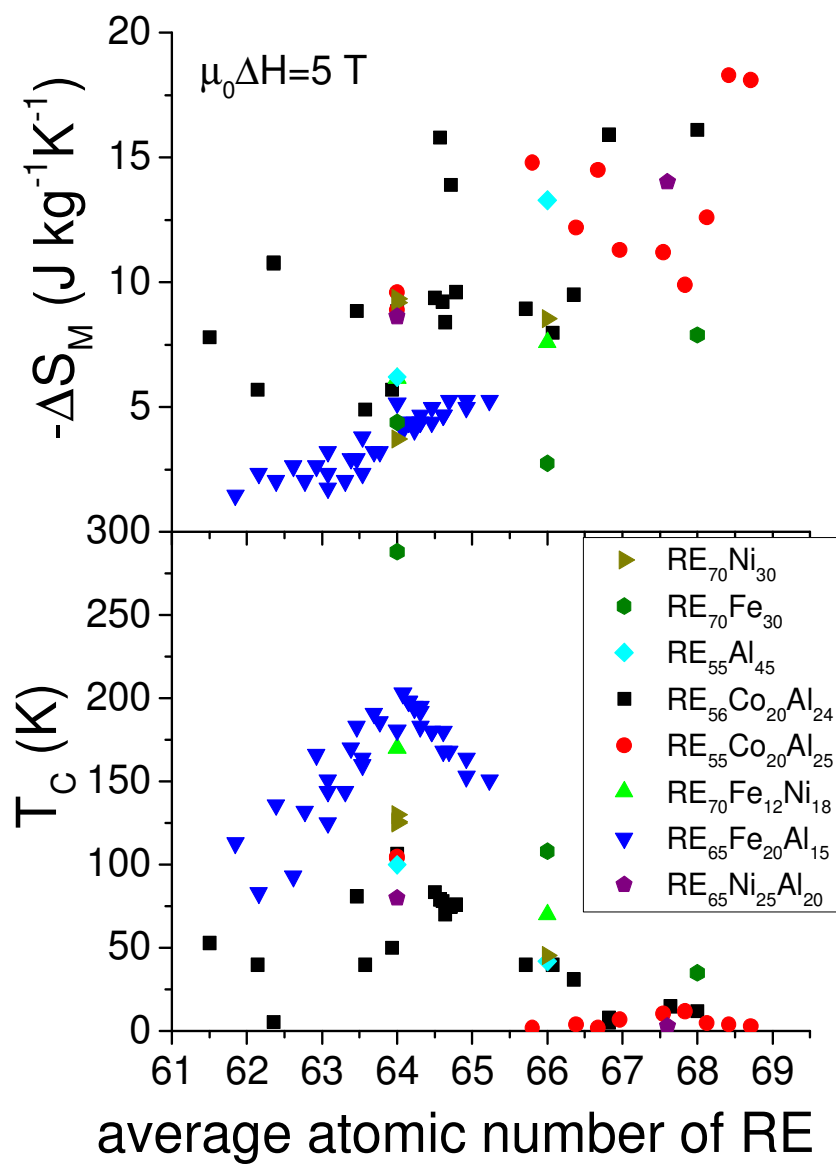


Figure 42

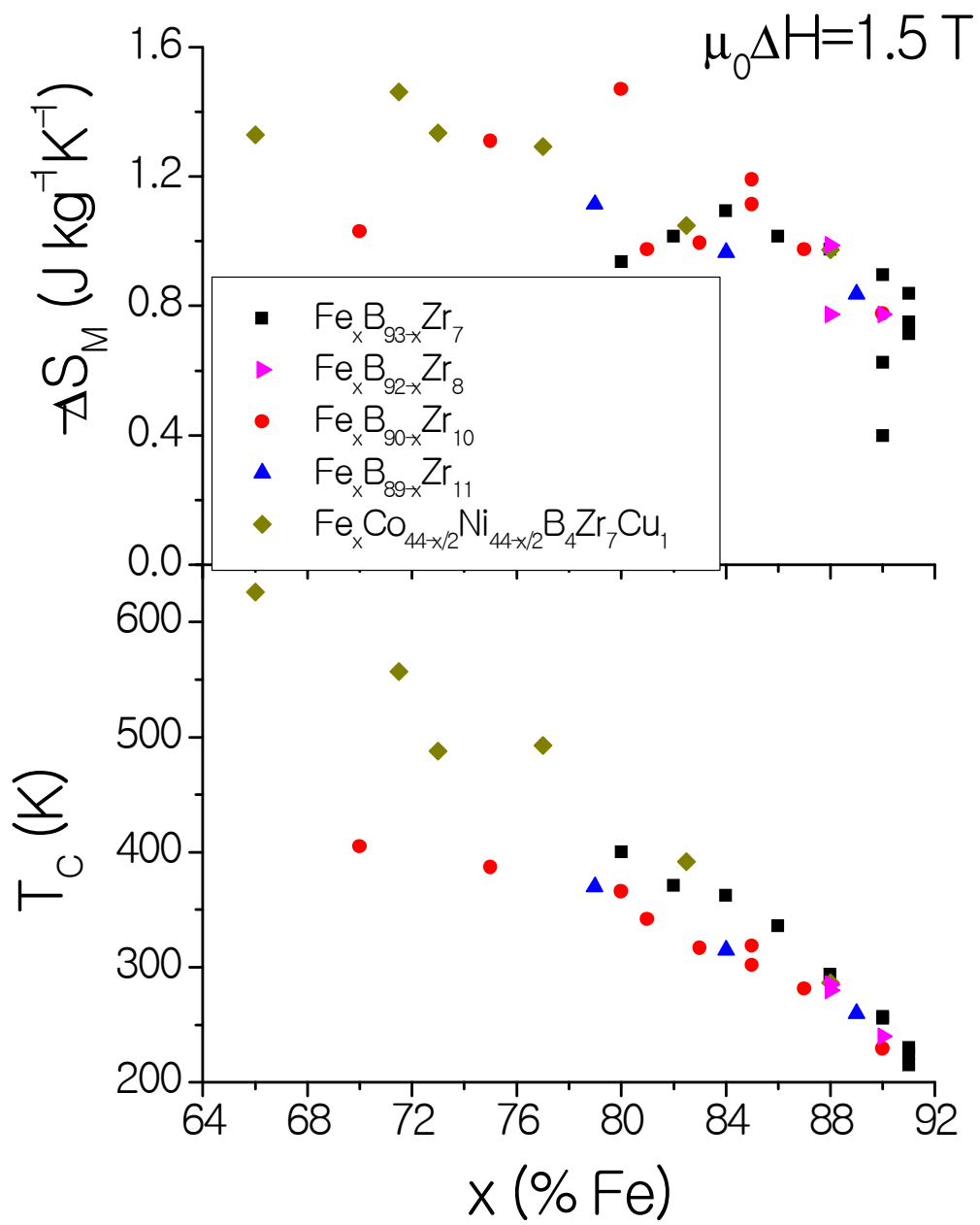


Figure 43